START

START Contents

ENN581 Robot Motion, Control & Planning: revision booklet

Written exam · 2 hours · any calculator · printed notes only. Print at 100 % (actual size), A4 landscape, double-sided (flip on the short edge).

How to use it. Find the question type on the **premises page** (what you are given, what you must find) or in the **lookup index** ("if the question says…"), then go to the week. Each week follows the slides' order and ends with *How this is examined: worked solutions*. The edge tab on the right shows where you are: START, RECIPES, BASICS (maths builders, tables), ROBOTS (standard robots), W1–W12, SAMPLE (the W13 sample questions solved). **Colour key.** Boxes: **blue** formula (with a SYMBOLS table), **green** steps, **purple** worked example, **grey** derivation, **teal** alternative approaches side by side, **red dashed** trap, **amber** slide check (a slide error and its correction), **slate** premise (given → find), **black** exam-style question with full solution. Tags: [p. 12] = slide page of that week's lecture; [added] = not on the slides; [checked] = number verified by script. Figures marked [W2 lec p. 28] are cut from the slides; [drawn for this booklet] figures were drawn from the same maths.

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START Premises: every problem type, what you are given and what you must find

Read the question, find its row, write the *given* and the *find* at the top of your answer (it earns method marks), then use the core relation. Symbols: $\mathbf{q} = (q_1, \ldots, q_n)$ joint variables ($q_i = \theta_i$ for a revolute joint in rad, $q_i = d_i$ for a prismatic joint in m); {0} or {S} base/station frame; {E} or {T} end-effector/tool frame; ${}^{A}\mathbf{T}_{B}$ pose of {B} expressed in {A}; subscript d = desired.

Problem	Given	Find	Core relation	Go to
Mechanisms and kinematics (W1–W6)				
Mobility (DoF) of a mechanism	a drawing: links N (count the ground), joints J, freedom f_i of each joint; planar ($D = 3$) or spatial ($D = 6$)	mobility M = number of independent inputs needed	$M = D(N - J - 1) + \sum_{i=1}^{J} f_i$; extended: $+R_C - R_M$	W1 p.9, W4 p.17
Forward kinematics (FK)	joint variables $\mathbf{q}$; geometry (link lengths, DH table or a drawing)	tool pose ${}^{0}\mathbf{T}_{E} = \begin{bmatrix} {}^{0}\mathbf{R}_{E} & {}^{0}\mathbf{p}_{E} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}$ (position and orientation)	${}^{0}\mathbf{T}_{E} = {}^{0}\mathbf{T}_{1}(q_1)\,{}^{1}\mathbf{T}_{2}(q_2)\cdots{}^{n-1}\mathbf{T}_{n}(q_n)\,{}^{n}\mathbf{T}_{E}$ (ETS or DH)	W2 p.11
Transform between frames	frames drawn (or described) relative to each other	${}^{A}\mathbf{T}_{B}$, a point in another frame ${}^{A}\mathbf{P} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{P}$, or the inverse ${}^{B}\mathbf{T}_{A}$	${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C}$; ${}^{B}\mathbf{T}_{A} = \begin{bmatrix} \mathbf{R}^{\top} & -\mathbf{R}^{\top}\mathbf{t} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}$	W2 p.11
Inverse kinematics (IK)	desired tool pose ${}^{0}\mathbf{T}_{E}$, or only position (x, y, z) / planar (x, y, ϕ)	*all* joint vectors $\mathbf{q}$ that reach it (0, 1, 2, … solutions; say which branch)	solve $\mathrm{FK}(\mathbf{q}) = {}^{0}\mathbf{T}_{E}$: geometric (cosine rule, atan2) or algebraic	W3 p.14
Workspace	link lengths, joint types (and limits)	reachable workspace (every position the tool can reach) / dexterous (reach with every orientation); sketch it	distance range from the shoulder $\lvert L_1 - L_2\rvert \ldots L_1 + L_2$; sweep about the base axis	W3 p.14
Parallel (closed-chain) IK	desired output pose of the platform / end link (e.g. four-bar output angle)	actuated (and passive) joint angles, every assembly mode	loop-closure: the vector sum round each loop = $\mathbf{0}$	W4 p.17
Parallel FK	actuated joint values	platform pose (usually several solutions)	same loop-closure equations, solved the other way	W4 p.17
Trajectory	start, goal (and via) values θ_0, θ_f (θ_v), duration t_f, boundary velocities (accelerations), or limits $\dot{\theta}_{\max}, \ddot{\theta}$	$\theta(t), \dot{\theta}(t), \ddot{\theta}(t)$: cubic coefficients $a_0..a_3$, or LSPB blend time t_b	cubic: 4 boundary conditions → 4 coefficients; LSPB: $s = ut + \frac{1}{2}at^2$ per segment	W5 p.20
Velocity kinematics	$\mathbf{q}$, joint rates $\dot{\mathbf{q}}$	tool velocity $\boldsymbol{\nu} = (\boldsymbol{\upsilon}, \boldsymbol{\omega})$ (linear, angular)	$\boldsymbol{\nu} = \mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}}$	W6 p.23
Inverse velocity (resolved rate)	desired tool velocity $\boldsymbol{\nu}$ (e.g. trace a line/circle), $\mathbf{q}$	joint rates $\dot{\mathbf{q}}$	$\dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\,\boldsymbol{\nu}$ (square, non-singular $\mathbf{J}$)	W6 p.23
Singularities	$\mathbf{J}(\mathbf{q})$	poses where the arm loses a direction of motion	$\det \mathbf{J}(\mathbf{q}) = 0$	W6 p.23
Statics	wrench at the tool $\mathbf{F}_E$ (force N, moment N m), $\mathbf{q}$	joint torques/forces $\boldsymbol{\tau}$ that hold it	$\boldsymbol{\tau} = \mathbf{J}^{\top}(\mathbf{q})\,\mathbf{F}_E$	W6 p.23
Dynamics and control (W7–W9)				
Equations of motion (EoM)	masses m_i, centre-of-mass positions, inertias I_i, geometry, gravity g	$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q})$ in full	Lagrange: $\mathcal{L} = K - P$, $Q_i = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i}$; or Newton–Euler	W7 p.26
Inverse dynamics	motion $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$	actuator torques/forces $\mathbf{Q}$	substitute into the EoM	W7 p.26, W9 p.31
Forward dynamics	torques $\mathbf{Q}$, state $\mathbf{q}, \dot{\mathbf{q}}$	accelerations $\ddot{\mathbf{q}}$ (then integrate)	$\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g})$	W7 p.26
Single-joint control (servo)	plant $m\ddot{x} + b\dot{x} + kx = f$, specification (critically damped, stiffness, disturbance)	gains k_p, k_v (k_i), closed-loop response, steady-state error	$f = \alpha f' + \beta$, $f' = \ddot{x}_d + k_v\dot{E} + k_p E$; critical: $k_v = 2\sqrt{k_p}$	W8 p.29
Model-based (computed-torque) control	robot model $\mathbf{M}, \mathbf{C}, \mathbf{g}$; desired trajectory $\mathbf{q}_d(t)$	control law $\mathbf{Q}$; error dynamics	$\mathbf{Q} = \boldsymbol{\alpha}\acute{\mathbf{Q}} + \boldsymbol{\beta}$, $\boldsymbol{\alpha} = \mathbf{M}(\mathbf{q})$, $\boldsymbol{\beta} = \mathbf{C} + \mathbf{g}$, $\acute{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$	W9 p.31
Cartesian / force control	desired Cartesian path $\mathbf{x}_d(t)$ or contact force $\mathbf{f}_d$	joint torques; which directions are position- vs force-controlled	$\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$; error $\mathbf{E} = \mathbf{x}_d - \mathbf{x}$ or $\mathbf{f}_d - \mathbf{f}_e$	W9 p.31
Wheeled robots (W10–W11)				
Wheeled FK	wheel speeds $\dot{\varphi}_i$ (and steering angles β_i), wheel layout	chassis velocity $\zeta = (u, v, \dot{\theta})$ in {C}; world velocity $(\dot{x}, \dot{y}, \dot{\theta})$ in {O}; ICR	invert the stacked wheel equations; $\nu = \mathbf{J}(\theta)\,\zeta$	W10 p.34
Wheeled IK	desired chassis (or world) velocity, e.g. drive a circle or a square	wheel speeds $\dot{\varphi}_i$ (and steering)	each wheel's rolling equation, one row per wheel	W10 p.34
Wheeled-robot dynamics	mass m, inertia ${}^{G}I$, wheel forces, COM offset	chassis accelerations; the forces/torques needed	Newton–Euler in {C}: $\mathbf{F} = m\,\mathbf{a}_G$, $N_z = {}^{G}I\ddot{\theta}$	W11 p.37
Planning (W12)				
Graph search	graph/grid, edge costs, start, goal (heuristic h)	order of expansion, path, its cost; is it optimal?	BFS (queue), DFS (stack), Dijkstra (cost g), A* ($f = g + h$, h admissible)	W12 p.39
Sampling-based planning	C-space, obstacles, start, goal	a collision-free path (not optimal)	PRM (roadmap, then search); RRT (grow a tree towards samples)	W12 p.39

START Fast lookup: if the question says …

If the question says… go to

"transformation matrix", "indicate frames", ETS ..... p. 4, 11
"DH table", "link parameters" ..... p. 4, 11
"sketch the workspace" ..... p. 4, 14
"derive the inverse kinematics" ..... p. 4, 14
"how many solutions" ..... p. 4
"mobility", "degrees of freedom", Grübler ..... p. 4, 9, 17
"parallel", "four-bar", "loop closure" ..... p. 4, 17
"cubic", "via point", "spline", "blend" ..... p. 4, 20
"Jacobian", "singularity", "joint torques for a force" ..... p. 4, 23
"equations of motion", Lagrange ..... p. 4, 26
"inertia tensor", "Newton–Euler" ..... p. 26
"critically damped", "gains", "steady-state error" ..... p. 4, 29
"computed torque", "partitioned", "force control" ..... p. 4, 31
"wheel speeds", "omni", "differential drive", ICR ..... p. 4, 34
"holonomic" ..... p. 34
"dynamics of a mobile robot" ..... p. 37
"A*", "Dijkstra", "PRM", "RRT", "C-space" ..... p. 4, 39
"what is given / what to find" ..... p. 2
"trace a circle / line / square" (joint rates) ..... p. 23
"drive a circle / rectangle" (wheel speeds) ..... p. 34
"block diagram", "model-based", "servo" ..... p. 29, 31
"BFS", "DFS", open/closed list ..... p. 39
RR, RP, RRP, SCARA, Puma, Panda: tables, ETS ..... p. 7
skew matrix, cross product, $s = ut + \frac{1}{2}at^2$**,** atan2 ..... p. 5
the W13 sample questions, solved ..... p. 42
trig identities, rotations, inertia of shapes ..... p. 5

Answer layout (method marks)

1. **Draw and define**: frames, axes, positive angle directions, assumptions (no joint limits; point masses; rest to rest; $\gamma = 0$).
2. **State the general tool** before using it: ETS, cosine rule, Euler–Lagrange, G–K, boundary conditions.
3. **Substitute and simplify**, one step per line, symbols first, numbers last, with units.
4. **Box the result**; list *every* branch (IK) or mode (four-bar).
5. **Sanity check** at $\theta = 0$, $t = 0$, or a symmetric case. Write "check:".

Time (2 hours)

Read all questions first (5 min). Split time by marks. Leave 10 min to check signs and units. If stuck: write the set-up and the equation you would solve, then move on.

Ten checks before you hand in

1. Calculator in **radians**? Angles labelled ° or rad?
2. atan2, not $\tan^{-1}(y/x)$ (quadrant).
3. Both IK branches (and "no solution if $|\cos| > 1$").
4. Order of matrix products (right = about the current frame).
5. Transposes: $\mathbf{R}^{-1} = \mathbf{R}^\top$; inverse $\mathbf{T}^{-1} = [\mathbf{R}^\top \quad -\mathbf{R}^\top \mathbf{t}]$.
6. $\mathcal{L} = K - P$; gravity sign; product rule in $\frac{d}{dt}$.
7. $k_v^2 = 4k_p$ for critical damping (unit mass).
8. Ground counted as a link; $D = 3$ planar, 6 spatial.
9. Wheel row: $-\dot{\theta}\,{}^C y$ in the x part, $+\dot{\theta}\,{}^C x$ in the y part.
10. Units: mm vs m; rad/s vs rpm; N m vs N.

Notation

${}^A\mathbf{T}_B$	pose of $\{B\}$ in $\{A\}$; ${}^A\mathbf{P} = {}^A\mathbf{T}_B\,{}^B\mathbf{P}$
${}^A\mathbf{R}_B$	columns = B's axes written in A
$R_x, R_y, R_z,$ t_x, t_y, t_z	elementary transforms (ETS)
$a_{i-1}, \alpha_{i-1}, d_i, \theta_i$	proximal (Craig) DH
c_i, s_i, c_{ij}	$\cos\theta_i$, $\sin\theta_i$, $\cos(\theta_i+\theta_j)$
$\mathbf{J}$	Jacobian: $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$
$\mathbf{M}, \mathbf{C}, \mathbf{g}$	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$
$E = q_d - q$	servo error (desired minus actual)
M, D, N, J, F_i	mobility; 3/6; links incl. ground; joints; joint DoF
R_C, R_M	over-constraints (add); local mobility (subtract)
$u, v, \dot{\theta}$	chassis velocity in $\{C\}$; $\dot{\varphi}_i$ wheel spin
β_i, γ_i	wheel drive angle from x_C; roller angle
g, h, f	A*: cost so far, heuristic, $f = g + h$

Elementary rotations

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Use: R_y's minus sign is bottom-left. A translation $t_x(a)$ is $\mathbf{I}$ with a in row 1, column 4.

Sample-paper map (W13 revision): solved in full on p. 42

Q1 FK of RRR (ETS)	W2	p. 4
Q2 workspace sketch	W3	p. 4
Q3 IK derivation	W3	p. 4
Q4 equations of motion	W7	p. 4
Q5 four-bar IK	W4	p. 4
Q6–Q7 mobility	W1, W4	p. 4
Q8–Q9 cubic splines	W5	p. 4
Q10 omni robot	W10	p. 4

RECIPES Answer recipes: one per question type (details on each week's page)

STEPS 1 · Forward kinematics by ETS (W13 Q1)

1. Draw {S}; mark each joint axis and its *drawn* positive sense.
2. Walk base → tool. Translations: t_x, t_y, t_z. Joint: $R_{axis}(\pm\theta_i)$; minus if the drawn arrow opposes the right-hand rule about your axis.
3. End with the constant rotation that makes your last frame match {T}.
4. Group per joint and write each ${}^{i-1}\mathbf{T}_i$ (= "indicate intermediate frames").
5. Multiply; give at least the position column. Check: all $\theta = 0$ reproduces the drawing.

STEPS 2 · Proximal DH table

1. z_i along joint axis i. Origin of {i} at the common normal to axis $i{+}1$ (or the intersection).
2. x_i along that normal towards axis $i{+}1$; $y_i = z_i \times x_i$.
3. Row i: a_{i-1} ($z_{i-1} \to z_i$ along x_{i-1}), α_{i-1} (about x_{i-1}), d_i ($x_{i-1} \to x_i$ along z_i), θ_i (about z_i).
4. ${}^{i-1}\mathbf{T}_i = R_x(\alpha_{i-1})t_x(a_{i-1})R_z(\theta_i)t_z(d_i)$; multiply rows.
5. Add a constant tool transform if {N} ≠ {tool}. Write "proximal (Craig)".

STEPS 3 · Sketch the workspace (W13 Q2)

1. Find the joints that only *spin the plane* (base rotation) and the planar sub-arm.
2. Planar 2-link sub-arm: distance from its first joint ranges $|L_a - L_b| \ldots L_a + L_b$ (cosine rule) → annulus.
3. Spin the annulus about the base axis. If the axis passes through the annulus's centre → spherical shell; otherwise → a torus-like solid. Joint limits cut wedges; a prismatic joint extrudes.
4. Draw side and top views; label radii, centre height. Say "no joint limits, collisions ignored".

STEPS 4 · Derive the inverse kinematics (W13 Q3)

1. Write FK first. Spherical wrist? Wrist centre $\mathbf{p}_W = \mathbf{p}_E - d_6\,\mathbf{R}\hat{z}$.
2. Top view: $\theta_1 = \operatorname{atan2}(y, x)$, second branch $\theta_1 + \pi$ (reach back, $r < 0$).
3. Side view: subtract base height; planar 2R in (r, z'):
$\cos\theta_3 = \dfrac{r^2 + z'^2 - L_a^2 - L_b^2}{2L_aL_b}$ (relative angle; interior angle is $\pi - \theta_3$).
4. $\theta_3 = \operatorname{atan2}(\pm\sqrt{1-c_3^2}, c_3)$; $\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_b s_3,\ L_a + L_b c_3)$.
5. Wrist: ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_6$ → Euler angles (two: wrist flip).
6. Count: shoulder × elbow (× flip). No solution if $|c_3| > 1$. Check one branch by FK.

STEPS 5 · Mobility (W13 Q6, Q7)

1. $D = 3$ planar, 6 spatial (S, U joints or "still 3D" → 6).
2. Number bodies; ground = 1; a prismatic pair = 2 bodies → N.
3. List joints and F_i (R, P = 1; U, C = 2; S = 3) → J, $\sum F_i$.
4. $M = D(N - J - 1) + \sum F_i$. Implausible? Extended: $+R_C - R_M$.
5. R_M: idle spins (rod with S at both ends → 1 each). $R_C = \sum F_i - \sum u_k$ (u_k: distinct platform DoF per loop, union).

Delta: $6(11 - 15 - 1) + 39 = 9$, then $9 + 0 - 6 = 3$. *4RRR:* $-6 + 7 - 0 = 1$.

STEPS 6 · Parallel robot: loop closure, IK, FK (W13 Q5)

1. Break the loop (at a point K, or both ends of a link) into serial chains.
2. Write the break point's pose from each chain; equate (3 equations at K; 1 length equation if broken at both ends).
3. **IK** (pose of K, ϕ given): $A = K - a(\cos\phi, \sin\phi)$, $\theta_1 = \operatorname{atan2}(A_y, A_x)$, $\varphi_2 = \phi - \theta_1$; $B = K + b(\cos\phi, \sin\phi)$, $\varphi_1 = \operatorname{atan2}(B_y, B_x - L_0)$, $\varphi_3 = \phi + \pi - \varphi_1$.
4. **FK** (θ_1 given): intersect circles (L_2 about A, L_3 about D): two assembly modes, or none.

STEPS 7 · Trajectories (W13 Q8, Q9)

1. Rest-to-rest cubic: $a_0 = \theta_0$, $a_1 = 0$, $a_2 = \dfrac{3\Delta}{t_f^2}$, $a_3 = -\dfrac{2\Delta}{t_f^3}$; peak speed $1.5\Delta/t_f$ at $t_f/2$.
2. Given end speeds: $a_2 = \frac{3\Delta}{t_f^2} - \frac{2\dot\theta_0}{t_f} - \frac{\dot\theta_f}{t_f}$, $a_3 = -\frac{2\Delta}{t_f^3} + \frac{\dot\theta_0 + \dot\theta_f}{t_f^2}$.
3. Two segments, continuous acceleration: list 8 conditions (pos at both ends ×2, rest at start and end, $\dot\theta$ and $\ddot\theta$ match at the via) and solve.
4. Blends (LSPB): need $\ddot\theta \geq 4\Delta/t_f^2$; $t_b = \dfrac{t_f}{2} - \dfrac{\sqrt{\ddot\theta^2 t_f^2 - 4\ddot\theta\Delta}}{2\ddot\theta}$; linear speed $\ddot\theta t_b$.
5. Check: positions and speeds at both ends.

STEPS 8 · Jacobian, singularity, statics

1. Position FK $\mathbf{x}(\mathbf{q}) \to \mathbf{J} = \partial\mathbf{x}/\partial\mathbf{q}$ (column j = how the tool moves for joint j). Geometric: revolute column $\hat{z}_j \times (\mathbf{p}_E - \mathbf{p}_j)$; prismatic $\hat{z}_j$.
2. Singular where $\det\mathbf{J} = 0$ (square) or rank drops. 2R: $\det = L_1L_2\sin q_2$.
3. $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}}$; near-singular → huge joint rates.
4. Statics: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ (same frame for J and F).

STEPS 9 · Equations of motion by Lagrange (W13 Q4)

1. Coordinates $\mathbf{q}$; gravity direction drawn.
2. COM positions from FK; differentiate → velocities.
3. $K = \sum \frac{1}{2} m_i v_{C_i}^2 + \frac{1}{2}\omega_i^\top I_i \omega_i$; $P = \sum m_i g h_i$; $\mathcal{L} = K - P$.
4. Per joint: $\partial\mathcal{L}/\partial\dot{q}_i \to \frac{d}{dt}$ (product rule!) → minus $\partial\mathcal{L}/\partial q_i$.
5. Collect M ($\ddot{q}$ terms), C (velocity products), g. Check M symmetric, g has no $\dot{q}$, C = 0 at rest.

2R point masses: $M_{11} = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2)$, $M_{12} = m_2(L_1L_2c_2 + L_2^2)$, $M_{22} = m_2L_2^2$.

STEPS 10 · Control design (W8, W9)

1. Plant $m\ddot{x} + b\dot{x} + kx = f$ (or $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} = \mathbf{Q}$).
2. Partition: $f = \alpha f' + \beta$, $\alpha = m$ (**M**), β = everything else.
3. Servo: $f' = \ddot{x}_d + k_v\dot{E} + k_pE$, $E = x_d - x \to \ddot{E} + k_v\dot{E} + k_pE = 0$.
4. Critical: $k_v = 2\sqrt{k_p}$ ($\omega_n = \sqrt{k_p}$, $\zeta = k_v/2\sqrt{k_p}$).
5. Constant disturbance: $E_{ss} = f_{dist}/(k_pm)$; add $k_i\int E$ to remove it (stable if $k_vk_p > k_i$).
6. Model error: hats on α, β; the error equation gets a right-hand side.

STEPS 11 · Wheeled robot kinematics (W13 Q10)

1. For each wheel: $({}^C x, {}^C y)$, β (drive direction from x_C), γ, r.
2. Row: $r\dot{\varphi}_i = [\cos\beta\ \ \sin\beta]\,[u - \dot\theta\,{}^C y,\ v + \dot\theta\,{}^C x]^\top$ (+ $\tan\gamma$ term for omni).
3. Standard wheels: add no-slide $[-\sin\beta\ \ \cos\beta]\,(\cdot) = 0$.
4. Stack → invert. Inertial: $\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$. ICR in {C}: $(-v/\dot\theta,\ u/\dot\theta)$.

STEPS 12 · Planning by hand (W12)

1. Table: node, g, h, $f = g + h$, parent; open and closed sets.
2. Expand the lowest f (Dijkstra: lowest g); relax neighbours; record improvements.
3. Stop when the goal is *expanded*, not when discovered. Trace parents.
4. A* optimal if h admissible ($h \leq$ true cost); consistent: $h(n) \leq c(n, m) + h(m)$.
5. PRM: sample, connect, search (multi-query). RRT: grow towards random samples (single query). Both probabilistically complete, not optimal.

BASICS Basic builders: maths and physics

1. Vectors

Dot, norm, angle, projection [added]

$$\mathbf{a}\cdot\mathbf{b} = \mathbf{a}^\top\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\varphi, \quad |\mathbf{a}| = \sqrt{\mathbf{a}\cdot\mathbf{a}}$$

$$\hat{\mathbf{a}} = \tfrac{\mathbf{a}}{|\mathbf{a}|}, \quad \varphi = \arccos\tfrac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \in [0,\pi], \quad \mathrm{proj}_{\mathbf{b}}\,\mathbf{a} = \tfrac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}\mathbf{b}$$

$\mathbf{a}\cdot\mathbf{b} = 0 \iff \perp$ (columns of $\mathbf{R}$ are unit and $\perp$).

Cross product: determinant (mind the $\hat{\mathbf{j}}$ sign) [added]

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix} = \begin{bmatrix}a_2b_3 - a_3b_2\\ a_3b_1 - a_1b_3\\ a_1b_2 - a_2b_1\end{bmatrix} \quad (\hat{\mathbf{j}}\text{ term has a }-)$$

$$\mathbf{a}\times\mathbf{b} = [\mathbf{a}]_\times\mathbf{b}, \quad [\mathbf{a}]_\times = \begin{bmatrix}0 & -a_3 & a_2\\ a_3 & 0 & -a_1\\ -a_2 & a_1 & 0\end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix}S_{32}\\ S_{13}\\ S_{21}\end{bmatrix} \text{ (vex)}$$

$|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\varphi$ = parallelogram area; direction by right-hand rule; $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$; $\mathbf{a}\times\mathbf{a} = \mathbf{0}$. Skew $\mathbf{S} = [\mathbf{a}]_\times$: $\mathbf{S}^\top = -\mathbf{S}$, $\mathbf{Sa} = \mathbf{0}$, $\det\mathbf{S} = 0$, $\mathbf{S}^2 = \mathbf{aa}^\top - |\mathbf{a}|^2\mathbf{I}$, $\mathbf{R}[\mathbf{a}]_\times\mathbf{R}^\top = [\mathbf{Ra}]_\times$.
$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \det[\mathbf{a}^\top;\mathbf{b}^\top;\mathbf{c}^\top]$ = signed volume (0 if coplanar) = $\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})$; $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$.
Trap check: $(1,2,2)\times(2,0,1) = (2,3,-4)$. [checked]

2. Matrices

2×2 and 3×3: det, inverse, eigenvalues [added]

$$\mathbf{A} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}, \ \det\mathbf{A} = ad - bc, \ \mathbf{A}^{-1} = \tfrac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$\lambda^2 - (a+d)\lambda + (ad-bc) = 0, \ \lambda_{1,2} = \tfrac{(a+d)\pm\sqrt{(a+d)^2-4(ad-bc)}}{2}, \ \mathbf{v} = \begin{bmatrix}b\\ \lambda - a\end{bmatrix}$$

$\lambda_1+\lambda_2 = a+d$, $\lambda_1\lambda_2 = ad-bc$; eigenvectors of a symmetric $\mathbf{A}$ are $\perp$.

$$\det\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix} = a(ei-fh) - b(di-fg) + c(dh-eg)$$

$$\mathbf{A}^{-1} = \tfrac{1}{\det\mathbf{A}}\,\mathrm{adj}\,\mathbf{A}, \quad \mathrm{adj}\,\mathbf{A} = \begin{bmatrix}ei-fh & ch-bi & bf-ce\\ fg-di & ai-cg & cd-af\\ dh-eg & bg-ah & ae-bd\end{bmatrix}$$

$(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$, $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ (order reverses). $\det = 0 \iff$ singular, rank drops. Check: $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$: $\lambda = (5\pm\sqrt{5})/2$. [checked]

Positive definite test; pseudo-inverse [added]

$\mathbf{M} = \mathbf{M}^\top > 0 \iff \mathbf{x}^\top\mathbf{Mx} > 0\ \forall\mathbf{x}\neq\mathbf{0} \iff$ all $\lambda > 0 \iff$ all leading minors > 0 ($M_{11} > 0$, $M_{11}M_{22} - M_{12}^2 > 0$, …). E.g. $\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}$: det $= -3$, $\lambda = 3, -1$, indefinite. [checked]

$\mathbf{J}\,(m\times n,\ \mathrm{rank}\ m < n)$: $\mathbf{J}^+ = \mathbf{J}^\top(\mathbf{JJ}^\top)^{-1}$, $\dot{\mathbf{q}} = \mathbf{J}^+\boldsymbol{\nu} \Rightarrow \mathbf{J}\dot{\mathbf{q}} = \boldsymbol{\nu}$, $|\dot{\mathbf{q}}|$ minimal

Tall $\mathbf{J}$ (rank n): $\mathbf{J}^+ = (\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$ (least squares). Used W6 (redundant arm), W10 (extra wheels). Check: $\mathbf{J} = \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}$, $\boldsymbol{\nu} = (1,2)$: $\dot{\mathbf{q}} = (0,1,1)$, $|\dot{\mathbf{q}}| = \sqrt{2}$. [checked]

Basic rotations (right-hand rule) [p. W2 10–12]

3. Rotations and homogeneous transforms

$$\mathbf{R}_x = \begin{bmatrix}1 & 0 & 0\\ 0 & c\theta & -s\theta\\ 0 & s\theta & c\theta\end{bmatrix} \ \mathbf{R}_y = \begin{bmatrix}c\theta & 0 & s\theta\\ 0 & 1 & 0\\ -s\theta & 0 & c\theta\end{bmatrix} \ \mathbf{R}_z = \begin{bmatrix}c\theta & -s\theta & 0\\ s\theta & c\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

$\mathbf{R}^\top\mathbf{R} = \mathbf{I}$, $\mathbf{R}^{-1} = \mathbf{R}^\top$, $\det\mathbf{R} = +1$, ${}^A\mathbf{P} = {}^A\mathbf{R}_B\,{}^B\mathbf{P}$; columns of ${}^A\mathbf{R}_B = \hat{x}_B, \hat{y}_B, \hat{z}_B$ in $\{A\}$. $\theta > 0$ anticlockwise looking down the axis to the origin. Same-axis angles add; different axes do not commute. Minus sign: upper right in $\mathbf{R}_x, \mathbf{R}_z$ but lower left in $\mathbf{R}_y$.

ALTERNATIVE APPROACHES

Current (moving) axes	Fixed (world) axes
Turn $\mathbf{R}_1$ then $\mathbf{R}_2$ about the *new* axes: $\mathbf{R} = \mathbf{R}_1\mathbf{R}_2$ (written order = done order; ETS, DH).	Turn $\mathbf{R}_1$ then $\mathbf{R}_2$ about the *original* axes: $\mathbf{R} = \mathbf{R}_2\mathbf{R}_1$ (last written = last done).

Check: $\mathbf{R}_z(90°)\mathbf{R}_x(90°) = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix} \neq \mathbf{R}_x(90°)\mathbf{R}_z(90°) = \begin{bmatrix}0 & -1 & 0\\ 0 & 0 & -1\\ 1 & 0 & 0\end{bmatrix}$. [checked]

Rodrigues; angles from $\mathbf{R}$ (r_{ij} = row i, col j) [p. W3 38]

$$\mathbf{R}(\hat{\mathbf{n}},\theta) = \mathbf{I} + s\theta\,[\hat{\mathbf{n}}]_\times + (1-c\theta)[\hat{\mathbf{n}}]_\times^2, \qquad \boldsymbol{\omega} = \dot\theta\hat{\mathbf{n}}$$

ZYZ $\mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_z(\gamma)$: $r_{13} = c_\alpha s_\beta$, $r_{23} = s_\alpha s_\beta$, $r_{33} = c_\beta$, $r_{31} = -s_\beta c_\gamma$, $r_{32} = s_\beta s_\gamma$

$$\beta = \mathrm{atan2}\big(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\big), \ \alpha = \mathrm{atan2}(r_{23}, r_{13}), \ \gamma = \mathrm{atan2}(r_{32}, -r_{31})$$

2nd solution: $-\beta$, $\alpha+\pi$, $\gamma+\pi$. If $s_\beta = 0$ only $\alpha\pm\gamma$ is fixed. Check $(30°, 40°, 50°)$ round-trips. [checked]
RPY $\mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi)$: $r_{11} = c_\psi c_\theta$, $r_{21} = s_\psi c_\theta$, $r_{31} = -s_\theta$, $r_{32} = c_\theta s_\phi$, $r_{33} = c_\theta c_\phi$

$$\theta = \mathrm{atan2}\big(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\big), \ \psi = \mathrm{atan2}(r_{21}, r_{11}), \ \phi = \mathrm{atan2}(r_{32}, r_{33})$$

2nd solution: $\pi-\theta$, $\psi+\pi$, $\phi+\pi$; singular at $\theta = \pm90°$.

Pose T, inverse, composition, elementary transforms [p. W2 12–16]

$${}^A\mathbf{T}_B = \begin{bmatrix}{}^A\mathbf{R}_B & {}^A\mathbf{P}_B\\ \mathbf{0}^\top & 1\end{bmatrix}, \ \begin{bmatrix}{}^A\mathbf{P}\\ 1\end{bmatrix} = {}^A\mathbf{T}_B\begin{bmatrix}{}^B\mathbf{P}\\ 1\end{bmatrix}, \ {}^A\mathbf{T}_C = {}^A\mathbf{T}_B\,{}^B\mathbf{T}_C$$

$$({}^A\mathbf{T}_B)^{-1} = {}^B\mathbf{T}_A = \begin{bmatrix}{}^A\mathbf{R}_B^\top & -{}^A\mathbf{R}_B^\top\,{}^A\mathbf{P}_B\\ \mathbf{0}^\top & 1\end{bmatrix}$$

$$t(a,b,c) = \begin{bmatrix}1 & 0 & 0 & a\\ 0 & 1 & 0 & b\\ 0 & 0 & 1 & c\\ 0 & 0 & 0 & 1\end{bmatrix}, \quad \mathbf{R}_{x,y,z}(\theta) \text{ as } 4\times4 = \begin{bmatrix}\mathbf{R} & \mathbf{0}\\ \mathbf{0}^\top & 1\end{bmatrix}$$

ETS = product read left to right, each step about the *current* axes. Proximal DH: ${}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(\theta_i)t_z(d_i)$. A direction uses $\mathbf{R}$ only (4th entry 0). Order trap: $t(2,1,0)\mathbf{R}_z(90°)$ has position $(2,1,0)$; $\mathbf{R}_z(90°)t(2,1,0)$ has $(-1,2,0)$. Check: $\mathbf{R}_z(90°)$, $\mathbf{P} = (2,1,0)$ maps $(1,0,0) \mapsto (2,2,0)$; inverse translation $(-1,2,0)$. [checked]

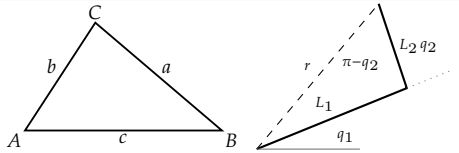
Identities

4. Trigonometry and algebra

$\sin(a\pm b) = \sin a\cos b \pm \cos a\sin b$ | $\cos(a\pm b) = \cos a\cos b \mp \sin a\sin b$
$\tan(a\pm b) = \frac{\tan a\pm\tan b}{1\mp\tan a\tan b}$ | $\cos^2 a + \sin^2 a = 1$
$\sin 2a = 2\sin a\cos a$ | $\cos 2a = \cos^2 a - \sin^2 a = 2\cos^2 a - 1 = 1 - 2\sin^2 a$
$\tan 2a = \frac{2\tan a}{1-\tan^2 a}$ | $\cos^2 a = \frac{1+\cos 2a}{2}$, $\sin^2 a = \frac{1-\cos 2a}{2}$
$\cos a\cos b = \frac{\cos(a-b)+\cos(a+b)}{2}$ | $\sin a\sin b = \frac{\cos(a-b)-\cos(a+b)}{2}$
$\sin a\cos b = \frac{\sin(a+b)+\sin(a-b)}{2}$ | $\sin(-a) = -\sin a$, $\cos(-a) = \cos a$
$\sin(\pi-a) = \sin a$, $\cos(\pi-a) = -\cos a$ | $\sin(\frac{\pi}{2}-a) = \cos a$, $\cos(\frac{\pi}{2}-a) = \sin a$

Arm forms: $c_1c_{12} + s_1s_{12} = c_2$, $s_1c_{12} - c_1s_{12} = -s_2$; $c_{12} = c_1c_2 - s_1s_2$, $s_{12} = s_1c_2 + c_1s_2$. Small angles (rad): $\sin\theta \approx \tan\theta \approx \theta$, $\cos\theta \approx 1 - \theta^2/2$. π rad $= 180°$; $1° = 0.017453$ rad; 1 rad $= 57.296°$. Set DEG/RAD before every trig key. [checked]

Cosine and sine rules (the arm triangle)



[drawn] left: side a opposite A; right: 2-link arm, interior elbow angle $\pi - q_2$

$$c^2 = a^2 + b^2 - 2ab\cos C, \quad \cos C = \tfrac{a^2+b^2-c^2}{2ab}$$

$$\tfrac{a}{\sin A} = \tfrac{b}{\sin B} = \tfrac{c}{\sin C}, \quad A + B + C = \pi$$

Arm: $r^2 = L_1^2 + L_2^2 - 2L_1L_2\cos(\pi - q_2) = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$ (sign flip is the classic trap). Check: $A = 40°, B = 60°, c = 10$: $a = 6.527$, $b = 8.794$; sides 5, 6, 7: $C = 78.46°$. [checked]

atan2; $a\cos\theta + b\sin\theta = c$; half-angle; quadratic

$\theta = \mathrm{atan2}(y,x) \in (-\pi,\pi]$, y **first**. $\arctan(y/x)$ cannot tell $(1,1)$ from $(-1,-1)$.

quad.	I	II	III	IV
x, y	$+,\geq 0$	$-,\geq 0$	$-,-$	$+,-$
atan2	$\arctan\frac{y}{x}$	$\arctan\frac{y}{x}+\pi$	$\arctan\frac{y}{x}-\pi$	$\arctan\frac{y}{x}$

$x = 0$: $\pm\pi/2$. Check: $(-1,1) \to 135°$ (not $-45°$); $(-1,-1) \to -135°$. [checked]

$$r = \sqrt{a^2+b^2}, \ \varphi = \mathrm{atan2}(b,a): \ a\cos\theta + b\sin\theta = r\cos(\theta-\varphi) = c$$

$$\Rightarrow \ \theta = \varphi \pm \arccos\tfrac{c}{r}$$

$|c| > r$ none, $|c| = r$ one, $|c| < r$ two. Half-angle $t = \tan\frac{\theta}{2}$: $\cos\theta = \frac{1-t^2}{1+t^2}$, $\sin\theta = \frac{2t}{1+t^2}$, so

$$(c+a)t^2 - 2bt + (c-a) = 0, \quad \theta = 2\arctan t \quad (\text{misses } \theta = \pi \text{ when } c + a = 0)$$

Check: $3\cos\theta + 4\sin\theta = 2.5$: $\theta = 113.13°$ or $-6.87°$ (both methods). [checked]

$$ax^2 + bx + c = 0: \ x = \tfrac{-b\pm\sqrt{\Delta}}{2a}, \ \Delta = b^2 - 4ac, \ x_1 + x_2 = -\tfrac{b}{a}, \ x_1x_2 = \tfrac{c}{a}$$

$\Delta > 0$ two solutions (elbow up/down, two assembly modes); $\Delta = 0$ one (reach boundary, singular); $\Delta < 0$ none (unreachable). IK: $\cos q_2 = \frac{x^2+y^2-L_1^2-L_2^2}{2L_1L_2}$; $|\cos q_2| > 1$ out of reach. RR $L_1 = L_2 = 1$, target $(1,1)$: $q_2 = \pm90°$. Servo $ms^2 + bs + k$: $\Delta = b^2 - 4mk$.
Cramer: $\alpha x + \beta y = e$, $\gamma x + \delta y = f$: $x = \frac{e\delta-\beta f}{\alpha\delta-\beta\gamma}$, $y = \frac{\alpha f-e\gamma}{\alpha\delta-\beta\gamma}$. Square and add removes an angle: $x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2c_2$. Complete the square: $x^2 + bx + c = (x + \frac{b}{2})^2 + c - \frac{b^2}{4}$. [checked]

BASICS

Chain rule, product rule, $\dot{\mathbf{R}}$, Lagrange partials [p. W6 18–19, W7 15–17]

5. Calculus of joint angles

$$\frac{d}{dt}f(q_1,q_2)=\sum_i \frac{\partial f}{\partial q_i}\dot{q}_i, \qquad \frac{d}{dt}(uv)=\dot{u}v+u\dot{v}$$

$$\frac{d}{dt}\cos q_1=-\sin q_1\,\dot{q}_1 \quad \frac{d}{dt}\sin q_1=\cos q_1\,\dot{q}_1$$
$$\frac{d}{dt}c_{12}=-s_{12}(\dot{q}_1+\dot{q}_2) \quad \frac{d}{dt}s_{12}=c_{12}(\dot{q}_1+\dot{q}_2)$$
$$\frac{d}{dt}\dot{q}^2=2\dot{q}\ddot{q} \qquad \frac{d}{dt}(\cos q\,\dot{q})=-\sin q\,\dot{q}^2+\cos q\,\ddot{q}$$

Inner rate: $\dot{q}_1+\dot{q}_2$ in c_{12}. RR tip: $\dot{x}=-L_1s_1\dot{q}_1-L_2s_{12}(\dot{q}_1+\dot{q}_2)$, $\dot{y}=L_1c_1\dot{q}_1+L_2c_{12}(\dot{q}_1+\dot{q}_2)$; $L_i=1$, $\mathbf{q}=(0,90°)$, $\dot{\mathbf{q}}=(1,0)$: $(\dot{x},\dot{y})=(-1,1)=\boldsymbol{\omega}\times\mathbf{r}$. [checked]

$$\dot{\mathbf{R}}=[\boldsymbol{\omega}]_\times\mathbf{R}, \quad [\boldsymbol{\omega}]_\times=\dot{\mathbf{R}}\mathbf{R}^\top, \quad \frac{d}{d\theta}\mathbf{R}_z=\begin{bmatrix}-s\theta & -c\theta & 0\\ c\theta & -s\theta & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$\mathcal{L}=K-P, \quad Q_i=\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}_i}-\frac{\partial\mathcal{L}}{\partial q_i}$$

Treat $q_i,\dot{q}_i$ as independent symbols for ∂; then differentiate $\partial\mathcal{L}/\partial\dot{q}_i$ in time. $K=\frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}}\Rightarrow\partial K/\partial\dot{\mathbf{q}}=\mathbf{M}\dot{\mathbf{q}}$. Q_i: N m (revolute), N (prismatic). Check, pendulum (q from downward vertical): $Q=mL^2\ddot{q}+mgL\sin q$. [checked]

6. Particle kinematics, dynamics, inertia

Constant acceleration; circular motion; rigid body

$$v=u+at, \quad s=ut+\tfrac{1}{2}at^2, \quad v^2=u^2+2as, \quad s=\tfrac{1}{2}(u+v)t, \quad s=\int v\,dt$$

Same for rotation: $s\to\theta$, $v\to\omega$, $a\to\alpha$. Omit the unknown you do not need. Check: $u=2$, $v=8$, $s=15$: $t=2s/(u+v)=3$ s, $a=2$ m/s^2. [checked]

$$v=\omega r, \quad a_c=\frac{v^2}{r}=\omega^2 r \text{ (to centre)}, \quad a_t=\alpha r, \quad |\mathbf{a}|=\sqrt{a_c^2+a_t^2}$$

$$\mathbf{v}=\boldsymbol{\omega}\times\mathbf{r}, \qquad \mathbf{v}_P=\mathbf{v}_O+\boldsymbol{\omega}\times\mathbf{r}_{OP}, \qquad \mathbf{v}_{B/A}=\mathbf{v}_B-\mathbf{v}_A$$

Rolling wheel (radius r, rate $\dot{\varphi}$): centre $r\dot{\varphi}$, top $2r\dot{\varphi}$, contact point 0 (no slip); diff-drive wheel at distance a: $u\pm a\dot{\theta}$. Check: $r=0.5$, $\omega=4$, $\alpha=2$: $a_c=8$, $a_t=1$, $|\mathbf{a}|=8.06$. [checked]

Newton, torque, energy, power

$$\mathbf{F}=m\mathbf{a}, \quad \boldsymbol{\tau}=\mathbf{r}\times\mathbf{F}, \quad \tau=I\alpha, \quad \mathbf{L}=\mathbf{I}\boldsymbol{\omega}, \quad \boldsymbol{\tau}=\dot{\mathbf{L}}, \quad W=\int\mathbf{F}\cdot d\mathbf{x}=\int\tau\,d\theta$$

$$K=\tfrac{1}{2}mv^2+\tfrac{1}{2}\boldsymbol{\omega}^\top\mathbf{I}\boldsymbol{\omega}\ (=\tfrac{1}{2}I\omega^2), \quad P=mgh, \quad \text{power}=\tau\omega=F v$$

Units: N, N m, kg m^2, J, W; $g=9.81$ m/s^2; h = height of the centre of mass. Frictionless swing: $K+P$ const. Check: $\mathbf{r}=(0.5,0,0)$, $\mathbf{F}=(0,10,0)$: $\boldsymbol{\tau}=(0,0,5)$ N m; at 20 rad/s: 100 W. [checked]

Inertia of standard bodies (about the centroid), parallel axis

Body (mass m)	Moment of inertia
Point mass at distance r	mr^2
Rod, length L, axis $\perp$ rod	centre $\frac{1}{12}mL^2$; end $\frac{1}{3}mL^2$
Solid cylinder/disc, radius r, own axis	$\frac{1}{2}mr^2$
Solid cylinder (r, length h), diameter axis	$\frac{1}{12}m(3r^2+h^2)$; thin disc $\frac{1}{4}mr^2$
Thin ring/tube, own axis	mr^2
Solid sphere; thin shell	$\frac{2}{5}mr^2$; $\frac{2}{3}mr^2$
Box $a\times b\times c$, axis along c	$\frac{1}{12}m(a^2+b^2)$; corner $\frac{1}{3}m(a^2+b^2)$

$$I=I_C+md^2 \quad \text{(parallel axes, } d=\text{distance between them)}$$

Never add inertias about different points. Gearbox ratio N (motor:load): load sees N^2I_m and torque $N\tau_m$. Check: $m=2$, $L=0.4$: centre 0.02667, end 0.10667 kg m^2. [checked]

Inertia tensor in full; tensor parallel axis [p. W7 34–39]

$$I_{xx}=\iiint(y^2+z^2)\rho\,dv, \quad I_{xy}=-\iiint xy\,\rho\,dv$$
$$^A\mathbf{I}=\begin{bmatrix}I_{xx} & I_{xy} & I_{xz}\\ I_{xy} & I_{yy} & I_{yz}\\ I_{xz} & I_{yz} & I_{zz}\end{bmatrix}, \quad I_{yy}=\iiint(x^2+z^2)\rho\,dv, \quad I_{yz}=-\iiint yz\,\rho\,dv$$
$$I_{zz}=\iiint(x^2+y^2)\rho\,dv, \quad I_{xz}=-\iiint xz\,\rho\,dv$$

$$^A\mathbf{I}={}^C\mathbf{I}+m(\mathbf{p}^\top\mathbf{p}\,\mathbf{1}_3-\mathbf{p}\mathbf{p}^\top)={}^C\mathbf{I}+m\begin{bmatrix}y^2+z^2 & -xy & -xz\\ -xy & x^2+z^2 & -yz\\ -xz & -yz & x^2+y^2\end{bmatrix}, \quad {}^B\mathbf{I}=\mathbf{R}\,{}^A\mathbf{I}\,\mathbf{R}^\top$$

$\mathbf{p}=(x,y,z)$: centroid $C\to$ new origin A, in $\{A\}$ axes. Moments ≥ 0; products carry the minus sign; symmetric. Check: box 0.2×0.3×0.1, $m=2$, corner frame: diag(0.01667, 0.00833, 0.02167) at the centroid becomes $\begin{bmatrix}0.06667 & -0.03 & -0.01\\ -0.03 & 0.03333 & -0.015\\ -0.01 & -0.015 & 0.08667\end{bmatrix}$. [checked]

7. Second-order systems and stability

$m\ddot{x}+b\dot{x}+kx=0$ [p. W8 16–26]

$$ms^2+bs+k=0, \quad \omega_n=\sqrt{\frac{k}{m}}, \quad \zeta=\frac{b}{2\sqrt{mk}}$$

$$s_{1,2}=\frac{-b\pm\sqrt{b^2-4mk}}{2m}=-\zeta\omega_n\pm\omega_n\sqrt{\zeta^2-1}$$

case	roots	$x(t)$ for $x(0)=x_0$, $\dot{x}(0)=v_0$
over $\zeta>1$	real $s_1\neq s_2$	$A_1e^{s_1t}+A_2e^{s_2t}$; $A_1+A_2=x_0$, $s_1A_1+s_2A_2=v_0$
crit. $\zeta=1$	double $-\omega_n$	$(A_1+A_2t)e^{-\omega_nt}$; $A_1=x_0$, $A_2=v_0+\omega_nx_0$
under $\zeta<1$	$\lambda\pm i\mu$	$e^{\lambda t}(A_1\cos\mu t+A_2\sin\mu t)$; $A_1=x_0$, $A_2=\frac{v_0-\lambda x_0}{\mu}$

$\lambda=-\zeta\omega_n$, $\mu=\omega_d=\omega_n\sqrt{1-\zeta^2}$. Unit-mass servo $\ddot{E}+k_v\dot{E}+k_pE=0$: $\omega_n=\sqrt{k_p}$, $\zeta=k_v/(2\sqrt{k_p})$, critical $k_v=2\sqrt{k_p}$ ($\omega_n=4$: $k_p=16$, $k_v=8$). Check: $m=1,b=2,k=5$: $\zeta=0.447$, $s=-1\pm2i$; $x_0=1,v_0=3$: $A_2=2$. $b=4,k=4$: $(1+2t)e^{-2t}$. [checked]

$$M_p=\exp\!\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right), \quad t_{\text{peak}}=\frac{\pi}{\omega_d}, \quad t_s\approx\frac{4}{\zeta\omega_n}\ (2\%)$$

1st order: $x=e^{-t/T_c}$ (63.2% at T_c). M_p: $\zeta=0.3$ 37.2%; 0.5 16.3%; 0.707 4.3%; ≥ 1 none. Check (m=1, b=2, k=5): $M_p=20.8\%$, $t_{\text{peak}}=1.571$ s, $t_s\approx 4$ s. [checked]

Routh. $s^2+a_1s+a_0$ stable iff $a_1,a_0>0$. Cubic $s^3+a_2s^2+a_1s+a_0$ stable iff $a_2,a_1,a_0>0$ **and** $a_2a_1>a_0$ (PID $s^3+k_vs^2+k_ps+k_i$: $k_vk_p>k_i$; equality: roots $\pm j\sqrt{k_p}$). Check: s^3+3s^2+2s+c stable iff $0<c<6$. [checked]

Conversions and conventions used in this booklet

8. Units and sign conventions

1 rpm = $\frac{2\pi}{60}=0.10472$ rad/s — 1 rad/s = 9.549 rpm; 1 Hz = 2π rad/s
1000 mm = 1 m — 1 W = 1 N m/s; N m = J/rad

Convert first: 120 rpm = 12.57 rad/s; 250 mm/s = 0.25 m/s. Gearbox $N=10$: motor 0.2 N m at 300 rad/s → load 2 N m at 30 rad/s. Jacobian row: m (revolute) or dimensionless (prismatic). [checked]

Frames	right-handed, z up; ${}^A\mathbf{T}_B$ = pose of $\{B\}$ in $\{A\}$
Rotation	anticlockwise about the axis looking to the origin; radians in formulas
Joints	revolute $q_i=\theta_i$ about z_i; prismatic $q_i=d_i$ along z_i
Force, gravity	$Q_i>0$ along $+q_i$; $\mathbf{g}=(0,0,-9.81)$ in $\{0\}$
Error	$\mathbf{E}=\mathbf{q}_d-\mathbf{q}$
Wheeled	u forward ($\hat{x}_C$), v left ($\hat{y}_C$), $\dot{\theta}>0$ anticlockwise; $\dot{\varphi}_i>0$ rolls forward

9. Graph-search vocabulary (W12)

Graph, cost, queue/stack/priority queue

$G=(V,E)$ nodes and edges; weight w_e; path cost = $\sum w_e$; tree: connected, no cycles, $|E|=|V|-1$. $g(n)$ = cheapest known cost start$\to n$; $h(n)$ = heuristic estimate $n\to$ goal (*admissible*: never above the true cost); $f(n)=g(n)+h(n)$. Grid: 4-connected step 1, 8-connected diagonal $\sqrt{2}=1.414$.

structure	removes next	used by
queue (FIFO)	oldest added	BFS (fewest edges, not cheapest)
stack (LIFO)	newest added	DFS (dives down a branch)
priority queue	smallest g or f	Dijkstra (g), A* (f)

Lists: open (frontier) and closed (expanded).

10. Notation

Symbols used across the booklet

symbol	meaning (week)
$\{0\},\{E\},\{i\}$	base/world, end-effector (tool), link frame i (W2)
$\{C\},\{O\}$	chassis frame; world (odometry) frame (W10)
${}^A\mathbf{T}_B$, ${}^A\mathbf{R}_B$	pose (4×4), orientation (3×3) of $\{B\}$ in $\{A\}$
${}^A\mathbf{P}$, ${}^A\mathbf{P}_B$	point in $\{A\}$; origin of $\{B\}$ in $\{A\}$ (m)
$\mathbf{q},\dot{\mathbf{q}},\ddot{\mathbf{q}}$	joint vector, rates, accelerations (rad or m)
$\mathbf{J}$, $\boldsymbol{\nu}=(\boldsymbol{v},\boldsymbol{\omega})$	${}^0\boldsymbol{\nu}={}^0\mathbf{J}\dot{\mathbf{q}}$; linear (m/s) above angular (rad/s) (W6)
$\mathbf{M},\mathbf{C},\mathbf{g}$	mass matrix (symmetric pos. def.), velocity terms, gravity vector (W7)
$\mathbf{Q}$	joint torques/forces, $\mathbf{Q}=\mathbf{M}\ddot{\mathbf{q}}+\mathbf{C}+\mathbf{g}$ (W7–W9)
$\mathbf{E}$, k_p,k_v,k_i	servo error $\mathbf{q}_d-\mathbf{q}$; gains (W8–W9)
$\boldsymbol{\zeta}=(u,v,\dot{\theta})$	chassis velocity in $\{C\}$ (W10)
$\dot{\varphi}_i$, r	wheel i speed (rad/s); wheel radius (W10)
β_i, γ_i	drive direction from $\hat{x}_C$ (bicycle: steering); roller direction from wheel-plane normal ($\gamma=0$ standard wheel) (W10)
g, h, f	path cost so far, heuristic, $f=g+h$ (W12)

BASICS

ROBOTS Standard robots: configurations, DH tables, ETS, FK, IK, Jacobian at a glance

Robot (joints)	n	reachable workspace (no joint limits)	IK solutions (position)	singular poses ($\det \mathbf{J} = 0$)	see
1. planar RR	2	annulus $\lvert L_1 - L_2\rvert \le r \le L_1 + L_2$; disc if $L_1 = L_2$	2 (elbow $\pm q_2$)	$q_2 = 0, \pi$: $\det \mathbf{J}_{xy} = L_1 L_2 \sin q_2$	W2 p. 11, W3 p. 14, W6 p. 23
2. planar RP (polar)	2	annulus $d_{\min} \le r \le d_{\max}$	1 (2 if $d_2 < 0$ allowed)	$d_2 = 0$: $\det \mathbf{J}_{xy} = -d_2$	W2 p. 11, W3 p. 14, W6 p. 23
3. planar PR (rail+rotary)	2	stadium: $D \times 2L$ rectangle + half-disc of radius L at each end	2 $(q_2, \pi - q_2)$	$q_2 = \pm 90°$: $\det \mathbf{J}_{xy} = L_2 \cos q_2$	W2 p. 11, W3 p. 14, W6 p. 23
4. planar RRR	3	disc $r \le \sum L_i$ (hole if $2L_{\max} > \sum L_i$); dexterous: $L_3 + \lvert L_1 - L_2\rvert \le r \le L_1 + L_2 - L_3$	2 for (x, y, φ); ∞ for (x, y)	$q_2 = 0, \pi$ for (x, y, ω_z): $\det = L_1 L_2 \sin q_2$	W2 p. 11, W3 p. 14, W6 p. 23
5. Cartesian PPP	3	box, edges = strokes	1	none: $\det \mathbf{J}_v = -1$	W1 p. 9
6. cylindrical RPP	3	hollow cylinder sector: radius $d_{3\,\min}..d_{3\,\max}$, height = stroke of d_2	1 (2 if $d_3 < 0$ allowed)	$d_3 = 0$: $\det \mathbf{J}_v = d_3$	W1 p. 9
7. spherical RRP	3	shell $d_{3\,\min} \le \rho \le d_{3\,\max}$ about $(0, 0, H)$	2 (4 if $d_3 < 0$ allowed)	$d_3 = 0$ or $\sin q_2 = 0$: $\det \mathbf{J}_v = -d_3^2 \sin q_2$	W3 p. 14, W6 p. 23
8. SCARA RRPR (position: RRP)	4	flat ring $\lvert a_1 - a_2\rvert \le r \le a_1 + a_2$ × stroke of d_3	2 (elbow); d_3, q_4 unique	$q_2 = 0, \pi$: $\det \mathbf{J}_v = -a_1 a_2 \sin q_2$	W1 p. 9, W2 p. 11, W3 p. 14
9. elbow arm RRR (W13 Q1–Q3)	3	shell $\lvert L_3 - L_4\rvert \le \rho \le L_3 + L_4$ about $(0, 0, L_1 + L_2)$	4 (shoulder × elbow)	$\sin \theta_3 = 0$ or $r = 0$	W2 p. 11, W3 p. 14, W6 p. 23
10. Puma 560 (6R, spherical wrist)	6	wrist centre in a spherical shell about the shoulder	up to 8	elbow $\sin q_3 = 0$; shoulder (W on waist axis); wrist $\sin q_5 = 0$	W2 p. 11, W3 p. 14, W6 p. 23
11. Franka Panda (7R)	7	thick sphere about $(0, 0, 0.333)$, flange distance 0.02–0.86 m (sampled)	∞ (1-D self-motion)	rank $\mathbf{J} < 6$, e.g. wrist centre on joint-1 axis	W2 p. 11, W3 p. 14
12. Furuta pendulum (RR in 3D)	2	tip on sphere of radius $\sqrt{L_1^2 + L_2^2}$ about $(0, 0, L_0)$, band $\lvert z - L_0\rvert \le L_2$	≤ 2 (only if the point is on the sphere)	$\sin \theta_2 = 0$	W2 p. 11

Mobile robots at a glance

Robot	wheels	holonomic?	δ_m	δ_s	δ_M	ICR (in $\{C\}$)	see
differential drive	2 fixed standard (driven, at $\pm a$) + castor	no	2	0	2	on the axle line: $y_{ICR} = u/\dot\theta = a\,\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1}$; $v = 0$	W10 p. 34 (sl. 43–47, 61)
unicycle model [added]	model of a diff. drive: inputs u, $\omega = \dot\theta$	no	2	0	2	$R = u/\omega$ on the axle line; $(\dot x, \dot y, \dot\theta) = (u\cos\theta, u\sin\theta, \omega)$	W10 p. 34
bicycle	rear fixed (driven) + front steered wheel at $(L, 0)$	no	1	1	2	rear-axle line at $R = L/\tan\beta$; $\dot\theta = u\tan\beta/L$	W10 p. 34 (sl. 69)
car (Ackermann)	2 rear fixed + 2 front steered	no	1	1	2	rear-axle line; $\tan\beta_{in} = \frac{L}{R - t/2}$, $\tan\beta_{out} = \frac{L}{R + t/2}$	W10 p. 34 (sl. 59–61)
tricycle	2 rear fixed + 1 front wheel driven and steered at $(b, 0)$	no	1	1	2	rear-axle line, $R = b/\tan\beta_1$; $\dot\varphi_1 = \frac{1}{r}\sqrt{u^2 + b^2\dot\theta^2}$, $\beta_1 = \operatorname{atan2}(b\dot\theta, u)$	W10 p. 34 (sl. 56–58)
3-wheel omni (Swedish triangle)	3 omni wheels at 120°, distance d from C	yes	3	0	3	anywhere: $(-v/\dot\theta,\ u/\dot\theta)$ for any $(u, v, \dot\theta)$	W10 p. 34 (sl. 53–54, W13 Q10)
4-wheel mecanum	4 wheels, roller angles $\gamma_i = \pm 45°$ at $(\pm\ell, \pm w)$	yes	3	0	3	anywhere (rank 3; 4 wheels for 3 unknowns: least squares)	W10 p. 34 (sl. 55)
skid steer	4 standard wheels, left pair and right pair	no (slide model: $v = 0$)	2 (model)	0	2	as the diff. drive; the 4 no-slide rows have rank 2 (also $\dot\theta = 0$): turning needs slip	W10 p. 34 (sl. 48–52)

Rules (W10). $\delta_M = \delta_m + \delta_s$; $\delta_m = 3 - \operatorname{rank} \mathbf{C}_1$ ($\mathbf{C}_1$ = no-slide rows of the standard wheels); a standard wheel ⇒ non-holonomic; no standard wheel (omni, mecanum) ⇒ holonomic, $\delta_m = 3$ (slide 64). [added]

Convention (all cards): DH row $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$, ${}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1}) t_x(a_{i-1}) \mathbf{R}_z(\theta_i) t_z(d_i)$; ETS read base to tool; $c_{12} = \cos(q_1 + q_2)$; z up; matrices checked by script. [added]

1. Planar RR [W2 sl. 15–16; W3 sl. 25–31; W6 sl. 33–39]

ETS ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1) t_x(L_1) \mathbf{R}_z(q_2) t_x(L_2)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $1\,(0, 0, 0, q_1)$ $2\,(L_1, 0, 0, q_2)$ $3\,(L_2, 0, 0, 0)$ (3 = tool)

FK $\mathbf{R} = \mathbf{R}_z(q_1 + q_2)$; position column $\mathbf{p} = (L_1 c_1 + L_2 c_{12},\ L_1 s_1 + L_2 s_{12},\ 0)$.

IK

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2 L_1 L_2}, \quad q_2 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2 q_2}, \cos q_2\right)$$

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin q_2,\ L_1 + L_2 \cos q_2)$$

J rows z, ω_x, ω_y zero, ω_z row $(1, 1)$; ${}^0\mathbf{J}_{xy} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, $\det = L_1 L_2 \sin q_2$ (singular: stretched/-folded).

Check: $L = (0.4, 0.3)$, $\mathbf{q} = (40°, 50°)$: tip $(0.3064, 0.5571)$; IK gives also $(82.38°, -50°)$. [checked]

2. Planar RP, polar [W2; W3 sl. 23–24; W6]

ETS ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1) t_x(d_2)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $1\,(0, 0, 0, q_1 - 90°)$ $2\,(0, -90°, d_2, 0)$ (prismatic axis must be z_2) plus a constant ${}^2\mathbf{T}_E$ (W2 part)

FK $\mathbf{R} = \mathbf{R}_z(q_1)$; $\mathbf{p} = d_2(c_1,\ s_1,\ 0)$: polar coordinates.

IK $d_2 = \sqrt{x^2 + y^2}$, $q_1 = \operatorname{atan2}(y, x)$; if $d_2 < 0$ allowed: also $(-d_2,\ q_1 + \pi)$.

J ${}^0\mathbf{J}_{xy} = \begin{bmatrix} -d_2 s_1 & c_1 \\ d_2 c_1 & s_1 \end{bmatrix}$, $\det = -d_2$; ω_z row $(1, 0)$, other rows 0. Singular only at $d_2 = 0$.

Check: $q_1 = 35°$, $d_2 = 0.6$: $(0.4915, 0.3441)$; target $(0.3, 0.4)$: $d_2 = 0.5$, $q_1 = 53.13°$. [checked]

3. Planar PR, rail + rotary [W2; W3; W6]

ETS ${}^S\mathbf{T}_E = t_x(d_1) \mathbf{R}_z(q_2) t_x(L_2)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $1\,(0, 0, d_1, 0)$ $2\,(0, 90°, 0, q_2 + 90°)$ $3\,(L_2, 0, 0, 0)$ with a constant ${}^S\mathbf{T}_0$ (slide along x_S needs $z_0 = x_S$; W2 part)

FK $\mathbf{R} = \mathbf{R}_z(q_2)$; $\mathbf{p} = (d_1 + L_2 c_2,\ L_2 s_2,\ 0)$.

IK $\sin q_2 = y/L_2$, $q_2 = \operatorname{atan2}\left(y, \pm\sqrt{L_2^2 - y^2}\right)$, $d_1 = x - L_2 \cos q_2$ (none if $\lvert y\rvert > L_2$).

J ${}^S\mathbf{J}_{xy} = \begin{bmatrix} 1 & -L_2 s_2 \\ 0 & L_2 c_2 \end{bmatrix}$, $\det = L_2 \cos q_2$; ω_z row $(0, 1)$. Singular at $q_2 = \pm 90°$.

Check: $d_1 = 0.3$, $q_2 = 60°$, $L_2 = 0.4$: $(0.5, 0.3464)$; $L_2 = 0.5$, target $(0.8, 0.3)$: $(0.4, 36.87°)$ or $(1.2, 143.13°)$. [checked]

4. Planar RRR [W2 sl. 6–9; W3; W6]

ETS ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1) t_x(L_1) \mathbf{R}_z(q_2) t_x(L_2) \mathbf{R}_z(q_3) t_x(L_3)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $1\,(0, 0, 0, q_1)$ $2\,(L_1, 0, 0, q_2)$ $3\,(L_2, 0, 0, q_3)$ $4\,(L_3, 0, 0, 0)$

FK $\mathbf{R} = \mathbf{R}_z(q_E)$, $q_E = q_1 + q_2 + q_3$; $\mathbf{p} = (L_1 c_1 + L_2 c_{12} + L_3 c_{123},\ L_1 s_1 + L_2 s_{12} + L_3 s_{123},\ 0)$.

IK pose (x, y, φ): $\mathbf{p}_W = (x - L_3 \cos\varphi,\ y - L_3 \sin\varphi)$; (q_1, q_2) = RR solution for $\mathbf{p}_W$ (card 1); $q_3 = \varphi - q_1 - q_2$. Two solutions; (x, y) only: ∞.

J ${}^0\mathbf{J}_{xy} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} - L_3 s_{123} & -L_2 s_{12} - L_3 s_{123} & -L_3 s_{123} \\ L_1 c_1 + L_2 c_{12} + L_3 c_{123} & L_2 c_{12} + L_3 c_{123} & L_3 c_{123} \end{bmatrix}$; ω_z row $(1, 1, 1)$. Pose block (x, y, ω_z): $\det = L_1 L_2 \sin q_2$.

Check: $L = (0.5, 0.4, 0.2)$, $\mathbf{q} = (30°, 45°, -60°)$: $(0.7297, 0.6881)$, $q_E = 15°$ (slide). $L = (1.0, 0.7, 0.3)$, $(1.2, 0.9)$, $\varphi = 60°$: $\mathbf{q} = (-3.32, 89.09, -25.77)°$ or $(66.06, -89.09, 83.02)°$. [checked]

5. Cartesian PPP [W1 sl. 56]

ETS ${}^0\mathbf{T}_E = t_z(d_1) t_y(d_2) t_x(d_3)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $1\,(0, 0, d_1, 0)$ $2\,(0, -90°, d_2, 90°)$ $3\,(0, 90°, d_3, 0)$; ${}^0\mathbf{T}_3 = {}^0\mathbf{T}_E \mathbf{R}_y(90°)$

$${}^0\mathbf{T}_E = \begin{bmatrix} 1 & 0 & 0 & d_3 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

IK $(d_1, d_2, d_3) = (z, y, x)$: one solution inside the box.

$${}^0\mathbf{J} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = -1$$

Orientation fixed; never singular. $M = 6(4 - 3 - 1) + 3 = 3$. [checked]

6. Cylindrical RPP [W1 sl. 55]

ETS ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)t_z(d_2)t_x(d_3)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, 0, q_1)$ 2 $(0, 0, d_2, 90°)$ 3 $(0, 90°, d_3, 0)$ with a constant ${}^3\mathbf{T}_E$ (tool x along the slide)

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 & -s_1 & 0 & d_3 c_1 \\ s_1 & c_1 & 0 & d_3 s_1 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

IK $q_1 = \operatorname{atan2}(y, x)$, $d_2 = z$, $d_3 = \sqrt{x^2 + y^2}$ (if $d_3 < 0$ allowed: $(q_1{+}\pi, d_2, -d_3)$).

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_3 s_1 & 0 & c_1 \\ d_3 c_1 & 0 & s_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = d_3 $$

Singular at $d_3 = 0$ (tool on the axis; q_1 arbitrary). Check: $q_1 = 40°$, $d_2 = 0.35$, $d_3 = 0.5$: $(0.3830, 0.3214, 0.35)$; $(0.3, 0.4, 0.6)$: $(53.13°, 0.6, 0.5)$. [checked]

7. Spherical RRP [W3; W6]

ETS ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)t_z(H)\mathbf{R}_y(q_2)t_z(d_3)$; q_2 = tilt of the boom from vertical.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, H, q_1)$ 2 $(0, -90°, 0, q_2)$ 3 $(0, 90°, d_3, 0)$

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & d_3 c_1 s_2 \\ s_1 c_2 & c_1 & s_1 s_2 & d_3 s_1 s_2 \\ -s_2 & 0 & c_2 & H + d_3 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

IK $q_1 = \operatorname{atan2}(y, x)$, $d_3 = \sqrt{r^2 + z'^2}$, $q_2 = \operatorname{atan2}(r, z')$, $r = \sqrt{x^2 + y^2}$, $z' = z - H$; 2nd: $(q_1{+}\pi, -q_2, d_3)$.

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_3 s_1 s_2 & d_3 c_1 c_2 & c_1 s_2 \\ d_3 c_1 s_2 & d_3 s_1 c_2 & s_1 s_2 \\ 0 & -d_3 s_2 & c_2 \\ 0 & -s_1 & 0 \\ 0 & c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix} $$

$\det \mathbf{J}_v = -d_3^2 \sin q_2$: singular if $d_3 = 0$ or $\sin q_2 = 0$. Check: $H = 0.5$, $(0.3, 0.4, 0.9)$: $(53.13°, 51.34°, 0.6403)$ or $(233.13°, -51.34°, 0.6403)$; $\det = -0.3202$. [checked]

8. SCARA RRPR [W1 sl. 54; W2; W3]

ETS ${}^0\mathbf{T}_E = t_z(H)\mathbf{R}_z(q_1)t_x(a_1)\mathbf{R}_z(q_2)t_x(a_2)t_z(-d_3)\mathbf{R}_z(q_4)$ (tool z up).

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, H, q_1)$ 2 $(a_1, 0, 0, q_2)$ 3 $(a_2, 180°, d_3, 0)$ 4 $(0, 180°, 0, q_4)$

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{124} & -s_{124} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{124} & c_{124} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & H - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

IK $d_3 = H - z$; (q_1, q_2) = RR solution for (x, y) (card 1, $L_i \to a_i$); $q_4 = \varphi - q_1 - q_2$.

$$ {}^0\mathbf{J} = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} & 0 & 0 \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} $$

$\det \mathbf{J}_v = -a_1 a_2 \sin q_2$ (the -1: d_3 is measured downwards). Check: $H = 0.4$, $a = (0.35, 0.25)$, $\mathbf{q} = (40°, -70°, d_3{=}0.12, 25°)$: $(0.4846, 0.1000, 0.28)$, yaw $-5°$. Target $(0.30, 0.25, 0.28)$, $\varphi = 60°$: $(0.83°, 100.70°, 0.12, -41.53°)$ or $(78.79°, -100.70°, 0.12, 81.92°)$. [checked]

9. Elbow arm RRR [W13 sl. 4–5, 7; W2; W3; W6]

ETS ${}^S\mathbf{T}_T = \mathbf{R}_z(\theta_1)t_z(L_1{+}L_2)\mathbf{R}_y(-\theta_2)t_x(L_3)\mathbf{R}_y(-\theta_3)t_x(L_4)\mathbf{R}_x(90°)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, L_1{+}L_2, \theta_1)$ 2 $(0, 90°, 0, \theta_2)$ 3 $(L_3, 0, 0, \theta_3)$ 4 $(L_4, 0, 0, 0)$

$$ {}^S\mathbf{T}_T = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1(L_3 c_2 + L_4 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1(L_3 c_2 + L_4 c_{23}) \\ s_{23} & c_{23} & 0 & L_1 + L_2 + L_3 s_2 + L_4 s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

IK 4 solutions: $\theta_1 = \operatorname{atan2}(y, x)$ $(r = +\sqrt{x^2 + y^2})$ or $\theta_1{+}\pi$ $(r = -\sqrt{\cdot})$; $z' = z - L_1 - L_2$

$$ \cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2 L_3 L_4}, \quad \theta_3 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3 \right) $$

$$ \theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4 \sin\theta_3,\ L_3 + L_4 \cos\theta_3) $$

$$ {}^S\mathbf{J} = \begin{bmatrix} -s_1(L_3 c_2 + L_4 c_{23}) & -c_1(L_3 s_2 + L_4 s_{23}) & -c_1 L_4 s_{23} \\ c_1(L_3 c_2 + L_4 c_{23}) & -s_1(L_3 s_2 + L_4 s_{23}) & -s_1 L_4 s_{23} \\ 0 & L_3 c_2 + L_4 c_{23} & L_4 c_{23} \\ 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix} $$

$$ \det \mathbf{J}_v = -L_3 L_4 \sin\theta_3 \, (L_3 c_2 + L_4 c_{23}) $$

Singular: $\sin\theta_3 = 0$ (boundary) or $r = 0$ (tool above the base axis). Check: $L = (0.6, 0.5, 0.4, 0.3)$, $\boldsymbol{\theta} = (30, 20, 40)°$: $(0.4554, 0.2629, 1.4966)$; IK also $(30, 54.047, -40)$, $(-150, 125.953, 40)$, $(-150, 160, -40)$. [checked]

10. Puma 560, 6R spherical wrist [W2; W3 sl. 32–38; W6]

Axes 4, 5, 6 meet at the wrist centre W; tool origin = W. **Zero-offset model** (home pose: arm straight up), L_b, L_u, L_f:

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, L_b, q_1)$ 2 $(0, -90°, 0, q_2{-}90°)$ 3 $(L_u, 0, 0, q_3{+}90°)$ 4 $(0, 90°, L_f, q_4)$ 5 $(0, -90°, 0, q_5)$ 6 $(0, 90°, 0, q_6)$

$$ {}^0\mathbf{T}_E = \begin{bmatrix} {}^0\mathbf{R}_3\, {}^3\mathbf{R}_6 & \mathbf{p}_W \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad {}^3\mathbf{R}_6 = \mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\mathbf{R}_z(q_6) $$

$$ {}^0\mathbf{R}_3 = \mathbf{R}_z(q_1)\mathbf{R}_y(q_2{+}q_3) = \begin{bmatrix} c_1 c_{23} & -s_1 & c_1 s_{23} \\ s_1 c_{23} & c_1 & s_1 s_{23} \\ -s_{23} & 0 & c_{23} \end{bmatrix} $$

$$ \mathbf{p}_W = \begin{bmatrix} c_1(L_u s_2 + L_f s_{23}) \\ s_1(L_u s_2 + L_f s_{23}) \\ L_b + L_u c_2 + L_f c_{23} \end{bmatrix} $$

IK 8 branches: $\rho = \pm\sqrt{x_W^2 + y_W^2}$, $q_1 = \operatorname{atan2}(y_W, x_W)$ $(\rho > 0)$ or $+\pi$; $z' = z_W - L_b$; $\cos q_3 = \dfrac{\rho^2 + z'^2 - L_u^2 - L_f^2}{2 L_u L_f}$, $q_3 = \operatorname{atan2}(\pm\sqrt{1 - \cos^2 q_3}, \cos q_3)$; $q_2 = \operatorname{atan2}(\rho, z') - \operatorname{atan2}(L_f \sin q_3, L_u + L_f \cos q_3)$. Wrist: ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_E$, $q_5 = \operatorname{atan2}(\pm\sqrt{r_{13}^2 + r_{23}^2}, r_{33})$, $q_4 = \operatorname{atan2}(\pm r_{23}, \pm r_{13})$, $q_6 = \operatorname{atan2}(\pm r_{32}, \mp r_{31})$ (same signs together; $q_5 = 0$: only $q_4{+}q_6$ fixed).

J $\det \mathbf{J} = -L_u L_f \sin q_3 \, (L_u s_2 + L_f s_{23}) \sin q_5$: elbow, shoulder ($W$ on waist axis), wrist ($\sin q_5 = 0$, rank 5).

Check: $L = (1.0, 0.5, 0.3)$, $\mathbf{q} = (30, 20, -40, 0, 0, 0)°$: $\mathbf{p}_W = (0.0592, 0.0342, 1.7518)$. [checked]

Course table (W3 practical, `puma560_fk_proximal_dh.py`), $\theta_i = q_i$:

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, 0, q_1)$ 2 $(0, -90°, 0, q_2)$ 3 $(0.4318, 0, 0.15005, q_3)$ 4 $(-0.0203, -90°, 0.4318, q_4)$ 5 $(0, 90°, 0, q_5)$ 6 $(0, -90°, 0, q_6)$ $\mathbf{q} = \mathbf{0}$: tool at $(0.4115, 0.1501, -0.4318)$ pointing down; $\mathbf{p}_W$ is independent of $q_{4,5,6}$; 8 IK solutions; W in the shell $0.150 \le |\mathbf{p}_W| \le 0.877$ m. [checked]

SLIDE CHECK W3 practical files: two different Puma tables

`puma560_fk_proximal_dh.py` and `..._distal_ets.py` differ: at $\mathbf{q} = \mathbf{0}$ the first gives $(0.4115, 0.1501, -0.4318)$ (tool down), the second $(0.4521, -0.1500, 0.4318)$; a_3 and α_1 signs differ. Use the table in the question.

11. Franka Panda, 7R redundant [W2; W3 sl. 40–52]

ETS ${}^0\mathbf{T}_E = \prod_{i=1}^{7} \mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(q_i)t_z(d_i) \cdot t_z(0.107)$ (flange).

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, 0.333, q_1)$ 2 $(0, -90°, 0, q_2)$ 3 $(0, 90°, 0.316, q_3)$ 4 $(0.0825, 90°, 0, q_4)$ 5 $(-0.0825, -90°, 0.384, q_5)$ 6 $(0, 90°, 0, q_6)$ 7 $(0.088, 90°, 0, q_7)$, then $t_z(0.107)$ (m).

$$ {}^0\mathbf{T}_E(\mathbf{q} = \mathbf{0}) = \begin{bmatrix} 1 & 0 & 0 & 0.088 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0.926 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$6{\times}7$ Jacobian: column $j = [\mathbf{z}_j \times (\mathbf{p}_E - \mathbf{o}_j); \mathbf{z}_j]$; ∞ IK solutions (self-motion, null space of $\mathbf{J}$): $\dot{\mathbf{q}} = \mathbf{J}^+\boldsymbol{\nu} + (\mathbf{I} - \mathbf{J}^+\mathbf{J})\dot{\mathbf{q}}_0$. Singular (rank 5): $q_2 = q_4 = 0$, or wrist centre on the joint-1 axis. No closed-form IK in the unit. [checked]

12. Furuta pendulum, RR in 3D [W2 sl. 17–19, 34]

ETS ${}^B\mathbf{T}_E = t_z(L_0)\mathbf{R}_z(\theta_1)t_x(L_1)\mathbf{R}_x(\theta_2)t_z(-L_2)\mathbf{R}_y(\pi)$.

DH $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1 $(0, 0, L_0, q_1{+}90°)$ 2 $(0, 90°, L_1, q_2)$ 3 $(0, 90°, L_2, 0)$; plus ${}^3\mathbf{T}_E = \mathbf{R}_z(-90°)$

$$ {}^B\mathbf{T}_E = \begin{bmatrix} -c_1 & -s_1 c_2 & -s_1 s_2 & L_1 c_1 - L_2 s_1 s_2 \\ -s_1 & c_1 c_2 & c_1 s_2 & L_1 s_1 + L_2 c_1 s_2 \\ 0 & s_2 & -c_2 & L_0 - L_2 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

IK $\cos\theta_2 = \frac{L_0 - z}{L_2}$, $\theta_2 = \operatorname{atan2}(\pm\sqrt{1 - \cos^2\theta_2}, \cos\theta_2)$; $\theta_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin\theta_2, L_1)$. Reachable only on the sphere $x^2{+}y^2{+}(z{-}L_0)^2 = L_1^2{+}L_2^2$.

$$ {}^B\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 c_1 s_2 & -L_2 s_1 c_2 \\ L_1 c_1 - L_2 s_1 s_2 & L_2 c_1 c_2 \\ 0 & L_2 s_2 \\ 0 & c_1 \\ 0 & s_1 \\ 1 & 0 \end{bmatrix} $$

Singular iff $\sin\theta_2 = 0$ (columns parallel). Check: $L = (0.5, 0.4, 0.3)$, $\boldsymbol{\theta} = (60°, 45°)$: $(0.0163, 0.4525, 0.2879)$; IK also $(115.88°, -45°)$. [checked]

ROBOTS

W1 Mechanisms, degrees of freedom, mobility, workspace

Definitions and chains [p. 18–26]

term	meaning
Kinematics [p. 18]	motion geometry *without* forces. Dynamics [p. 19]: forces needed to move; statics = at rest, kinetics = accelerating
Mechanism [p. 21]	transfers/modifies motion and/or force from sources to outputs
Kinematic chain [p. 22]	rigid bodies connected by joints; a mechanism = chain(s) from base to end effector
Link / joint [p. 23]	link = rigid body; joint = connection allowing *relative* motion. Base = link 0 (ground); end effector = tool link
Open / serial [p. 25]	open: no closed loop. Serial: every link but first and last joins two others. $N = J + 1$, $M = \sum F_i$
Closed / parallel	closed: has a loop ($\ell \geq 1$). Parallel: end effector tied to base by several chains (delta, Stewart) [added]

Serial vs parallel [p. 26]: serial = big workspace, easy FK, motors ride on arm. Parallel = motors on base, light, load shared, small workspace, easy IK, hard FK (W4). Six-joint arm: open, serial. Four-bar: closed, *not* serial (no first/last link). Two fingers on one palm: open, not serial [added].

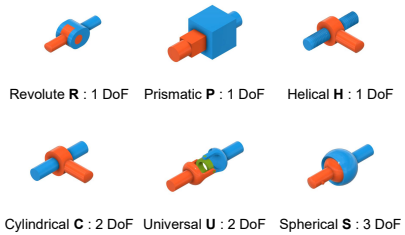
DoF of a body; joint types [p. 27–36]

Degrees of freedom [p. 27]

DoF = independent parameters fixing the configuration. Free body: $D = 3$ (plane: x, y, θ), $D = 6$ (3D: x, y, z + 3 rotations). Point in 2D: 2. Pendulum: 1 ($x^2 + y^2 = L^2$). Joint removes $D - F$:

constraints removed = $D - F$ (F = DoF the joint allows)

Why [added]: plane: 2 points = 4 coords, 1 fixed distance: $4 - 1 = 3$. Space: 3 points = 9, 3 distances: $9 - 3 = 6$.



Revolute **R** : 1 DoF Prismatic **P** : 1 DoF Helical **H** : 1 DoF

Cylindrical **C** : 2 DoF Universal **U** : 2 DoF Spherical **S** : 3 DoF

[W01 lec p. 34] — joints, slide 34 (top R, P, H; bottom C, U, S)

joint [p. 34]	sym.	F	cuts 3D	cuts 2D
Revolute (pin)	R	1	5	2
Prismatic (slider)	P	1	5	2
Helical (screw)	H	1	5	–
Cylindrical	C	2	4	–
Universal	U	2	4	–
Spherical (ball)	S	3	3	–

Planar joint E (3 in 3D) is not on the slides [added]. Planar mechanisms: R, P only.

Example [p. 36]. Two planar bodies $3 + 3 = 6$; pin makes 2 points coincide (x and y): $6 - 2 = 4$. [checked]

TRAP

Body in 2D has 3 DoF, a *point* 2. Joint DoF $F \neq$ mobility M.

Controllable vs total DoF; actuation [p. 37–38]

Controllable DoF = actuated. controllable = total: **fully actuated**; <: **under-actuated** (cheaper, lighter [p. 38]); >: **redundant** (slide wording). Needed motors = M [p. 39].

example [added]	motors	M	name
6R arm; 4 motors	6; 4	6	fully; under
3-RPR, 3 P motors	3	3	fully
4-RPR, 4 motors	4	3	redundant
Stewart, 6 linear	6	6	fully
delta, 3 base motors	3	3 (not 9)	fully

SLIDE CHECK

Slide 37: controllable > total ⇒ redundant, but notebook cell 8 calls a 7-DoF arm (7 motors) redundant. Usual meaning [added]: kinematic redundancy (M > task DoF, e.g. 7R for a 6-DoF pose). State which you use.

Mobility: Grübler–Kutzbach [p. 39–41]

GIVEN drawing: every link (ground included), every joint with type, planar or spatial FIND mobility M = DoF = minimum actuators [p. 39]

G–K, form 1 [p. 40]

$$M_{3D} = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - P_5, \qquad M_{2D} = 3(N-1) - 2P_1 - P_2$$

where N = links incl. ground; P_i = number of joints with i DoF (coefficient = $D - i$).

G–K, form 2 [p. 41]: W13 Q6, Q7

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i = D(N-1) - DJ + \sum F_i$$

where $D = 3$ planar, 6 spatial; J = joints; F_i = DoF of joint i (1 for R, P, H; 2 C, U; 3 S). Handles C, U, S. Same as Lynch & Park $m(N - 1 - J) + \sum f_i$ [added].

DERIVATION form 2 from form 1; loop view [added]

$N - 1$ moving links free: $D(N - 1)$; joint i removes $D - F_i$: $M = D(N-1) - \sum(D - F_i) = D(N - J - 1) + \sum F_i$; grouping by i gives form 1. With $\ell = J - N + 1$ loops: $M = \sum F_i - D\ell$ (four-bar $4 - 3 = 1$; 3-RPR $9 - 3 \cdot 2 = 3$; Stewart $36 - 6 \cdot 5 = 6$). [checked]

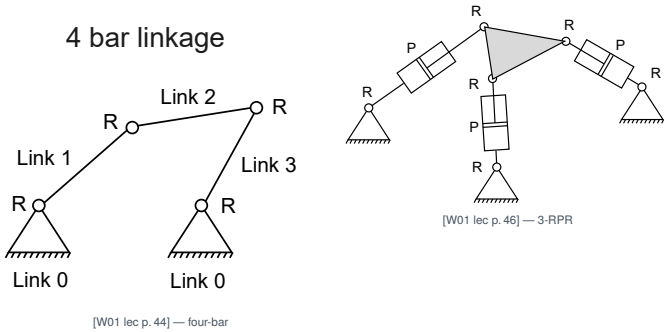
STEPS Mobility count

1. $D = 3$ if all axes $\perp$ page; any S, U, C or "still 3D" ⇒ $D = 6$.
2. Number links, **ground** = 0 (all ground pivots = one link); a P joint joins two links (cylinder, rod), so R–P–R leg has 2 links.
3. List joints with F_i: get J, $\sum F_i$.
4. $M = D(N - J - 1) + \sum F_i$, show substitution; planar cross-check with form 1.
5. Check: $M \geq 1$ for a machine; serial $M = \sum F_i$; $M \leq 0$ ⇒ dependent constraints?

Extended G–K (W13 p.16; W4)

$$M = D(N - J - 1) + \sum F_i + R_C - R_M$$

where $R_C = \sum F_i - \sum_k u_k$ over-constraints (u_k = unique DoFs of loop k, as printed); R_M = local mobility (DoF that do not move the output, e.g. rod spins).



[W01 lec p. 44] — four-bar [W01 lec p. 46] — 3-RPR

Slide examples (all [checked])

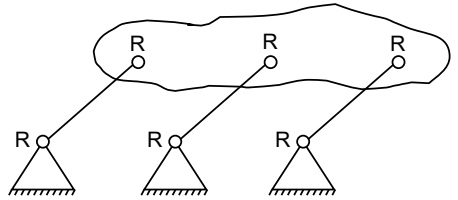
example	D	N	J	$\sum F$	M
RPR serial [p. 42]	3	4	3	3	3
four-bar [p. 44]	3	4	4	4	1
3-RPR [p. 46], Q6	3	8	9	9	3
Stewart 6-UPS [p. 48] [added]	6	14	18	36	6
delta, Q7 (W13 p.15)	6	11	15	39	9; 3
3 equal legs [p. 49]	3	5	6	6	0 (true 1)
Hoberman [p. 51]	3	24	36	36	−3

RPR: $3(4 - 3 - 1) + 3 = 3$; form 1 $3 \cdot 3 - 2 \cdot 3 = 3$. Four-bar: ground drawn twice is one link; $3(4 - 4 - 1) + 4 = 1$ (8 constraints on 9 freedoms). 3-RPR: $N = 1 + 1 + 3 \cdot 2$; $3(8 - 9 - 1) + 9 = 3$. Stewart: per leg U, P, S ($2 + 1 + 3$); $6(14 - 18 - 1) + 36 = 6$; six SPS give 12 (legs spin). Delta: $N = 2 + 3 \cdot 3$, per leg R + 4 S; $9 + 0 - 6 = 3$, $R_M = 6$ rod spins; platform translates x, y, z. Hoberman: $6(24 - 36 - 1) + 36 = -42$ if $D = 6$.

Task DoF m: 2 point in plane, 3 planar pose or point in 3D, 6 3D pose. $M < m$ cannot do it, $M = m$ exact, $M > m$ redundant. 3-RPS ($D = 6$, $N = 8$, $J = 9$, $\sum F = 15$): $M = 3$ [added].

Where G–K fails [p. 49–51]

G–K assumes *independent* constraints (p.49), sees only N, J, F_i, never lengths. Slide 49: body on three equal parallel legs (R both ends), $M = 0$ but it moves, 1 DoF: delete one middle leg, $N = 4$, $J = 4$, $M = 1$. Unequal legs: rank 6, locks, 0 is right. It also over-counts (delta spins).



[W01 lec p. 49] — three equal parallel legs

STEPS "Did you notice anything odd?"

1. Count, show $M = 0$ (or < 0). 2. State: it moves, 1 DoF.
2. Reason: constraints dependent, one leg redundant. 4. Remove it, recount: $M = 1$. 5. Caveat: equal parallel legs only.

TRAP
False: "$M \le 0$ means it cannot move" and "G–K always gives true DoF".

Workspace and robot configurations [p. 52–57]

Set by geometry (link lengths) and joint limits [p. 52].

Reachable and dexterous [p. 57]

$$W_D \subseteq W_R, \qquad 2\text{R}: \ |L_1 - L_2| \le r \le L_1 + L_2$$
$$3\text{R (equal } L): \ W_R: \ r \le 3L, \quad W_D: \ r \le L$$

W_R: points reached in one orientation; W_D: in all orientations; r = tool distance from base axis. Answer $W_R \supseteq W_D$ (strict $\subset$ is a trap: both can be empty).

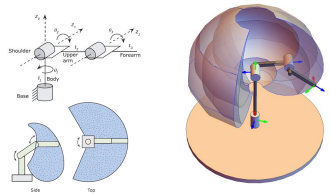
W_D derivation [added]: wrist $\mathbf{w} = \mathbf{p} - L(\cos\varphi, \sin\varphi)$ must satisfy $|\mathbf{w}| \le 2L$ for every φ: $r + L \le 2L$; for some φ: $r - L \le 2L$. 2R: W_D empty (2 DoF cannot set point and angle). $L = (0.5, 0.3)$: annulus $0.2 \le r \le 0.8$ m. [checked]

STEPS Sketch a workspace

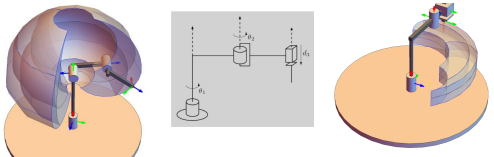
1. Centre on first joint axis. 2. Fold (min r), stretch (max r). 3. Sweep spinning joints: annulus → ring/shell; P adds stroke. 4. Limits cut wedges. 5. State W_R, W_D (empty if DoF < pose).

arm [p. 53–56]	joints	workspace
PUMA	RRR	thick spherical shell (waist limit cuts wedge)
SCARA	RRP	flat ring, thick by vertical stroke
cylindrical	RPP	thick-walled sector (R angle, P height, P radius)
Cartesian	PPP	box, edges = strokes
spherical [added]	RRP	thick shell (P sets radius range)

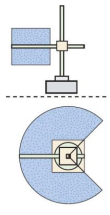
Each has $M = 6(4-3-1)+3 = 3$ in 3D: position only. [checked]



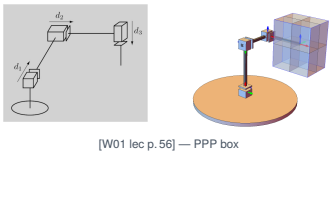
[W01 lec p. 53] — RRR shell



[W01 lec p. 54] — RRP ring



[W01 lec p. 55] — RPP sector



[W01 lec p. 56] — PPP box

Next topics; notebook Q1–Q7

Manipulation problems [p. 59–62]: FK joints→pose (W2), IK pose→joints (W3), velocity (W6), dynamics (W7–W9), trajectories (W5). Pose = 3 + 3 numbers, 4×4 [p. 62]; frames {B} base, {W} wrist, {T} tool, {S} station, {G} goal. Pose has 6 numbers: 6-DoF arm minimum.

nb: Q1 rigid body 2D: 3 (point: 2). Q2 four-bar: $3(4-1) - 2 \cdot 4 = 1$. Q3 planar parallel: $3(8-1) - 2 \cdot 9 = 3$. Q4 (assumed p.49 system) $N = 5$, $P_1 = 6$: 0. Q5 odd: moves, 1 DoF, dependent constraints. Q6 workspaces: table above. Q7 $W_R \supseteq W_D$.

W13 sample Q6 (3-RPR mobility) and **Q7** (delta mobility): solved on p. 42.

Exam-style questions

Marks not published; about 1 per step: right D, N with ground, every F_i, substitution, answer, one sentence on plain vs real DoF.

EXAM-STYLE QUESTION Derive and count a Watt six-bar [marks assumed 6]

(a) Derive G–K form 2. (b) Planar Watt six-bar: 6 links (ground incl.), 7 R: find M. (c) Which slip gives −5?

SOLUTION (a) $D(N-1) - \sum(D - F_i) = D(N-J-1) + \sum F_i$. (b) $3(6-7-1)+7 =$ $\boxed{1}$; form 1 $3 \cdot 5 - 2 \cdot 7 = 1$. (c) $D = 6$ on a planar drawing; ($N = 5$, ground forgotten: −2). [checked]

EXAM-STYLE QUESTION Odd mechanism, actuation [marks assumed 4]

(a) Define fully/under-actuated/redundant. (b) Body on three equal parallel R–R links: G–K gives 0 but it moves; explain.

SOLUTION (a) controllable =, <, > total; 6R arm with 6 motors fully, 4 motors under. (b) $N = 5$, $J = 6$: $M =$ $\boxed{0}$; constraints dependent; remove one leg: $M =$ $\boxed{1}$. [checked]

EXAM-STYLE QUESTION Workspace of planar 3R [marks assumed 6]

$L = (0.5, 0.3, 0.2)$ m, no limits: find W_R, W_D.

SOLUTION Wrist $\mathbf{w} = \mathbf{p} - L_3(\cos\varphi, \sin\varphi)$ in 2R annulus $[0.2, 0.8]$. W_R: $r \le 0.8 + 0.2 =$ $\boxed{1.0}$ m disc. W_D: $r + 0.2 \le 0.8$ and $r - 0.2 \ge 0.2$: ring $\boxed{0.4 \le r \le 0.6}$ m. Check: equal links give $r \le L$. [checked]

W1

W2 Forward kinematics: frames, ETS, proximal DH

FK: answer layout [p. 5, 10–11]

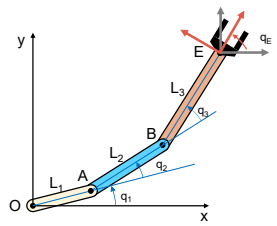
GIVEN joint variables $\mathbf{q}$ (θ_i rad revolute, d_i m prismatic), link constants FIND ${}^0\mathbf{T}_E(\mathbf{q})$: position + orientation of tool $\{E\}$ in base $\{0\}$ (one answer; IK may have 0, 1, many)

STEPS any FK question

1. Draw zero pose; name $\{S\}/\{0\}$, $\{E\}/\{T\}$, joint axes, *drawn* sign of each angle.
2. Rule: ${}^0\mathbf{T}_E = {}^0\mathbf{T}_1\,{}^1\mathbf{T}_2 \cdots {}^{n-1}\mathbf{T}_E$ (inner indices cancel).
3. ETS or DH table, one line per joint; matrix of each intermediate frame if asked.
4. Multiply symbolically (position column first), numbers last, units.
5. Box ${}^0\mathbf{T}_E$. Check at $\mathbf{q} = \mathbf{0}$ and at one pose with non-zero angles.

W13 sample Q1 (RRR arm by ETS, intermediate frames): solved on p. 42.

Planar RRR by trigonometry [p. 6–9]



[W02 lec p. 9] — q_k measured from link $k-1$

Planar RRR [p. 7–9]

$$x = L_1c_1 + L_2c_{12} + L_3c_{123}, \quad q_E = q_1 + q_2 + q_3$$
$$y = L_1s_1 + L_2s_{12} + L_3s_{123}$$

where $c_{12} = \cos(q_1+q_2)$ etc.; $\phi_k = q_1 + \cdots + q_k$ = absolute direction of link k (joint k turns everything beyond it); q_k ccw positive.

WORKED numbers

$L = (0.5, 0.4, 0.2)$ m, $q = (30, 45, -60)°$, $\phi = (30, 75, 15)°$: $A = (0.4330, 0.2500)$, $B = (0.5365, 0.6364)$, $E = (0.7297, 0.6881)$ m, $q_E = 15°$. [checked]

TRAP

Using q_k instead of the running sums ϕ_k (agree at $\mathbf{q} = 0$, so test a non-zero pose); $q_E \neq q_3$.

Rotation matrices, homogeneous T, inverse [p. 10–12]

${}^A\mathbf{R}_B$ and elementary rotations [added]

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}, \quad \mathbf{R}_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

$$\mathbf{R}_z(\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A\mathbf{R}_B = \begin{bmatrix} {}^A\hat{x}_B & {}^A\hat{y}_B & {}^A\hat{z}_B \end{bmatrix}, \quad \mathbf{R}^\top\mathbf{R} = \mathbf{I}, \ \det\mathbf{R} = +1, \ {}^B\mathbf{R}_A = {}^A\mathbf{R}_B^\top$$

θ: right-hand rule about the subscript axis. Reading a pose: write the 3 axes of $\{B\}$ as columns in $\{A\}$; check unit length, orthogonal, $\hat{x}_B \times \hat{y}_B = \hat{z}_B$.

Homogeneous transform and full inverse [p. 12]

$${}^A\mathbf{T}_B = \begin{bmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad {}^A\mathbf{P} = {}^A\mathbf{R}_B\,{}^B\mathbf{P} + {}^A\mathbf{t}, \quad {}^B\mathbf{T}_A = \begin{bmatrix} {}^A\mathbf{R}_B^\top & -{}^A\mathbf{R}_B^\top\,{}^A\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

${}^A\mathbf{t}$ = origin of $\{B\}$ in $\{A\}$ (m). Inverse is *not* $-\mathbf{t}$ in the last column.

WORKED point and inverse

${}^A\mathbf{R}_B = \mathbf{R}_z(90°)$, ${}^A\mathbf{t} = (2,1,3)^\top$, ${}^B\mathbf{P} = (1, 0.5, 0)^\top$: ${}^A\mathbf{P} = (-0.5, 1, 0)^\top + (2,1,3)^\top = (1.5, 2, 3)^\top$. $-\mathbf{R}^\top\mathbf{t} = (-1, 2, -3)^\top$:

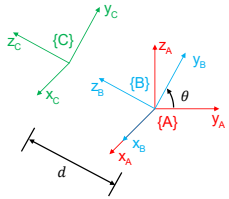
$${}^B\mathbf{T}_A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Check ${}^B\mathbf{T}_A(1.5, 2, 3, 1)^\top = (1, 0.5, 0, 1)^\top$. [checked]

TRAP

Last column $-\mathbf{t} = (-2, -1, -3)$ instead of $-\mathbf{R}^\top\mathbf{t}$ (hidden when $\mathbf{R} = \mathbf{I}$).

Compounding: fixed vs current frame [p. 13–14]



[W02 lec p. 13] — ${}^A\mathbf{T}_C = \mathbf{R}_x(\theta)\,t_z(d)$

Compounding [p. 13]

$${}^A\mathbf{T}_C = {}^A\mathbf{T}_B\,{}^B\mathbf{T}_C, \qquad \mathbf{R}_x(\theta)\,t_z(d)$$

rotate about x_A, then slide along the *new* z. Inner letters must cancel; "note the order".

WORKED slide: $\theta = 30°$, $d = 2$ m

$${}^A\mathbf{T}_C = \mathbf{R}_x(30°)t_z(2) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.5 & -1 \\ 0 & 0.5 & 0.866 & 1.732 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Origin $(0, -1, 1.732)$: moved along the *tilted* z_B. Reversed $t_z(2)\mathbf{R}_x(30°)$: origin $(0, 0, 2)$. [checked]

ALTERNATIVE APPROACHES

Left to right = CURRENT frame	Right to left = FIXED frame
Each factor on the right is a move in the frame just reached (joint i turns about its *own* axis).	Each factor on the left is a move about the original axes; apply the last factor first.

WORKED order matters

$$\mathbf{R}_z(90°)t_x(1) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad t_x(1)\mathbf{R}_z(90°) = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Origins $(0,1,0)$ vs $(1,0,0)$. Link turned by its own joint: $\mathbf{R}_z(q)t_x(L)$. [checked]

Elementary Transform Sequence (ETS) [p. 15–19]

Six elementary transforms and the product [p. 16]

$$t_{x,y,z}(\eta) = \begin{bmatrix} \mathbf{I}_3 & \eta\hat{\mathbf{e}} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad \mathbf{R}_{x,y,z}(\eta) = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

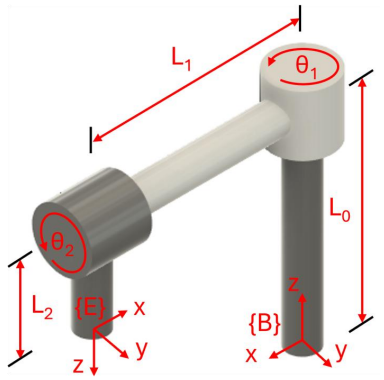
$${}^B\mathbf{T}_E = E_1(\eta_1)E_2(\eta_2)\cdots E_m(\eta_m)$$

η_i = constant (offset, fixed rotation) or joint variable (θ_j revolute, d_j prismatic); each E_i acts along/about the *current* frame's axis.

STEPS ETS from a drawing [p. 18]

1. Redraw with every joint displaced from rest.
2. From $\{B\}$: t_x, t_y, t_z (signed) to the first joint; $\mathbf{R}_{axis}(\pm q)$ (minus if the drawn arrow opposes the right-hand rule).
3. Repeat joint by joint; coincident axes: no translation between rotations.
4. End with the constant rotation that makes the last frame the drawn $\{E\}$.
5. Group per joint = ${}^{i-1}\mathbf{T}_i$ (intermediate frames), then multiply.

Example: Furuta pendulum [p. 17–19]



[W02 lec p. 17] — $\{B\}$, $\{E\}$ (z_E down); displaced pose p. 19

Furuta ETS and result [p. 19]

$$ {}^B\mathbf{T}_E = t_z(L_0)\mathbf{R}_z(\theta_1)\,t_x(L_1)\mathbf{R}_x(\theta_2)\,t_z(-L_2)\mathbf{R}_y(\pi) $$

$$ {}^B\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\quad {}^1\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & L_1 \\ 0 & c_2 & -s_2 & 0 \\ 0 & s_2 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^2\mathbf{T}_E = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & -L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^B\mathbf{T}_E = \begin{bmatrix} -c_1 & -s_1c_2 & -s_1s_2 & L_1c_1 - L_2s_1s_2 \\ -s_1 & c_1c_2 & c_1s_2 & L_1s_1 + L_2c_1s_2 \\ 0 & s_2 & -c_2 & L_0 - L_2c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

L_0 post, L_1 arm, L_2 pendulum (m); θ_1 about z_B, θ_2 about the arm's x. $\mathbf{R}_y(\pi)$ flips z down.

WORKED Furuta numbers

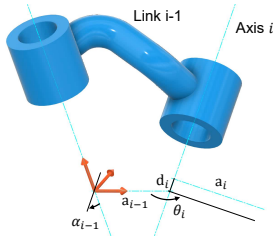
$L = (0.5, 0.4, 0.3)$, $\theta = (60, 45)°$:

$$ {}^B\mathbf{T}_E = \begin{bmatrix} -0.5 & -0.6124 & -0.6124 & 0.0163 \\ -0.8660 & 0.3536 & 0.3536 & 0.4525 \\ 0 & 0.7071 & -0.7071 & 0.2879 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

At $\theta = 0$: tip $(L_1, 0, L_0 - L_2)$. Workspace ([added]): sphere $x^2 + y^2 + (z - L_0)^2 = L_1^2 + L_2^2$, band $|z - L_0| \le L_2$. [checked]

Proximal DH (Craig) [p. 20–36]

Frame $\{i\}$ on joint axis i at the *near* end of link i; link 0 = base; joint i joins link $i-1$ to i ([p. 23]).



[W02 lec p. 28] — parameters between axes $i-1$ and i

Four parameters [p. 24–27, 30]

a_{i-1}	link length: $z_{i-1} \to z_i$ along x_{i-1}, ≥ 0	m
α_{i-1}	twist: $z_{i-1} \to z_i$ about x_{i-1}	rad
d_i	offset: $x_{i-1} \to x_i$ along z_i (variable if prismatic)	m
θ_i	angle: $x_{i-1} \to x_i$ about z_i (variable if revolute)	rad

Proximal link transform [p. 28, 35]

$$ {}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(\theta_i)t_z(d_i) $$

$$ = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

${}^0\mathbf{T}_n = {}^0\mathbf{T}_1 \cdots {}^{n-1}\mathbf{T}_n$. Operator order $\mathbf{R}_x\ t_x\ \mathbf{R}_z\ t_z$ (distal differs).

DERIVATION link transform

$$ \mathbf{R}_x(\alpha)t_x(a) \text{ (the } x\text{-shift is on the rotation axis)} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & c\alpha & -s\alpha & 0 \\ 0 & s\alpha & c\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, $$

$$ \mathbf{R}_z(\theta)t_z(d) = \begin{bmatrix} c\theta & -s\theta & 0 & 0 \\ s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}. $$

Product rotation rows: $(c\theta, -s\theta, 0)$, $(c\alpha s\theta, c\alpha c\theta, -s\alpha)$, $(s\alpha s\theta, s\alpha c\theta, c\alpha)$; translation $\mathbf{R}_x(\alpha)(0,0,d)^\top + (a,0,0)^\top = (a, -s\alpha d, c\alpha d)^\top$. [added]

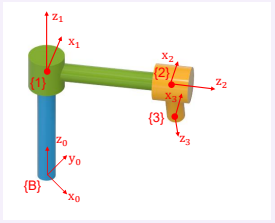
STEPS Proximal frame assignment [p. 29]

1. Joint axes; for axes $i, i{+}1$ find the common normal (or their intersection).
2. Origin of $\{i\}$ = where axis i meets that normal; z_i along axis i.
3. x_i along the normal towards axis $i{+}1$ (intersecting axes: $x_i \perp$ their plane, either sign); $y_i = z_i \times x_i$.
4. $\{0\} = \{1\}$ at $q_1 = 0$; $\{N\}$: origin, x_N free so most parameters vanish.
5. Read $a_{i-1}, \alpha_{i-1}, d_i, \theta_i$ from the box above.

TRAP

${}^0\mathbf{T}_n$ is the pose of the *last joint frame*, not the tool: add a row (planar) or a constant ${}^n\mathbf{T}_E$. Only $a_{i-1} \ge 0$; the others are signed.

WORKED Furuta DH table [p. 31–36]



[W02 lec p. 32] — frames $\{0\}..\{3\}$; $\{3\} \ne \{E\}$

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_0	$q_1 + 90°$
2	0	90°	L_1	q_2
3	0	90°	L_2	0

$$ {}^0\mathbf{T}_3 = \begin{bmatrix} -s_1c_2 & c_1 & -s_1s_2 & L_1c_1 - L_2s_1s_2 \\ c_1c_2 & s_1 & c_1s_2 & L_1s_1 + L_2c_1s_2 \\ s_2 & 0 & -c_2 & L_0 - L_2c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix},\quad {}^3\mathbf{T}_E = \mathbf{R}_z(-90°) $$

${}^0\mathbf{T}_3\ {}^3\mathbf{T}_E = {}^B\mathbf{T}_E$ (ETS) for all q_1, q_2 (slide 36 homework). ETS: 6 moves, no rules; DH: rules fix frames, tool needs a constant. [checked]

Simple robots: DH rows, ETS, ${}^0\mathbf{T}_E$ [added]

All $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$; $c_{12} = \cos(q_1{+}q_2)$. Figures: see Standard robots, p. 7.

RR	$(0,0,0,q_1)\ (L_1,0,0,q_2)\ (L_2,0,0,0)$
RRR	$(0,0,0,q_1)\ (L_1,0,0,q_2)\ (L_2,0,0,q_3)\ (L_3,0,0,0)$
RP	$(0,0,0,q_1{-}90°)\ (0,-90°,d_2,0)$; then ${}^2\mathbf{T}_E = \mathbf{R}_x(90°)\mathbf{R}_z(90°)$ (slider along z_2, tool x_E along it)
PR	$(0,0,d_1,0)\ \ (0,90°,0,q_2{+}90°)\ \ (L_2,0,0,0)$, with ${}^S\mathbf{T}_0 = \mathbf{R}_y(90°)\mathbf{R}_z(-90°)$ ($z_0 = x_S$ along the slide)
SCARA RRP	$(0,0,H,q_1)\ (a_1,0,0,q_2)\ (a_2,180°,d_3,0)$

ETS and ${}^0\mathbf{T}_E$ (equal to the DH product)

RR: $\mathbf{R}_z(q_1)t_x(L_1)\mathbf{R}_z(q_2)t_x(L_2)$

$$ \begin{bmatrix} c_{12} & -s_{12} & 0 & L_1c_1 + L_2c_{12} \\ s_{12} & c_{12} & 0 & L_1s_1 + L_2s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

L=(0.4, 0.3), q=(40, 50)°: tip (0.3064, 0.5571) m. **RRR:** $\mathbf{R}_z(q_1)t_x(L_1)\mathbf{R}_z(q_2)t_x(L_2)\mathbf{R}_z(q_3)t_x(L_3)$

$$ \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1c_1 + L_2c_{12} + L_3c_{123} \\ s_{123} & c_{123} & 0 & L_1s_1 + L_2s_{12} + L_3s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Numbers of p. 6–9: $\mathbf{R} = \begin{bmatrix} 0.9659 & -0.2588 \\ 0.2588 & 0.9659 \end{bmatrix}$, tip (0.7297, 0.6881); q_E = atan2(T_{21}, T_{11}) = 15°.

RP: $\mathbf{R}_z(q_1)t_x(d_2)$

$$ \begin{bmatrix} c_1 & -s_1 & 0 & d_2c_1 \\ s_1 & c_1 & 0 & d_2s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

q_1=35°, d_2=0.6: tip (0.4915, 0.3441) m. **PR:** $t_x(d_1)\mathbf{R}_z(q_2)t_x(L_2)$

$$ {}^S\mathbf{T}_E = \begin{bmatrix} c_2 & -s_2 & 0 & d_1 + L_2c_2 \\ s_2 & c_2 & 0 & L_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\quad {}^S\mathbf{T}_0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

d_1=0.3, q_2=60°, L_2=0.4: tip (0.5, 0.3464) m. Vertical slide: rows $(0,0,d_1,0)$ $(0,0,0,q_2)$ $(L_2,0,0,0)$.

SCARA: $t_z(H)\mathbf{R}_z(q_1)t_x(a_1)\mathbf{R}_z(q_2)t_x(a_2)t_z(-d_3)\mathbf{R}_x(180°)$

$$ \begin{bmatrix} c_{12} & s_{12} & 0 & a_1c_1 + a_2c_{12} \\ s_{12} & -c_{12} & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & -1 & H - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

H=0.5, a=(0.4, 0.3), q=(40, 50)°, d_3=0.15: tip (0.3064, 0.5571, 0.35) m. $\mathbf{R}_x(180°)$ points z_E down: positive d_3 lowers the tool. [checked]

Puma 560 and Franka Panda [added]

Not on the slides (photos only, [p. 10]); notebook Q1, Q2, Q5.

ALTERNATIVE APPROACHES

Puma-like ETS (L_b=1.0, L_u=0.5, L_f=0.3 m)

$t_z(L_b)\mathbf{R}_z(q_1)\mathbf{R}_y(q_2)t_z(L_u)$
$\mathbf{R}_y(q_3)t_z(L_f)\mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\mathbf{R}_z(q_6)$;
coincident axes: no translation.

proximal DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_b	q_1
2	0	−90°	0	q_2−90°
3	L_u	0	0	q_3+90°
4	0	90°	L_f	q_4
5	0	−90°	0	q_5
6	0	90°	0	q_6

Home: ${}^0\mathbf{T}_E = \begin{bmatrix} \mathbf{I}_3 & (0,0,1.8)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}$. $q = (30, 20, -40, 0, 0, 0)°$: wrist centre $(c_1\rho, s_1\rho, L_b + L_uc_2 + L_fc_{23})$, $\rho = L_us_2 + L_fs_{23}$ = (0.0592, 0.0342, 1.7518); $q_4..q_6$ do not move it. [checked]

Franka Panda proximal DH

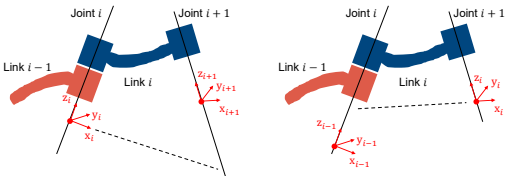
i	1	2	3	4	5	6	7
a_{i-1}	0	0	0	0.0825	−0.0825	0	0.088
α_{i-1}	0	−90°	90°	90°	−90°	90°	90°
d_i	0.333	0	0.316	0	0.384	0	0
θ_i	q_1	q_2	q_3	q_4	q_5	q_6	q_7

Flange $t_z(0.107)$. At $\mathbf{q}=0$: $\begin{bmatrix}1&0&0&0.088\\0&-1&0&0\\0&0&-1&0.926\\0&0&0&1\end{bmatrix}$. [checked]

SLIDE CHECK notebook cell 42, 35

Printed Panda ETS gives flange $(0.316, 0, 0.9945)$ at $\mathbf{q}=0$, not $(0.088, 0, 0.926)$: use the DH rows. Cell 35: `tz(-30)` should be `tz(-0.30)`. [checked]

Distal (standard) DH [p. 37–41]



[W02 lec p. 38] — proximal (left) vs distal (right) frame placement

ALTERNATIVE APPROACHES

Proximal (Craig) [p. 40]	Distal (standard) [p. 39–40]
Frame $\{i\}$ on axis i, near end of link i.	Frame $\{i-1\}$ on axis i, far end of link $i-1$.
a_{i-1}, α_{i-1} at x_{i-1}; d_i, θ_i at z_i.	a_i, α_i at x_i; d_i, θ_i at z_{i-1}.
$\mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(\theta_i)t_z(d_i)$	$\mathbf{R}_z(\theta_i)t_z(d_i)t_x(a_i)\mathbf{R}_x(\alpha_i)$

Distal link transform [p. 40]

$$ {}^{i-1}\mathbf{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Toolbox: `RevoluteDH` = distal; `RevoluteMDH/PrismaticMDH` = proximal (this unit).

WORKED proximal → distal [added]

Distal row i: same (θ_i, d_i), but (a_i, α_i) from the *next* proximal row; no tool row. RR: $(\theta, d, a, \alpha) = (q_1, 0, L_1, 0), (q_2, 0, L_2, 0)$. SCARA: $(q_1, H, a_1, 0), (q_2, 0, a_2, 180°), (0, d_3, 0, 0)$. Furuta: $(q_1{+}90°, L_0, 0, 90°), (q_2, L_1, 0, 90°), (0, L_2, 0, 0)$. All reproduce the proximal results. [checked]

TRAP

Mixing conventions: slide-40 numbers $(a, \alpha, d, \theta) = (0.2, 90°, 0.05, -20°)$ in the wrong formula move $\{2\}$ from $(0.1982, 0.0567, 0.30)$ to $(0.1970, 0.0347, 0.35)$ m. [checked]

Other conventions [p. 41–47]. Variants exist (Handbook of Robotics), plus screw/POE and URDF: state yours. *Why 4 not 6?* DH places frames by rule ($x_i \perp$ and intersecting z_{i+1}), removing 2 freedoms; price: frame need not lie on the link, and the common normal jumps for near-parallel axes (not continuous). Good model: complete, continuous, minimal [p. 44]. *URDF* [p. 45–47]: `<joint>` with `type`, parent, child, `origin xyz` (constant t), `axis xyz`: $\mathbf{R}_z(q_1)\,t_x(1)\mathbf{R}_z(q_2)\,t_x(1)$ for the planar example.

Notebook exercises, solved

Q1/Q2 Panda: $\mathbf{R}_x(\alpha)t_x(a)\mathbf{R}_z(q)t_z(d)$ for the 7 rows, then $t_z(0.107)$; flange $(0.088, 0, 0.926)$ at $\mathbf{q}=0$. [checked]

Q3 two frames: $\{B\}$ at $(3, 5, 2)$, $x_B = -z_A$, $z_B = y_A$, $y_B = -x_A$ $(= t(3,5,2)\mathbf{R}_y(90°)\mathbf{R}_x(-90°))$:

$$ {}^A\mathbf{T}_B = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 0 & 0 & 1 & 5 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Q4: the transform is the ordered product $E_1 E_2 \cdots E_m$ of the six elementary matrices.

Q5 Puma: $t_z(L_b)\mathbf{R}_z(q_1)\mathbf{R}_y(q_2)t_z(L_u)\mathbf{R}_y(q_3)t_z(L_f)\mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\mathbf{R}_z(q_6)$; ${}^0\mathbf{T}_1 = t_z(L_b)\mathbf{R}_z(q_1)$, ${}^1\mathbf{T}_2 = \mathbf{R}_y(q_2)$, ${}^2\mathbf{T}_3 = t_z(L_u)\mathbf{R}_y(q_3)$, ${}^3\mathbf{T}_4 = t_z(L_f)\mathbf{R}_z(q_4)$, ${}^4\mathbf{T}_5 = \mathbf{R}_y(q_5)$, ${}^5\mathbf{T}_6 = \mathbf{R}_z(q_6)$. [checked]

Q6 workspaces: Furuta: sphere zone above. Puma wrist centre: shell of radii 0.2–0.8 m about $(0, 0, 1.0)$ (2R annulus $|L_u - L_f|..L_u + L_f$ spun by q_1). [checked]

SLIDE CHECK notebook Q3 answer, cell 46

`ty(5) tx(3) Ry(90) Rx(90)` misses $t_z(2)$; last rotation must be $\mathbf{R}_x(-90°)$ (it gives origin $(3, 5, 0)$ and flips y_B, z_B). [checked]

TRAP W2: top five lost marks

(1) Multiplication order. (2) DH frame taken as tool. (3) Proximal vs distal mix. (4) Drawn arrow opposite to right-hand rule. (5) Inverse as $-\mathbf{t}$. Calculator in degrees vs radians.

EXAM-STYLE QUESTION Elbow arm: proximal DH table [marks not published]

Post L_0; joint 1 vertical; joint 2 horizontal axis at the post top; upper arm L_2; joint 3 parallel to 2 (elbow); forearm L_3 to the tool. At $q_2{=}q_3{=}0$ both links point along $+x_1$; $+q_2$ raises the arm. (a) Table incl. tool row. (b) ETS. (c) Tool point for $L_0, L_2, L_3 = 0.45, 0.40, 0.30$ m, $q = (35, 50, -30)°$.

SOLUTION (a) Axes 1,2 perpendicular: $\alpha_1 = 90°$; 2,3 parallel: $\alpha_2 = 0$. Rows: $(0, 0, L_0, q_1)$, $(0, 90°, 0, q_2)$, $(L_2, 0, 0, q_3)$, tool $(L_3, 0, 0, 0)$. (b) $t_z(L_0)\mathbf{R}_z(q_1)\mathbf{R}_x(90°)\mathbf{R}_z(q_2)t_x(L_2)\mathbf{R}_z(q_3)t_x(L_3)$. (c) $r = L_2c_2 + L_3c_{23} = 0.5390$, $z = L_0 + L_2s_2 + L_3s_{23} = 0.8590$; spin by q_1:

$$ \boxed{\mathbf{p} = (0.4415,\ 0.3092,\ 0.8590)\ \text{m}} $$ [checked]

Check: omitting the 90° twist leaves $z = 0.45$.

EXAM-STYLE QUESTION Camera on the wrist [marks not published]

${}^B\mathbf{T}_W = t(0.5, 0.2, 0.6)\mathbf{R}_z(90°)$, ${}^W\mathbf{T}_C = t(0.05, 0, 0.1)\mathbf{R}_y(90°)$ (m). An object is 0.8 m along z_C. Find it in $\{B\}$.

SOLUTION ${}^B\mathbf{P} = {}^B\mathbf{T}_W\,{}^W\mathbf{T}_C\,{}^C\mathbf{P}$, ${}^C\mathbf{P} = (0, 0, 0.8)^\top$. $\mathbf{R}_y(90°)$ sends $z \to x$: ${}^W\mathbf{P} = (0.85, 0, 0.10)$. $\mathbf{R}_z(90°)$: $(0, 0.85, 0.10)$, plus $(0.5, 0.2, 0.6)$:

$$ \boxed{{}^B\mathbf{P} = (0.50,\ 1.05,\ 0.70)\ \text{m}} $$ [checked]

Reversed order gives $(1.45, 0.2, -0.4)$: inner subscripts do not cancel.

EXAM-STYLE QUESTION Find the frame from the transform [added] [marks not published]

${}^A\mathbf{T}_B = \begin{bmatrix}0&0&1&0.4\\1&0&0&0.2\\0&1&0&0.5\\0&0&0&1\end{bmatrix}$. (a) Origin and axes of $\{B\}$; valid? (b) ETS. (c) ${}^A\mathbf{P}$ for ${}^B\mathbf{P} = (0, 0, 0.1)^\top$. (d) ${}^B\mathbf{T}_A$.

SOLUTION (a) Origin $(0.4, 0.2, 0.5)$; $\hat{x}_B = \hat{y}_A$, $\hat{y}_B = \hat{z}_A$, $\hat{z}_B = \hat{x}_A$; $\hat{y}_A \times \hat{z}_A = \hat{x}_A$ so $\det = +1$. (b) $t(0.4, 0.2, 0.5)\mathbf{R}_y(90°)\mathbf{R}_z(90°)$. (c) $(0.1, 0, 0) + (0.4, 0.2, 0.5) = (0.5, 0.2, 0.5)$ m. (d) $-\mathbf{R}^\top\mathbf{t} = -(0.2, 0.5, 0.4)^\top$:

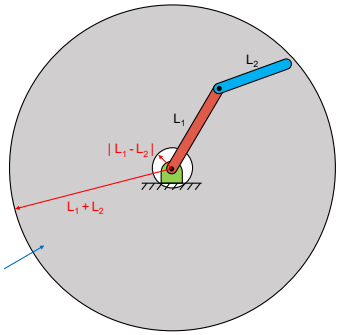
$$ \boxed{{}^B\mathbf{T}_A = \begin{bmatrix}0&1&0&-0.2\\0&0&1&-0.5\\1&0&0&-0.4\\0&0&0&1\end{bmatrix}} $$ [checked]

W3 Inverse kinematics and workspace

Which recipe, how many solutions, workspace [p. 7–20] [added]

robot (given)	method	#sol.	reachable workspace W_R
RP (x, y)	polar	1 (2 if $S<0$)	annulus $S_{\min} \le r \le S_{\max}$
PR (x, y)	$\sin\theta = y/L$	2	stadium: $D \times 2L$ + half-disc L each end
RR (x, y)	cosine rule, ±	2	annulus $\lvert L_1 - L_2\rvert \le r \le L_1 + L_2$ (L_1=L_2: disc)
RRR (x, y, φ)	wrist point, RR	2	disc $r \le \sum L_i$; hole if $2L_{\max} > \sum L_i$
RRR (x, y) only	–	∞	(1 DoF spare)
RRP (x, y, φ) / 3D	quadratic / polar	2	3D: shell $d_{3\min} \le \rho \le d_{3\max}$
RRR 3D (x, y, z)	top, side view	4	shell about shoulder, $\lvert L_3 - L_4\rvert \le \rho \le L_3 + L_4$
6-DoF, sph. wrist	wrist centre	8	–
general 6R	numerical	≤ 16	–

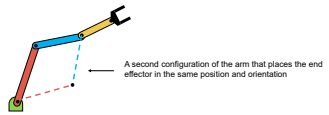
Reachable W_R: some orientation. **Dexterous** $W_D \subseteq W_R$: every orientation (empty for RR, RP, PR, 3D RRR; planar RRR: $L_3 + \lvert L_1 - L_2\rvert \le r \le L_1 + L_2 - L_3$). A full pose needs W_D, a position only needs W_R. Boundary: two branches merge, 1 solution ($\sin q_2 = 0$). Outside: none. Axis through target (x=y=0): θ_1 arbitrary. *Slide numbers:* $L = (1, 0.7)$: $0.3 \le r \le 1.7$; $(0, 0.2)$: $\cos q_2 = -1.036$, unreachable. [checked]
Planar RRR $L = (1, 0.7, 0.3)$: W_R $r \le 2.0$; W_D $0.6 \le r \le 1.4$. [checked]



[W03 lec p. 15] — RR, $L = (1, 0.7)$, no limits: annulus; W_D empty. With $q_{1,2} \in [0, \pi/2]$ (p. 16) a crescent with arcs $r = 1.7$, $r = 1.221$ and two radius-0.7 circles

STEPS Sketch a workspace (W13 Q2) [added]
1. Name frame and centre (shoulder if joint 1 is a pure spin).
2. Fold: min distance; stretch: max ($\lvert L_a - L_b\rvert$, $L_a + L_b$).
3. Spin about an axis through the centre: annulus → ring (planar) or shell (3D).
4. Draw two views (side, top), radii and centre height to scale.
5. State W_R, W_D, what a limit removes; reason: cosine rule.

W13 sample Q2 (workspace sketch) and **Q3** (3D RRR IK): solved on p. 42.



[W03 lec p. 17] — dashed arm: second configuration, same tool pose (elbow flip)

Choosing a branch [p. 18–19]: closest to current **q** (weighted; prefer moving small distal joints), obstacle-free (p. 19), within joint limits. List all branches *then* apply limits. IK is closed form (algebra/geometry, all solutions) or numerical (iterative, long chains).

TRAP
Giving one branch. List both, then say which you choose and why.

Closed-form recipe; RP and PR [p. 22–24]

STEPS Closed-form IK
1. Sketch, frame {0}, positive direction of each joint.
2. Write FK (position, plus orientation if a pose is given).
3. Isolate joints by geometry/algebra (top and side views; planar 2R triangles).
4. atan2, keep every ± branch; state when no solution ($\lvert\cos\rvert > 1$) and how many.
5. Check one branch by FK.

GIVEN tip (x, y) in {0} FIND S, q_1

RP: polar [p. 24]
$x = S\cos q_1,\ y = S\sin q_1 \implies S = \sqrt{x^2 + y^2},\ q_1 = \operatorname{atan2}(y, x)$
Use: slide writes $\tan^{-1}(y/x)$*: use atan2. If* $S < 0$ *allowed:* $(-S, q_1 + \pi)$.

Ex: $(0.3, 0.4)$: $S = 0.5$, $q_1 = 53.130°$; $(-0.3, 0.4)$: $126.870°$ ($\tan^{-1}$ gives $-53.13°$, wrong). [checked]

PR (slider d_1 along x_0, then link L at angle θ): $x = d_1 + L\cos\theta$, $y = L\sin\theta$ gives $\theta = \operatorname{atan2}(y, \pm\sqrt{L^2 - y^2})$, $d_1 = x - L\cos\theta$. None if $\lvert y\rvert > L$. *Ex:* $L = 0.5$, $(0.8, 0.3)$: $(36.870°, 0.4)$ or $(143.130°, 1.2$ m). [checked]

Planar RR: algebraic and geometric [p. 25–31]

GIVEN tip (x, y) in {0}, L_1, L_2 FIND q_1 (from x_0), q_2 (relative to link 1), both anticlockwise

DERIVATION q_2: square and add [p. 26,28]
$x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$. Square, add, use $c_1c_{12} + s_1s_{12} = \cos(q_{12} - q_1)$:

$$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$$

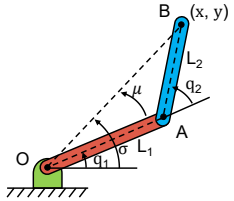
Planar RR [p. 28,30]

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2},\quad q_2 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2 q_2}, \cos q_2\right)$$

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2\sin q_2,\ L_1 + L_2\cos q_2)$$

Use: $\lvert\cos q_2\rvert > 1$: *unreachable. One* q_1 *per* q_2 *branch. General tool [p. 30]:* $A\cos\varphi + B\sin\varphi = C \Rightarrow \varphi = \operatorname{atan2}(C, \pm\sqrt{A^2 + B^2 - C^2}) - \operatorname{atan2}(A, B)$.

DERIVATION q_1: rotated vector [p. 30]
Expand c_{12}, s_{12}: $x = k_1c_1 - k_2s_1$, $y = k_2c_1 + k_1s_1$ with $k_1 = L_1 + L_2c_2$, $k_2 = L_2s_2$. (x, y) is (k_1, k_2) rotated by q_1, so polar angles subtract: $\operatorname{atan2}(y, x) = q_1 + \operatorname{atan2}(k_2, k_1)$.



[W03 lec p. 31] — triangle OAB: O base, A elbow, B tip; $\sigma = \operatorname{atan2}(y, x)$, μ at O

DERIVATION geometric: triangle OAB [p. 31]
Interior angle at A is $\pi - q_2$. Cosine rule: $x^2 + y^2 = L_1^2 + L_2^2 - 2L_1L_2\cos(\pi - q_2)$, same as above. Sine rule: $\sin\mu/L_2 = \sin q_2/\sqrt{x^2 + y^2}$; $q_1 = \sigma - \mu$. [added] μ has the sign of q_2; $\mu = \operatorname{atan2}(L_2\sin q_2, L_1 + L_2\cos q_2)$ gives both branches and equals the algebraic result.

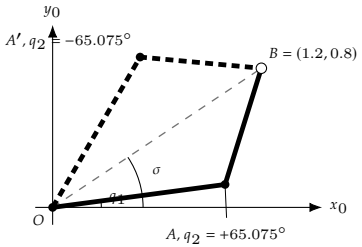
ALTERNATIVE APPROACHES

A. Algebraic	B. Geometric
Needs only FK.	Needs a clean figure.
$\cos q_2$ from $x^2 + y^2$.	Interior angle $\pi - q_2$.
$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(k_2, k_1)$.	$q_1 = \sigma - \mu$.
Any chain reducing to $A\cos\varphi + B\sin\varphi = C$.	Mirror about OB for the other branch. Full marks if the triangle is drawn and labelled.

SLIDE CHECK $\tan^{-1}$ on pp. 24, 30; W13 p. 7
Slides write $\tan^{-1}(y/x)$ (range ±90°): cannot tell $(1, 1)$ from $(-1, -1)$. Use atan2 [p. 29]: $\operatorname{atan2}(y, x) \in [-\pi, \pi]$, first argument is the sine-like one, $\operatorname{atan2}(-1, -1) = -135°$; calculator Pol$(x, y)$.

TRAP cosine rule
$x^2 + y^2 = L_1^2 + L_2^2 - 2L_1L_2\cos q_2$ is wrong (gives the supplement, 114.925° below). Spurious root: solving only the x-equation gives $-59.804°$, which has $y = -0.8$. [checked]

WORKED RR, both elbows, IK→FK
$L = (1.0, 0.7)$, target $(1.2, 0.8)$, degrees. $\cos q_2 = \dfrac{2.08 - 1 - 0.49}{1.4} = 0.421429$, $q_2 = \pm 65.075°$; $\sigma = \operatorname{atan2}(0.8, 1.2) = 33.690°$.
$+65.075°$: $\mu = \operatorname{atan2}(0.6348, 1.2950) = 26.114°$, $q_1 = \mathbf{7.576°}$.
$-65.075°$: $\mu = -26.114°$, $q_1 = \mathbf{59.804°}$.
FK ($q_2 > 0$): $x = \cos 7.576° + 0.7\cos 72.651° = 1.2000$ ✓, $y = 0.1318 + 0.6682 = 0.8000$ ✓. Other: $0.5030 + 0.6970$, $0.8643 - 0.0643$ ✓. [checked]



[drawn for this booklet] both branches reach B; solid $(7.576°, 65.075°)$, dashed $(59.804°, -65.075°)$.

Planar RRR and RRP with tool angle; RRP in 3D [added]

GIVEN (x, y), tool angle φ from x_0; L_1, L_2, L_3 FIND $\theta_{1,2,3}$ (relative)

STEPS Planar RRR (notebook Q10)
1. Wrist point $\mathbf{p}_W = (x - L_3\cos\varphi,\ y - L_3\sin\varphi)$.
2. RR to $\mathbf{p}_W$ (formulas above): 2 branches.
3. $\theta_3 = \varphi - \theta_1 - \theta_2$. FK check of (x, y, φ).

Ex: $L = (1, 0.7, 0.3)$, $(1.2, 0.9)$, $\varphi = 60°$: $\mathbf{p}_W = (1.0500, 0.6402)$; $\theta_2 = \pm 89.085°$; $\theta_1 = -3.319°$ or $66.061°$; $\theta_3 = -25.766°$ or $83.024°$. Without φ: ∞. [checked]

RRP planar (slider direction $\varphi = \theta_1 + \theta_2$, $d_3 \ge 0$): $\mathbf{p} = L_1(c_1, s_1) + d_3(c_\varphi, s_\varphi)$; $\lvert\mathbf{p} - d_3\mathbf{u}\rvert^2 = L_1^2$ gives

$$d_3 = p \pm \sqrt{p^2 - (x^2 + y^2 - L_1^2)},\quad p = xc_\varphi + ys_\varphi$$

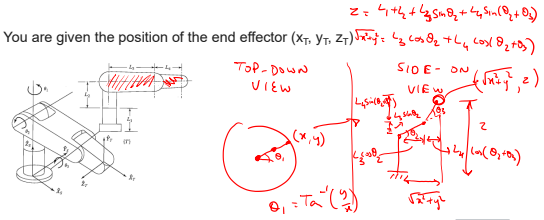
W3

$$\theta_1 = \operatorname{atan2}(y - d_3 s_\varphi,\ x - d_3 c_\varphi),\quad \theta_2 = \varphi - \theta_1$$

Ex: $L_1 = 0.5$, $(0.7, 0.4)$, $\varphi = 30°$: $p = 0.8062$, $d_3 = 1.3062$ or 0.3062; $(29.589°, 0.411°)$ for 0.3062. [checked]

RRP 3D (height H, boom tilted θ_2 from vertical): $\mathbf{p} = (c_1 d_3 s_2,\ s_1 d_3 s_2,\ H + d_3 c_2)$ gives $\theta_1 = \operatorname{atan2}(y, x)$, $d_3 = \sqrt{r^2 + z'^2}$, $\theta_2 = \operatorname{atan2}(r, z')$ ($r = \sqrt{x^2 + y^2}$, $z' = z - H$); second: $(\theta_1 + \pi, -\theta_2, d_3)$. *Ex:* $H = 0.5$, $(0.3, 0.4, 0.9)$: $53.130°$, $d_3 = 0.6403$, $\theta_2 = 51.340°$. [checked]

3D RRR arm (W13 Q3): top view, side view [p. W13 7]



[W13 lec p. 7] — lecturer's sketch: top view gives θ_1; side view is a planar RR in $(\sqrt{x^2 + y^2}, z)$

IK, four branches

FK: $x = c_1 r,\ y = s_1 r,\ z = L_1 + L_2 + L_3 s_2 + L_4 s_{23},\ r = L_3 c_2 + L_4 c_{23}$.

$$\theta_1 = \operatorname{atan2}(y, x)\ (r > 0) \text{ or } \theta_1 + \pi\ (r < 0)$$

$$z' = z - L_1 - L_2$$

$$\cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2 L_3 L_4}$$

$$\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4 \sin\theta_3,\ L_3 + L_4 \cos\theta_3)$$

Use: first axis vertical, rest in one plane. $4 = 2 \times 2$ solutions; none if $|\cos\theta_3| > 1$. $\theta_{2,3}$ positive lifts the arm. Numbers: $L = (0.6, 0.5, 0.4, 0.3)$, $\mathbf{p}_T = (0.4554, 0.2629, 1.4966)$ gives $(30, 20, 40)$, $(30, 54.047, -40)$, $(-150, 125.953, 40)$, $(-150, 160, -40)$ degrees; worked on p. 42. [checked]

Six-DoF arm with spherical wrist [p. 32–38]

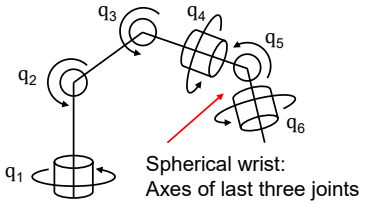
GIVEN pose ${}^0\mathbf{T}_E = \begin{bmatrix} \mathbf{R} & \mathbf{p}_E \\ \mathbf{0} & 1 \end{bmatrix}$ FIND $q_1 \dots q_6$

Wrist joints rotate about one point W, so: **position of** $W \to q_1, q_2, q_3$; **orientation** $\to q_4, q_5, q_6$ [p. 33–34].

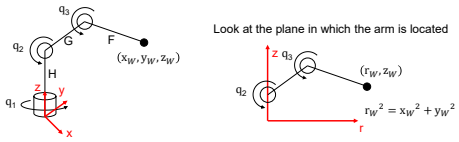
Wrist centre [added]; no slide formula

$${}^0\mathbf{p}_W = {}^0\mathbf{p}_E - d_6\, {}^0\mathbf{R}_E\, [0\ 0\ 1]^\top$$

Use: d_6 = wrist centre to tool point along $\hat{z}_E$. Skip if (x_W, y_W, z_W) is given.



[W03 lec p. 32] — 6-DoF arm; last three axes meet at the wrist centre



[W03 lec p. 36] — step 2: the arm plane, (r_W, z_W), $r_W^2 = x_W^2 + y_W^2$ (top view, p. 35: $q_1 = \operatorname{atan2}(y_W, x_W)$)

DERIVATION position joints [p. 35–37]

q_1: arm lies in the vertical plane through z_0 and W. Triangle joint2–joint3–W (sides G, F, $\sqrt{r_W^2 + (z_W - H)^2}$), interior angle $\pi - q_3$: $r_W^2 + (z_W - H)^2 = G^2 + F^2 + 2GF\cos q_3$.

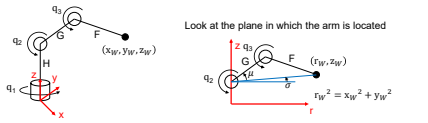
Position joints, signed angles

$$q_1 = \operatorname{atan2}(y_W, x_W),\quad \cos q_3 = \frac{r_W^2 + (z_W - H)^2 - F^2 - G^2}{2FG}$$

$$q_3 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2 q_3}, \cos q_3\right)$$

$$\sigma = \operatorname{atan2}(z_W - H, r_W),\quad \mu = \operatorname{atan2}(F\sin q_3,\ G + F\cos q_3),\quad q_2 = \sigma - \mu$$

Use: H joint-2 height; G, F link 2, link 3 to W. Branches: $q_1 + \pi$ (with $r_W \to -r_W$) and $\pm q_3$.



[W03 lec p. 37] — step 3: σ = elevation of the line to W, μ = angle between it and link G

SLIDE CHECK pp. 36–37

p. 36's $\cos(\pi - q_3)$ equals $\cos q_3$ on the right by the cosine rule; p. 37's $q_2 = \sigma + \mu$ holds only for a clockwise elbow bend. Use signed anticlockwise angles as above and fix the sign on your sketch. [checked]

WORKED position part

$H = 0.5$, $G = 0.4$, $F = 0.3$, $W = (0.4813, 0.4039, 0.5974)$. $q_1 = 40°$; $r_W = 0.6283$, $z_W - H = 0.0974$; $\cos q_3 = \dfrac{0.3948 + 0.0095 - 0.09 - 0.16}{0.24} = 0.6428$, $q_3 = \pm 50°$.
$-50°$: $\sigma = 8.811°$, $\mu = -21.189°$, $q_2 = \mathbf{30.0}°$. $+50°$: $\mu = 21.189°$, $q_2 = -12.378°$.
$q_1 + \pi = 220°$: $(q_2, q_3) = (150.0°, 50°)$, $(192.378°, -50°)$. 4 position solutions; with wrist flip $4 \times 2 = 8$. [checked]

DERIVATION wrist [p. 38]

${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\, {}^0\mathbf{R}_6$ (known once $q_{1..3}$ are). For a z, y, z wrist [added]: ${}^3\mathbf{R}_6 = R_z(q_4) R_y(q_5) R_z(q_6)$, so $r_{13} = c_4 s_5$, $r_{23} = s_4 s_5$, $r_{33} = c_5$, $r_{31} = -s_5 c_6$, $r_{32} = s_5 s_6$.

Z–Y–Z wrist angles [added]

$$q_5 = \operatorname{atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$

$$q_4 = \operatorname{atan2}(r_{23}, r_{13}),\quad q_6 = \operatorname{atan2}(r_{32}, -r_{31})$$

Use: r_{ij}: row i, column j of ${}^3\mathbf{R}_6$. Flip: $q_5 \to -q_5$, $q_4 + \pi$, $q_6 + \pi$ (take $-\sqrt{}$). $q_5 = 0$: singular, only $q_4 + q_6$ fixed. Other wrist types: derive from the same product.

WORKED wrist angles from ${}^3\mathbf{R}_6$

${}^3\mathbf{R}_6 = \begin{bmatrix} 0.4963 & -0.1684 & 0.8517 \\ -0.4014 & 0.8253 & 0.3971 \\ -0.7698 & -0.5390 & 0.3420 \end{bmatrix}$: $q_5 = \operatorname{atan2}(0.9397, 0.3420) = 70°$, $q_4 = \operatorname{atan2}(0.3971, 0.8517) = 25°$, $q_6 = \operatorname{atan2}(-0.5390, 0.7698) = -35°$. Flip $(-155°, -70°, 145°)$ gives the same matrix. [checked]

STEPS Decoupled 6-DoF IK (W13-style)

1. ${}^0\mathbf{p}_W = {}^0\mathbf{p}_E - d_6\, {}^0\mathbf{R}_E [0\ 0\ 1]^\top$.
2. $q_1 = \operatorname{atan2}(y_W, x_W)$ (also $+\pi$).
3. $\cos q_3$, $\pm q_3$, $q_2 = \sigma - \mu$.
4. ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_6$; $q_{4,5,6}$ as Euler angles (both q_5 signs).
5. $2 \times 2 \times 2 = 8$ (PUMA-type). FK check.

Closed form needs three neighbouring axes meeting at a point (Pieper) [added], [p. 55–56]; the slides give no PUMA table in W3.

Numerical IK [p. 40–54]

Newton–Raphson [p. 45–52]

Goal: minimise $G(\mathbf{q}) = (x^* - x)^2 + (y^* - y)^2$ [p. 41–42]. Scalar root: $q^{k+1} = q^k - G(q^k)/G'(q^k)$, stop when $|G(q^{k+1}) - G(q^k)|/|G(q^k)| \le \epsilon$. For IK use the residual [p. 50]:

$$\mathbf{e} = \mathbf{x}^* - \mathcal{F}(\mathbf{q}^i),\qquad \mathbf{q}^{i+1} = \mathbf{q}^i + \mathbf{J}^\dagger(\mathbf{q}^i)\,\mathbf{e},\qquad \mathbf{J} = \partial\mathcal{F}/\partial\mathbf{q} \in \mathbb{R}^{m \times n}$$

Use: $\mathbf{J}$ *in rad.* 2×2 *inverse:* $\frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. *Planar 2R:* $\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, $\det\mathbf{J} = L_1 L_2 \sin q_2$.

Which inverse? [p. 51]

shape	$\mathbf{J}^\dagger$	meaning
square $n = m$	$\mathbf{J}^{-1}$	exact
fat $n > m$	$\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$	smallest $\Delta\mathbf{q}$
tall $n < m$	$(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$	least squares

STEPS One Newton iteration

1. FK; $\mathbf{e} = \mathbf{x}^* - \mathcal{F}$; stop if $|\mathbf{e}| \le \epsilon$.
2. $\mathbf{J}(\mathbf{q}^i)$ in full, radians.
3. $\Delta\mathbf{q} = \mathbf{J}^\dagger\mathbf{e}$; $\mathbf{q}^{i+1} = \mathbf{q}^i + \Delta\mathbf{q}$; confirm $|\mathbf{e}|$ fell.

WORKED one step, RR

$L = (1.0, 0.7)$, target $(1.2, 0.8)$, $\mathbf{q}^0 = (0.5, 0.5)$: FK $= (1.2558, 1.0685)$, $\mathbf{e} = (-0.0558, -0.2685)$. $\mathbf{J} = \begin{bmatrix} -1.0685 & -0.5890 \\ 1.2558 & 0.3782 \end{bmatrix}$, $\det = 0.3355$.
$\Delta\mathbf{q} = \frac{1}{0.3355}\begin{bmatrix} 0.3782 & 0.5890 \\ -1.2558 & -1.0685 \end{bmatrix}\mathbf{e} = (-0.5341, 1.0635)$; $\mathbf{q}^1 = (-0.0341, 1.5635)$; $|\mathbf{e}|$: $0.2742 \to 0.2181$; converges to $(7.576°, 65.075°)$ in 5 steps. [checked]
Fat: planar RRR, $\mathbf{q} = (0.3, 0.5, 0.4)$, $\mathbf{e} = (0.05, -0.02)$: $\Delta\mathbf{q} = (0.0186, -0.0740, -0.0437)$, $\mathbf{J}\Delta\mathbf{q} = \mathbf{e}$ ✓. [checked]

TRAP

$\mathbf{J}^\top$ for $\mathbf{J}^{-1}$ is gradient descent (needs a step size). Degrees in $\mathbf{J}$. For tall $\mathbf{J}$, $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$ is wrong.

Why numerical IK fails (notebook Q4–Q6): (1) target outside W_R ($(2, 0.5)$, $L = (1, 0.7)$: stops 0.362 m short); (2) local minimum, or blow-up at $\det\mathbf{J} = 0$ ($q_2 = 0$); (3) result depends on the initial guess. [checked] Closed form: all solutions, fast, hard for long chains; numerical: one solution near the guess, suits long chains [p. 54].

SLIDE CHECK p. 54

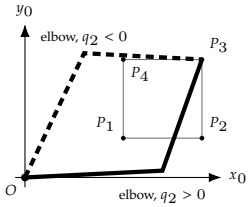
"unique solution a priori" contradicts pp. 17, 20; read "complete set of solutions".

Scenario: corners of a square; edge cases [added]

GIVEN RR, $L = (0.7, 0.6)$; corners $(0.5, 0.2)$, $(0.9, 0.2)$, $(0.9, 0.6)$, $(0.5, 0.6)$
FIND (q_1, q_2) per corner, one branch throughout

corner	$\cos q_2$	$q_2 > 0$: (q_1, q_2)	$q_2 < 0$: (q_1, q_2)
P_1	-0.6667	$(-34.34°, 131.81°)$	$(77.95°, -131.81°)$
P_2	0.0000	$(-28.07°, 90.00°)$	$(53.13°, -90.00°)$
P_3	0.3810	$(2.83°, 67.61°)$	$(64.55°, -67.61°)$
P_4	-0.2857	$(2.79°, 106.60°)$	$(97.60°, -106.60°)$

$r \in [0.538, 1.082] \subset [0.1, 1.3]$: reachable. $|q_2| \in [67.6°, 131.8°]$ never hits 0 or 180°, so no singularity and no branch switch; choose by start pose, limits ($q_2 \in [0, 150°] \Rightarrow q_2 > 0$) or obstacle. [checked]



[drawn for this booklet] arm at P_3, both branches.

RR target	result
$r > L_1+L_2$ or $r < \|L_1-L_2\|$	$\|\cos q_2\| > 1$: none
$r = L_1 + L_2$	$q_2 = 0$, 1 solution ($(1,0) \to (0,0)$)
$r = \|L_1 - L_2\|$	$q_2 = \pm 180°$, 1 solution
$L_1 = L_2$, target = base	q_1 arbitrary: ∞

Why (explain): at the boundary the branches coincide and $\det \mathbf{J} = L_1 L_2 \sin q_2 = 0$: a small target move needs a huge joint move; Newton steps blow up there.

EXAM-STYLE QUESTION SCARA RRP with tool yaw [marks assumed 6]

Column height $H = 0.40$, $L_1 = 0.35$, $L_2 = 0.25$ m, vertical axes q_1, q_2, q_4, slider d_3 down. Tool at $(0.30, 0.25, 0.28)$, yaw 60°. (a) Derive all branches and the existence condition. (b) Evaluate. (c) How many solutions?

SOLUTION (a) $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$, $z = H - d_3$, $\varphi = q_1 + q_2 + q_4$: $d_3 = H - z$, RR for q_1, q_2, $q_4 = \varphi - q_1 - q_2$. Exists if $|L_1 - L_2| \le r \le L_1 + L_2$, $0 \le d_3 \le$ stroke.

(b) $r = 0.3905 \in [0.10, 0.60]$ ✓. $\cos q_2 = \dfrac{0.1525 - 0.1225 - 0.0625}{0.175} = -0.1857$, $q_2 = \pm 100.70°$; $\operatorname{atan2}(0.25, 0.30) = 39.806°$.

$+100.70°$: $\mu = \operatorname{atan2}(0.2457, 0.3036) = 38.980°$, $q_1 = 0.826°$, $q_4 = -41.53°$.

$-100.70°$: $q_1 = 78.785°$, $q_4 = 81.917°$. $d_3 =$ **0.12** m both.

(c) Two (elbow); d_3, q_4 follow; one on the outer boundary, none beyond. FK ✓. [checked]

EXAM-STYLE QUESTION Joint limits choose the branch [marks assumed 4]

RR, $L_1 = 0.6$, $L_2 = 0.45$, $q_2 \in [0°, 150°]$. Reach $(0.70, -0.30)$: find (q_1, q_2); why is the other solution rejected?

SOLUTION $\cos q_2 = \dfrac{0.58 - 0.36 - 0.2025}{0.54} = 0.03241$, $q_2 = \pm 88.143°$; $\operatorname{atan2}(-0.3, 0.7) = -23.20°$.

$+88.143°$: $\mu = \operatorname{atan2}(0.4498, 0.6146) = 36.20°$, $q_1 = -59.40°$.

$-88.143°$: $q_1 = 13.00°$.

Only $q_2 = +88.143°$ is in range: $(-59.40°, 88.14°)$. List all branches, then apply limits. FK ✓. [checked]

EXAM-STYLE QUESTION Spot the errors [marks assumed 4]

RR ($L_1 = 0.5$, $L_2 = 0.4$), target $(-0.45, 0.30)$: "$\cos q_2 = -0.29375$, $q_2 = \cos^{-1}(\cdot) = 107.08°$; $q_1 = \tan^{-1}(y/x) - \operatorname{atan2}(L_2 \sin q_2, L_1 + L_2 \cos q_2) = -78.68°$. Unique." Find the errors; correct q_1 for $q_2 = +107.08°$.

SOLUTION (1) $\tan^{-1}(0.30/-0.45) = -33.69°$ loses the quadrant; $\operatorname{atan2}(0.30, -0.45) = 146.31°$. (2) $\cos^{-1}$ gives one sign; $q_2 = -107.08°$ also works, not unique. The student's pose reaches $(0.45, -0.30)$.

$\mu = \operatorname{atan2}(0.4 \sin 107.08°, 0.5 + 0.4 \cos 107.08°) = 44.99°$: $q_1 =$ **101.32**°; other branch $q_1 = 191.30°$. FK ✓. [checked]

W4 Parallel manipulators: mobility, loop closure, four-bar, delta

Serial vs parallel; G–K reminder

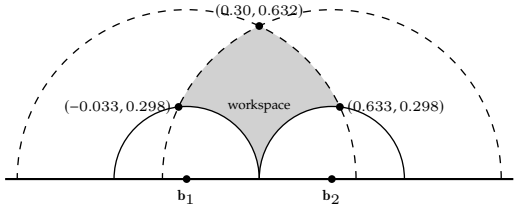
Serial vs parallel [p. 74]

	Serial	Parallel
Chains	open	closed
Joints	all active	active + passive
FK; IK	simple; complex	**complex**; **simple**
Joint errors	accumulate	do not
Work volume	large	small
Payload/weight; inertia	low; high	very high; low

G–K: $M = D(N - J - 1) + \sum F_i$; $D = 3$ planar, 6 spatial; N counts the ground; R, P: 1; U: 2; S: 3. Full recipe and joint table: W1, p. 9.

TRAP
"Parallel FK is easy": wrong way round. Planar mechanism with $D = 6$ (slider-crank gives −2, not 1). [checked]

Workspace [p. 21–23]: reachable (any orientation), dexterous (all orientations), constant-orientation. Parallel: each leg's $[\ell_{min}, \ell_{max}]$ gives an annulus/shell about its base pivot; the workspace is their **intersection**. Planar 2 legs, pivots $(0,0), (0.6,0)$, $0.3 \le \ell \le 0.7$: apex $(0.30, 0.632)$, area $y \ge 0$ about 0.175 m^2. [checked]



[drawn for this booklet] solid: ℓ_{min}; dashed: ℓ_{max}; mirror below the axis if collisions ignored.

Extended Grübler–Kutzbach [p. 35–43]

Extended G–K [p. 35]

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M, \qquad R_C = \sum_{i} F_i - \sum_{k=1}^{L} u_k$$

R_C: over-constraints (constraints repeated by several joints; add back).
R_M: local mobility (internal DoF that do not move the output; subtract).
L loops; u_k = platform DoF loop k allows (union, shared DoF once).

STEPS extended count
1. Plain G–K; write the number.
2. R_M: a rod with S joints at both ends spins about its axis: +1 each (U ends: 0).
3. R_C: per loop list the platform DoF allowed; take the union; $R_C = \sum F_i - \sum u_k$.
4. M = plain $+R_C - R_M$. Check against the motion you expect.

Plain vs extended

Mechanism	D	N	J	$\sum F$	plain	R_C	R_M	M
3-PRR [p. 27]	3	8	9	9	3	0	0	3
Simplest delta [p. 30]	6	8	9	15	3	0	0	3
Common delta [p. 33]	6	11	15	39	9	0	6	3
4RRR [p. 38]	6	10	12	12	−6	7	0	1
3-RPS [added]	6	8	9	15	3	0	0	3
3-SPS [added]	6	8	9	21	9	0	3	6
Four-bar, slider-crank [added]	3	4	4	4	1	0	0	1
Planar five-bar [added]	3	5	5	5	2	0	0	2
Stewart 6-UPS [added]	6	14	18	36	6	0	0	6

WORKED counting bodies and joints [checked]
3-PRR: ground, platform, 3 cylinders, 3 pistons ($N = 8$); 3R + 3P + 3R ($J = 9$). Simplest delta: base, 3 arms, 3 rods, platform ($N = 8$); 3R + 3U + 3U ($J = 9$), $\sum F = 3(1+2+2)$. Common delta: base, 3 arms, 6 parallelogram bars, platform ($N = 11$); 3R + 12S ($J = 15$), $\sum F = 3(1+4\cdot3) = 39$. Plain $6(11-15-1)+39 = 9$, but the platform only translates: 6 bars spin about themselves, $R_M = 6$, $R_C = 0$: $\mathbf{M = 3}$. Stewart: ground, platform, 6×2 leg bodies ($N = 14$); 6U + 6P + 6S ($J = 18$). 4RRR: legs 4×2 links, 4 legs $\times$ 3R. Loop 1 allows $\{T_y, T_z, R_x\}$, loop 2 $\{T_x, T_z, R_y\}$ [p. 39–41]: union has 5, so $R_C = 12 - 5 = 7$, $M = -6 + 7 - 0 = 1$ (platform slides along z only).

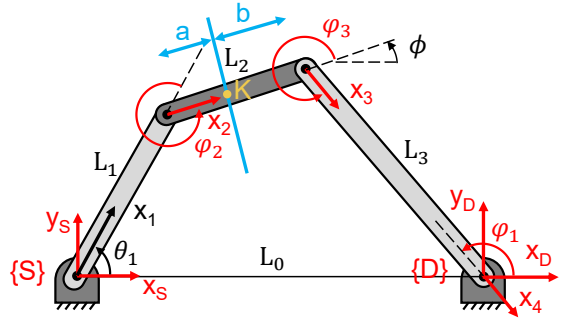
SLIDE CHECK slides 29, 35–36
Handwritten $D(N - J + 1)$ on slide 29: wrong; the printed equation and all examples use -1 (the $+1$ would give 15 for the simplest delta). For the delta the slides just state $R_C = 0$; the u_k formula does not give that, so argue "no repeated constraint". [checked]

TRAP
Treating G–K as exact: it assumes independent constraints. Equal parallel links (3 hanging links, $N = 5$, $J = 6$) give $M = 0$ but the body really moves.

Loop closure and four-bar [p. 45–58, 71]

GIVEN geometry and active joints $\boldsymbol{\theta}$ (n of them) FIND m equations $\eta_i(\mathbf{q}) = 0$ for the passive joints $\boldsymbol{\varphi}$

STEPS loop closure [p. 49]
1. Cut into serial chains; pick the cut with the fewest passive unknowns.
2. For each chain write the pose of the cut point.
3. Equate the two descriptions at the cut.
4. Count: n active, m passive, m equations. Four-bar: cut at K $\Rightarrow n = 1$, $m = 3$; cut link 2 (U, V) $\Rightarrow m = 1$. Stewart: 6 active, 30 passive, not hand-solvable.



[W04 lec p. 71] — four-bar: S left pivot, D at $(L_0, 0)$; K on link 2, a, b its distances to the ends, $L_2 = a + b$, output angle ϕ

Four-bar, cut link 2 at K [p. 54]

$$L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2) = L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3)$$
$$L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2) = L_3\sin\varphi_1 + b\sin(\varphi_1+\varphi_3)$$
$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

θ_1 active (crank angle at S); φ_1 at D to link 3; φ_2, φ_3 the coupler joints; all absolute, CCW, rad. Third equation: B→K points back along the coupler ($+\pi$). [checked]

Freudenstein equation, cut both ends [p. 57]

$$\eta_1 = (L_1\cos\theta_1 - L_0 - L_3\cos\varphi_1)^2 + (L_1\sin\theta_1 - L_3\sin\varphi_1)^2 - L_2^2 = 0$$

$n = m = 1$. Given θ_1: two roots φ_1 = two assembly modes. With $u = L_1\cos\theta_1 - L_0$, $v = L_1\sin\theta_1$, $d^2 = u^2 + v^2$:

$$k = \frac{d^2 + L_3^2 - L_2^2}{2L_3}, \qquad \varphi_1 = \operatorname{atan2}(v,u) \pm \arccos\frac{k}{d}$$

$$\varphi_2 = \delta - \theta_1, \qquad \varphi_3 = \theta_1 + \varphi_2 - \varphi_1 - \pi$$

$|k/d| > 1$: no assembly. ($\delta = \theta_1 + \varphi_2$ = coupler direction.)

Four-bar IK (easy): K and ϕ given [p. 68–73]

GIVEN K= (x, y) in S, output angle ϕ; L_0, L_1, L_3, a, b FIND $\theta_1, \varphi_1, \varphi_2, \varphi_3$

Easy because each leg is a serial chain with known end pose: solve legs separately, one answer per leg.

DERIVATION [p. 70–72]
Left S→K: $x = L_1\cos\theta_1 + a\cos\phi$, $y = L_1\sin\theta_1 + a\sin\phi$ (using $\theta_1 + \varphi_2 = \phi$). Right D→K: $\varphi_1 + \varphi_3 = \phi + \pi$, $\cos(\varphi_1+\varphi_3) = -\cos\phi$: $L_3\cos\varphi_1 = x - L_0 + b\cos\phi$, $L_3\sin\varphi_1 = y + b\sin\phi$.

STEPS four-bar IK
1. $\theta_1 = \operatorname{atan2}(y - a\sin\phi,\ x - a\cos\phi)$.
2. $\varphi_2 = \phi - \theta_1$ (add 2π if positive wanted).
3. $\varphi_1 = \operatorname{atan2}(y + b\sin\phi,\ x + b\cos\phi - L_0)$.
4. $\varphi_3 = \phi + \pi - \varphi_1 \pmod{2\pi}$.
5. Check: $|\mathbf{A}| = L_1$, $|\mathbf{B} - \mathbf{D}| = L_3$, and $\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$.

WORKED IK numbers

$L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$, $a = 0.2$, $b = 0.25$ m; K= $(0.3710, 0.4678)$, $\phi = -0.600°$. $\theta_1 = \text{atan2}(0.4699, 0.1710) = 70.0°$; $\varphi_2 = 289.4°$ ($\equiv -70.6°$); $\varphi_1 = \text{atan2}(0.4652, -0.3790) = 129.17°$; $\varphi_3 = -0.6 + 180 - 129.175 = 50.2°$. Check: $359.4° = 129.2 + 50.2 + 180$. [checked]

SLIDE CHECK W13 p. 11 (Q5)

Printed $\varphi_3 = \varphi_1 - \phi + \pi$ puts K at $(0.669, 0.711)$, not $(0.371, 0.468)$. W4 p. 72 ($\phi = \varphi_1 + \varphi_3 - \pi$) is right: use $\varphi_1 + \varphi_3 = \phi + \pi$. [checked]

TRAP

Radians vs degrees; atan (not atan2) when $x + b\cos\phi - L_0 < 0$ (second quadrant); the printed φ_3.

W13 sample Q5 (four-bar IK), **Q6** (planar mobility), **Q7** (delta mobility): solved on p. 42.

Four-bar FK (hard) [p. 59–67]

GIVEN θ_1 and geometry FIND φ_i, then the output pose; up to 2 assembly modes

Steps [p. 60]: solve the non-linear loop equations for φ_i, then compute the pose. No general closed form for other parallel robots; numerics [p. 67].

ALTERNATIVE APPROACHES

A. Slide route (algebra) [p. 61–66]

Eliminate angles by squaring and adding: two equations $A_i\cos\delta + B_i\sin\delta + C_i = 0$; Cramer or half-angle; then $\varphi_2 = \delta - \theta_1$, φ_1 from Freudenstein.

B. Geometry (circle intersection) [added]

B on circle L_2 about A *and* circle L_3 about D; two intersections = two modes. Fastest by hand: use this unless told otherwise.

Route B: geometric FK [added]

$$\mathbf{A} = L_1\begin{bmatrix}\cos\theta_1\\ \sin\theta_1\end{bmatrix},\ \mathbf{D} = \begin{bmatrix}L_0\\ 0\end{bmatrix},\ d = \|\mathbf{D}-\mathbf{A}\|,\ \hat{\mathbf{u}} = \frac{\mathbf{D}-\mathbf{A}}{d},\ \hat{\mathbf{n}} = \begin{bmatrix}-u_y\\ u_x\end{bmatrix}$$

$$x_a = \frac{L_2^2 - L_3^2 + d^2}{2d},\quad h = \sqrt{L_2^2 - x_a^2},\quad \mathbf{B} = \mathbf{A} + x_a\hat{\mathbf{u}} \pm h\hat{\mathbf{n}}$$

$$\phi = \text{atan2}(B_y - A_y, B_x - A_x),\quad \varphi_2 = \phi - \theta_1,\quad \mathbf{K} = \mathbf{A} + a(\cos\phi, \sin\phi)$$

$$\varphi_1 = \text{atan2}(B_y, B_x - L_0),\quad \varphi_3 = \phi + \pi - \varphi_1 \pmod{2\pi}$$

+: B left of $\mathbf{A}\to\mathbf{D}$ (above); −: below. Needs $|L_2 - L_3| \le d \le L_2 + L_3$.

DERIVATION slide route [p. 61–65]

x, y = K in S; $\delta = \theta_1 + \varphi_2$; $\cos(\varphi_1 + \varphi_3) = -\cos\delta$. Right chain: $L_3\cos\varphi_1 = (x - L_0) + b\cos\delta$, $L_3\sin\varphi_1 = y + b\sin\delta$; square, add: $A_1 = x - L_0$, $B_1 = y$, $C_1 = \frac{1}{2b}[(x-L_0)^2 + y^2 + b^2 - L_3^2]$. Left chain: $x - a\cos\delta = L_1\cos\theta_1$, $y - a\sin\delta = L_1\sin\theta_1$; square, add: $A_2 = x$, $B_2 = y$, $C_2 = \frac{1}{2a}[L_1^2 - a^2 - x^2 - y^2]$. Cramer, $\Delta = A_1B_2 - A_2B_1$:

$$\cos\delta = \frac{B_1C_2 - B_2C_1}{\Delta},\quad \sin\delta = \frac{A_2C_1 - A_1C_2}{\Delta},\quad \delta = \text{atan2}(\sin\delta, \cos\delta)$$

Half-angle $t = \tan\frac{\delta}{2}$: $(C_i - A_i)t^2 + 2B_it + (A_i + C_i) = 0$; eliminating t^2:

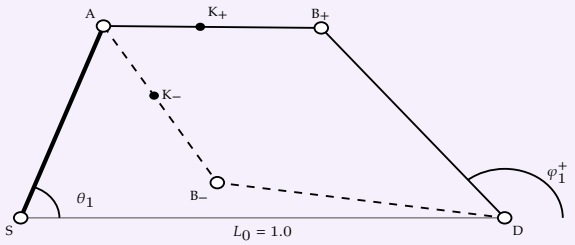
$$t = \frac{A_2C_1 - A_1C_2}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)},\qquad \delta = 2\tan^{-1}t \quad \text{[checked]}$$

SLIDE CHECK slide 65

Printed $\delta = \tan^{-1}\frac{A_1C_2 - A_2C_1}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}$: sign flipped and factor 2 missing (mode + test: slide 0.30°, true −0.60°). Use Cramer/atan2. Slide's quartic holds for every K on the coupler curve and has no θ_1. [checked]

WORKED four-bar FK, $\theta_1 = 70°$, both modes

Same lengths as the IK example; $L_2 = 0.45$. $\mathbf{A} = (0.1710, 0.4698)$; $d^2 = 1.25 - \cos 70° = 0.9080$, $d = 0.9529$; $\hat{\mathbf{u}} = (0.8700, -0.4931)$, $\hat{\mathbf{n}} = (0.4931, 0.8700)$; $x_a = \frac{0.2025 - 0.36 + 0.9080}{1.9058} = 0.3938$, $h = \sqrt{0.2025 - 0.1551} = 0.2178$.



[drawn for this booklet] solid: mode +; dashed: mode −.

WORKED four-bar FK results [checked]

	mode +	mode −
B (m)	$(0.6210, 0.4651)$	$(0.4062, 0.0862)$
ϕ	$-0.600°$	$-58.486°$
φ_2	$289.400°$	$231.514°$
φ_1	$129.175°$	$171.739°$
φ_3	$50.225°$	$309.775°$
K (m)	$(0.3710, 0.4678)$	$(0.2756, 0.2993)$

Freudenstein route: $u = -0.8290$, $v = 0.4698$, $k = 0.8879$, $\arccos(k/d) = 21.282°$, $\text{atan2}(v, u) = 150.457°$: $\varphi_1 = 129.175°$ or $171.739°$. Closure (mode +): $0.1710 + 0.2\cos 359.4° = 0.3710 = 1 - 0.3790 + 0.25\cos 179.4°$ ✓.

STEPS crank range and full turn [added]

1. Closes iff $|L_2 - L_3| \le d \le L_2 + L_3$, $d^2 = L_0^2 + L_1^2 - 2L_0L_1\cos\theta_1$.
2. $\frac{L_0^2 + L_1^2 - (L_2 + L_3)^2}{2L_0L_1} \le \cos\theta_1 \le \frac{L_0^2 + L_1^2 - (L_2 - L_3)^2}{2L_0L_1}$; if it contains $[-1, 1]$ the crank turns fully, else it rocks (modes merge at the limits).
3. Grashof: shortest + longest ≤ other two ⇒ link next to the shortest turns fully.

Example above: $\cos\theta_1 \ge 0.1475$, $|\theta_1| \le 81.52°$; $0.45 + 1.0 > 0.5 + 0.6$: non-Grashof, crank rocks. [checked]

Delta robot IK (physical branch) [added]

GIVEN platform centre $\mathbf{p}$ in B (z up, $p_z < 0$); R_b, R_p, upper arm L_a, rod L_r FIND $\theta_1, \theta_2, \theta_3$

Platform stays level: solve leg i alone, rod joint $\mathbf{p} + R_p\hat{\mathbf{u}}_i$.

DERIVATION leg i, $\gamma_i = 0°, 120°, 240°$

$\hat{\mathbf{u}}_i = (\cos\gamma_i, \sin\gamma_i, 0)$; θ_i measured *down* from horizontal; elbow $\mathbf{E}_i = R_b\hat{\mathbf{u}}_i + L_a(\cos\theta_i\hat{\mathbf{u}}_i - \sin\theta_i\hat{\mathbf{z}})$. With $\mathbf{d}_i = \mathbf{p} + (R_p - R_b)\hat{\mathbf{u}}_i$, rod length L_r:

$$L_a^2 + \|\mathbf{d}_i\|^2 - 2L_a(d_u\cos\theta_i - d_z\sin\theta_i) = L_r^2,\quad d_u = \mathbf{d}_i\cdot\hat{\mathbf{u}}_i,\ d_z = \mathbf{d}_i\cdot\hat{\mathbf{z}}$$

$A = d_u$, $B = -d_z$, $K = \frac{\|\mathbf{d}_i\|^2 + L_a^2 - L_r^2}{2L_a}$, $\rho = \sqrt{A^2 + B^2}$:

$$A\cos\theta_i + B\sin\theta_i = K \ \Rightarrow\ \theta_i = \text{atan2}(B, A) - \arccos(K/\rho),\ \ |K| \le \rho$$

Physical root: the minus root. The + root folds the arm back above the base (elbow inside base circle, $z > 0$). Pick elbow outside the base circle and below it: $\|\mathbf{E}_{xy}\| > R_b$, $E_z < 0$.

WORKED delta numbers

$R_b = 0.20$, $R_p = 0.05$, $L_a = 0.20$, $L_r = 0.45$ m; $\mathbf{p} = (0.03, -0.02, -0.38)$. Leg 1: $\mathbf{d}_1 = (-0.12, -0.02, -0.38)$, $\|\mathbf{d}_1\|^2 = 0.1592$, $K = -0.00825$, $A = -0.12$, $B = 0.38$, $\rho = 0.3985$; $\text{atan2}(B, A) = 107.526°$, $\arccos(-0.0207) = 91.186°$: $\theta_1 = 16.339°$ (other root 198.7° unphysical). $\theta_2 = 30.871°$, $\theta_3 = 22.996°$; rods all 0.45 m. Home $\mathbf{p} = (0, 0, -0.4)$: $\theta_i = 27.277°$. [checked]

Five-bar IK (2 DoF) [added]

GIVEN tip $\mathbf{P}$; pivots $\mathbf{O}_1, \mathbf{O}_2$; proximal L_p, distal L_d FIND θ_1, θ_2

Each side is a 2R problem: $\mathbf{v}_j = \mathbf{P} - \mathbf{O}_j$, $r_j = \|\mathbf{v}_j\|$, cosine rule in $\mathbf{O}_j\mathbf{E}_j\mathbf{P}$:

$$\cos\beta_j = \frac{r_j^2 + L_p^2 - L_d^2}{2r_jL_p},\qquad \theta_j = \text{atan2}(v_{j,y}, v_{j,x}) \pm \beta_j$$

Reachable iff $|L_p - L_d| \le r_j \le L_p + L_d$; 4 configurations, "elbows outward": $+\beta_1, -\beta_2$.

WORKED five-bar numbers

$\mathbf{O}_{1,2} = (\mp 0.1, 0)$, $L_p = 0.3$, $L_d = 0.4$; $\mathbf{P} = (0.05, 0.45)$. Left: $r_1 = 0.4743$, $\cos\beta_1 = 0.5446$, $\beta_1 = 57.00°$, atan2 = $71.565°$: $\theta_1 = 128.567°$. Right: $r_2 = 0.4528$, $\beta_2 = 60.20°$, atan2 = $96.340°$: $\theta_2 = 36.138°$. Check $\|\mathbf{P} - \mathbf{E}_j\| = 0.4$. FK: intersect the two L_d circles about $\mathbf{E}_1, \mathbf{E}_2$ (2 points). [checked]

Singularities [added]: none on slides (W6: $\det\mathbf{J} = 0$). Four-bar modes merge at $d = L_2 \pm L_3$ (dead centre); a leg is singular stretched/folded ($r_j = L_p \pm L_d$).

Slider-crank and Stewart leg IK [added]

Slider-crank (crank L_1, rod $L_2 > L_1$, slider on the pivot line)

$$x = L_1\cos\theta + \sqrt{L_2^2 - L_1^2\sin^2\theta},\qquad \beta = -\arcsin\frac{L_1\sin\theta}{L_2}$$

Closure: tip height $L_1\sin\theta = -L_2\sin\beta$. $L_1 = 0.1$, $L_2 = 0.3$: $x(0°) = 0.4000$, $x(60°) = 0.3372$, $x(90°) = 0.2828$, $x(180°) = 0.2$; stroke $2L_1$. [checked]

Stewart leg i [p. 73]; formulas [added]

$$^B\mathbf{l}_i = \mathbf{t} + \mathbf{R}\mathbf{p}_i - \mathbf{b}_i,\quad d_i = \|^B\mathbf{l}_i\|,\quad ^B\hat{\mathbf{n}}_i = \frac{^B\mathbf{l}_i}{d_i},\quad ^P\hat{\mathbf{n}}_i = \mathbf{R}^\top\,^B\hat{\mathbf{n}}_i$$

$$\psi_i = \text{atan2}(n_y, n_x),\quad \varepsilon_i = \arcsin n_z$$

$\mathbf{b}_i$ in B, $\mathbf{p}_i$ in P, $\mathbf{t}, \mathbf{R}$ pose of P in B; d_i active length. U-axis convention not fixed by the slide: state yours. Base circle 1.0 m ($-10, 10, 110, 130, 230, 250°$), platform circle 0.5 m ($-50, 50, 70, 170, 190, 290°$), home $\mathbf{t} = (0, 0, 1)$: $\mathbf{l}_1 = (-0.6634, -0.2094, 1.0)$, $d_1 = 1.2182$ m (all six equal). [checked]

EXAM-STYLE QUESTION assembly modes, crank test, mobility [marks assumed]

Four-bar $L_0 = 0.8$, crank $L_1 = 0.3$ at S= $(0, 0)$, coupler $L_2 = 0.7$, rocker $L_3 = 0.5$ at $\mathbf{D} = (0.8, 0)$ m. (a) At $\theta_1 = 60°$ find both B and ϕ. (b) Does the crank turn fully? (c) M?

SOLUTION (a) $\mathbf{A} = (0.1500, 0.2598)$; $d^2 = 0.64 + 0.09 - 0.24 = 0.49$, $d = 0.7$; $x_a = \frac{0.49 - 0.25 + 0.49}{1.4} = 0.5214$, $h = \sqrt{0.49 - 0.2719} = 0.4670$, $\hat{\mathbf{u}} = (0.9286, -0.3712)$, $\hat{\mathbf{n}} = (0.3712, 0.9286)$. B+ = $\boxed{(0.8075, 0.4999)}$, $\phi = 20.06°$; B− = $\boxed{(0.4608, -0.3674)}$, $\phi = -63.64°$.

(b) $0.3 + 0.8 = 1.1 \le 0.5 + 0.7 = 1.2$, shortest is the crank: crank-rocker, turns fully (d runs 0.5 to 1.1 ∈ [0.2, 1.2]).

(c) $3(4 - 4 - 1) + 4 = \boxed{1}$.

EXAM-STYLE QUESTION spot the errors: simplest delta [marks assumed]

"$D = 6$, $N = 8$, $J = 9$, $\sum F = 15$, $M = 6(8 - 9 + 1) + 15 = 15$. The rod spins, $R_M = 6$, so $M = 9$." Find both errors.

SOLUTION (1) Sign: $D(N - J - \mathbf{1})$: $M = -12 + 15 = \boxed{3}$. (2) Rods have a U joint at each end; U transmits spin about the rod axis, so $R_M = 0$; the student invented R_M to patch error 1. Contrast common delta (ball joints): plain 9, $R_M = 6$, true 3.

EXAM-STYLE QUESTION planar 3-RPR IK [marks assumed]

$\mathbf{b}_1 = (0, 0)$, $\mathbf{b}_2 = (1.0, 0)$, $\mathbf{b}_3 = (0.5, 0.9)$ m; platform points $\mathbf{a}_i = 0.2(\cos\gamma_i, \sin\gamma_i)$, $\gamma_i = 0°, 120°, 240°$. Leg lengths at $(x, y, \varphi) = (0.5, 0.35, 10°)$? Why is FK harder?

SOLUTION $\mathbf{p}_i = \mathbf{t} + \mathbf{R}(\varphi)\mathbf{a}_i$, $\ell_i = \|\mathbf{p}_i - \mathbf{b}_i\|$: $\mathbf{p}_1 = (0.6970, 0.3847)$, $\mathbf{p}_2 = (0.3714, 0.5032)$, $\mathbf{p}_3 = (0.4316, 0.1621)$.

$$\boxed{\ell_1 = 0.7961,\ \ell_2 = 0.8052,\ \ell_3 = 0.7411 \text{ m}}$$

Wrong: not rotating $\mathbf{a}_i$ ($\ell_1 = 0.7826$). FK: three coupled non-linear loop equations, several solutions; IK: none. [checked]

W5 Trajectory generation

Joint vs Cartesian space [p. 6–14, 42–47]

GIVEN start/goal pose of $\{T\}$ (+ via points), t_F *or* speed/accel limits FIND $q_i(t), \dot{q}_i, \ddot{q}_i$ for *every* joint

Trajectory = path + timing [p. 7]. Motors always get joint $q, \dot{q}, \ddot{q}$ [p. 68]. Teach mode [p. 9]: record and replay, no kinematics. Constant-acceleration formulas ($s = ut + \frac{1}{2}at^2$, $v = u + at$, area under speed graph = distance): basics p. 5.

	Joint space [p. 12]	Task space [p. 43]
Give	IK at both ends; a curve $q_i(t)$ per joint	path of $\{T\}$ in $\{S\}$ (line, ellipse), ${}^S\mathbf{T}_T(t)$
Cost	cheap (IK twice)	IK at every sample, fit, resample
Tool path	curved, uncontrolled [p. 46]	as asked
Check	joint limits at ends	workspace, singularity, IK branch *all along* [p. 47]

Linear interpolation [p. 14, 31]

$q(t) = q_0 + \dfrac{q_F - q_0}{t_F - t_0}(t - t_0)$, $\dot{q}$ = const. Velocity jumps $0 \to \dot{q}$ at the ends (infinite acceleration), so use cubics/blends.

Cubic polynomial [p. 15–20]

$N{+}1$ unknowns need $N{+}1$ conditions [p. 15]: position and velocity at both ends ⇒ cubic.

Cubic, $t_0 = 0$ [p. 16]

$$q = a_0 + a_1 t + a_2 t^2 + a_3 t^3, \quad \dot{q} = a_1 + 2a_2 t + 3a_3 t^2, \quad \ddot{q} = 2a_2 + 6a_3 t$$

Conditions $q(0) = q_0$, $q(t_F) = q_F$, $\dot{q}(0) = \dot{q}_0$, $\dot{q}(t_F) = \dot{q}_F$. a_k in unit/s^k.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & t_F & t_F^2 & t_F^3 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2t_F & 3t_F^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_F \\ \dot{q}_0 \\ \dot{q}_F \end{bmatrix}$$

$$\mathbf{M}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{3}{t_F^2} & \frac{3}{t_F^2} & -\frac{2}{t_F} & -\frac{1}{t_F} \\ \frac{2}{t_F^3} & -\frac{2}{t_F^3} & \frac{1}{t_F^2} & \frac{1}{t_F^2} \end{bmatrix}$$

Use: $t_0 \neq 0$: use local clock $\tau = t - t_0$, $T = t_F - t_0$ (via segments, any move not at $t = 0$).

DERIVATION four coefficients; $\Delta q = q_F - q_0$

$t = 0$: $a_0 = q_0$, $a_1 = \dot{q}_0$. At t_F:

$$a_2 t_F^2 + a_3 t_F^3 = \Delta q - \dot{q}_0 t_F, \quad 2a_2 t_F + 3a_3 t_F^2 = \dot{q}_F - \dot{q}_0$$

Eliminate a_2 (= $(\Delta q - \dot{q}_0 t_F - a_3 t_F^3)/t_F^2$): $a_3 t_F^2 = (\dot{q}_F + \dot{q}_0) - 2\Delta q/t_F$.

$$\boxed{a_3 = \frac{-2\Delta q + (\dot{q}_0 + \dot{q}_F)t_F}{t_F^3}}, \quad \boxed{a_2 = \frac{3\Delta q - (2\dot{q}_0 + \dot{q}_F)t_F}{t_F^2}}$$

Rest to rest: $a_2 = 3\Delta q/t_F^2$, $a_3 = -2\Delta q/t_F^3$; peak speed $1.5\Delta q/t_F$ at $t_F/2$; end accelerations $\pm 6\Delta q/t_F^2$ (not zero). [checked]

STEPS cubic between two points

1. List the 4 conditions with units; local clock, $T = t_F - t_0$.
2. $a_0 = q_0$, $a_1 = \dot{q}_0$; a_2, a_3 from the boxes.
3. Write $q, \dot{q}, \ddot{q}$ with numbers.
4. Check $q(T) = q_F$, $\dot{q}(T) = \dot{q}_F$.

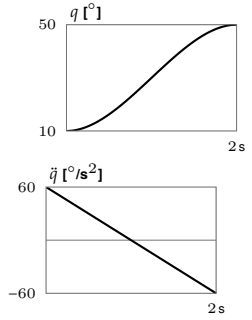
TRAP

a_3 divides by T^3 ($20° \to 80°$, 3 s: $a_3 = -2(60)/27 = -4.44$, not -13.3). Do not mix degrees and radians.

WORKED slide pp. 17–19: $10° \to 50°$ in 2 s, rest to rest

$\Delta q = 40$: $a_2 = 3(40)/4 = 30$, $a_3 = -2(40)/8 = -10$. $q = 10 + 30t^2 - 10t^3$, $\dot{q} = 60t - 30t^2$, $\ddot{q} = 60 - 60t$. $q(2) = 10 + 120 - 80 = 50$ ✓, $\dot{q}(2) = 0$ ✓; peak $\dot{q}(1) = 30°/\text{s}$. [checked]

$\dot{q}_F = 10, 20° \to 80°$, 3 s: $a_2 = \frac{180-30}{9} = 16.667$, $a_3 = \frac{-120+30}{27} = -3.333$ [checked]. Several joints [p. 20]: one cubic each, same T ($\Delta q = 40, 90, -20°$, $T = 2$: $a_2 = 30, 67.5, -15$; $a_3 = -10, -22.5, 5$) [checked].



W13 sample Q9 (prismatic cubic, 0.02 → 0.15 m in 0.8 s) and **Q8** (via-point cubics): solved on p. 42.

Via points [p. 21–28]

GIVEN points q_k at times t_k, end speeds (usually 0) FIND cubic per segment, own clock $\tau \in [0, T_k]$

Cubic between adjacent points; each via point needs a speed [p. 24]. Rest-to-rest at every point = stops everywhere [p. 25]. Options [p. 26]: (1) user gives speeds; (2) simple formula; (3) speeds chosen for **continuous acceleration**.

Option 2: simple rule [p. 27]

Chord slope $s_k = \dfrac{q_{k+1} - q_k}{t_{k+1} - t_k}$. At a via point between segments $k{-}1, k$:

$$\dot{q}_k = \begin{cases} 0 & s_{k-1}s_k \leq 0 \\ \frac{1}{2}(s_{k-1} + s_k) & \text{same sign} \end{cases} \qquad \dot{q} = 0 \text{ at start, end}$$

Gives continuous $q, \dot{q}$; acceleration jumps.

WORKED notebook Q2: $q = 10, 50, 20, 40, 45°$ at $t = 0, 5, 10, 15, 20$ s

$s = (8, -6, 4, 1)$; speeds $(0, 0, 0, 2.5, 0)$, $T_k = 5$.

seg.	a_0	a_1	a_2	a_3
$q_0 \to q_A$	10	0	4.8	−0.64
$q_A \to q_B$	50	0	−3.6	0.48
$q_B \to q_C$	20	0	1.9	−0.22
$q_C \to q_F$	40	2.5	−0.4	0.02

Check $q_C \to q_F$ at 5 s: $40 + 12.5 - 10 + 2.5 = 45$ ✓. $\ddot{q}$ jumps +2.4, −3.4, +2.0 at the joins. [checked]

Option 3: continuous acceleration [p. 28], notebook Q3

Slide: "constant acceleration a_{via}"; read as $\ddot{\theta}_1 = \ddot{\theta}_2$ **at the via point** (W13 Q8). Unknowns $a_0..a_3, b_0..b_3$ (8); $\theta_1 = \sum a_k t^k$, $\theta_2 = \sum b_k t^k$ (W13 p. 18).

8 conditions

1	$\theta_1(0) = \theta_0$	5	$\theta_2(T_2) = \theta_g$
2	$\dot{\theta}_1(0) = 0$	6	$\dot{\theta}_2(T_2) = 0$
3	$\theta_1(T_1) = \theta_v$	7	$\dot{\theta}_1(T_1) = \dot{\theta}_2(0)$
4	$\theta_2(0) = \theta_v$	8	$\ddot{\theta}_1(T_1) = \ddot{\theta}_2(0)$

ALTERNATIVE APPROACHES

A. Brute force

8 linear equations; 1, 2, 4 give $a_0 = \theta_0, a_1 = 0, b_0 = \theta_v$; solve 5 left.

B. Via-speed shortcut

Via speed w unknown (7 holds); each segment an ordinary cubic with known end speeds; 8 fixes w.

DERIVATION via speed w; $d_1 = \theta_v - \theta_0, d_2 = \theta_g - \theta_v$

Seg. 1 (end speeds $0, w$): $a_2 = \frac{3d_1 - wT_1}{T_1^2}$, $a_3 = \frac{-2d_1 + wT_1}{T_1^3}$. Seg. 2 $(w, 0)$: $b_1 = w$, $b_2 = \frac{3d_2 - 2wT_2}{T_2^2}$, $b_3 = \frac{-2d_2 + wT_2}{T_2^3}$.

$$\ddot{\theta}_1(T_1) = 2a_2 + 6a_3 T_1 = \frac{-6d_1 + 4wT_1}{T_1^2}, \quad \ddot{\theta}_2(0) = 2b_2 = \frac{6d_2 - 4wT_2}{T_2^2}$$

Equate: $\boxed{w = \dfrac{3[T_1^2 d_2 + T_2^2 d_1]}{2T_1T_2(T_1 + T_2)}}$; $T_1 = T_2$: $\boxed{w = \dfrac{3(\theta_g - \theta_0)}{4T}}$. Shared $\ddot{\theta} = 3(\theta_0 + \theta_g - 2\theta_v)/T^2$. [checked]

STEPS two cubics through a via point

1. Own clocks; list the 8 conditions.
2. $a_0 = \theta_0$, $a_1 = 0$, $b_0 = \theta_v$.
3. Cubics with end speeds $(0, w)$, $(w, 0)$.
4. Equate accelerations at the via point; solve w.
5. Check positions, speeds, accelerations match.

W5

WORKED $\theta = 10°, 40°, 20°, T_1 = T_2 = 1\,\text{s}$

$w = 3(10)/4 = 7.5$. Seg.1 $10 + 82.5t^2 - 52.5t^3$; seg.2 $40 + 7.5t - 75t^2 + 47.5t^3$. $\theta_1(1) = 40$, $\dot\theta_1(1) = 165 - 157.5 = 7.5$, $\ddot\theta_1(1) = -150 = 2b_2$, $\theta_2(1) = 20$, $\dot\theta_2(1) = 0$ ✓. $T_2 = 2$: $w = 25$; seg.1 $10 + 65t^2 - 35t^3$, seg.2 $40 + 25t - 40t^2 + 11.25t^3$. [checked]

Quintic and time scaling [p. 29–30]

Quintic [p. 29]

$$q = \sum_{k=0}^{5} a_k t^k, \quad \dot q = \sum k a_k t^{k-1}, \quad \ddot q = \sum k(k-1) a_k t^{k-2}$$

6 conditions: $q, \dot q, \ddot q$ at 0 and t_F.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & t_F & t_F^2 & t_F^3 & t_F^4 & t_F^5 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2t_F & 3t_F^2 & 4t_F^3 & 5t_F^4 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6t_F & 12t_F^2 & 20t_F^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_F \\ \dot q_0 \\ \dot q_F \\ \ddot q_0 \\ \ddot q_F \end{bmatrix}$$

Use: zero acceleration at start and stop.

DERIVATION rest to rest, zero end acc. [added]

$a_0 = q_0$, $a_1 = a_2 = 0$. With $u = t_F$: $a_3u^3 + a_4u^4 + a_5u^5 = \Delta$, $3a_3u^2 + 4a_4u^3 + 5a_5u^4 = 0$, $6a_3u + 12a_4u^2 + 20a_5u^3 = 0$. Third: $a_3 = -2a_4u - \frac{10}{3}a_5u^2$; second: $a_4 = -\frac{5}{2}a_5u$, $a_3 = \frac{5}{3}a_5u^2$; first: $a_5u^5/6 = \Delta$.

$$\boxed{a_3 = \frac{10\Delta}{t_F^3}, \quad a_4 = -\frac{15\Delta}{t_F^4}, \quad a_5 = \frac{6\Delta}{t_F^5}}$$

Peak speed $1.875\Delta/t_F$; $|\ddot q|_{max} = 5.77\Delta/t_F^2$. $0 \to 15°$, 1 s: $(150, -225, 90)$, $\dot q(1) = \ddot q(1) = 0$ ✓. [checked]

Time scaling [added]: $q = q_0 + \Delta q\, s(\tau)$, $\tau = (t - t_0)/T$; $s_{cubic} = 3\tau^2 - 2\tau^3$, $s_{quintic} = 10\tau^3 - 15\tau^4 + 6\tau^5$; same s on all joints ⇒ they finish together.

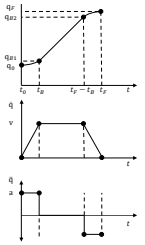
	lin.	cubic	quint.	LSPB*
peak $\dot q/(\Delta q/T)$	1	1.5	1.875	1.17
end $\ddot q \cdot T^2/\Delta q$	∞	±6	0	±8

*$a = 2a_{min}$. Polynomials [p. 30]: speed is a hump, so a higher peak than the trapezoid for the same $\Delta q, T$. [checked]

Linear segment with parabolic blends (LSPB) [p. 31–37]

GIVEN $\Delta q = q_F - q_0$, t_F ($t_0 = 0$), blend acceleration a FIND blend time t_B, coast speed v, $q(t)$

Speed up at a for t_B, coast at v, slow down: speed graph = trapezoid [p. 32].



Linear with parabolic blends

$q_{B1} = q_0 + \frac{a(t_B - t_0)^2}{2}$ ← Second order polynomial in time

$q_{B2} = q_{B1} + v(t_F - 2t_B)$ ← Linear in time

$q_F = q_{B2} + vt_B - \frac{a(t_B)^2}{2}$ ← Second order polynomial in time

$t_B = \frac{v}{a}$

[W05 lec p. 33] — pp. 33–34: $q, \dot q, \ddot q$ and the three position equations

Slide equations [p. 33–34], $t_0 = 0$

$q_{B1} = q_0 + \frac{1}{2}at_B^2$, $q_{B2} = q_{B1} + v(t_F - 2t_B)$, $q_F = q_{B2} + vt_B - \frac{1}{2}at_B^2$, $t_B = \frac{v}{a}$

v coast speed; $a > 0$ if $q_F > q_0$, $-a$ otherwise. Segments: $0 \le t \le t_B$: $q = q_0 + \frac{1}{2}at^2$, $\dot q = at$; coast: $q = q_0 + v(t - \frac{1}{2}t_B)$, $\dot q = v$; last: $q = q_F - \frac{1}{2}a(t_F - t)^2$, $\dot q = a(t_F - t)$.

DERIVATION blend time and existence [p. 36]

Area under trapezoid: $\Delta q = v(t_F - t_B)$ (sum of the 3 slide equations agrees). $v = at_B$:

$$\boxed{at_B^2 - at_Ft_B + \Delta q = 0} \Rightarrow t_B = \frac{t_F}{2} - \frac{\sqrt{a^2t_F^2 - 4a\Delta q}}{2a}$$

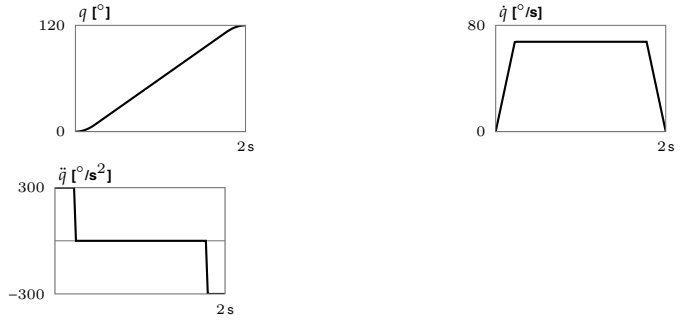
Real root needs $\boxed{a \ge 4\Delta q/t_F^2}$. Blends must not overlap, $2t_B \le t_F$: take the **minus** root. Equality: $t_B = t_F/2$, triangle, $v = 2\Delta q/t_F$. If v is given: $t_B = t_F - \Delta q/v$, $a = v/t_B$, needs $\Delta q/t_F < v \le 2\Delta q/t_F$. [checked]

STEPS LSPB, given $\Delta q, t_F, a$

1. Existence: $a \ge 4\Delta q/t_F^2$, else "not possible".
2. t_B (minus root); $v = at_B$; coast time $t_F - 2t_B$.
3. Check $v(t_F - t_B) = \Delta q$.

WORKED slide p. 37: $120°$ in 2 s, $a = 300°/\text{s}^2$

$4(120)/4 = 120 \le 300$ ✓. $t_B = 1 \pm \frac{\sqrt{300^2 \cdot 4 - 4 \cdot 300 \cdot 120}}{600} = 1 \pm 0.7746 = 0.2254$ (reject 1.7746). $v = 300(0.2254) = 67.62°/\text{s}$. Check $67.62(1.7746) = 120$ ✓; $q_{B1} = 7.62°$. $a = 120$: $t_B = 1$ triangle; $a = 100$: disc. < 0; $v = 70$ given: $t_B = 0.2857$ s, $a = 245$. [checked]



Minimum time, rest to rest, $|\dot q| \le v_m, |\ddot q| \le a_m$ [added]

$\Delta \ge v_m^2/a_m$ (trapezoid): $T_{min} = \frac{\Delta}{v_m} + \frac{v_m}{a_m}$. Else triangle: $T_{min} = 2\sqrt{\Delta/a_m}$, $v_{pk} = \sqrt{a_m\Delta}$.

Use: $v_m = 60$, $a_m = 120$: $\Delta = 90$: $1.5 + 0.5 = 2.0\,s$; $\Delta = 20$: $0.816\,s$, *peak* 49.0. *Cubic needs* $T \ge \max(1.5\Delta/v_m, \sqrt{6\Delta/a_m}) = 2.25\,s$ *for* 90°. *[checked]*

TRAP

Plus root ($t_B = 1.7746$); t_F for t_F^2 in a_{min}; "bigger a gives faster coast" (it lowers v, toward $\Delta q/t_F$); sign of a when $q_F < q_0$.

SLIDE CHECK pp. 33–34

General-t_0 form $q_{B1} = q_0 + a(t_B - t_0)^2/2$ is only consistent for $t_0 = 0$ ($t_0 = 1, t_F = 3$, 120° gives roots 2.049, −0.049, not 0.2254). Use $T = t_F - t_0$ and a local clock.

Several segments with blends [p. 38–41]

Path passes *near* the via points (rounded corners) [p. 40]. Given θ_k, durations t_{djk}, $|\ddot\theta_j|$; find blend times t_j (Craig §7.3, [added]).

Blend times, $\theta_1 \ldots \theta_n$

First: $t_1 = t_{d12} - \sqrt{t_{d12}^2 - \frac{2(\theta_2 - \theta_1)}{\ddot\theta_1}}$, $v_{12} = \frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1}$, $\ddot\theta_1 = \text{sgn}(\theta_2 - \theta_1)|\ddot\theta_1|$.

Interior: $v_{jk} = \frac{\theta_k - \theta_j}{t_{djk}}$, $\ddot\theta_j = \text{sgn}(v_{jk} - v_{ij})|\ddot\theta_j|$, $t_j = \frac{v_{jk} - v_{ij}}{\ddot\theta_j}$.

Last: $t_n = t_{d,n-1\,n} - \sqrt{t_{d,n-1\,n}^2 + \frac{2(\theta_n - \theta_{n-1})}{\ddot\theta_n}}$, $v_{n-1\,n} = \frac{\theta_n - \theta_{n-1}}{t_{d,n-1\,n} - \frac{1}{2}t_n}$, $\ddot\theta_n = \text{sgn}(\theta_{n-1} - \theta_n)|\ddot\theta_n|$.

Linear parts: first $t_{12} = t_{d12} - t_1 - \frac{1}{2}t_2$; interior $t_{jk} = t_{djk} - \frac{1}{2}t_j - \frac{1}{2}t_k$; last $t_{n-1\,n} = t_{d,n-1\,n} - \frac{1}{2}t_{n-1} - t_n$. Corner miss $|\ddot\theta_j|t_j^2/8$.

Example $0°, 40°, 10°$, $t_d = 2, 2$, $|\ddot\theta| = 50$: $t_1 = 0.4508$, $v_{12} = 22.54$; $t_3 = 0.3267$, $v_{23} = -16.33$; $t_2 = 0.7775$; reaches 36.22° (miss 3.78°). [checked]

Cartesian paths, RR arm [p. 42–68]

Use for a prescribed path or obstacles [p. 43]. Describe ${}^S\mathbf{T}_T(t)$; for IK ${}^B\mathbf{T}_T = {}^B\mathbf{T}_S\,{}^S\mathbf{T}_T(t)$ [added]. Do not interpolate rotation matrices entrywise [p. 65]. **Joint-linear ≠ Cartesian-linear.** RR L_1=4, L_2=3, $(0, 4.75) \to (4.75, 0)$: joint-linear q_1 : 51.07° → −38.93°, q_2 = 95.83° const; tool on an arc, midpoint (3.359, 3.359) vs line midpoint (2.375, 2.375), 1.39 m off. [checked]

STEPS check a Cartesian line first [p. 47]

1. Workspace: $|L_1 - L_2| \le r \le L_1 + L_2$ on every point (L_1=1, L_2=0.7, line $y = 0.2$: ends r=1.02 ok, middle 0.2 < 0.3 hole). [checked]
2. Singularity: $\det \mathbf{J} = L_1L_2 \sin q_2$; joint rates $\propto 1/\sin q_2$ (L_1=L_2=1, 1 m/s on y=0: max rate 1.15, 3.20, 10.0, 31.6 rad/s at x = 1, 1.9, 1.99, 1.999). [checked]
3. Same IK branch at both ends, else the arm jumps.

STEPS task-space trajectory [p. 48]

1. Path as a function ($y = mx + c$).
2. Sample: t_F/N steps; constant speed v: $x = x_0 + \frac{vt}{\sqrt{1+m^2}}$, $y = y_0 + \frac{vmt}{\sqrt{1+m^2}}$.
3. IK each sample, **same branch**.
4. Fit a cubic (quintic if needed [p. 60]) per joint.
5. Resample the fit; FK to see the tool path.

Planar RR FK/IK [p. 56]

$$x = L_1c_1 + L_2c_{12}, \quad y = L_1s_1 + L_2s_{12}, \quad \cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$q_2 = \text{atan2}(\pm\sqrt{1 - \cos^2 q_2}, \cos q_2)$$

$$q_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \sin q_2, L_1 + L_2 \cos q_2)$$

Non-constant speed: $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}}$.

W5

WORKED slide pp. 49–60: $L_1{=}4, L_2{=}3, (0, 4.75) \to (4.75, 0)$ **at 3 m/s, branch** $q_2 > 0$

Line $y = -x + 4.75$, length $4.75\sqrt{2} = 6.7175$, $T = 2.2392$ s; $x = 3t/\sqrt{2}$, $y = 4.75 - x$; 10 sections $t_i = 0.2239i$. IK table (computed, not transcribed):

i	t_i	x	y	q_1 [°]	q_2 [°]	
0	0	0	4.750	51.07	95.83	
1	0.224	0.475	4.275	41.48	105.71	
2	0.448	0.950	3.800	31.44	113.73	
3	0.672	1.425	3.325	20.75	119.76	
4	0.896	1.900	2.850	9.44	123.56	[checked]
5	1.120	2.375	2.375	−2.13	124.86	
6	1.344	2.850	1.900	−13.18	123.56	
7	1.567	3.325	1.425	−22.85	119.76	
8	1.791	3.800	0.950	−30.48	113.73	
9	2.015	4.275	0.475	−35.84	105.71	
10	2.239	4.750	0	−38.93	95.83	

$t_i = T/2$: $\cos q_2 = \frac{2(2.375)^2 - 25}{24} = -0.5716$, $q_2 = 124.86°$, $q_1 = 45° - \text{atan2}(2.4616, 2.2852) = -2.13°$ (branch −: 92.13°, −124.86°); FK: $(2.375, 2.375)$ ✓. Cubic fits [p. 59]: $q_1 = 0.1346t^3 - 0.3375t^2 - 0.6182t + 0.8858$, $q_2 = 0.0000t^3 - 0.4032t^2 + 0.9029t + 1.6674$ (rad); fit error 0.46°, tool wanders 37 mm (quintic 7.7 mm). Other branch: $(128.93°, -95.83°) \to (38.93°, -95.83°)$, same path. Midpoint velocity $(3, -3)/\sqrt{2}$: $\det \mathbf{J} = 12 \sin q_2 = 9.846$, $\dot{\mathbf{q}} = (-0.893, 0)$ rad/s. [checked]

SLIDE CHECK p. 59

Slide prints $q_2 = 0.0004t^3 - 0.4043t^2 + 0.9036t + 1.6674$; symmetry forces the cubic term to vanish (least squares: $0.0000t^3 - 0.4032t^2 + 0.9029t$). Method matters, not last digits.

Exam-style questions

Marks not published; ~1 mark per step (conditions, formula, substitution, units, check).

EXAM-STYLE QUESTION LSPB [marks not published]

75° in $t_F = 1.5$ s, $a = 200°/\text{s}^2$. (a) Minimum a. (b) t_B, v. (c) Is $a = 130$ possible?

SOLUTION (a) $a_{min} = 4(75)/2.25 = 133.33$. (b) $\sqrt{90000 - 60000} = 173.21$; $t_B = 0.75 - 173.21/400 = 0.317$ s; $v = at_B = \boxed{63.40°/\text{s}}$; coast 0.866 s; check $63.40(1.183) = 75.0$ ✓. (c) $130^2(2.25) - 4(130)(75) = -975 < 0$: not possible. [checked]

EXAM-STYLE QUESTION Via points, simple rule [marks not published]

10, 30, 70, 80, 60° at $t = 0, 2, 5, 9, 12$ s. (a) Speed at 5 s. (b) a_2, a_3 for $5 \to 9$ s. (c) Acceleration jump at 5 s.

SOLUTION $s = (10, 13.333, 2.5, -6.667)$. (a) $\frac{1}{2}(13.333 + 2.5) = 7.917°/\text{s}$. (b) $T = 4$, $\Delta = 10$, $(\dot{q}_0, \dot{q}_F) = (7.917, 0)$: $a_2 = \frac{30 - 15.833(4)}{16} = -2.0833$, $a_3 = \frac{-20 + 7.917(4)}{64} = 0.18229$; $q(4) = 80$ ✓. (c) Seg. $2 \to 5$ ($T = 3$, $11.667 \to 7.917$): $a_2 = 2.9167$, $a_3 = -0.7870$, end acc. -8.333; next start $2a_2 = -4.167$; jump $\boxed{4.167°/\text{s}^2}$. [checked]

EXAM-STYLE QUESTION RR tool on a circle [added] [marks not published]

RR $L_1{=}0.6$, $L_2{=}0.4$ m; tool once CCW round a circle $r{=}0.2$ centred $(0.5, 0.3)$ in $T{=}4$ s at constant speed, from $(0.7, 0.3)$. (a) Reachable? (b) q at $T/2$ ($q_2 > 0$). (c) Joint speeds at $t{=}0$.

SOLUTION $x = 0.5 + 0.2\cos\frac{2\pi t}{4}$, $y = 0.3 + 0.2\sin\frac{2\pi t}{4}$, speed 0.314 m/s. (a) Centre at 0.583; points 0.383 to 0.783 from the base, inside $[0.2, 1.0]$ ✓. (b) $(0.3, 0.3)$: $\cos q_2 = \frac{0.18 - 0.52}{0.48} = -0.7083$, $q_2 = 135.10°$, $q_1 = 45° - \text{atan2}(0.2824, 0.3167) = \boxed{3.28°}$. Others: $t{=}0$ (−8.21°, 82.82°); $t{=}1$ (10.58°, 92.39°); $t{=}3$ (−29.95°, 122.80°). (c) $\dot{\mathbf{x}} = (0, 0.3142)$; $\mathbf{J} = \begin{bmatrix} -0.3000 & -0.3857 \\ 0.7000 & 0.1061 \end{bmatrix}$, $\det = 0.2381$; $\dot{\mathbf{q}} = \boxed{(0.509, -0.396)}$ rad/s. [checked]

W6 Velocity kinematics, Jacobian, singularities, statics

Jacobian: joint rates → tool velocity [p. 6–7, 20–21]

Manipulator Jacobian [p. 20–21]

$$^0\boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_v(\mathbf{q}) \\ \mathbf{J}_\omega(\mathbf{q}) \end{bmatrix} \dot{\mathbf{q}} = {}^0\mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}} \qquad (\mathbf{J}:\ 6 \times n)$$

$\boldsymbol{v}$ (m/s), $\boldsymbol{\omega}$ (rad/s): tool linear/angular velocity *in frame* {0} (state the frame, p.21); $\dot{q}_j = \dot{\theta}_j$ or $\dot{d}_j$. Column j = tool twist for $\dot{q}_j = 1$; the joints' effects add. Units: linear rows m per rad (revolute), angular rows dimensionless.

Which formula: $\dot{\mathbf{q}} \to \boldsymbol{\nu}$: $\boldsymbol{\nu} = \mathbf{J}\dot{\mathbf{q}}$. $\boldsymbol{\nu} \to \dot{\mathbf{q}}$: $\mathbf{J}^{-1}$ or $\mathbf{J}^{+}$. Singular: $\det \mathbf{J} = 0$. Ellipse: $\mathbf{J}\mathbf{J}^\top$. Statics: $\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F}_E$.

Rotation rate: $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times \mathbf{R}$, vex [p. 9–12, 18–19]

Skew matrix, vex, angular velocity [p. 10, 18]

$$[\mathbf{s}]_\times = \begin{bmatrix} 0 & -s_3 & s_2 \\ s_3 & 0 & -s_1 \\ -s_2 & s_1 & 0 \end{bmatrix}, \qquad \vee_\times(\mathbf{S}) = \begin{bmatrix} S_{32} \\ S_{13} \\ S_{21} \end{bmatrix}$$

$$\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times \mathbf{R} \iff [\boldsymbol{\omega}]_\times = \dot{\mathbf{R}}\mathbf{R}^\top, \quad \boldsymbol{\omega} = \vee_\times(\dot{\mathbf{R}}\mathbf{R}^\top)$$

$\dot{\mathbf{R}}$ is 3×3 (entrywise derivative), $\boldsymbol{\omega}$ is 3×1: convert through the skew matrix. Fixed axis $\hat{\mathbf{n}}$: $\boldsymbol{\omega} = \dot{\theta}\hat{\mathbf{n}}$.

DERIVATION why $\dot{\mathbf{R}}\mathbf{R}^\top$ is skew [added]

$\mathbf{R}\mathbf{R}^\top = \mathbf{I} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^\top + (\dot{\mathbf{R}}\mathbf{R}^\top)^\top = \mathbf{0}$, so $\mathbf{S} = \dot{\mathbf{R}}\mathbf{R}^\top$ is skew; call it $[\boldsymbol{\omega}]_\times$, multiply right by $\mathbf{R}$.

WORKED $\mathbf{R}_z(\theta)$, $\theta = 30°$, $\dot{\theta} = 2$ rad/s

$$\dot{\mathbf{R}}_z\mathbf{R}_z^\top = \dot{\theta}\begin{bmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} = \dot{\theta}\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \vee_\times \Rightarrow \boldsymbol{\omega} = (0, 0, 2)\,\text{rad/s}.$$ [checked]

SLIDE CHECK cross-product note, w06-cross

Expansion prints the $\hat{\mathbf{j}}$ term as $+(\omega_x P_z - \omega_z P_x)$; row-1 cofactor expansion gives $-(\omega_x P_z - \omega_z P_x) = \omega_z P_x - \omega_x P_z$ (= middle row of $[\boldsymbol{\omega}]_\times \mathbf{P}$). $\boldsymbol{\omega} = (1, 2, 3)$, $\mathbf{P} = (4, 5, 6)$: printed gives -6, right $+6$; $\boldsymbol{\omega} \times \mathbf{P} = (-3, 6, -3)$. Follow this page. [checked]

Jacobian from the ETS: the slides' method [p. 13–19, 25–32]

Columns from $\partial \mathbf{T}/\partial q_j$ [p. 17–19]

$$^0\mathbf{T}_E = \prod_{i=1}^{M} \mathbf{E}_i(\eta_i), \quad \dot{\mathbf{T}} = \sum_{j=1}^{n} \frac{\partial \mathbf{T}}{\partial q_j}\dot{q}_j, \quad \frac{\partial \mathbf{T}}{\partial q_j} = \begin{bmatrix} \mathbf{J}_{R_j} & \mathbf{J}_{t_j} \\ \mathbf{0} & 0 \end{bmatrix}$$

$$\mathbf{J}_{v,j} = \tau\left(\frac{\partial \mathbf{T}}{\partial q_j}\right), \qquad \mathbf{J}_{\omega,j} = \vee_\times\left(\rho\left(\frac{\partial \mathbf{T}}{\partial q_j}\right)\rho(\mathbf{T})^\top\right)$$

$\tau(\mathbf{M})$: rows 1–3 of column 4; $\rho(\mathbf{M})$: upper-left 3×3; $\rho(\mathbf{T}) = \mathbf{R}$. Stack $\mathbf{J}_v$ over $\mathbf{J}_\omega$.

DERIVATION of the rotation rows [p. 18–19]

$\dot{\mathbf{R}} = \sum_j \mathbf{J}_{R_j}\dot{q}_j = [\boldsymbol{\omega}]_\times \mathbf{R}$. Post-multiply by $\mathbf{R}^\top$: $[\boldsymbol{\omega}]_\times = \sum_j (\mathbf{J}_{R_j}\mathbf{R}^\top)\dot{q}_j$; $\vee_\times$ is linear. Translation: $\boldsymbol{v} = \dot{\mathbf{t}} = \sum_j \mathbf{J}_{t_j}\dot{q}_j$.

Derivative of one elementary transform [p. 29, 31, 32]

$$\frac{\partial \mathbf{T}_{R_k}(\theta)}{\partial \theta} = \hat{\mathbf{R}}_k \mathbf{T}_{R_k}(\theta), \qquad \frac{\partial \mathbf{T}_{t_k}(d)}{\partial d} = \hat{\mathbf{t}}_k$$

$$\hat{\mathbf{R}}_x = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \hat{\mathbf{R}}_y = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \hat{\mathbf{R}}_z = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\mathbf{t}}_x = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \hat{\mathbf{t}}_y = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \hat{\mathbf{t}}_z = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

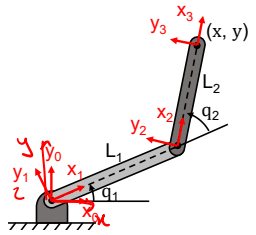
$$\frac{\partial \mathbf{T}}{\partial q_j} = \Big(\prod_{i<\mu(j)} \mathbf{E}_i\Big)\frac{\partial \mathbf{E}_{\mu(j)}}{\partial q_j}\Big(\prod_{i>\mu(j)}^{M} \mathbf{E}_i\Big)$$

$\mu(j)$ = position of the factor holding q_j. Freeze all other factors, keep the order. Sign flip if the joint's positive sense opposes the axis: $\frac{d}{dq}R_y(-q) = -\hat{\mathbf{R}}_y\mathbf{T}_{R_y}(-q)$, $\frac{d}{dd}t_z(-d) = -\hat{\mathbf{t}}_z$.

STEPS Jacobian by the slides' method [p. 25–26]

1. ETS base to tool, $^0\mathbf{T}_E = \mathbf{E}_1 \cdots \mathbf{E}_M$; mark $\mu(j)$.
2. Joint j: replace only $\mathbf{E}_{\mu(j)}$ by its derivative; sign flip if needed.
3. Multiply out $\partial \mathbf{T}/\partial q_j$ (4×4).
4. Column j: $\tau(\cdot)$ on top, $\vee_\times(\rho(\cdot)\rho(\mathbf{T})^\top)$ below.
5. Stack into $6 \times n$, state the frame.
6. Check at $\mathbf{q} = \mathbf{0}$: each column = velocity from turning that joint by hand.

Worked: planar RR, two ways [p. 33–39]



[W06 lec p. 35] — RR frames {0}, {1}, {2}, {3} (slide 35)

ETS $\mathbf{T} = \mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2) = \mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4$, $\mu(1) = 1$, $\mu(2) = 3$.

ALTERNATIVE APPROACHES

A. Direct differentiation [p. 33–34]

$x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$.
$\dot{x} = -L_1s_1\dot{q}_1 - L_2s_{12}(\dot{q}_1 + \dot{q}_2)$
$\dot{y} = L_1c_1\dot{q}_1 + L_2c_{12}(\dot{q}_1 + \dot{q}_2)$
coefficients of $\dot{q}_j$ = $\mathbf{J}_{xy}$; $\omega_z = \dot{q}_1 + \dot{q}_2$ by hand. Planar only.

B. ETS method [p. 35–39]

$\partial_1\mathbf{T} = \hat{\mathbf{R}}_z\mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4$
$\partial_2\mathbf{T} = \mathbf{E}_1\mathbf{E}_2\hat{\mathbf{R}}_z\mathbf{E}_3\mathbf{E}_4$
then τ, $\vee_\times(\rho\rho^\top)$ per column. Any arm, 3D, all six rows.

DERIVATION method B, lines (slides 36–39)

$$\mathbf{T} = \begin{bmatrix} c_{12} & -s_{12} & 0 & x \\ s_{12} & c_{12} & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \ \hat{\mathbf{R}}_z\mathbf{T}: \text{row 1} = -\text{row 2 of } \mathbf{T}, \text{ row 2} = \text{row 1}.$$

$$\frac{\partial \mathbf{T}}{\partial q_1} = \begin{bmatrix} -s_{12} & -c_{12} & 0 & -y \\ c_{12} & -s_{12} & 0 & x \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \frac{\partial \mathbf{T}}{\partial q_2} = \begin{bmatrix} -s_{12} & -c_{12} & 0 & -L_2s_{12} \\ c_{12} & -s_{12} & 0 & L_2c_{12} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\mathbf{J}_{t_1} = (-y, x, 0)^\top$, $\mathbf{J}_{t_2} = L_2(-s_{12}, c_{12}, 0)^\top$. Rotation, joint 1 (same for 2):

$$\rho\Big(\frac{\partial \mathbf{T}}{\partial q_1}\Big)\mathbf{R}^\top = \begin{bmatrix} -s_{12} & -c_{12} & 0 \\ c_{12} & -s_{12} & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\vee_\times} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Planar RR in {0}, full 6×2 [p. 34–39]

$$^0\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{matrix} \leftarrow v_x \\ \leftarrow v_y \\ \leftarrow v_z \\ \leftarrow \omega_x \\ \leftarrow \omega_y \\ \leftarrow \omega_z \end{matrix}, \qquad \det \mathbf{J}_{xy} = L_1L_2 \sin q_2$$

q_2 relative to link 1; $s_{12} = \sin(q_1 + q_2)$ is the absolute angle of link 2. Rows 3–5 zero in the plane (slide 45).

WORKED notebook Q1: $L = (3, 4)$ m, $q_1 = 30°$, $q_2 = 15°$

$q_1 + q_2 = 45°$: $J_{11} = -1.5 - 2.8284 = -4.3284$, $J_{12} = -2.8284$, $J_{21} = 2.5981 + 2.8284 = 5.4265$, $J_{22} = 2.8284$.

$$^0\mathbf{J} = \begin{bmatrix} -4.3284 & -2.8284 \\ 5.4265 & 2.8284 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \quad \det \mathbf{J}_{xy} = 3.1058 = 12 \sin 15°$$

$\dot{\mathbf{q}} = (0.2, -0.1)$ rad/s: $\boldsymbol{v} = (-0.5828, 0.8025)$ m/s, $\omega_z = 0.1$ rad/s. [checked]

W6

PUMA 560 numeric 6×6 and reading a Jacobian [p.22–24]

WORKED slide 24, assumed nominal pose $\mathbf{q}=(0,45^\circ,180^\circ,0,45^\circ,0)$, frame $\{0\}$

$$^0\mathbf{J}=\begin{bmatrix}0.13&0&0.306&0&0&0\\0.664&0&0&-0.038&0&0\\0&0.664&0.359&0&0.053&0\\0&0&0&0.707&0&1\\0&-1&-1&0&-1&0\\1&0&0&-0.707&0&0\end{bmatrix}$$

Slide gives no pose/frame/units (and prints j_{76} for j_{66}). Rows 4–6 = joint axes in $\{0\}$ (cols 2,3,5 parallel $(0,-1,0)$; cols 4,6 at $45^\circ=q_5$). Rows 1–3 of a revolute column = $\hat{\mathbf{z}}_j\times(\mathbf{p}_E-\mathbf{o}_j)$: col 1 = $(-p_y,p_x,0)$, so tool at $p_x=0.664$, $p_y=-0.130$ m. $\det\approx-0.081$, rank 6. [checked]

Changing the frame of the Jacobian [p. 40, 45]

Express a Jacobian in another frame [p. 40]

$$^A\mathbf{J}(\mathbf{q})=\begin{bmatrix}^A\mathbf{R}_B&\mathbf{0}_{3\times3}\\\mathbf{0}_{3\times3}&^A\mathbf{R}_B\end{bmatrix}{}^B\mathbf{J}(\mathbf{q}),\qquad {}^E\mathbf{R}_0=({}^0\mathbf{R}_E)^\top$$

Rotate *both* halves (a 3×3 cannot multiply $6\times n$). det, singular values, singular poses: frame-independent.

WORKED RR of Q1 in tool frame $\{3\}$ [p. 45]

$^3\mathbf{R}_0=\mathbf{R}_z(-45^\circ)$:

$$^3\mathbf{J}=\begin{bmatrix}L_1s_2&0\\L_1c_2+L_2&L_2\\0&0\\0&0\\0&0\\1&1\end{bmatrix}=\begin{bmatrix}0.7765&0\\6.8978&4\\0&0\\0&0\\0&0\\1&1\end{bmatrix}$$

det = 3.1058 as in $\{0\}$. $\dot{\mathbf{q}}=(0.2,-0.1)$: $^3\boldsymbol{\upsilon}=(0.1553,0.9796)$ m/s; speed 0.9918 m/s in both frames. [checked]

Alternative: velocity propagation, geometric rule [added]

ALTERNATIVE APPROACHES

A. Craig propagation (result in last frame)

Revolute: $^{i+1}\boldsymbol{\omega}_{i+1}={}^{i+1}_{i}\mathbf{R}\,{}^i\boldsymbol{\omega}_i+\dot\theta_{i+1}\hat{\mathbf{Z}}$

$^{i+1}\mathbf{v}_{i+1}={}^{i+1}_{i}\mathbf{R}({}^i\mathbf{v}_i+{}^i\boldsymbol{\omega}_i\times{}^i\mathbf{P}_{i+1})$

Prismatic: $\boldsymbol{\omega}$ unchanged; add $\dot d_{i+1}\hat{\mathbf{Z}}$ to $\mathbf{v}$. Start $\mathbf{0}$. Coefficients of $\dot q_j={}^n\mathbf{J}$; rotate to $\{0\}$ if asked.

B. Geometric column rule (all in $\{0\}$)

$\hat{\mathbf{z}}_j$ axis of joint j, $\mathbf{o}_j$ a point on it, $\mathbf{p}_E$ tool.

revolute: $\begin{bmatrix}\hat{\mathbf{z}}_j\times(\mathbf{p}_E-\mathbf{o}_j)\\\hat{\mathbf{z}}_j\end{bmatrix}$, prismatic: $\begin{bmatrix}\hat{\mathbf{z}}_j\\\mathbf{0}\end{bmatrix}$

Fast for planar/simple arms.

Jacobians of the simple robots [added]

ETS, J in $\{0\}$, det, singular poses (all checked against finite differences) [checked]

Rows v_x,v_y,ω_z shown for planar arms (rows $v_z,\omega_x,\omega_y\equiv0$). $s_{12}=\sin(q_1+q_2)$, $s_{123}=\sin(q_1+q_2+q_3)$.

RR: $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$, J as above. $\det\mathbf{J}_{xy}=L_1L_2\sin q_2$: $q_2=0$ stretched, π folded; rank 1, columns $\perp$ arm, lost direction *along* the arm.

RP: $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(d_2)$, $x=d_2c_1$, $y=d_2s_1$.

$$\begin{bmatrix}-d_2s_1&c_1\\d_2c_1&s_1\\1&0\end{bmatrix},\quad\det\mathbf{J}_{xy}=-d_2$$

Singular at $d_2=0$ (tool on the joint axis): tangential direction lost. $q_1=40^\circ$, $d_2=0.6$: det $=-0.6$; tangential 0.3 m/s needs $\dot q_1=0.5$, $\dot d_2=0$.

PR: $\mathbf{T}_{t_x}(d_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$, $x=d_1+L_2c_2$, $y=L_2s_2$.

$$\begin{bmatrix}1&-L_2s_2\\0&L_2c_2\\0&1\end{bmatrix},\quad\det\mathbf{J}_{xy}=L_2\cos q_2$$

Singular at $q_2=\pm90^\circ$ (link $\perp$ rail): columns $(1,0)$ and $(\mp L_2,0)$ parallel, y lost.

Planar RRR [checked]

$\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)\mathbf{T}_{R_z}(q_3)\mathbf{T}_{t_x}(L_3)$.

$$\begin{bmatrix}-L_1s_1-L_2s_{12}-L_3s_{123}&-L_2s_{12}-L_3s_{123}&-L_3s_{123}\\L_1c_1+L_2c_{12}+L_3c_{123}&L_2c_{12}+L_3c_{123}&L_3c_{123}\\1&1&1\end{bmatrix}$$

Column j sums links j outwards. (x,y) only: 2×3, rank 1 only if all links collinear. (x,y,ω_z): det $=L_1L_2\sin q_2$ (indep. of q_3). $L=(0.4,0.3,0.2)$, $\mathbf{q}=(20^\circ,40^\circ,-30^\circ)$: $\mathbf{J}_{xy}=\begin{bmatrix}-0.4966&-0.3598&-0.1000\\0.6991&0.3232&0.1732\end{bmatrix}$, $\det_{(x,y,\omega_z)}=0.0771$; $\dot{\mathbf{q}}=(0.3,-0.2,0.5)$: $\boldsymbol{\upsilon}=(-0.1270,0.2317)$, $\omega_z=0.6$.

RRP spherical (W3 arm) [checked]

$\mathbf{T}_{R_z}(\theta_1)\mathbf{T}_{t_z}(H)\mathbf{T}_{R_y}(\theta_2)\mathbf{T}_{t_z}(d_3)$; tip $(c_1d_3s_2,\ s_1d_3s_2,\ H+d_3c_2)$. Axes: $(0,0,1)$, $(-s_1,c_1,0)$, slide (c_1s_2,s_1s_2,c_2).

$$^0\mathbf{J}=\begin{bmatrix}-d_3s_1s_2&d_3c_1c_2&c_1s_2\\d_3c_1s_2&d_3s_1c_2&s_1s_2\\0&-d_3s_2&c_2\\0&-s_1&0\\0&c_1&0\\1&0&0\end{bmatrix},\ \det\mathbf{J}_v=-d_3^2\sin\theta_2$$

Singular: $d_3=0$ (tool on joint-2 axis), $\sin\theta_2=0$ (boom vertical, tool on z_0; direction $(-s_1,c_1,0)$ lost). $H=0.5$, tip $(0.3,0.4,0.9)$ ($\theta_1=53.13^\circ$, $\theta_2=51.34^\circ$, $d_3=0.6403$): $\mathbf{J}_v=\begin{bmatrix}-0.4&0.24&0.4685\\0.3&0.32&0.6247\\0&-0.5&0.6247\end{bmatrix}$, det $=-0.3202$; $\dot{\mathbf{q}}=(0.2,-0.1,0.05)$: $\boldsymbol{\upsilon}=(-0.0806,0.0592,0.0812)$, $\boldsymbol{\omega}=(0.08,-0.06,0.2)$.

Inverse velocity: tool velocity → joint rates [p. 41–42]

Inverse Jacobian and pseudo-inverses [p. 41–42]

$$\dot{\mathbf{q}}=\mathbf{J}^{-1}\boldsymbol{\nu},\qquad\mathbf{J}_{RR}^{-1}=\frac{1}{L_1L_2\sin q_2}\begin{bmatrix}L_2c_{12}&L_2s_{12}\\-L_1c_1-L_2c_{12}&-L_1s_1-L_2s_{12}\end{bmatrix}$$

$$\text{tall }(m>n):\ \mathbf{J}^+=(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top\text{ (least sq.)}$$

$$\text{fat }(n>m):\ \mathbf{J}^+=\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\text{ (min norm)}$$

m rows (task components), n joints. Slide's 3 questions (p.42): square? non-singular? invertible for all $\mathbf{q}$? Resolved rate: $\mathbf{q}_{k+1}=\mathbf{q}_k+\mathbf{J}^{-1}(\mathbf{q}_k)\boldsymbol{\nu}\,\Delta t$, recompute $\mathbf{J}$ each step. Damped LS $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top+\lambda^2\mathbf{I})^{-1}$ [added].

WORKED notebook Q4: $L=(3,4)$, $\mathbf{q}=(30^\circ,60^\circ)$, $\boldsymbol{\nu}=(0.2,-0.3,-0.2)$

$\mathbf{J}=\begin{bmatrix}-5.5&-4\\2.5981&0\\1&1\end{bmatrix}$ (v_x,v_y,ω_z), $\det\mathbf{J}_{xy}=10.3923$. 2 joints, 3 components: over-determined. **A** (follow position): $v_y\Rightarrow\dot q_1=-0.1155$; $v_x\Rightarrow\dot q_2=0.1088$ rad/s ($\omega_z=-0.0067$). **B** (least squares) $\mathbf{J}^+\boldsymbol{\nu}=(-0.1056,0.0840)$, residual $(0.0446,0.0258,0.1784)$. State which reading. [checked]

Singularities [p. 43–47]

Definition and types [p. 44]

$\det\mathbf{J}(\mathbf{q})=0$ (non-square: rank drop, $\det\mathbf{J}\mathbf{J}^\top=0$, $\sigma_{\min}=0$). Then some $\mathbf{u}\neq\mathbf{0}$ with $\mathbf{u}^\top\mathbf{J}=\mathbf{0}$: the tool cannot move along $\mathbf{u}$; $\mathbf{J}^{-1}\propto1/\det\mathbf{J}$ blows up.

Boundary: arm stretched/folded (RR $q_2=0,\pi$). **Interior**: axes align (wrist $q_5=0$; tool on base axis).

RR: $\det\begin{bmatrix}L_1s_2&0\\L_1c_2+L_2&L_2\end{bmatrix}=L_1L_2\sin q_2$ (slide 45), indep. of q_1 and the frame.

WORKED RR $L=(3,4)$, $q_1=30^\circ$, singular poses (notebook cell 46)

Stretched $q_2=0$: $\{0\}$ block $\begin{bmatrix}-3.5&-2\\6.0622&3.4641\end{bmatrix}$, col 1 = 1.75×col 2, both $\perp$ arm: rank 1. $^3\mathbf{J}_{xy}=\begin{bmatrix}0&0\\7&4\end{bmatrix}$ (row along the arm is zero). Folded $q_2=180^\circ$: $^3\mathbf{J}_{xy}=\begin{bmatrix}0&0\\1&4\end{bmatrix}$. [checked]

WORKED 3D arms

W13-type RRR: $\det\mathbf{J}_v=-L_3L_4\sin\theta_3\,(L_3c_2+L_4c_{23})$: $\sin\theta_3=0$ boundary; $r=L_3c_2+L_4c_{23}=0$ interior (tool on shoulder vertical). $L_3=0.4$, $L_4=0.3$, $\theta_2=60^\circ$: $\theta_3=71.81^\circ$. **PUMA 560 (zero-offset), notebook Q3**: $\det\mathbf{J}=-L_uL_f\sin q_3\,(L_u\sin q_2+L_f\sin(q_2+q_3))\sin q_5$: elbow $\sin q_3=0$; shoulder $L_u\sin q_2+L_f\sin(q_2+q_3)=0$ (wrist centre on waist axis; $L_u=0.5$, $L_f=0.3$, $q_2=22.92^\circ$, $q_3=-63.39^\circ$); wrist $\sin q_5=0$ (axes 4, 6 aligned, rank 5). Real PUMA has offsets: same types, shifted [added]. [checked]

STEPS Find and explain singularities

1. Write $\mathbf{J}$ (the square block that matters; say which rows).
2. det, factorise (RR: $L_1L_2\sin q_2$).
3. Solve each factor, list every angle in range.
4. Name (boundary/interior), draw the pose.
5. State what is lost; rates $\to\infty$ for a velocity with a component there.
6. Check: two columns parallel at the pose.

Why rates blow up near $\det \mathbf{J} = 0$ [added]

$\mathbf{J}^{-1} = \operatorname{adj}\mathbf{J}/\det\mathbf{J}$. RR: $\dot{q}_1 = v_\perp/(L_1 \sin q_2) \to \infty$ as $q_2 \to 0$. $L = (3, 4)$, 0.1 m/s $\perp$ arm:

q_2	90°	30°	15°	5°	1°
$\dot{q}_1$	0.0333	0.0667	0.1288	0.3825	1.9100
$\dot{q}_2$	−0.0333	−0.1100	−0.2221	−0.6682	−3.3422

30° → 5°: $\sin 30°/\sin 5° = 5.7\times$. [checked]

Scenarios: line, circle, rectangle [added]

RR $L = (0.5, 0.4)$ m, $q_2 > 0$, frame {0}, pose $\mathbf{q} = (30°, 60°)$, tool $\mathbf{p} = (0.4330, 0.6500)$:

$$\mathbf{J} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}, \quad \det\mathbf{J} = 0.1732, \quad \mathbf{J}^{-1} = \begin{bmatrix} 0 & 2.3094 \\ -2.5 & -3.7528 \end{bmatrix}$$

(a) Line. $v = 0.2$ m/s along $+x_0$: $\dot{\mathbf{q}} = \mathbf{J}^{-1}(0.2, 0) = (0, -0.5)$ rad/s. Along $+y_0$: $(0.4619, -0.7506)$. 0.1 m takes 0.5 s. Recompute $\mathbf{J}$ each step: after 0.5 s $\mathbf{q} = (32.03°, 42.14°)$, tip $(0.5330, 0.6500)$. Reach limit on $y = 0.65$: $x = \sqrt{0.9^2 - 0.65^2} = 0.6225$ (0.1895 m, 0.947 s).

x	q_1, q_2 (°)	$\det\mathbf{J}$	$\dot{q}_1$	$\dot{q}_2$
0.433	30.00, 60.00	0.1732	0	−0.500
0.533	32.03, 42.14	0.1342	0.163	−0.794
0.600	37.81, 21.37	0.0729	0.562	−1.647
0.620	43.18, 7.14	0.0249	2.055	−4.990

(b) Circle $\rho = 0.1$ m, $v = 0.2$ m/s, CCW from rightmost point, centre $\mathbf{c} = (0.3330, 0.6500)$.

STEPS circle
1. $\Omega = v/\rho = 2$ rad/s; $T = 2\pi/\Omega = 3.1416$ s.
2. $\mathbf{p} = \mathbf{c} + \rho(\cos\Omega t, \sin\Omega t)$; $\boldsymbol{\upsilon} = 0.2(-\sin 2t, \cos 2t)$, $|\boldsymbol{\upsilon}| = v$. ω_z not controlled.
3. In workspace? farthest $0.83 < 0.9$; smallest $q_2 = 45.7°$.
4. Each sample: IK ($q_2 > 0$), $\mathbf{J}(\mathbf{q})$, $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\upsilon}$.
5. Check: $\mathbf{q}$ returns to $(30°, 60°)$ after T.

Ωt	$\mathbf{p}$ (m)	$\boldsymbol{\upsilon}$	$\mathbf{q}$ (°)	$\dot{\mathbf{q}}$ (rad/s)
0	(0.433, 0.650)	(0, 0.2)	(30.0, 60.0)	(0.462, −0.751)
$\pi/2$	(0.333, 0.750)	(−0.2, 0)	(44.5, 48.8)	(0.031, 0.443)
π	(0.233, 0.650)	(0, −0.2)	(35.4, 80.4)	(−0.365, 0.659)
$3\pi/2$	(0.333, 0.550)	(0.2, 0)	(20.3, 89.5)	(−0.136, −0.333)

$t = 0$: $\dot{\mathbf{q}} = (0.2 \cdot 2.3094,\ 0.2 \cdot (-3.7528)) = (0.462, -0.751)$. Largest rate on the lap 0.78 rad/s. [checked]

(c) Rectangle $A(0.45, 0.45)$, $B(0.65, 0.45)$, $C(0.65, 0.55)$, $D(0.45, 0.55)$, $v = 0.1$ m/s: sides AB 2 s, BC 1 s, CD 2 s, DA 1 s (6 s). $\boldsymbol{\upsilon}$ constant per side, $\mathbf{J}$ not: evaluate at the midpoint (IK, $q_2 > 0$).

side	midpoint	$\boldsymbol{\upsilon}$	$\mathbf{q}$ (°)	$\dot{\mathbf{q}}$ (rad/s)
AB	(0.55, 0.45)	(0.1, 0)	(6.1, 76.3)	(0.027, −0.283)
BC	(0.65, 0.50)	(0, 0.1)	(16.0, 49.0)	(0.240, −0.331)
CD	(0.55, 0.55)	(−0.1, 0)	(18.3, 60.8)	(−0.043, 0.315)
DA	(0.45, 0.50)	(0, −0.1)	(11.8, 83.9)	(−0.200, 0.251)

End of BC (corner C, 0.8515 m, $q_2 = 38.0°$): $(0.285, -0.446)$. At a corner $\boldsymbol{\upsilon}$ jumps: infinite acceleration, stop or blend (W5). [checked]

(d) Moving along the arm near the boundary. $r^2 = L_1^2 + L_2^2 + 2L_1L_2c_2 \Rightarrow \dot{q}_2 = -r\dot{r}/(L_1L_2 \sin q_2) \to \infty$ as $q_2 \to 0$. $\dot{r} = 0.1$ m/s: $\dot{q}_2 = -0.302, -0.681, -4.745$ rad/s at $r = 0.60, 0.85, 0.899$ m. Rate limit 2 rad/s at $r = 0.85$: $\dot{r} \le 2 \cdot 0.2 \sin 38.6°/0.85 = 0.294$ m/s. [checked]

TRAP
One $\mathbf{J}$ (start pose) for a whole path; end point outside the workspace (0.6225, not 0.633).

Manipulability and the velocity ellipsoid [p. 49–55]

DERIVATION ellipsoid equation [p. 51–53]

Unit rates $\dot{\mathbf{q}}^\top\dot{\mathbf{q}} = 1$, $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$: $\boldsymbol{\nu}^\top(\mathbf{J}^{-1})^\top\mathbf{J}^{-1}\boldsymbol{\nu} = \boldsymbol{\nu}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{\nu} = 1$. $\mathbf{A} = \mathbf{J}\mathbf{J}^\top$: semi-axes along eigenvectors $\mathbf{v}_i$, length $\sqrt{\lambda_i}$ (for $\boldsymbol{\nu} = \sqrt{\lambda_i}\mathbf{v}_i$: $\lambda_i\mathbf{v}_i^\top\mathbf{A}^{-1}\mathbf{v}_i = 1$ [added]).

Velocity ellipsoid, manipulability [p. 52–55]

$$\boldsymbol{\nu}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{\nu} \le 1, \quad \mathbf{A} = \mathbf{J}\mathbf{J}^\top$$

$$m = \frac{\sqrt{\lambda_{\min}}}{\sqrt{\lambda_{\max}}} \in [0, 1], \quad w = \sqrt{\det\mathbf{J}\mathbf{J}^\top} = |\det\mathbf{J}|$$

$\sqrt{\lambda_i} = \sigma_i$: tool speed per unit joint-rate norm along $\mathbf{v}_i$. $m = 1$ isotropic, 0 singular. $w = \sigma_1\sigma_2\cdots$ (RR $L_1L_2|\sin q_2|$) [added]. Some books use the reciprocal for m: state yours. 2×2: $\lambda = \frac{1}{2}(\operatorname{tr}\mathbf{A} \pm \sqrt{\operatorname{tr}^2\mathbf{A} - 4\det\mathbf{A}})$, $\det\mathbf{A} = (\det\mathbf{J})^2$, $\operatorname{tr}\mathbf{A} = L_1^2 + 2L_2^2 + 2L_1L_2c_2$ (RR).

WORKED RR $L = (3, 4)$: σ, m, w

q_2	σ_1	σ_2	m	w
15°	8.002	0.3881	0.0485	3.1058
60°	7.1328	1.4570	0.2043	10.3923
90°	6.0927	1.9696	0.3233	12.0000
150°	4.2711	1.4048	0.3289	6.0000

2R never reaches $m = 1$ (best 0.4267 at $q_2 \approx 125.8°$); w peaks at 90°. [checked]

WORKED $L = (0.5, 0.4)$, $\mathbf{q} = (30°, 60°)$: what to draw

$\mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 0.5825 & -0.2815 \\ -0.2815 & 0.1875 \end{bmatrix}$, $\lambda = 0.7288,\ 0.04116$; semi-axes 0.8537 (along 152.5°) and 0.2029 (along 62.5°), $m = 0.2376$: thin ellipse centred on the tool. [checked]

TRAP
λ for the semi-axis (it is $\sqrt{\lambda}$); $m = \lambda_{\min}$ alone; "bigger ellipse near a singularity": false, $\sigma_{\min} \to 0$.

Statics: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}_E$ [p. 57–59]

Joint torques from tool wrench [p. 58]

$$\boldsymbol{\tau} = \mathbf{J}^\top(\mathbf{q})\,\mathbf{F}_E, \quad \mathbf{F}_E = \begin{bmatrix} F_x & F_y & F_z & n_x & n_y & n_z \end{bmatrix}^\top, \quad \boldsymbol{\tau}^\top\dot{\mathbf{q}} = \mathbf{F}_E^\top\boldsymbol{\nu}$$

$\mathbf{F}_E$ (N; N m) in the frame of $\mathbf{J}$; $\boldsymbol{\tau}$ (N m; N). Planar: $\boldsymbol{\tau} = \mathbf{J}_{xy}^\top\mathbf{F} + n_z(1, 1, \ldots)^\top$. Reverse: $\mathbf{F} = \mathbf{J}^{-\top}\boldsymbol{\tau}$ (square, non-singular). **Convention** [added]: $\mathbf{F}_E$ = force the tool applies to the environment, $\boldsymbol{\tau}$ = actuator torque to hold it; a load on the tool flips signs. State frame and sign.

DERIVATION virtual work [added]

$\boldsymbol{\tau}^\top\dot{\mathbf{q}} = \mathbf{F}_E^\top\boldsymbol{\nu} = \mathbf{F}_E^\top\mathbf{J}\dot{\mathbf{q}}$ for all $\dot{\mathbf{q}} \Rightarrow \boldsymbol{\tau}^\top = \mathbf{F}_E^\top\mathbf{J} \Rightarrow \boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}_E$.

Force ellipsoid [p. 59]

$\boldsymbol{\tau}^\top\boldsymbol{\tau} \le 1$, $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$: $\mathbf{F}^\top\mathbf{J}\mathbf{J}^\top\mathbf{F} \le 1$ (no inverse). Same axes as the velocity ellipsoid, semi-axes $1/\sqrt{\lambda_i}$. Fast direction = weak force. At a singularity the force axis → ∞: structure carries load with zero torque.

WORKED $L = (0.5, 0.4)$, $\mathbf{q} = (30°, 60°)$; $\mathbf{J}^\top = \begin{bmatrix} -0.65 & 0.4330 \\ -0.4 & 0 \end{bmatrix}$

Torques. $\mathbf{F} = (0, -15)$ N: $\boldsymbol{\tau} = (-6.495, 0)$ N m ($\tau_2 = 0$: link 2 vertical, force through the elbow). $\mathbf{F} = (10, -20)$: $(-15.160, -4.000)$ (wrong **JF**: $(1.5, 4.330)$).

Force from torques. $\boldsymbol{\tau} = (2, 1)$: $-0.4F_x = 1 \Rightarrow F_x = -2.5$; $-0.65(-2.5) + 0.4330F_y = 2 \Rightarrow F_y = 0.866$; $|\mathbf{F}| = 2.646$ N.

Stretched $\mathbf{q}$ = (30°, 0): 50 N along the arm: $\boldsymbol{\tau} = (0, 0)$; 50 N $\perp$: (45, 20) N m. **Frame:** ${}^0\mathbf{J}^{\top 0}\mathbf{F} = {}^3\mathbf{J}^{\top 3}\mathbf{F}$ with ${}^3\mathbf{F} = \mathbf{R}_z(45°)^\top\mathbf{F}$. Axes: velocity 0.2029, 0.8537; force 4.929, 1.171. [checked]

STEPS Statics question
1. Sketch arm and force; frame {0}; sign convention.
2. $\mathbf{J}_{xy}$ in that frame (add ω_z row if a moment is given).
3. $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$: torque j = column j dotted with $\mathbf{F}$.
4. Check by moments: $\tau_1 = xF_y - yF_x$. Units N m.
5. Singular pose: forces needing zero torque.

SLIDE CHECK slides 21/45, 32, 51, 58

Slide 21 stresses {0} but slide 45 prints RR in the tool frame {3}: both right (rotation above). Slide 32 ends the product at $\mathbf{E}_n$; it should be $\mathbf{E}_M$. Slide 51: $\dot{\mathbf{q}}\dot{\mathbf{q}}^\top = 1$ should be $\dot{\mathbf{q}}^\top\dot{\mathbf{q}} = 1$. Slide 58 gives no frame or sign convention for $\mathbf{F}_E, \boldsymbol{\tau}$: write both.

Exam-style questions (marks assumed)

No W13 sample question is on W6. Typical lost marks: running sums, mixed frames, **JF** for $\mathbf{J}^\top\mathbf{F}$, radians, one **J** for a path.

EXAM-STYLE QUESTION Q1: Jacobian, statics, rates [assumed 7]

RR $L_1 = 0.5$, $L_2 = 0.4$ m. (a) 2×2 **J** in {0}, singular poses [3]. (b) $\mathbf{q} = (30°, 60°)$: torques holding tip force $(0, -10)$ N; explain τ_2 [2]. (c) $\dot{\mathbf{q}}$ for tip velocity $(0.1, 0)$ m/s [2].

SOLUTION (a) $\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$, $\det = L_1L_2(s_{12}c_1 - c_{12}s_1) = L_1L_2 \sin q_2$; zero at $q_2 = 0, \pi$ [3].

(b) $\mathbf{J} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$, $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F} = (-4.330, 0)$ N m [1]; $\tau_2 = 0$: link 2 vertical, force along it, no moment about joint 2 (load on tip: $(+4.330, 0)$) [1].

(c) $\det = 0.1732$; $\dot{\mathbf{q}} = \frac{1}{0.1732}\begin{bmatrix} 0 & 0.4 \\ -0.433 & -0.65 \end{bmatrix}\begin{bmatrix} 0.1 \\ 0 \end{bmatrix} = (0, -0.25)$ rad/s [2]. Check $-0.4(-0.25) = 0.1$ ✓. [checked]

EXAM-STYLE QUESTION Q2: spot the errors [assumed 5]

Student: $\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_2 & -L_2s_2 \\ L_1c_1 + L_2c_2 & L_2c_2 \end{bmatrix}$, "$\det = L_1L_2 \sin q_2$ only when $q_1 = 0$", "$\omega_z = \dot{q}_2$". $L = (0.5, 0.3)$, $\mathbf{q} = (30°, 60°)$: correct **J**; mark the claims.

SOLUTION Running sum: $s_2, c_2 \to s_{12}, c_{12}$ ($q_1 + q_2 = 90°$): $J_{11} = -0.55$, $J_{12} = -0.3$, $J_{21} = 0.4330$, $J_{22} = 0$ (student: −0.5098, 0.5830) [2].
$\det = L_1L_2 \sin q_2 = 0.1299$ for every q_1, frame-independent [2].
$\omega_z = \dot{q}_1 + \dot{q}_2$ [1]. [checked]

EXAM-STYLE QUESTION Q3: planar 3R, redundant [assumed 6]

$L = (0.4, 0.3, 0.2)$ m, $\mathbf{q} = (20°, 40°, -30°)$ ($\mathbf{J}_{xy}$ in the RRR box). (a) Tool velocity for $\dot{\mathbf{q}} = (0.3, -0.2, 0.5)$ [2]. (b) Smallest-norm $\dot{\mathbf{q}}$ for $(0.05, 0.02)$ m/s [3]. (c) When is $\mathbf{J}_{xy}$ rank-deficient? [1]

SOLUTION (a) $\boldsymbol{\upsilon} = (-0.1270, 0.2317)$ m/s, $\omega_z = 0.3 - 0.2 + 0.5 = 0.6$ rad/s [2].

(b) Fat: $\dot{\mathbf{q}} = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{\upsilon}$; $\mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 0.3861 & -0.4808 \\ -0.4808 & 0.6232 \end{bmatrix}$, $\det = 0.00945$, $\mathbf{y} = (4.3169, 3.3626)$, $\dot{\mathbf{q}} = \mathbf{J}^\top\mathbf{y} = (0.2069, -0.4664, 0.1507)$, $\|\dot{\mathbf{q}}\| = 0.5321$ [3]. Check $\mathbf{J}\dot{\mathbf{q}} = (0.05, 0.02)$ ✓.

(c) Rank 1 only if all links collinear (one straight elbow is fine for (x, y)); (x, y, φ): $\det = L_1L_2 \sin q_2$ [1]. [checked]

W6

W7 Manipulator dynamics: Lagrange and Newton–Euler

Forward and inverse dynamics [p. 6–12]

GIVEN $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$ (planned motion), mass data FIND joint torques/forces $\mathbf{Q}$ (control, motor sizing; no integration)

GIVEN $\mathbf{Q}$, state $\mathbf{q}, \dot{\mathbf{q}}$, mass data FIND $\ddot{\mathbf{q}}$ (needs $\mathbf{M}^{-1}$; integrate; simulation)

Torque balances [p. 11]: inertial ($\propto \ddot{q}$), centripetal, Coriolis (two rotating links), static load, gravity, friction.

Rigid body: COM, inertia tensor, Newton–Euler [p. 34–39]

COM $\mathbf{p}_C = \frac{1}{m}\iiint \mathbf{r}\rho\, dv$; $\{C\}$ at the COM, axes parallel to the link frame.

Inertia tensor in $\{A\}$, density ρ [p. 36]

$$ {}^{A}\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}, \quad I_{xx} = \iiint (y^2+z^2)\rho\, dv, \quad I_{xy} = -\iiint xy\rho\, dv $$

$$ I_{yy} = \iiint (x^2+z^2)\rho\, dv, \quad I_{yz} = -\iiint yz\rho\, dv $$

$$ I_{zz} = \iiint (x^2+y^2)\rho\, dv, \quad I_{xz} = -\iiint xz\rho\, dv $$

kg m^2; moments ≥ 0; products carry the minus sign; tensor changes with the frame [p. 39]. Parallel axis, standard bodies: p. 5.

DERIVATION cuboid W, L, H (x,y,z), frame $\{A\}$ at a corner [p. 37–38]

$\rho = m/(WLH)$: $I_{xx} = \rho\int_0^W\int_0^L\int_0^H (y^2+z^2)dz\,dy\,dx = \rho W\left(\frac{L^3}{3}H + L\frac{H^3}{3}\right) = \frac{m(L^2+H^2)}{3}$; $I_{xy} = -\rho\frac{W^2}{2}\frac{L^2}{2}H = -\frac{mWL}{4}$. Others by cyclic permutation.

$$ \text{corner: } {}^{A}\mathbf{I} = \begin{bmatrix} \frac{m(L^2+H^2)}{3} & -\frac{mWL}{4} & -\frac{mHW}{4} \\ -\frac{mWL}{4} & \frac{m(W^2+H^2)}{3} & -\frac{mHL}{4} \\ -\frac{mHW}{4} & -\frac{mHL}{4} & \frac{m(L^2+W^2)}{3} \end{bmatrix} $$

$$ \text{centroid: } {}^{C}\mathbf{I} = \operatorname{diag}\left(\frac{m(L^2+H^2)}{12}, \frac{m(W^2+H^2)}{12}, \frac{m(L^2+W^2)}{12}\right) $$

Ex.: $m = 6, (W, L, H) = (0.2, 0.3, 0.1)$: centroid $\operatorname{diag}(0.050, 0.025, 0.065)$; corner $I_{xx} = 0.200$, $I_{xy} = -0.090$, $I_{xz} = -0.030$, $I_{yz} = -0.045$ (parallel axis $0.05 + 6(0.15^2 + 0.05^2) = 0.200$). [checked]

SLIDE CHECK p. 37

Slide 37 prints $I_{xy} = +mWL/4$; slides 36, 38 give $-mWL/4$. Use the minus sign.

Newton and Euler [p. 34]

$$ \mathbf{F} = m\dot{\mathbf{v}}_C, \qquad \mathbf{N} = {}^{C}\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times {}^{C}\mathbf{I}\boldsymbol{\omega} $$

$\mathbf{F}, \mathbf{N}$ net force/moment about the COM; $\dot{\mathbf{v}}_C$ COM acceleration; all in one frame. $\boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} = \mathbf{0}$ for planar motion.

WORKED W8 practical Q1: one link

$m = 1$, ${}^{1}\mathbf{P}_C = (3,0,0)$, ${}^{C}\mathbf{I} = \operatorname{diag}(1,2,3)$, $q = bt + ct^2$. $\boldsymbol{\omega} = (0,0,b+2ct)$, $\dot{\boldsymbol{\omega}} = (0,0,2c)$. $\dot{\mathbf{v}}_C = \dot{\boldsymbol{\omega}} \times \mathbf{P}_C + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{P}_C) = (0,6c,0) + (-3\dot{q}^2, 0, 0)$:

$$ \boxed{\dot{\mathbf{v}}_C = (-3(b+2ct)^2,\ 6c,\ 0)} $$

$\mathbf{N} = (0,0,6c)$; $\tau = N_z + (\mathbf{P}_C \times \mathbf{F})_z = 6c + 18c = \boxed{24c}$ N m $= (I_{zz} + m3^2)\ddot{q}$. ($b$=1, c=0.5, t=1: $\dot{\mathbf{v}}_C = (-12, 3, 0)$, $\tau = 12$.) [checked]

Lagrange's method [p. 15–17]

Energy of link i [added]

$$ K_i = \tfrac{1}{2} m_i \mathbf{v}_{C_i}^{\top} \mathbf{v}_{C_i} + \tfrac{1}{2}\, {}^{i}\boldsymbol{\omega}_i^{\top}\, {}^{C_i}\mathbf{I}_i\, {}^{i}\boldsymbol{\omega}_i, \qquad P_i = m_i g\, h_{C_i} $$

Point mass: drop the rotation term. Planar link: rotation term $\frac{1}{2} I_{zz}\dot{\theta}^2$ with $\dot{\theta}$ the *absolute* rate ($\dot{q}_1 + \dot{q}_2$ for link 2). Zero of h is free.

Euler–Lagrange [p. 16–17]

$$ \mathcal{L} = K - P, \qquad Q_i = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} $$

Q_i: torque (R, N m) or force (P, N). Internal reactions never appear [p. 14].

STEPS Lagrange (W13 Q4)

1. Draw: $\{0\}$ (x right, y up), q_i and where measured from, gravity.
2. COM positions; differentiate; $v^2 = \dot{x}^2 + \dot{y}^2$.
3. K (translation + rotation), $P = \sum m_i g h_i$ in full; $\mathcal{L} = K - P$.
4. Per joint: $\partial\mathcal{L}/\partial\dot{q}_i$; d/dt (product rule); $\partial\mathcal{L}/\partial q_i$; subtract.
5. $\ddot{q}_j$ coefficients $\rightarrow \mathbf{M}$; $\dot{q}_j\dot{q}_k$ terms $\rightarrow \mathbf{C}$; no-$\dot{q}$ terms $\rightarrow \mathbf{g}$.
6. Check: $\mathbf{M} = \mathbf{M}^{\top}$, $M_{ii} > 0$; at rest only $\mathbf{g}$; arm horizontal.

Shortcut [added]: $K = \frac{1}{2}\dot{\mathbf{q}}^{\top}\mathbf{M}\dot{\mathbf{q}}$, so $Q_i = \sum_j M_{ij}\ddot{q}_j + \sum_{j,k}\left(\frac{\partial M_{ij}}{\partial q_k} - \frac{1}{2}\frac{\partial M_{jk}}{\partial q_i}\right)\dot{q}_j\dot{q}_k + \frac{\partial P}{\partial q_i}$.

Single link: pendulum [added]

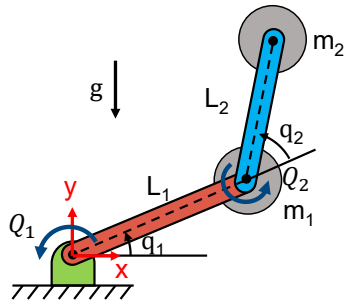
DERIVATION q from x_0, gravity $-y_0$, point mass m at ℓ / rod with COM ℓ_c

$K = \frac{1}{2}m\ell^2\dot{q}^2$, $P = mg\ell s_q$: $\partial\mathcal{L}/\partial\dot{q} = m\ell^2\dot{q}$, $\partial\mathcal{L}/\partial q = -mg\ell c_q$.

$$ Q = m\ell^2\ddot{q} + mg\ell\cos q; \qquad \text{rod: } Q = (I_c + m\ell_c^2)\ddot{q} + mg\ell_c\cos q $$

Uniform rod: $I_c + m\ell_c^2 = m\ell^2/12 + m\ell^2/4 = m\ell^2/3$ (not $m\ell^2$, not I_c). Ex.: rod $m = 2$, $\ell = 0.6$, $q = 30°$, $\ddot{q} = 4$: $Q = 0.240(4) + 5.886\cos 30° = 6.057$ N m. Release from horizontal: $\dot{q}_{bottom} = \sqrt{2(5.886)/0.240} = 7.004$ rad/s. [checked]

Planar 2R arm, point masses [p. 18–25]



[W07 lec p. 18] — 2R: m_1, m_2 at link tips; q_1 from x_0, q_2 relative; gravity $-y$

GIVEN L_i, m_i (tips), q_1 from x_0, q_2 relative, g along $-y$ FIND Q_1, Q_2; $\mathbf{M}, \mathbf{C}, \mathbf{g}$

DERIVATION 1. energies [p. 19–22]

$x_1 = L_1c_1$, $y_1 = L_1s_1$: $K_1 = \frac{1}{2}m_1L_1^2\dot{q}_1^2$, $P_1 = m_1gL_1s_1$.
$x_2 = L_1c_1 + L_2c_{12}$, $y_2 = L_1s_1 + L_2s_{12}$; $\dot{x}_2 = -L_1s_1\dot{q}_1 - L_2s_{12}(\dot{q}_1+\dot{q}_2)$, $\dot{y}_2 = L_1c_1\dot{q}_1 + L_2c_{12}(\dot{q}_1+\dot{q}_2)$.
$s_1s_{12}+c_1c_{12} = c_2 \Rightarrow \dot{x}_2^2+\dot{y}_2^2 = L_1^2\dot{q}_1^2+L_2^2(\dot{q}_1+\dot{q}_2)^2+2L_1L_2c_2\dot{q}_1(\dot{q}_1+\dot{q}_2)$.

$$ K_2 = \tfrac{1}{2}m_2\left[(L_1^2+2L_1L_2c_2+L_2^2)\dot{q}_1^2 + 2(L_2^2+L_1L_2c_2)\dot{q}_1\dot{q}_2 + L_2^2\dot{q}_2^2\right] $$

$P_2 = m_2g(L_1s_1+L_2s_{12})$. $A = m_1L_1^2 + m_2(L_1^2+2L_1L_2c_2+L_2^2)$, $B = m_2(L_1L_2c_2+L_2^2)$, $D = m_2L_2^2$: $K = \frac{1}{2}A\dot{q}_1^2 + B\dot{q}_1\dot{q}_2 + \frac{1}{2}D\dot{q}_2^2$, $P = (m_1+m_2)gL_1s_1 + m_2gL_2s_{12}$.

DERIVATION 2. Euler–Lagrange, joint 1

$\partial\mathcal{L}/\partial\dot{q}_1 = A\dot{q}_1 + B\dot{q}_2$; $\dot{A} = -2m_2L_1L_2s_2\dot{q}_2$, $\dot{B} = -m_2L_1L_2s_2\dot{q}_2$:

$$ \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}_1} = A\ddot{q}_1 + B\ddot{q}_2 - 2m_2L_1L_2s_2\dot{q}_1\dot{q}_2 - m_2L_1L_2s_2\dot{q}_2^2 $$

$\partial\mathcal{L}/\partial q_1 = -\partial P/\partial q_1 = -[(m_1+m_2)gL_1c_1 + m_2gL_2c_{12}]$ (K has no q_1).

$Q_1 = A\ddot{q}_1 + B\ddot{q}_2 - m_2L_1L_2s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) + (m_1+m_2)gL_1c_1 + m_2gL_2c_{12}$

DERIVATION 3. Euler–Lagrange, joint 2

$\partial\mathcal{L}/\partial\dot{q}_2 = B\dot{q}_1 + D\dot{q}_2$; $\frac{d}{dt} = B\ddot{q}_1 + D\ddot{q}_2 - m_2L_1L_2s_2\dot{q}_1\dot{q}_2$.
$\partial\mathcal{L}/\partial q_2 = \frac{1}{2}\frac{\partial A}{\partial q_2}\dot{q}_1^2 + \frac{\partial B}{\partial q_2}\dot{q}_1\dot{q}_2 - \frac{\partial P}{\partial q_2} = -m_2L_1L_2s_2(\dot{q}_1^2 + \dot{q}_1\dot{q}_2) - m_2gL_2c_{12}$.
The $\dot{q}_1\dot{q}_2$ terms cancel:

$$ Q_2 = B\ddot{q}_1 + D\ddot{q}_2 + m_2L_1L_2s_2\dot{q}_1^2 + m_2gL_2c_{12} $$

Result $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$ [p. 24–25]

$$ \mathbf{M} = \begin{bmatrix} m_1L_1^2+m_2(L_1^2+2L_1L_2c_2+L_2^2) & m_2(L_1L_2c_2+L_2^2) \\ m_2(L_1L_2c_2+L_2^2) & m_2L_2^2 \end{bmatrix} $$

$$ \mathbf{C} = \begin{bmatrix} -m_2L_1L_2s_2(2\dot{q}_1\dot{q}_2+\dot{q}_2^2) \\ m_2L_1L_2s_2\dot{q}_1^2 \end{bmatrix} $$

$$ \mathbf{g} = \begin{bmatrix} (m_1+m_2)gL_1c_1 + m_2gL_2c_{12} \\ m_2gL_2c_{12} \end{bmatrix} $$

c_{12} in $\mathbf{g}$: gravity on mass 2 acts at the absolute angle q_1+q_2.

WORKED $m = (2,1), L = (1, 0.8), q = (30°, 45°), \dot{q} = (1, -0.5), \ddot{q} = (0.5, 2)$

$c_2 = s_2 = 0.7071$, $h = m_2L_1L_2s_2 = 0.5657$. $M_{11} = 4.7714$, $M_{12} = 1.2057$, $M_{22} = 0.64$. $C_1 = -0.5657(2(1)(-0.5) + 0.25) = 0.4243$, $C_2 = 0.5657$. $g_1 = 3(9.81)\cos 30° + 0.8(9.81)\cos 75° = 27.518$, $g_2 = 2.031$.

$$ Q_1 = 2.3857 + 2.4114 + 0.4243 + 27.5183 = \boxed{32.740} $$

$$ Q_2 = 0.6028 + 1.28 + 0.5657 + 2.0312 = \boxed{4.480}\ \text{N m} $$

Power: $\dot{\mathbf{q}}^{\top}\mathbf{Q} = 30.500$ W $= \dot{K} + \dot{P} = 3.997 + 26.503$. [checked]

TRAP

$\mathcal{L} = K + P$; missing cross term $2L_1L_2c_2$; g_2 with sin; calling $\dot{q}_1\dot{q}_2$ centripetal (it is Coriolis); no product rule on $\cos q_2\,\dot{q}$.

Structure of M, C, g and properties [p. 25–31]

General equation [p. 25–26]

$$\underbrace{\mathbf{Q}}_{n\times 1} = \underbrace{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}}}_{n\times n} + \underbrace{\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})}_{\text{vel. products}} + \underbrace{\mathbf{g}(\mathbf{q})}_{\text{gravity}} \quad \left(+\,\mathbf{f}(\dot{\mathbf{q}}) + \mathbf{J}^{\top}(\mathbf{q})\mathbf{w}\right)$$

$\mathbf{f}$ friction; $\mathbf{J}^{\top}\mathbf{w}$ torques balancing tool wrench $\mathbf{w}$. Forward: $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g} - \mathbf{f} - \mathbf{J}^{\top}\mathbf{w})$ [p. 48]. Task space: $\mathcal{F} = \mathbf{M}_x\ddot{\mathbf{x}} + \mathbf{C}_x + \mathbf{g}_x$, $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ [p. 32].

$$\mathbf{C} = C(\mathbf{q})[\dot{\mathbf{q}}^2] + B(\mathbf{q})[\dot{\mathbf{q}}\dot{\mathbf{q}}] \text{ [p. 30]}$$

$$2R:\ \mathbf{C} = \begin{bmatrix} 0 & -h \\ h & 0 \end{bmatrix}\begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix} + \begin{bmatrix} -2h \\ 0 \end{bmatrix}[\dot{q}_1\dot{q}_2] \text{ [added]}$$

Squares $\dot{q}_i^2$ = centripetal; products $\dot{q}_i\dot{q}_j$ ($i<j$, $n(n-1)/2$ of them) = Coriolis.

Properties [p. 28], [added]

$\mathbf{M} = \mathbf{M}^{\top} > 0$ (since $K = \frac{1}{2}\dot{\mathbf{q}}^{\top}\mathbf{M}\dot{\mathbf{q}} > 0$), so $\mathbf{M}^{-1}$ exists; $\mathbf{M}, \mathbf{g}$ depend on $\mathbf{q}$ only; $\mathbf{C}$ is quadratic in $\dot{\mathbf{q}}$, zero at rest. With $\mathbf{C} = \mathbf{C}_m\dot{\mathbf{q}}$: $\dot{\mathbf{M}} - 2\mathbf{C}_m$ skew-symmetric $\Rightarrow$ ($\mathbf{Q} = \mathbf{0}$, no friction) $K + P$ const. 2R: $\det\mathbf{M} = m_2L_1^2L_2^2(m_1 + m_2s_2^2) > 0$; $\mathbf{C}_m = \begin{bmatrix} -h\dot{q}_2 & -h(\dot{q}_1+\dot{q}_2) \\ h\dot{q}_1 & 0 \end{bmatrix}$. [checked]

PUMA 560 $\mathbf{g}$ (N m) [p. 27, 29]: rest $[0, 31.64, 6.035, 0, 0.02825, 0]$; stretched $[0, 46.01, 8.772, 0, 0.02825, 0]$; up $[0, -0.7752, 0.2489, 0, 0, 0]$ (largest at full reach).

Scenarios: hold, accelerate, release [added]

situation	torque
held still	$\mathbf{Q} = \mathbf{g}$ [p. 27]
constant speed	$\mathbf{Q} = \mathbf{C} + \mathbf{g}$
from rest ($\dot{\mathbf{q}} = \mathbf{0}$)	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}$
tool presses with $\mathbf{w}$	$+\mathbf{J}^{\top}\mathbf{w}$ (state sign reading)
$\mathbf{Q}$ given	$\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g})$

WORKED slide 2R above

Hold: $\mathbf{g} = (27.518, 2.031)$; stretched ($q$=0): $(37.278, 7.848)$. From rest, $\ddot{\mathbf{q}} = (0.5, 2)$: $\mathbf{Q} = (32.315, 3.914)$. Release, $\mathbf{Q} = \mathbf{0}$ ($\det\mathbf{M} = 1.600$): $\ddot{\mathbf{q}} = -\mathbf{M}^{-1}\mathbf{g} = (-9.477, 14.679)$. Press $\mathbf{w} = (-5, 12)$ N: $\mathbf{J} = \begin{bmatrix} -1.2727 & -0.7727 \\ 1.0731 & 0.2071 \end{bmatrix}$, $\mathbf{Q} = \mathbf{g} + \mathbf{J}^{\top}\mathbf{w} = (46.759, 8.380)$. [checked]

TRAP

$\mathbf{Jw}$ for $\mathbf{J}^{\top}\mathbf{w}$; keeping $\mathbf{C}$ in a "hold still" answer.

Prismatic joints: RP arm, PR arm [added]

DERIVATION RP: θ_1 from x_0, slider d_2 along the arm; m_1 at l_1 (I_{zz1}), m_2 at d_2 (I_{zz2})

$v_2^2 = \dot{d}_2^2 + d_2^2\dot{\theta}_1^2$. $K = \frac{1}{2}(m_1l_1^2 + I_{zz1} + I_{zz2} + m_2d_2^2)\dot{\theta}_1^2 + \frac{1}{2}m_2\dot{d}_2^2$, $P = (m_1l_1 + m_2d_2)gs_1$.

θ_1: $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\theta}_1} = (\cdot)\ddot{\theta}_1 + 2m_2d_2\dot{d}_2\dot{\theta}_1$; $\frac{\partial\mathcal{L}}{\partial\theta_1} = -(m_1l_1 + m_2d_2)gc_1$. d_2: $m_2\ddot{d}_2$; $\frac{\partial\mathcal{L}}{\partial d_2} = m_2d_2\dot{\theta}_1^2 - m_2gs_1$.

$$\begin{bmatrix} \tau_1 \\ f_2 \end{bmatrix} = \mathbf{M}\begin{bmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \end{bmatrix} + \mathbf{C} + \mathbf{g}, \quad \mathbf{M} = \begin{bmatrix} m_1l_1^2 + I_{zz1} + I_{zz2} + m_2d_2^2 & 0 \\ 0 & m_2 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 2m_2d_2\dot{\theta}_1\dot{d}_2 \\ -m_2d_2\dot{\theta}_1^2 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} (m_1l_1 + m_2d_2)g\cos\theta_1 \\ m_2g\sin\theta_1 \end{bmatrix}$$

$\mathbf{M}$ diagonal yet $\mathbf{C} \neq \mathbf{0}$ (Coriolis $2m_2d_2\dot{\theta}_1\dot{d}_2$; centrifugal $-m_2d_2\dot{\theta}_1^2$).

WORKED $m = (2,1)$, $l_1 = 0.3$, $I_{zz} = (0.05, 0.02)$, $\theta_1 = 0.7$, $d_2 = 0.6$, $\dot{\theta}_1 = 1.3$, $\dot{d}_2 = -0.8$, $\ddot{\theta}_1 = 2$, $\ddot{d}_2 = 1.5$

$M_{11} = 0.61$: $\tau_1 = 1.220 - 1.248 + 9.004 = \boxed{8.976}$ N m; $f_2 = 1.500 - 1.014 + 6.320 = \boxed{6.806}$ N. Hold at 30°, $d_2 = 0.6$: (10.195 N m, 4.905 N). [checked]

TRAP

$K_2 = \frac{1}{2}m_2\dot{d}_2^2$ only; $\partial P/\partial d_2$ with $\cos\theta_1$; "M diagonal so no Coriolis".

Vertical slider (z up): $K = \frac{1}{2}m\dot{z}^2$, $P = mgz$, $f = m(\ddot{z} + g)$ ($m = 2$: hold 19.62 N; $\ddot{z} = 1.5$: 22.62 N). [checked]

PR arm (cart d_1 mass m_1, massless link L, tip m_2, φ from x_0): $K = \frac{1}{2}(m_1+m_2)\dot{d}_1^2 - m_2Ls_{\varphi}\dot{d}_1\dot{\varphi} + \frac{1}{2}m_2L^2\dot{\varphi}^2$, $P = m_2gLs_{\varphi}$; the $\dot{d}_1\dot{\varphi}$ terms cancel in joint 2:

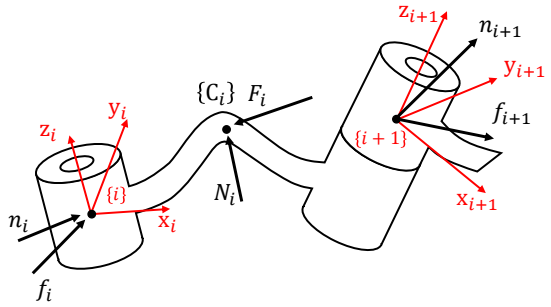
$$\begin{bmatrix} F_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} m_1+m_2 & -m_2Ls_{\varphi} \\ -m_2Ls_{\varphi} & m_2L^2 \end{bmatrix}\begin{bmatrix} \ddot{d}_1 \\ \ddot{\varphi} \end{bmatrix} + \begin{bmatrix} -m_2Lc_{\varphi}\dot{\varphi}^2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ m_2gLc_{\varphi} \end{bmatrix}$$

Ex.: $m = (3,1)$, $L = 0.5$, $\varphi = 40°$, $\dot{d}_1 = 1$, $\dot{\varphi} = 2$, $\ddot{d}_1 = 0.5$, $\ddot{\varphi} = 1$: $F_1 = 0.147$ N, $\tau_2 = 3.847$ N m. [checked]

Iterative Newton–Euler [p. 40–46]

Per link, force and moment balance [p. 14, 41]. **Outward** (base→tool): $\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}, \dot{\boldsymbol{v}}$, then $\mathbf{F}_i, \mathbf{N}_i$ [p. 43]. **Inward** (tool→base): force/moment balance, then Q_i [p. 44–45].

GIVEN $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$; frames ${}^{i}\mathbf{P}_{i+1}$, ${}^{i}_{i+1}\mathbf{R}$; m_i, ${}^{i}\mathbf{P}_{C_i}$, ${}^{C_i}\mathbf{I}_i$ FIND $\mathbf{Q}$



[W07 lec p. 45] — link i: $\mathbf{f}_i, \mathbf{n}_i$ from link $i-1$ at $\{i\}$; $\mathbf{f}_{i+1}, \mathbf{n}_{i+1}$ from link $i+1$; $\mathbf{F}_i, \mathbf{N}_i$ required by the motion

Outward, $i = 0 \to n-1$, revolute $i+1$ [added]; $\hat{\mathbf{Z}} = (0,0,1)^{\top}$

$\mathbf{R} = {}^{i}_{i+1}\mathbf{R}$, $\mathbf{P} = {}^{i}\mathbf{P}_{i+1}$; left side in $\{i+1\}$:

$${}^{i+1}\boldsymbol{\omega}_{i+1} = \mathbf{R}^{\top}{}^{i}\boldsymbol{\omega}_i + \dot{q}_{i+1}\hat{\mathbf{Z}}$$

$${}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} = \mathbf{R}^{\top}{}^{i}\dot{\boldsymbol{\omega}}_i + \mathbf{R}^{\top}{}^{i}\boldsymbol{\omega}_i \times \dot{q}_{i+1}\hat{\mathbf{Z}} + \ddot{q}_{i+1}\hat{\mathbf{Z}}$$

$${}^{i+1}\dot{\boldsymbol{v}}_{i+1} = \mathbf{R}^{\top}\left({}^{i}\dot{\boldsymbol{\omega}}_i \times \mathbf{P} + {}^{i}\boldsymbol{\omega}_i \times ({}^{i}\boldsymbol{\omega}_i \times \mathbf{P}) + {}^{i}\dot{\boldsymbol{v}}_i\right)$$

$$\dot{\boldsymbol{v}}_C = \dot{\boldsymbol{\omega}} \times \mathbf{P}_C + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{P}_C) + \dot{\boldsymbol{v}}$$

$$\mathbf{F}_{i+1} = m_{i+1}\dot{\boldsymbol{v}}_C, \quad \mathbf{N}_{i+1} = {}^{C}\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times {}^{C}\mathbf{I}\boldsymbol{\omega}$$

Start ${}^{0}\boldsymbol{\omega}_0 = \dot{\boldsymbol{\omega}}_0 = \mathbf{0}$. Gravity trick: ${}^{0}\dot{\boldsymbol{v}}_0 = (0, g, 0)^{\top}$ (base accelerates up).

Prismatic joint $i+1$: lines that change [added]

$${}^{i+1}\boldsymbol{\omega}_{i+1} = \mathbf{R}^{\top}{}^{i}\boldsymbol{\omega}_i, \quad {}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} = \mathbf{R}^{\top}{}^{i}\dot{\boldsymbol{\omega}}_i$$

$${}^{i+1}\dot{\boldsymbol{v}}_{i+1} = \mathbf{R}^{\top}\left({}^{i}\dot{\boldsymbol{\omega}}_i \times \mathbf{P} + {}^{i}\boldsymbol{\omega}_i \times ({}^{i}\boldsymbol{\omega}_i \times \mathbf{P}) + {}^{i}\dot{\boldsymbol{v}}_i\right) + 2\,{}^{i+1}\boldsymbol{\omega}_{i+1} \times \dot{d}_{i+1}\hat{\mathbf{Z}} + \ddot{d}_{i+1}\hat{\mathbf{Z}}$$

Inward unchanged, but $Q_i = \mathbf{f}_i^{\top}\hat{\mathbf{Z}}$.

Inward, $i = n \to 1$, $\mathbf{f}_{n+1} = \mathbf{n}_{n+1} = \mathbf{0}$ [p. 44–45]

$\mathbf{R}' = {}^{i}_{i+1}\mathbf{R}$; all in $\{i\}$:

$$\mathbf{f}_i = \mathbf{R}'\mathbf{f}_{i+1} + \mathbf{F}_i$$

$$\mathbf{n}_i = \mathbf{N}_i + \mathbf{R}'\mathbf{n}_{i+1} + \mathbf{P}_{C_i} \times \mathbf{F}_i + \mathbf{P}_{i+1} \times \mathbf{R}'\mathbf{f}_{i+1}$$

$$Q_i = \mathbf{n}_i^{\top}\hat{\mathbf{Z}}\ (\text{R}), \quad Q_i = \mathbf{f}_i^{\top}\hat{\mathbf{Z}}\ (\text{P})$$

Slide balances (p. 45): $\mathbf{F}_i = \mathbf{f}_i - \mathbf{R}'\mathbf{f}_{i+1}$, $\mathbf{N}_i = \mathbf{n}_i - \mathbf{R}'\mathbf{n}_{i+1} - \mathbf{P}_{C_i} \times \mathbf{f}_i - (\mathbf{P}_{i+1} - \mathbf{P}_{C_i}) \times \mathbf{R}'\mathbf{f}_{i+1}$; substitute $\mathbf{f}_i$: the $\mathbf{P}_{C_i} \times \mathbf{R}'\mathbf{f}_{i+1}$ terms cancel [added].

SLIDE CHECK NE, p. 45

Slide writes ${}^{i}\mathbf{T}_{i+1}$ in the force balance (a rotation ${}^{i}_{i+1}\mathbf{R}$ is meant); outward equations (p. 43) are in words only; boxes are Craig's.

STEPS Newton–Euler by hand

1. Frames with joint axes z; list ${}^{i}\mathbf{P}_{i+1}$, ${}^{i}_{i+1}\mathbf{R}$, ${}^{i}\mathbf{P}_{C_i}$, m_i, $\mathbf{I}_i$.
2. ${}^{0}\boldsymbol{\omega}_0 = \dot{\boldsymbol{\omega}}_0 = \mathbf{0}$, ${}^{0}\dot{\boldsymbol{v}}_0 = (0, g, 0)^{\top}$.
3. Outward $i = 1..n$: $\boldsymbol{\omega}_i, \dot{\boldsymbol{\omega}}_i, \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{v}}_{C_i}, \mathbf{F}_i, \mathbf{N}_i$ (point mass: $\mathbf{N}_i = \mathbf{0}$).
4. Inward $i = n..1$: $\mathbf{f}_i, \mathbf{n}_i$.
5. $Q_i = \mathbf{n}_i \cdot \hat{\mathbf{Z}}$ (P: $\mathbf{f}_i \cdot \hat{\mathbf{Z}}$).
6. Check: static case leaves only $\mathbf{g}$.

DERIVATION slide homework p. 46: NE on the 2R point-mass arm

${}^{0}_{1}\mathbf{R} = R_z(q_1)$, ${}^{1}_{2}\mathbf{R} = R_z(q_2)$, ${}^{1}\mathbf{P}_2 = {}^{1}\mathbf{P}_{C_1} = (L_1, 0, 0)$, ${}^{2}\mathbf{P}_{C_2} = (L_2, 0, 0)$, $\mathbf{N}_i = \mathbf{0}$. $a = -L_1\dot{q}_1^2 + gs_1$, $b = L_1\ddot{q}_1 + gc_1$.

Out. ${}^{1}\dot{\boldsymbol{v}}_1 = (gs_1, gc_1, 0)$; ${}^{1}\dot{\boldsymbol{v}}_{C_1} = (a, b, 0)$, $\mathbf{F}_1 = m_1(a, b, 0)$. ${}^{2}\dot{\boldsymbol{v}}_2 = R_z(q_2)^{\top}(a, b, 0) = (c_2a + s_2b, -s_2a + c_2b, 0)$; ${}^{2}\boldsymbol{\omega}_2 = (0, 0, \dot{q}_1 + \dot{q}_2)$; ${}^{2}\dot{\boldsymbol{v}}_{C_2} = (-L_2(\dot{q}_1+\dot{q}_2)^2, L_2(\ddot{q}_1+\ddot{q}_2), 0) + {}^{2}\dot{\boldsymbol{v}}_2$; $\mathbf{F}_2 = m_2\dot{\boldsymbol{v}}_{C_2}$.

In. $\mathbf{f}_2 = \mathbf{F}_2$, $\mathbf{n}_2 = \mathbf{P}_{C_2} \times \mathbf{F}_2 = (0, 0, L_2F_{2y})$:

$$Q_2 = m_2L_2^2(\ddot{q}_1 + \ddot{q}_2) + m_2L_1L_2c_2\ddot{q}_1 + m_2L_1L_2s_2\dot{q}_1^2 + m_2gL_2c_{12}$$

($=$ Lagrange Q_2.) $\mathbf{f}_1 = R_z(q_2)\mathbf{f}_2 + \mathbf{F}_1$, $Q_1 = Q_2 + L_1F_{1y} + L_1(s_2F_{2x} + c_2F_{2y})$ $=$ Lagrange Q_1 [checked].

WORKED NE numbers, same state as the Lagrange example

Link 1: $\boldsymbol{\omega}_1 = (0, 0, 1)$, $\dot{\boldsymbol{\omega}}_1 = (0, 0, 0.5)$, $\dot{\boldsymbol{v}}_1 = (4.905, 8.496, 0)$, $\dot{\boldsymbol{v}}_{C_1} = (3.905, 8.996, 0)$, $\mathbf{F}_1 = (7.810, 17.991, 0)$. Link 2: $\boldsymbol{\omega}_2 = (0, 0, 0.5)$, $\dot{\boldsymbol{\omega}}_2 = (0, 0, 2.5)$, $\dot{\boldsymbol{v}}_2 = (9.122, 3.600, 0)$, $\dot{\boldsymbol{v}}_{C_2} = (8.922, 5.600, 0) = \mathbf{f}_2$; $\mathbf{n}_2 = (0, 0, 4.480)$. $\mathbf{f}_1 = (10.159, 28.260, 0)$, $\mathbf{n}_1 = (0, 0, 32.740)$: $Q = (32.740, 4.480)$, equal to Lagrange. [checked]

Lagrange versus Newton–Euler [p. 12–14, 47]

ALTERNATIVE APPROACHES

A. Lagrange (energy)	B. Newton–Euler (forces)
Whole robot; no reactions [p. 14]. Needs COM positions, K, P. Closed-form $\mathbf{M}, \mathbf{C}, \mathbf{g}$: analysis, control design. Mult. $32n^4 + 86n^3 + 171n^2 + 53n - 128$.	Link by link, out then in [p. 41]. Needs frames, $\mathbf{P}_i$, $\mathbf{I}_i$. Numerical, gives reactions. Real-time. Mult. $104n - 77$.

Cost [p. 47]: $n = 2$: 131 vs 1 862; $n = 3$: 235 vs 6 484; $n = 6$: 547 vs 66 394 (adds 482 vs 51 456). Same torques either way.

W7

Exam questions

W13 sample Q4 (derive the equations of motion, 3D RRR arm): solved on p. 42. If time runs out: give $\mathbf{M}$ and $\mathbf{g}$ correctly, set up $\mathbf{C}$. Marks not published; indicative, 1 per step.

EXAM-STYLE QUESTION Uniform links, 2R [indicative 8]

Rods $m_1 = 3$, $\ell_1 = 0.6$; $m_2 = 2$, $\ell_2 = 0.4$ m ($I_c = m\ell^2/12$); q_1 from horizontal, q_2 relative, gravity $-y$. (a) $\mathbf{M}, \mathbf{g}$. (b) At rest, $q = (45°, -30°)$, $\ddot{\mathbf{q}} = (1, 2)$: torques.

SOLUTION $I_1 = m_1\ell_1^2/12 = 0.09$, $I_2 = 0.0267$. Link 1 about its end: $K_1 = \frac{1}{2}(I_1 + m_1\ell_1^2/4)\dot{q}_1^2$. Link 2: $v_{C_2}^2 = \ell_1^2\dot{q}_1^2 + \frac{\ell_2^2}{4}(\dot{q}_1+\dot{q}_2)^2 + \ell_1\ell_2 c_2\dot{q}_1(\dot{q}_1+\dot{q}_2)$, $K_2 = \frac{1}{2}m_2 v_{C_2}^2 + \frac{1}{2}I_2(\dot{q}_1+\dot{q}_2)^2$.

$$M_{11} = I_1 + \tfrac{m_1\ell_1^2}{4} + I_2 + m_2\left(\ell_1^2 + \tfrac{\ell_2^2}{4} + \ell_1\ell_2 c_2\right)$$

$$M_{12} = I_2 + m_2\left(\tfrac{\ell_2^2}{4} + \tfrac{\ell_1\ell_2}{2}c_2\right), \quad M_{22} = I_2 + \tfrac{m_2\ell_2^2}{4}$$

$$g_1 = \left(\tfrac{m_1\ell_1}{2} + m_2\ell_1\right)g c_1 + \tfrac{m_2\ell_2}{2} g c_{12}, \qquad g_2 = \tfrac{m_2\ell_2}{2} g c_{12}$$

(b) $\mathbf{M} = \begin{bmatrix} 1.6024 & 0.3145 \\ 0.3145 & 0.1067 \end{bmatrix}$, $\mathbf{g} = (18.357, 3.790)$, $\mathbf{C} = \mathbf{0}$: $\boxed{\mathbf{Q} = (20.589,\ 4.318)\ \text{N m}}$. Check $M_{22} = m_2\ell_2^2/3 = 0.1067$. [checked]

EXAM-STYLE QUESTION Spot the errors: RP arm [indicative 5]

Revolute θ (from horizontal) then slider d, point mass $m_2 = 1.5$ at d, gravity $-y$. Student: "$K = \frac{1}{2}m_2\dot{d}^2$; $f_2 = m_2\ddot{d} + m_2 g\cos\theta$; $\tau_1 = m_2 g d\cos\theta$; $\mathbf{M}$ diagonal, so no Coriolis." Correct f_2, τ_1 at $d = 0.4$, $\dot{d} = 0.3$, $\ddot{d} = 0.5$, $\theta = 0.9$, $\dot{\theta} = 1.1$, $\ddot{\theta} = 0.6$.

SOLUTION (1) $K = \frac{1}{2}m_2(\dot{d}^2 + d^2\dot{\theta}^2)$. (2) $\partial P/\partial d = m_2 g\sin\theta$. (3) τ_1 misses $m_2 d^2\ddot{\theta} + 2m_2 d\dot{d}\dot{\theta}$. (4) Diagonal $\mathbf{M}$ removes only acceleration coupling. $f_2 = 1.5(0.5 - 0.4(1.21)) + 1.5(9.81)\sin 0.9 = \boxed{11.551}$ N; $\tau_1 = 0.144 + 0.396 + 3.659 = \boxed{4.199}$ N m (student: 9.897 N, 3.659 N m). [checked]

EXAM-STYLE QUESTION Short explanations [indicative 2 each]

(a) Forward vs inverse dynamics, one use each. (b) Why is $\mathbf{M}$ invertible? (c) Why NE for real time? (d) Non-zero terms held still / at constant speed? (e) Mult. for $n = 5$.

SOLUTION (a) Inverse: $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \to \mathbf{Q}$ (computed torque); forward: $\mathbf{Q} \to \ddot{\mathbf{q}}$ (simulation). (b) $K = \frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}} > 0$, so $\mathbf{M} > 0$, $\det \neq 0$. (c) NE is $O(n)$; Lagrange $O(n^4)$ but gives symbolic structure; same torques. (d) Still: $\mathbf{g}$; constant speed: $\mathbf{C} + \mathbf{g}$. (e) NE 443; Lagrange 35 162. [checked]

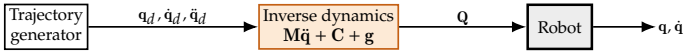
W8 Control 1: linear, partitioned and PID joint control

Which controller for which situation [added]

Law	Signal	Fixes / cannot fix
P	$f=-k_p x$	stiffness only; rings; offset under constant disturbance
PD	$-k_p x - k_v \dot{x}$	stiffness $k{+}k_p$, damping $b{+}k_v$; gains need m,b,k; offset remains
Partitioned	$f=\alpha f'+\beta$, $f'=-k_v\dot{x}-k_p x$	cancels plant (unit mass): $k_p=\omega_n^2, k_v=2\omega_n$; regulation
+ feed-forward	$f'=\ddot{x}_d+k_v\dot{E}+k_pE$	tracking: $\ddot{E}+k_v\dot{E}+k_pE=0$
PID	$f'=\ddot{x}_d+k_v\dot{E}+k_pE+k_i\int E$	removes constant-disturbance offset; stable iff $k_vk_p>k_i$
PI [added]	$f'=k_pE+k_i\int E$	unstable on unit mass ($s^3+k_ps+k_i$, no s^2 term)

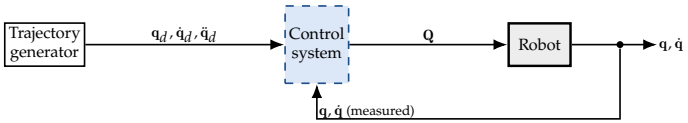
Open loop, closed loop, servo error [p. 8–14]

GIVEN $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$ (W5); dynamics $\mathbf{Q}=\mathbf{M}\ddot{\mathbf{q}}+\mathbf{C}+\mathbf{g}$ (W7) FIND torques $\mathbf{Q}(t)$ (N m R, N P) so that $\mathbf{q}$ follows $\mathbf{q}_d$



no feedback: errors never corrected

[drawn for this booklet] (1) open loop [p. 11]



[drawn for this booklet] (2) closed loop [p. 12]; the control-system block is this week's topic. Slide p. 11 names the trajectory $\mathbf{q}(t)$; p. 12–13 $\mathbf{q}_D$, $\mathbf{q}_d$ (same).

Open loop, exact model: $\ddot{E}=0 \Rightarrow E(t)=E(0)+\dot{E}(0)t$: a start error or push is never removed. $E(0)=0.1, \dot{E}(0)=0.02$: $E(5)=0.2$ rad [checked]

Servo error [p. 13]

$$E=\mathbf{q}_d-\mathbf{q}, \quad \dot{E}=\dot{\mathbf{q}}_d-\dot{\mathbf{q}} \quad \text{(one joint: } E=x_d-x\text{)}$$

Use: desired − actual; with $E=q-q_d$ every sign flips. Independent joint control [p. 14]: n single-joint loops; coupling in $\mathbf{M}, \mathbf{C}$ treated as disturbance.

Spring–mass–damper: poles and three cases [p. 16–26]

Characteristic equation, ω_n, ζ [p. 16, 17]

$$m\ddot{x}+b\dot{x}+kx=0 \Rightarrow ms^2+bs+k=0, \quad s_{1,2}=\frac{-b\pm\sqrt{b^2-4mk}}{2m}$$

$$\omega_n=\sqrt{k/m}, \quad \zeta=\frac{b}{2\sqrt{mk}}, \quad s_{1,2}=-\zeta\omega_n\pm\omega_n\sqrt{\zeta^2-1} \text{ [added]}$$

Use: try $x=e^{st}$. $\zeta>1$ over, $=1$ critical, <1 under, $=0$ undamped (same test as b^2-4mk). Joint: $m\to I$. Slides never name ζ, ω_n.

Case	Poles	$x(t)$; constants from $x(0), \dot{x}(0)$
$b^2-4mk>0$ over [p. 18]	real, unequal	$A_1e^{s_1t}+A_2e^{s_2t}$; $s_1A_1+s_2A_2=\dot{x}_0$ — $A_1+A_2=x_0$,
<0 under [p. 21]	$\lambda\pm i\mu$	$e^{\lambda t}(A_1\cos\mu t+A_2\sin\mu t)$; $\lambda A_1+\mu A_2=\dot{x}_0$ — $A_1=x_0$,
$=0$ critical [p. 24]	$s=-\frac{b}{2m}$ twice	$(A_1+A_2t)e^{st}$; $A_1=x_0$, $sA_1+A_2=\dot{x}_0$

$\lambda=-\frac{b}{2m}, \mu=\frac{\sqrt{4mk-b^2}}{2m}$ [added]. Slips: $\lambda=-b/m$; $A_2=\dot{x}_0/\mu$ (forgets λA_1).

STEPS Classify and solve $x(t)$

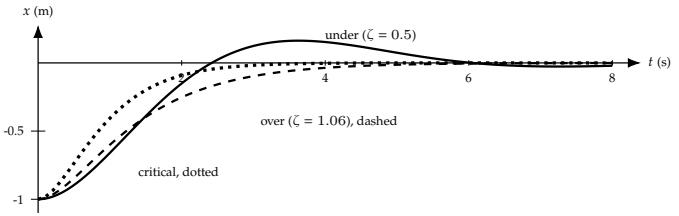
1. b^2-4mk (or ζ): name the case; poles.
2. Write the matching $x(t)$; two equations from $x_0, \dot{x}_0$.
3. Check: returns $x_0, \dot{x}_0$; $x\to 0$.

WORKED slide examples, $x(0)=-1, \dot{x}(0)=0$ [p. 19–26]

Over $1,1,1$: $s=-1,-2$; $A_1{+}A_2{=}-1$, $-A_1{-}2A_2{=}0 \Rightarrow x=-2e^{-t}+e^{-2t}$, $x(1)=-0.6004$; $\zeta=1.061$.

Under $1,1,1$: $s=-\frac{1}{2}\pm i\frac{\sqrt{3}}{2}$; $A_1{=}-1$, $\frac{1}{2}+\frac{\sqrt{3}}{2}A_2{=}0 \Rightarrow A_2{=}-\frac{1}{\sqrt{3}}$; $x=-e^{-t/2}[\cos\frac{\sqrt{3}}{2}t+\frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t]$; peak $+0.1630$ at 3.628 s; $\zeta=0.5$.

Critical $1,4,4$: $s=-2$; $A_1{=}-1$, $2{+}A_2{=}0 \Rightarrow x=(-1-2t)e^{-2t}$, $x(1)=-0.4060$. [checked]



[drawn for this booklet] the three examples from the exact formulas. Real poles: farther left = faster; complex: Re sets decay, Im the frequency.

SLIDE CHECK p.26 sketch

Red hand sketch (large late bump for "complex"; "real, unequal" slowest) is not to scale. Exact 2% settling: critical 2.92 s < over 4.60 s < under 8.08 s. Only "critical is fastest without overshoot" is safe in general. [checked]

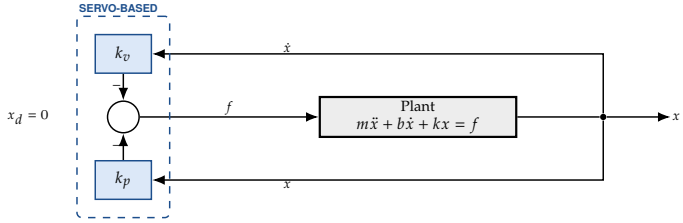
Actuated system and PD control [p. 27–32]

GIVEN $m\ddot{x}+b\dot{x}+kx=f$; $x, \dot{x}$ measured FIND k_p, k_v for critical damping

PD law [p. 28–32]

$$f=-k_px-k_v\dot{x} \Rightarrow m\ddot{x}+\underbrace{(b+k_v)}_{b'}\dot{x}+\underbrace{(k+k_p)}_{k'}x=0$$

Use: critical iff $b'^2=4mk'$, i.e. $(b+k_v)^2=4m(k+k_p)$. Always include the plant's own b, k. k_p [N/m], k_v [N s/m].



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}$, $\boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] PD regulation (after slide p. 30): both signs minus, target $x=0$.

WORKED critical damping of the raw plant [p. 32]

$m{=}2, b{=}3, k{=}8$; choose $k_p{=}10$: $k'=18$, $b'=\sqrt{4mk'}=12$, $k_v=b'-b=9$; double pole -3. Trap: $k_v=2\sqrt{k_p}=6.32$ stays under-damped (needs partitioning). [checked]

Control-law partitioning [p. 33–37]

GIVEN model m, b, k; measured $x, \dot{x}$ FIND $f=\alpha f'+\beta$ with $\ddot{x}=f'$ (unit mass) and servo gains independent of m, b, k

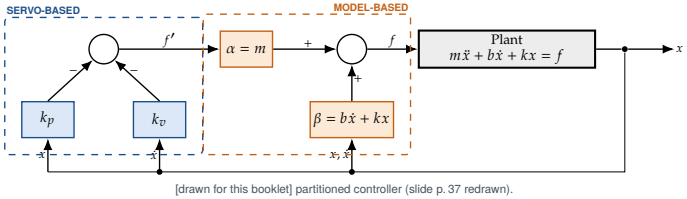
Partitioned law [p. 33–36]

$$f=\alpha f'+\beta, \quad \alpha=m, \ \beta=b\dot{x}+kx \Rightarrow \ddot{x}=f'$$

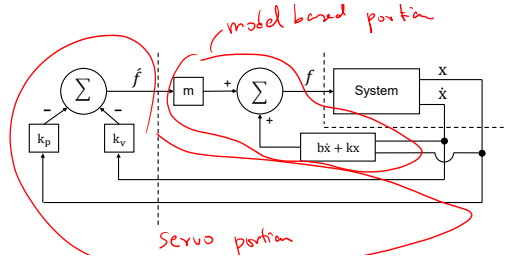
$$f'=-k_v\dot{x}-k_px \Rightarrow \ddot{x}+k_v\dot{x}+k_px=0, \quad k_v=2\sqrt{k_p} \text{ (critical)}$$

where f' = servo output, a desired *acceleration* [m/s^2]; α model gain [kg]; β model force [N]; k_p, k_v in s^{-2}, s^{-1}.

Use: model part cancels the physics, servo part sees a double integrator. Poles $p_{1,2}$: $k_v=-(p_1{+}p_2)$, $k_p=p_1p_2$ [added].



[drawn for this booklet] partitioned controller (slide p. 37 redrawn).



[W08 lec p. 37] — the lecturer's annotation: "servo portion" (k_p, k_v, f'), "model-based portion" ($m, b\dot{x}+kx$)

SLIDE CHECK p.36 critical damping

Slide prints $k_p^2=4k_v$ (swapped). From $s^2+k_vs+k_p$: $k_v^2=4k_p$. Slide's rule: $k_p{=}16 \Rightarrow k_v{=}64$, roots $-63.75, -0.25$; correct $k_v{=}8$, double root -4. [checked]

STEPS Partitioned force at one instant

1. $\alpha=m, \beta=b\dot{x}+kx$ at the *measured* $x, \dot{x}$.
2. $f'=-k_v\dot{x}-k_px$ (regulation) or $\ddot{x}_d+k_v\dot{E}+k_pE$ (tracking).
3. Critical: $k_p=\omega_n^2, k_v=2\omega_n$. Then $f=mf'+b\dot{x}+kx$.
4. Check: $(f-b\dot{x}-kx)/m=f'$.

WORKED force at a state

$m, b, k{=}2, 3, 8$; $k_p{=}25, k_v{=}10$; $x{=}0.5, \dot{x}{=}1$: $f'=-10(1)-25(0.5)=-22.5$; $f=2(-22.5)+3(1)+8(0.5)=-38$ N. Check $(-38-3-4)/2=-22.5$. [checked]

ALTERNATIVE APPROACHES

A. PD on raw plant

$f=-k_px-k_v\dot{x}$; $m\ddot{x}+(b{+}k_v)\dot{x}+(k{+}k_p)x=0$; $k_p=m\omega_n^2-k$, $k_v=2m\omega_n-b$.

B. Partitioned

$f=mf'+b\dot{x}+kx$; $\ddot{x}+k_v\dot{x}+k_px=0$; $k_p=\omega_n^2$, $k_v=2\omega_n$.

WORKED scenario: same plant, both ways

$m{=}0.5, b{=}1, k{=}18$, critical with $\omega_n{=}12$. **A:** $k_p=72-18=54$, $k_v=12-1=11$. **B:** $k_p=144$, $k_v=24$, $f=0.5f'+\dot{x}+18x$. Both: double pole -12. B's gains on the raw plant: $s^2+50s+324$, poles $-42.35, -7.65$ (not critical). [checked]

Gains from the poles [added]

Gains ↔ poles (unit-mass servo)

$$\omega_n = \sqrt{k_p},\quad \zeta = \frac{k_v}{2\sqrt{k_p}},\quad k_p = p_1 p_2,\quad k_v = -(p_1+p_2)$$

$$\text{overshoot } (\zeta<1) = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

$$t_{s,2\%} \approx \frac{4}{\zeta\omega_n}\ (\zeta<1),\quad \approx \frac{4}{|s_{slow}|}\ (\zeta>1),\quad = \frac{5.83}{\omega_n}\ (\zeta=1)$$

Use: given ζ: $k_v = 2\zeta\omega_n$. 5.83 = root of $(1+u)e^{-u} = 0.02$. Rules approximate. Re > 0 *unstable;* $\pm i\omega$ (k_v=0) *never settles.*

WORKED scenario: response from the poles [checked]

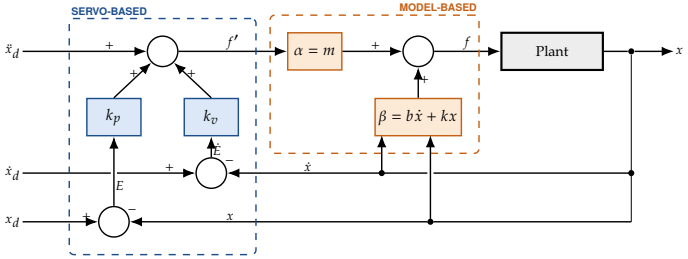
(a) $-3 \pm 4i$: ω_n=5, ζ=0.6, k_p=25, k_v=6; overshoot 9.48%; $t_s \approx 1.33$ s (exact 1.19).
(b) $-2, -8$: k_p=16, k_v=10, ζ=1.25; $x = (8e^{-2t} - 2e^{-8t})/6$; $t_s \approx 2.0$ s (exact 2.10).
(c) $-5, -5$: k_p=25, k_v=10; $t_s = 5.83/5 = 1.17$ s. (d) k_v=0, k_p=9: $\pm 3i$, period 2.09 s.

Trajectory following [p. 38–40]

Servo law with feed-forward [p. 39]

$$f' = \ddot{x}_d + k_v\dot{E} + k_p E,\quad f = mf' + b\dot{x} + kx \Rightarrow \ddot{E} + k_v\dot{E} + k_p E = 0$$

Use: from $\ddot{x} = f'$ and $\ddot{E} = \ddot{x}_d - \ddot{x}$; the error obeys the regulation equation for any smooth trajectory. Without $\ddot{x}_d$, constant $\ddot{x}_d = a$ leaves $E_{ss} = a/k_p$.



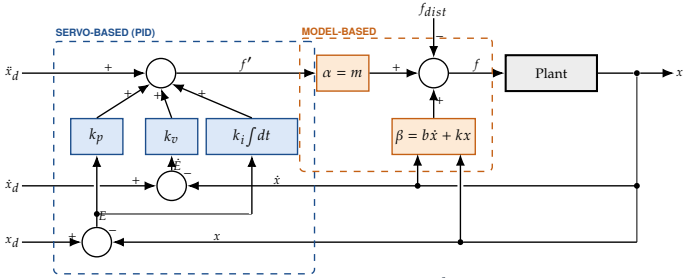
[drawn for this booklet] trajectory-following controller (slide p. 40 redrawn): x, $\dot{x}$ enter the error junctions with − and the model part β.

WORKED tracking numbers [checked]

k_p=100, k_v=20 (critical, ω_n=10), $E(0)$=0.5, $\dot{E}(0)$=0: $E = E_0(1+10t)e^{-10t}$, $E(0.5) = 0.5(6)e^{-5} = 0.0202$ m. $E(0) = \dot{E}(0) = 0$: stays 0. No feedforward, $x_d = t^2$: $E_{ss} = 2/100 = 0.02$ m.

Disturbance, steady-state error and PID [p. 41–45]

GIVEN partitioned tracking controller; unmodelled constant force f_{dist} FIND error equation, E_{ss}, cure



[drawn for this booklet] PID with disturbance (slides p. 41, 44 combined). Without $k_i\int dt$: the PD loop of p. 41. f_{dist} enters with −.

Disturbance and steady-state error [p. 41–43]

$$m\ddot{x} + b\dot{x} + kx = f - f_{dist},\ f = mf' + \beta \Rightarrow \ddot{E} + k_v\dot{E} + k_p E = \frac{f_{dist}}{m}$$

$$f_{dist} \text{ const},\ \ddot{E}=\dot{E}=0:\quad E_{ss} = \frac{f_{dist}}{k_p\, m}$$

Use: bounded f_{dist} $\Rightarrow$ bounded E (BIBO). Only k_p, m matter (k_v, b, k do not). Tolerance E_{max}: $k_p \geq f_{dist}/(mE_{max})$. Joint sags in the direction of the push.

WORKED steady-state error [checked]

m=1.5, f_{dist}=6, k_p=64, k_v=16: $E_{ss} = 6/(64 \cdot 1.5) = 0.0625$ m (same for $k_v = 40$). For $E_{ss} \leq 5$ mm: $k_p \geq 800$, $k_v = 2\sqrt{800} = 56.57$ (big gains: use integral action).

TRAP

"$E_{ss} = f/k_v$"; "raising k_v cuts it" (only changes speed); "independent of mass"; "PD removes it".

PID law and stability [p. 44, 45]

$$f' = \ddot{x}_d + k_v\dot{E} + k_p E + k_i\int E\,dt \Rightarrow \dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_i E = \frac{\dot{f}_{dist}}{m}$$

$$s^3 + k_v s^2 + k_p s + k_i = 0:\quad \text{stable} \iff k_v, k_p > 0,\ 0 < k_i < k_v k_p \text{ [added]}$$

Use: constant f_{dist}: $\dot{f}_{dist}$=0, steady state $k_i E = 0 \Rightarrow E = 0$. k_i [s^{-3}]. Routh: s^1 entry $(k_v k_p - k_i)/k_v > 0$. At $k_i = k_v k_p$: $(s + k_v)(s^2 + k_p)$, oscillation at $\sqrt{k_p}$.

STEPS PID stability

1. Write $s^3 + k_v s^2 + k_p s + k_i$; compare k_i with $k_v k_p$: below stable, equal marginal, above unstable.
2. Roots if asked: calculator polynomial solver.

WORKED scenario: is it stable? [checked]

k_p=64, k_v=16 (limit 1024): k_i=900: $-15.60, -0.20 \pm 7.59i$ stable; 1024: $-16, \pm 8i$ marginal; 1100: $-16.23, +0.12 \pm 8.23i$ unstable. k_p=30, k_v=5, k_i=200 > 150: unstable ($-5.79, +0.39 \pm 5.87i$); k_i=100 stable.

TRAP

"integral never hurts" (large k_i destabilises); "PI is enough" (unstable on unit mass).

DC motor, friction and gearing [p. 46–50]

GIVEN motor (k_4, I_m, c_m), gear ratio η, link I, c; planned $\ddot{q}, \dot{q}$ at the axle FIND axle torque Q, motor torque Q_m, armature current i_a

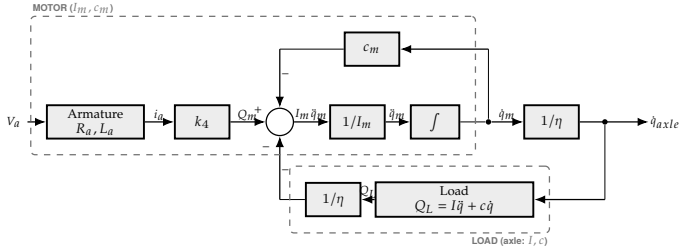
Motor and gearing [p. 49, 50]

$$Q_m = k_1 B i_a = k_4 i_a,\quad I_m\ddot{q}_m = k_4 i_a - c_m\dot{q}_m$$

$$Q = \eta Q_m,\quad \dot{q}_{axle} = \frac{1}{\eta}\dot{q}_m,\quad Q = (I + \eta^2 I_m)\ddot{q}_{axle} + (c + \eta^2 c_m)\dot{q}_{axle}$$

where k_4 [N m/A] torque constant, I_m rotor inertia, c_m motor friction, $\eta > 1$: motor faster than axle. Then $Q_m = Q/\eta$, $i_a = Q_m/k_4$.

Use: motor inertia and friction count η^2 (not η). Electrical side (R_a, L_a, V_b) is only drawn on p. 48.



[drawn for this booklet] torque balance at the motor shaft, $Q_m = I_m\ddot{q}_m + c_m\dot{q}_m + Q_L/\eta$, $q_m = \eta q_{axle}$ [added](slide p. 48 draws only the circuit). With $Q_L = I\ddot{q} + c\dot{q}$, $\ddot{q}_m = \eta\ddot{q}$, multiply by η: the η^2 result.

WORKED gear numbers [checked]

η=50, I_m=2×10^{-5}, I=0.04, c=0.1, c_m=10^{-4}, k_4=0.04; $\ddot{q}$=10, $\dot{q}$=2. $I_{eff} = 0.04 + 2500(2\times10^{-5}) = 0.09$; $c_{eff} = 0.1 + 2500(10^{-4}) = 0.35$; $Q = 0.09(10) + 0.35(2) = 1.6$ N m; $Q_m = 0.032$; $i_a = 0.8$ A; motor speed 100 rad/s. Wrong: 1.6 (no $\div\eta$), 0.012 (no η^2).

SLIDE CHECK p.50, 51

$Q = \eta Q_m$ is the torque at the axle. Slide p. 51 ("Bringing it all together") is blank: no combined PID + motor formula exists.

Exam-style questions

W13 has no control question. **Practical sheet Q1** (one link, $\dot{\mathbf{v}}_C = (-3(b+2ct)^2, 6c, 0)$, $\tau = 24c$): worked in W7, p. 26. **Practical Q2** (3D RRR equations of motion = W13 Q4): solved on p. 42.

EXAM-STYLE QUESTION Practical sheet Q3: planar 3-link arm along a cubic [marks not published]

$L = (3, 2, 1)$ m. Rest at tool pose $(5, 1, 90°)$ to rest at $(-3, 3, 180°)$, $T = 2$ s; point masses 3, 2, 1 kg at the link ends, vertical plane, gravity $-y$. Joint torques at $t = 0, 1, 2$ s.

SOLUTION Wrist = $(x - L_3\cos\varphi,\ y - L_3\sin\varphi)$; $\cos q_2 = (w^2 - L_1^2 - L_2^2)/(2L_1L_2)$. Start wrist $(5, 0)$: stretched, $q_0 = (0, 0, 90°)$. Goal wrist $(-2, 3)$, $\cos q_2 = 0$, take $q_2 = +90°$: $q_1 = \text{atan2}(3, -2) - \text{atan2}(2, 3) = 90°$, $q_3 = 0$: $q_f = (90°, 90°, 0)$ (other elbow (157.38°, −90°, 112.62°)).

Cubic rest-to-rest: $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$, $\Delta = (90, 90, -90)°$; $\ddot{q}(0) = 6\Delta/T^2 = \pm 2.356$, peak $\dot{q} = 1.5\Delta/T = 1.178$ rad/s at $t = 1$.

$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{V} + \mathbf{G}$ from $K = \sum \frac{1}{2} m_i |\dot{\mathbf{p}}_i|^2$, $P = \sum m_i g y_i$ ($\mathbf{p}_i$ = link ends); $G_1(0) = 9.81(9 + 10 + 5) = 235.44$.

t (s)	$\mathbf{q}$ (°)	Q_1	Q_2	Q_3 (N m)
0	(0, 0, 90)	548.814	160.176	2.356
1	(45, 45, 45)	70.817	20.781	5.078
2	(90, 90, 0)	−268.947	−141.712	−21.591

Check: $Q_1(0) = 235.44$ (gravity) + $\mathbf{M}\ddot{\mathbf{q}}$ term. [checked]

EXAM-STYLE QUESTION Mock 1 Q7: partitioned control of one joint [marks assumed 7]

m=2, b=0.5, k=4. (a) Partitioned law and error equation. (b) k_p, k_v critical, ω_n=8. (c) E_{ss} for constant 5 N; removal. (d) f at x=0.10, $\dot{x} = -0.20$, x_d=0.15, $\dot{x}_d = \ddot{x}_d$=0.

SOLUTION (a) $\alpha = m$, $\beta = b\dot{x} + kx$, $f' = \ddot{x}_d + k_v\dot{E} + k_p E$; $\ddot{E} + k_v\dot{E} + k_p E = 0$ [3]
(b) $k_p = 64$, $k_v = 16$ [1]
(c) $E_{ss} = 5/(64 \cdot 2) = 0.0391$ m, in the direction of the push; add $k_i < k_v k_p = 1024$ [2]
(d) E=0.05, $\dot{E}$=0.20: $f' = 16(0.2)+64(0.05) = 6.4$; $f = 2(6.4)+0.5(-0.2)+4(0.10) =$ **13.1 N** [1]. Check $(13.1 + 0.1 - 0.4)/2 = 6.4$. [checked]

EXAM-STYLE QUESTION Practice Q8.4: motor, gearbox, current [marks assumed 6]

η=40; I_m=3×10^{-5}, c_m=2×10^{-4}; I=0.05, c=0.2; k_4=0.05. Axle $\ddot{q}$=8, $\dot{q}$=3. Find Q, i_a.

SOLUTION $\eta^2 = 1600$: $I_{eff} = 0.098$, $c_{eff} = 0.52$. $Q = 0.098(8) + 0.52(3) =$ **2.344 N m**; $Q_m = 0.0586$; $i_a =$ **1.172 A**; motor speed 120 rad/s. Check: $Q_m = I_m\eta\ddot{q} + c_m\eta\dot{q} + (I\ddot{q} + c\dot{q})/\eta = 0.0096 + 0.0240 + 0.0250 = 0.0586$. η instead of η^2 gives 0.517 A (wrong). [checked]

W9 Control 2: computed torque, Cartesian, force and hybrid control

Which law for which problem? [added]

Law	needs	gives / use
joint PD/PID (W8)	none or $\mathbf{g}$; $\mathbf{q}$, $\dot{\mathbf{q}}$	stable; slow/geared; steady error without integral
computed torque [p. 18–21]	exact $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}$	$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}$, decoupled
inexact model [p. 27–28]	$\hat{\mathbf{M}}, \hat{\mathbf{C}}, \hat{\mathbf{g}}, \hat{\mathbf{f}}$	forced error eq.: small, not zero
Cartesian [p. 32–33]	$\mathbf{M}_x, \mathbf{c}_x, \mathbf{g}_x, \mathcal{K}, \mathbf{J}$	same form in $\mathbf{x}$; no IK or $\mathbf{J}^{-1}$; fails at singularity
force [p. 36–43]	m (k_e); f_e, $\dot{x}$	Opt. 2 $E_{ss} = f_{dist}/(1 + mk_e^{-1}k_p)$
hybrid [p. 47]	Cartesian + force law	each axis obeys one law

Go to: write law p. 31; K_p, K_v p. 31; "mass 10 % off" p. 31; Cartesian p. 32; force/wall p. 32; which axes p. 32; practical sheet p. 33.

Scalar partitioned law: cubic spring, single link [p. 7–15]

GIVEN $m\ddot{x} + b\dot{x} + kx^3 = f$; $x_d, \dot{x}_d, \ddot{x}_d$; measured $x, \dot{x}$ FIND f giving $\ddot{E} + k_v\dot{E} + k_pE = 0$

Partitioned (computed-torque) law, scalar [p. 10]

$$f = \alpha\acute{f} + \beta, \quad \alpha = m, \quad \beta = b\dot{x} + kx^3, \quad \acute{f} = \ddot{x}_d + k_v\dot{E} + k_pE, \quad E = x_d - x$$

where α = coefficient of $\ddot{x}$ (kg); β = the rest, with the *measured* $x, \dot{x}$ (N); $\acute{f}$ = commanded acceleration (m/s^2); k_p, k_v in s^{-2}, s^{-1}.

DERIVATION closed loop, every line [p. 10]

Insert f: $m\ddot{x} + b\dot{x} + kx^3 = m\acute{f} + b\dot{x} + kx^3 \Rightarrow \ddot{x} = \acute{f}$. With $\ddot{E} = \ddot{x}_d - \ddot{x}$:

$$\boxed{\ddot{E} + k_v\dot{E} + k_pE = 0}$$

linear whatever the non-linearity; stable for $k_p, k_v > 0$.

STEPS Write a partitioned law [p. 15]

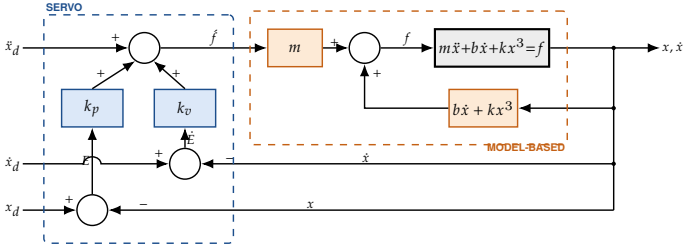
1. EoM as (coeff)$\ddot{q}$ + (rest) = input (Lagrange).
2. α = coeff of $\ddot{q}$; β = rest, signs kept, measured state.
3. $\acute{q} = \ddot{q}_d + k_v\dot{E} + k_pE$, $E = q_d - q$.
4. input = $\alpha\acute{q} + \beta$; closed loop $\ddot{E} + k_v\dot{E} + k_pE = 0$.
5. Check units; at rest on the path law = β (e.g. $mgL\cos q$).

WORKED cubic spring; $m = 2, b = 4, k = 50, k_p = 100, k_v = 20$

$x = 0.10$, $\dot{x} = 0.30$; $x_d = 0.12$, $\dot{x}_d = 0.40$, $\ddot{x}_d = 0.50$. $E = 0.02$, $\dot{E} = 0.10$. $\acute{f} = 0.50 + 20(0.10) + 100(0.02) = 4.5$; $\beta = 4(0.30) + 50(0.10)^3 = 1.25$ N; $f = 2(4.5) + 1.25 = 10.25$ N. Check $\ddot{x} = (10.25 - 1.25)/2 = 4.5 = \acute{f}$ ✓. [checked]

TRAP

$f = \acute{f} + \beta$ (no α) mixes m/s^2 and N; $\ddot{x}_d$ goes in $\acute{f}$, not β; $E = x_d - x$.



[drawn for this booklet] non-linear SISO partitioned controller (slide p. 11). **Orange** = model-based (m, β), **blue** = servo (k_p, k_v on the error), grey = plant. $\ddot{x}_d$ feed-forward; β from measured $x, \dot{x}$.

Single link, viscous + Coulomb friction [p. 12–14]

GIVEN link L, tip mass m, angle q from horizontal, friction b_v, b_c; $q_d, \dot{q}_d, \ddot{q}_d$ FIND torque τ

DERIVATION EoM, slide p. 13

$\tau_v = b_v\dot{q}$, $\tau_c = b_c\,\mathrm{sgn}\,\dot{q}$. $x = L\cos q$, $y = L\sin q$: $K = \frac{1}{2}mL^2\dot{q}^2$, $P = mgL\sin q$. $Q = \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}} - \frac{\partial\mathcal{L}}{\partial q}$ with $\frac{\partial\mathcal{L}}{\partial\dot{q}} = mL^2\dot{q}$, $\frac{\partial\mathcal{L}}{\partial q} = -mgL\cos q$:

$$\boxed{\tau = mL^2\ddot{q} + b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q}$$

Control law of the single link [p. 14]

$$\tau = \alpha\acute{\tau} + \beta, \quad \alpha = mL^2, \quad \beta = b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q, \quad \acute{\tau} = \ddot{q}_d + k_v\dot{E} + k_pE$$

α = inertia about the pivot (kg m^2); at rest on the path $\tau = mgL\cos q$.

WORKED $m = 2, L = 0.5, b_v = 0.3, b_c = 0.2, k_p = 64, k_v = 16$

$q = 30°$, $\dot{q} = 1$, $\ddot{q}_d = 2$, $E = 0.05$, $\dot{E} = -0.1$. $\alpha = 0.5$; $\acute{\tau} = 2 + 16(-0.1) + 64(0.05) = 3.6$; $\beta = 0.3 + 0.2 + 2(9.81)(0.5)\cos 30° = 8.9957$; $\tau = 0.5(3.6) + 8.9957 = 10.7957$ N m. Check $\ddot{q} = (\tau - \beta)/\alpha = 3.6 = \acute{\tau}$ ✓. [checked]

TRAP

Degrees in $\cos q$; sgn of $\dot{q}$ (not q, E); $\alpha = mL$; $\cos q$ dropped. Real laws smooth sgn [added]. Limit [p. 15]: needs the structure and parameters of the model.

MIMO computed torque: n joints [p. 16–21]

GIVEN $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}(\dot{\mathbf{q}})$; $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$; measured $\mathbf{q}, \dot{\mathbf{q}}$ FIND $\mathbf{Q}$ ($n \times 1$)

Computed-torque law of a manipulator [p. 18–19]

$$\mathbf{Q} = \boldsymbol{\alpha}\acute{\mathbf{Q}} + \boldsymbol{\beta}, \quad \boldsymbol{\alpha} = \mathbf{M}(\mathbf{q}), \quad \boldsymbol{\beta} = \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}(\dot{\mathbf{q}}),$$
$$\acute{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}, \quad \mathbf{E} = \mathbf{q}_d - \mathbf{q}$$

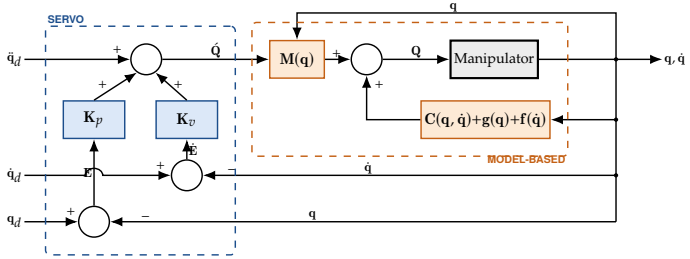
where $\mathbf{M}$ $n \times n$ symmetric positive definite (kg m^2); $\mathbf{C}$ Coriolis/centripetal, $\mathbf{g}$ gravity, $\mathbf{f}$ friction (N m); $\mathbf{K}_p, \mathbf{K}_v$ *diagonal* (s^{-2}, s^{-1}). $\boldsymbol{\alpha} = \mathbf{M}$ is **not** diagonal: it decouples.

DERIVATION closed loop [p. 21]

Insert $\mathbf{Q}$ in the plant: $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} = \mathbf{M}\acute{\mathbf{Q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} \Rightarrow \mathbf{M}\ddot{\mathbf{q}} = \mathbf{M}\acute{\mathbf{Q}} \Rightarrow \ddot{\mathbf{q}} = \acute{\mathbf{Q}}$ ($\mathbf{M}$ invertible). With $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d - \ddot{\mathbf{E}}$:

$$\boxed{\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}} \Rightarrow \ddot{e}_i + k_{vi}\dot{e}_i + k_{pi}e_i = 0 \quad (i = 1..n)$$

n decoupled linear equations; the off-diagonals of $\mathbf{M}$ cancel the coupling $M_{ij}\ddot{q}_j$.



[drawn for this booklet] MIMO computed torque (slide p. 20). Slide's red box "linearises" = orange group; green "stabilises" = blue group. Arrow $\mathbf{q} \to \mathbf{M}(\mathbf{q})$ added.

WORKED $\boldsymbol{\alpha}$ not diagonal: practical 3-link arm, $\mathbf{q}_0 = (0, 0, 90°)$

$\mathbf{M} = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (p. 33). $\acute{\mathbf{Q}} = (1, 0, 0)$ needs $\boldsymbol{\alpha}\acute{\mathbf{Q}} = (103, 31, 1)$ N m: the 31, 1 hold joints 2, 3 still. Diagonal $\boldsymbol{\alpha}$ gives only 103. [checked]

STEPS Choose $\mathbf{K}_p, \mathbf{K}_v$, critical damping [added]

1. Per joint $s^2 + k_vs + k_p$; double root $-\omega_n$: $k_p = \omega_n^2$, $k_v = 2\omega_n$ ($k_v^2 = 4k_p$).
2. ω_n from settling: $e = e_0(1 + \omega_nt)e^{-\omega_nt}$ hits 2 % at $\omega_nt = 5.83$, 5 % at 4.74. [checked]
3. $\mathbf{K}_p = \mathrm{diag}(\omega_i^2)$, $\mathbf{K}_v = \mathrm{diag}(2\omega_i)$ (act on accelerations, independent of inertia).
4. Check: equal negative roots; no overshoot.

WORKED gains from settling time

2 % in 0.5 s: $\omega_n \geq 5.83/0.5 = 11.67$; take 12: $k_p = 144$, $k_v = 24$ ($24^2 = 4 \cdot 144$ ✓). [checked] $\omega = (10, 15, 20)$: $\mathbf{K}_p = \mathrm{diag}(100, 225, 400)$, $\mathbf{K}_v = \mathrm{diag}(20, 30, 40)$. [checked] $k_p = 100$, $k_v = 20$, $e(0) = 0.1$: $e(0.1\,\mathrm{s}) = 0.0736$. [checked]

TRAP

"$\mathbf{K}$ diagonal $\Rightarrow$ $\boldsymbol{\alpha}$ diagonal": false. Error equation holds only for an exact model and a continuous law. Outer-loop gains (p. 25) are torque gains.

Imprecise model, discretisation, outer loop [p. 22–28]

Model may be wrong (friction, wear, grasped load); the law runs in discrete time.

Law and closed loop with hats [p. 27–28]

$$\mathbf{Q} = \hat{\mathbf{M}}\acute{\mathbf{Q}} + \hat{\mathbf{C}}(\mathbf{q}, \dot{\mathbf{q}}) + \hat{\mathbf{g}}(\mathbf{q}) + \hat{\mathbf{f}}(\mathbf{q}, \dot{\mathbf{q}}), \quad \acute{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$$
$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \hat{\mathbf{M}}^{-1}\left[(\mathbf{M} - \hat{\mathbf{M}})\ddot{\mathbf{q}} + (\mathbf{C} - \hat{\mathbf{C}}) + (\mathbf{g} - \hat{\mathbf{g}}) + (\mathbf{f} - \hat{\mathbf{f}})\right]$$

Hat = controller's estimate. The right side is a *forcing*: error does not decay to zero unless every hatted term is exact.

DERIVATION the slide prints only the result [added]

Plant = law: $\hat{\mathbf{M}}\acute{\mathbf{Q}} = \mathbf{M}\ddot{\mathbf{q}} + \boldsymbol{\Delta}$, $\boldsymbol{\Delta} = (\mathbf{C} - \hat{\mathbf{C}}) + (\mathbf{g} - \hat{\mathbf{g}}) + (\mathbf{f} - \hat{\mathbf{f}})$. So $\acute{\mathbf{Q}} - \ddot{\mathbf{q}} = \hat{\mathbf{M}}^{-1}(\mathbf{M} - \hat{\mathbf{M}})\ddot{\mathbf{q}} + \hat{\mathbf{M}}^{-1}\boldsymbol{\Delta}$, and $\acute{\mathbf{Q}} - \ddot{\mathbf{q}} = \ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$. [checked]

Single link, only the mass wrong [added]

$\hat{\alpha} = \hat{m}L^2$, $\hat{\beta} = \hat{m}gL\cos q$:

$$\ddot{e} + k_v\dot{e} + k_pe = \frac{m - \hat{m}}{\hat{m}}\left[\ddot{q} + \frac{g}{L}\cos q\right], \qquad e_{ss} \approx \frac{m - \hat{m}}{\hat{m}}\frac{g}{L}\frac{\cos q}{k_p}$$

$\hat{m} < m$: link sags, $e > 0$; $e_{ss} = 0$ at 90°. [checked]

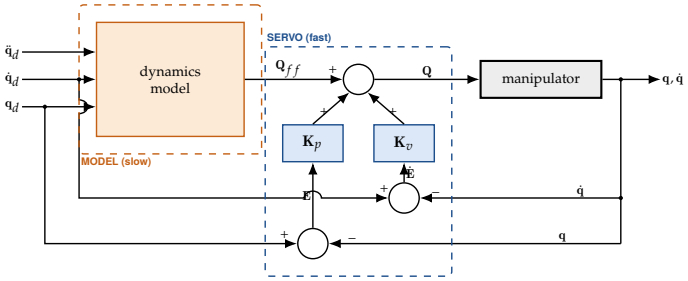
WORKED scenario: mass 10 % off; $m = 2, L = 0.5, k_p = 100, k_v = 20, g/L = 19.62$

$\hat{m} = 1.8$: ratio $0.2/1.8 = 1/9$. Held horizontal: forcing $\frac{1}{9}(19.62) = 2.18$; $e_{ss} = 0.0218$ rad = 1.25°. At 30°: 0.0189 rad (simulation 0.0191). At 90°: 0. $\hat{m} = 2.2$: ratio $-1/11$, $e_{ss} = -0.0178$ rad = $-1.02°$ (link rises). [checked] Error $\approx$ forcing/k_p: not zero; k_p ↑ shrinks it but sampling limits k_p; integral action (W8) removes the constant part [added].

SLIDE CHECK p. 19 vs 27

Friction is $\mathbf{f}(\dot{\mathbf{q}})$ on p. 19, $\hat{\mathbf{f}}(\mathbf{q}, \dot{\mathbf{q}})$ on pp. 27–28: treat as $\mathbf{f}(\mathbf{q}, \dot{\mathbf{q}})$. Notation only.

Discretisation [p. 23–25]. Computing $\mathbf{M}, \mathbf{C}, \mathbf{g}$ is slow, the PD part fast. **Fix:** evaluate the model on the *planned* $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$ at a slow rate (even tabulated); the fast loop adds only $\mathbf{K}_p\mathbf{E} + \mathbf{K}_v\dot{\mathbf{E}}$.



[drawn for this booklet] model in the outer loop (slide p. 25). $\mathbf{Q}_{ff}$ is my name for the model output; the model reads only the plan.

SLIDE CHECK p. 25: gain units

Servo terms are added *to the torque*: $\mathbf{K}_p, \mathbf{K}_v$ are torque gains (N m/rad, N m s/rad), not s^{-2}, s^{-1}; only approximately decoupled [added]. Practical link 1 ($\alpha = 27$): $27 \cdot 100 = 2700$, $27 \cdot 20 = 540$. [checked]

Numbers: link 1 ($\alpha = 27$, $\beta = 88.29 \cos q$, $k_p = 100$, $k_v = 20$), cubic $0 \to \pi/2$ in 2 s: continuous law $|e| < 10^{-10}$; refresh every 10 ms peak error 0.26 mrad; 50 ms 1.13 mrad. [checked]

Cartesian-space control [p. 29–33]

GIVEN tool path $\mathbf{x}_d, \dot{\mathbf{x}}_d, \ddot{\mathbf{x}}_d$ in $\{0\}$; FK $\mathcal{K}$, $\mathbf{J}(\mathbf{q})$, dynamics FIND $\mathbf{Q}$ so the *tool* follows the path

Joint-space route [p. 30–31]: $\mathbf{q} = \mathcal{K}^{-1}(\mathbf{x})$, $\dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\dot{\mathbf{x}}$, $\ddot{\mathbf{q}} = \frac{d}{dt}\left(\mathbf{J}^{-1}(\mathbf{q})\right)\dot{\mathbf{x}} + \mathbf{J}^{-1}(\mathbf{q})\ddot{\mathbf{x}}$: IK and $\mathbf{J}^{-1}$ at every point, fails at a singularity.

SLIDE CHECK p. 31

Slide prints $\dot{J}^{-1}\dot{x}$. Correct: $\frac{d}{dt}\mathbf{J}^{-1} = -\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}$, so $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}})$. [checked]

Cartesian computed-torque law [p. 32–33]

$$\mathcal{F} = \mathbf{M}_x(\mathbf{q})\ddot{\mathbf{x}} + \mathbf{c}_x(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}_x(\mathbf{q}) \text{ (plant)}, \qquad \mathbf{Q} = \mathbf{J}^\top(\mathbf{q})\mathcal{F}$$

$$\mathcal{F} = \mathbf{M}_x\left(\ddot{\mathbf{x}}_d + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x\right) + \mathbf{c}_x + \mathbf{g}_x, \quad \mathbf{E}_x = \mathbf{x}_d - \mathbf{x}$$

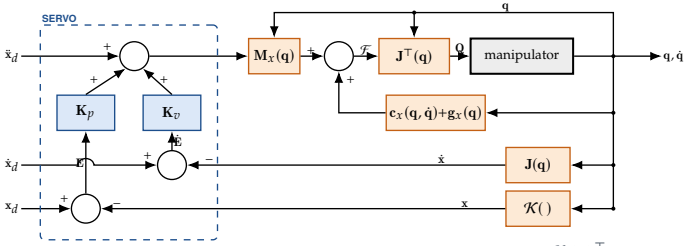
where $\mathcal{F}$ = tool force/moment in $\{0\}$ (N, N m); $\mathbf{x} = \mathcal{K}(\mathbf{q})$, $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; $\mathbf{M}_x$ tool inertia (kg), $\mathbf{c}_x, \mathbf{g}_x$ (N); $\mathbf{K}_p, \mathbf{K}_v$ (s^{-2}, s^{-1}). Closed loop $\ddot{\mathbf{E}}_x + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x = \mathbf{0}$; no IK, no $\mathbf{J}^{-1}$ on the path; $\mathbf{M}_x$ blows up near a singularity.

DERIVATION $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$ and $\mathbf{M}_x, \mathbf{c}_x, \mathbf{g}_x$ [added]

Virtual work: $\delta\mathbf{x} = \mathbf{J}\delta\mathbf{q}$, $\mathbf{Q}^\top\delta\mathbf{q} = \mathcal{F}^\top\mathbf{J}\delta\mathbf{q}\ \forall\delta\mathbf{q} \Rightarrow \mathbf{Q} = \mathbf{J}^\top\mathcal{F}$. Put into $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} = \mathbf{Q}$ with $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}})$ (no friction):

$$\mathcal{F} = \underbrace{\mathbf{J}^{-\top}\mathbf{M}\mathbf{J}^{-1}}_{\mathbf{M}_x}\ddot{\mathbf{x}} + \underbrace{\mathbf{J}^{-\top}\left(\mathbf{C} - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\dot{\mathbf{q}}\right)}_{\mathbf{c}_x} + \underbrace{\mathbf{J}^{-\top}\mathbf{g}}_{\mathbf{g}_x} \quad (\mathbf{J} \text{ square, invertible})$$

[checked]



[drawn for this booklet] Cartesian-based controller (slide p. 33). Orange includes the kinematic model $\mathcal{K}, \mathbf{J}, \mathbf{J}^\top$; feedback $\mathbf{q} \to \mathcal{K} \to \mathbf{x}, \dot{\mathbf{q}} \to \mathbf{J} \to \dot{\mathbf{x}}$; $\mathbf{q}$ lines into $\mathbf{M}_x, \mathbf{J}^\top$ added.

WORKED $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$: planar 2R, $l_1 = l_2 = 1, \mathbf{q} = (30°, 60°)$

$\mathbf{J} = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} = \begin{bmatrix} -1.5 & -1 \\ 0.8660 & 0 \end{bmatrix}$. $\mathcal{F} = (3, -4)$ N: $\mathbf{Q} = \mathbf{J}^\top\mathcal{F} = (-7.9641, -3.0)$ N m. Moments: tool $(0.866, 1.5)$: $0.866(-4) - 1.5(3) = -7.9641$; about joint 2 $(0, 1$ away$)$: $0(-4) - 1(3) = -3$. [checked]

STEPS Cartesian controller on paper

1. Frame: $\mathbf{x}, \mathbf{J}, \mathcal{F}$ in $\{0\}$. 2. $\mathbf{x} = \mathcal{K}(\mathbf{q})$, $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; $\mathbf{E}_x, \dot{\mathbf{E}}_x$.
2. $\mathcal{F} = \mathbf{M}_x(\ddot{\mathbf{x}}_d + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x) + \mathbf{c}_x + \mathbf{g}_x$.
3. $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$ (transpose, $\mathbf{J}$ at the current $\mathbf{q}$).
4. Check: static $\Rightarrow$ $\mathbf{Q} = \mathbf{g}$; torque $= \mathbf{r} \times \mathcal{F}$; singular poses (6-DoF: $\mathbf{J}$ is $6 \times n$).

Force control [p. 34–44]

GIVEN mass m on a wall (stiffness k_e), unknown f_{dist}; $f_d(t)$; measured $f_e, \dot{x}$ FIND f so $f_e \to f_d$

DERIVATION plant and partitioned law [p. 36–38]

$f_e = k_e x \Rightarrow \ddot{x} = k_e^{-1}\ddot{f}_e$. Forces [p. 36]: $f - f_e - f_{dist} = m\ddot{x}$, so $f = mk_e^{-1}\ddot{f}_e + f_{dist} + f_e$. Controlled variable f_e: $\alpha = mk_e^{-1}$, $\beta = f_{dist} + f_e$, $E = f_d - f_e$:

$$f = mk_e^{-1}\left[\ddot{f}_d + k_v\dot{E} + k_pE\right] + f_{dist} + f_e \Rightarrow \ddot{E} + k_v\dot{E} + k_pE = 0.$$

Infeasible: contains the unknown f_{dist} [p. 38].

The two feasible options [p. 39–40]; $\alpha = mk_e^{-1}$, $X = \alpha k_p$

Opt. 1 (drop f_{dist}): $f = \alpha[\ddot{f}_d + k_v\dot{E} + k_pE] + f_e$;

$$\ddot{E} + k_v\dot{E} + k_pE = \frac{f_{dist}}{\alpha}, \quad E_{ss} = \frac{f_{dist}}{\alpha k_p}$$

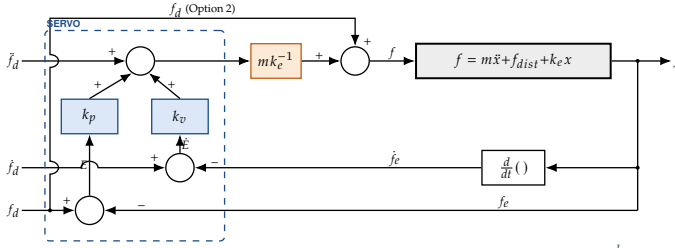
Opt. 2 (f_d for f_{dist}+f_e): $f = \alpha[\ddot{f}_d + k_v\dot{E} + k_pE] + f_d$;

$$\ddot{E} + k_v\dot{E} + \left(\frac{1}{\alpha} + k_p\right)E = \frac{f_{dist}}{\alpha}, \quad E_{ss} = \frac{f_{dist}}{1 + \alpha k_p}$$

$f_e = k_e x$ measured contact force; α in s^2. $E_2/E_1 = X/(1+X) < 1$: Opt. 2 always better. Both non-zero unless $f_{dist} = 0$. Opt. 2 adds k_e/m to the stiffness term.

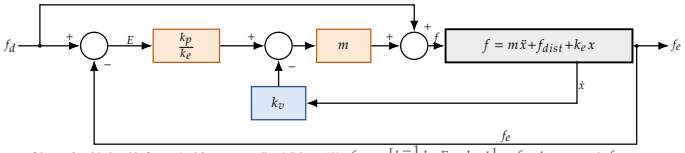
DERIVATION the two loops [added]

Plant $\alpha\ddot{f}_e + f_{dist} + f_e = f$. Opt. 1: $\alpha(\ddot{f}_d - \ddot{f}_e) + \alpha k_v\dot{E} + \alpha k_pE = f_{dist}$. Opt. 2: $\alpha\ddot{E} + \alpha k_v\dot{E} + \alpha k_pE = f_{dist} + f_e - f_d = f_{dist} - E \Rightarrow$ static $E = \frac{f_{dist}/\alpha}{k_p + 1/\alpha} = \frac{f_{dist}}{1 + \alpha k_p}$.



[drawn for this booklet] force controller (slide p. 41). Top path adds f_d = Option 2 (Option 1: measured f_e). The $\frac{d}{dt}$ block is flagged "not a good idea" (p. 42): it amplifies force noise.

Practical law [p. 43–44]: constant f_d, $\dot{E} = -k_e\dot{x}$, so Opt. 2 with $\dot{f}_e \to k_e\dot{x}$. k_e is chosen high; design robust to small changes in k_e.



[drawn for this booklet] practical force controller (slide p. 43): $f = m\left[k_e^{-1}k_pE - k_v\dot{x}\right] + f_d$; $\dot{x}$ measured, f_e never differentiated.

WORKED $m = 2, f_{dist} = 5, k_p = 100, k_v = 20$

$k_e = 1000$: $\alpha = 0.002$, $X = 0.2$, $E_1 = 25$ N, $E_2 = 5/1.2 = 4.17$ N. Soft $k_e = 20$: $X = 10$, $E_1 = 0.5$, $E_2 = 0.455$. $E_2 = 1\%$ of f_{dist} (stiff): $1 + X = 100$, $k_p = 49\,500$. Practical law, $f_d = 10$, $f_e = 7$, $\dot{x} = 0.003$: $E = 3$, $f = 2[0.001(100)(3) - 20(0.003)] + 10 = 10.48$ N = Opt. 2. Opt. 2 loop $\omega_n^2 = k_p + k_e/m = 600$, critical $k_v = 49.0$. [checked]

TRAP

Opt. 1 E_{ss} used for Opt. 2 (missing 1+); "stiffer wall helps" (both grow with k_e; Opt. 2 saturates at f_{dist}); differentiating f_e; $E = f_e - f_d$; $k_v = 2\sqrt{k_p}$ for Opt. 2.

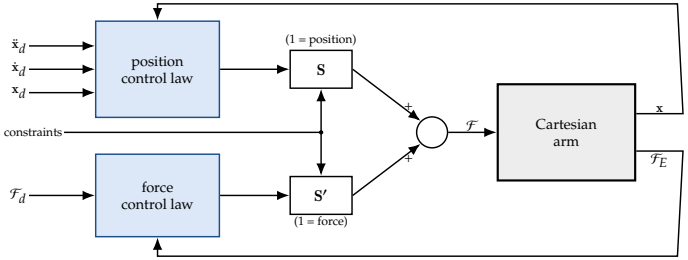
Hybrid force/position control [p. 45–47]

GIVEN some directions blocked (force), others free (position); $\ddot{\mathbf{x}}_d, \dot{\mathbf{x}}_d, \mathbf{x}_d, \mathcal{F}_d$; constraints FIND $\mathcal{F}$, then $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$

Selection matrices [p. 47]

$$\mathcal{F} = \mathbf{S}\mathcal{F}_{pos} + \mathbf{S}'\mathcal{F}_{force}, \qquad \mathbf{S}' = \mathbf{I} - \mathbf{S}, \qquad \mathbf{Q} = \mathbf{J}^\top\mathcal{F}$$

$\mathbf{S}$ diagonal, entry **1 = position-controlled**; $\mathbf{S}'$ entry **1 = force-controlled**; $\mathcal{F}_{pos}$ from the position law (feedback $\mathbf{x}$), $\mathcal{F}_{force}$ from the force law ($\mathcal{F}_d$, measured $\mathcal{F}_E$). $\mathbf{S}' = \mathbf{I} - \mathbf{S}$ is my reading of "complementary". Write in the *constraint frame* (rotate $\mathcal{F}$ if the surface is tilted) [added].



[drawn for this booklet] hybrid controller (slide p. 47, after Craig). Both laws are servo (blue); $\mathbf{S}, \mathbf{S}'$ pick per axis which law reaches the arm.

Entries of S: blocked $\to$ force ($\mathbf{S}$ 0), free $\to$ position (1). Table, z normal (p. 46): diag(1, 1, 0); wall normal x: diag(0, 1, 1); drawer along x [added]: diag(1, 0, 0); peg in vertical hole $(x, y, z, \theta_x, \theta_y, \theta_z)$ [added]: diag(0, 0, 1, 0, 0, 1).

WORKED wipe a table, z normal; $\mathcal{F}_{pos} = (12, -3, 9), \mathcal{F}_{force} = (1, 2, 10)$ N

$\mathbf{S}$ = diag(1, 1, 0), $\mathbf{S}'$ = diag(0, 0, 1): $\mathcal{F}$ = (12, −3, 0) + (0, 0, 10) = (12, −3, 10) N; position 9 and force 1, 2 discarded. Normal x: $\mathbf{S}$ = diag(0, 1, 1) gives (1, −3, 9). Peg: position law (1, 2, −15, 0.3, −0.2, 0.8), force law (−3, 4, 2, 0.5, −0.6, 0.1), $\mathbf{S}$ = diag(0, 0, 1, 0, 0, 1): $\mathcal{F}$ = (−3, 4, −15, 0.5, −0.6, 0.8). [checked]

TRAP

$\mathbf{S}, \mathbf{S}'$ swapped (eraser ignores the press); both 1 on one axis (laws fight); world axes on a tilted surface; forgetting $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$.

Exam questions

Only the practical sheet is lecturer-written (W13 Q4 EoM is the stepping stone: derive the EoM, then partition). Marks not published; indicative 1 per step.

EXAM-STYLE QUESTION Practical sheet (a)–(c) [marks assumed]

Arm masses $m = (3, 2, 1)$ kg at the distal link ends; rest at $(x, y, \varphi) = (5, 1, 90°)$ to rest at $(-3, 3, 180°)$. (a) Joint angles, cubics ($T = 2$ s assumed). (b) Lagrange $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$. (c) Evaluate at start.

SOLUTION **(a)** $\varphi = \theta_1 + \theta_2 + \theta_3$; wrist $\mathbf{w} = (x - L_3\cos\varphi, y - L_3\sin\varphi)$, then planar 2R. Start: $\mathbf{w} = (5, 0)$, $\cos\theta_2 = (25 - 9 - 4)/12 = 1$, $\theta_2 = 0$, $\theta_1 = 0$, $\theta_3 = 90°$: $\mathbf{q}_0 = (0, 0, 90°)$; FK $(5, 1)$ ✓. Goal: $\mathbf{w} = (-2, 3)$, $|\mathbf{w}|^2 = 13$, $\cos\theta_2 = 0$, $\theta_2 = 90°$; $\theta_1 = \operatorname{atan2}(3, -2) - \operatorname{atan2}(2, 3) = 123.69° - 33.69° = 90°$; $\theta_3 = 0$: $\mathbf{q}_f = (90°, 90°, 0)$. Other branch $(157.38°, -90°, 112.62°)$. [checked]

Cubic $q = q_0 + a_2t^2 + a_3t^3$, $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$, $\Delta = (\frac{\pi}{2}, \frac{\pi}{2}, -\frac{\pi}{2})$: joints 1, 2: $a_2 = 1.1781$, $a_3 = -0.3927$; joint 3: -1.1781, $+0.3927$. Peak speed $1.5\Delta/T = 1.1781$ at $t = 1$; peak acceleration $6\Delta/T^2 = 2.3562$ at $t = 0$. Joint cubic is not a straight tool line: $t = 1$ gives $(45°, 45°, 45°)$, tool $(1.414, 4.828)$. [checked]

EXAM-STYLE QUESTION Practical sheet (b), (c)

Positions $\mathbf{p}_1 = L_1(c_1, s_1)$, $\mathbf{p}_2 = \mathbf{p}_1 + L_2(c_{12}, s_{12})$, $\mathbf{p}_3 = \mathbf{p}_2 + L_3(c_{123}, s_{123})$; $K = \sum \frac{1}{2} m_i |\dot{\mathbf{p}}_i|^2$, $P = g\sum m_i y_i$; $M_{ij} = \partial^2 K/\partial\dot{q}_i\partial\dot{q}_j$, $g_i = \partial P/\partial q_i$. $A = 18$, $B = 3$, $D = 2$ kg m^2:

$$M_{11} = 67 + 36c_2 + 6c_{23} + 4c_3, \quad M_{12} = 13 + 18c_2 + 3c_{23} + 4c_3,$$
$$M_{13} = 1 + 3c_{23} + 2c_3, \quad M_{22} = 13 + 4c_3,$$
$$M_{23} = 1 + 2c_3, \quad M_{33} = 1$$

$$C_1 = -As_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) - Bs_{23}\big(2\dot{q}_1(\dot{q}_2+\dot{q}_3) + (\dot{q}_2+\dot{q}_3)^2\big) - Ds_3\big(2(\dot{q}_1+\dot{q}_2)\dot{q}_3 + \dot{q}_3^2\big)$$
$$C_2 = (As_2 + Bs_{23})\dot{q}_1^2 - Ds_3\big(2(\dot{q}_1+\dot{q}_2)\dot{q}_3 + \dot{q}_3^2\big)$$
$$C_3 = Bs_{23}\dot{q}_1^2 + Ds_3(\dot{q}_1 + \dot{q}_2)^2$$

$g_1 = g[18c_1 + 6c_{12} + c_{123}]$, $g_2 = g[6c_{12} + c_{123}]$, $g_3 = g\,c_{123}$ [checked]

(c) $\mathbf{q}_0 = (0, 0, 90°)$, $\dot{\mathbf{q}} = \mathbf{0}$:

$$\mathbf{M} = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{C} = \mathbf{0}, \quad \mathbf{g} = 9.81\begin{bmatrix} 24 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 235.44 \\ 58.86 \\ 0 \end{bmatrix}$$

Check $g_1 = 9.81(3\cdot3 + 2\cdot5 + 1\cdot5)$ ✓; $\mathbf{Q}(0) = \mathbf{M}\ddot{\mathbf{q}}_d + \mathbf{g} = (548.8, 160.2, 2.4)$. [checked]

EXAM-STYLE QUESTION Practical sheet (d)–(i) [marks assumed]

(d)(e) Partitioned law, link 1. (g) Error along the path. (h) $\hat{m} = 2.8$. (i) All joints.

SOLUTION **(e)** Link 1 alone: $\tau = m_1L_1^2\ddot{q} + m_1gL_1\cos q$; $\alpha = 27$, $\beta = 88.29\cos q$; $k_p = 100$, $k_v = 20$: $\tau = 27(\ddot{q}_d + 20\dot{E} + 100E) + 88.29\cos q$. At $t = 0$: $\tau = 27(2.3562) + 88.29 = 151.91$ N m (joint 1 of the full arm: $\alpha = M_{11} = 103$, $\beta = g_1 = 235.44$). [checked]

(g) Exact model: $\ddot{E} + 20\dot{E} + 100E = 0$, roots $-10, -10$; on the path $E \equiv 0$; 0.1 rad behind: $E(0.1) = 0.0736$, no overshoot. Refresh 10/50 ms: peak 0.26/1.13 mrad. [checked]

(h) $m = 3$, $\hat{m} = 2.8$, $L = 3$: $\ddot{E} + 20\dot{E} + 100E = \frac{1}{14}[\ddot{q} + 3.27\cos q]$. Horizontal: $E_{ss} = \frac{1}{14}(3.27)/100 = 2.34$ mrad (0.134°); vertical: 0. On the cubic: forcing at $t = 0$ 0.402; simulated peak 3.35 mrad at $t \approx 0.47$ s. Full arm ($m_1 = 2.8$): $\hat{M}_{11} = 101.2$, $\hat{g}_1 = 229.554$. [checked]

(i) $\mathbf{Q} = \mathbf{M}(\ddot{\mathbf{q}}_d + 20\dot{\mathbf{E}} + 100\mathbf{E}) + \mathbf{C} + \mathbf{g}$, $\mathbf{M}$, $\mathbf{C}$, $\mathbf{g}$ from (b) each step: three decoupled critically damped loops. New vs one joint: $\boldsymbol{\alpha}$ full; recompute each step (outer loop). With diag($\mathbf{M}$): row 1 forcing $(31(2.3562) - 2.3562)/103 = 0.6863$ rad/s^2 (29 % of 2.36). [checked]

EXAM-STYLE QUESTION EQ3: computed torque, 2R arm [marks assumed: 8]

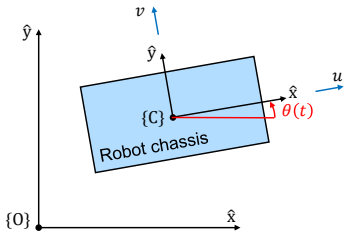
2R, $l_1 = l_2 = 1$, masses 1 kg at link ends: $\mathbf{M} = \begin{bmatrix} 3 + 2c_2 & 1 + c_2 \\ 1 + c_2 & 1 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} -s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ s_2\dot{q}_1^2 \end{bmatrix}$, $\mathbf{g} = g\begin{bmatrix} 2c_1 + c_{12} \\ c_{12} \end{bmatrix}$. At $\mathbf{q} = (30°, 60°)$, $\dot{\mathbf{q}} = (1, -0.5)$: $\ddot{\mathbf{q}}_d = (0.5, 0)$, $\mathbf{E} = (0.02, -0.01)$, $\dot{\mathbf{E}} = \mathbf{0}$, $\mathbf{K}_p = 100\mathbf{I}$, $\mathbf{K}_v = 20\mathbf{I}$. (a) Law [2]. (b) $\mathbf{M}$, $\mathbf{C}$, $\mathbf{g}$ [3]. (c) $\mathbf{Q}$ [2]. (d) $\dot{\mathbf{Q}} = (0.5, 0)$: what does joint 2 still receive, why? [1]

SOLUTION (a) $\mathbf{Q} = \mathbf{M}\dot{\mathbf{Q}} + \mathbf{C} + \mathbf{g}$, $\dot{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$. (b) $c_2 = 0.5$, $s_2 = 0.866$, $c_1 = 0.866$, $c_{12} = 0$: $\mathbf{M} = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 1 \end{bmatrix}$; $C_1 = -0.866(2(1)(-0.5) + 0.25) = 0.6495$, $C_2 = 0.866$; $\mathbf{g} = 9.81(1.732, 0) = (16.991, 0)$. (c) $\dot{\mathbf{Q}} = (2.5, -1.0)$; $\mathbf{M}\dot{\mathbf{Q}} = (8.5, 2.75)$; $\mathbf{Q} = \boxed{(26.14,\ 3.62)\ \text{N m}}$. (d) $\mathbf{M}(0.5, 0)^\top = (2.0, 0.75)$: joint 2 gets 0.75 N m, cancelling $M_{21}\ddot{q}_1$ ($\boldsymbol{\alpha}$ not diagonal). [checked]

W10 Wheeled robot kinematics

Frames $\{O\}$, $\{C\}$ and $\mathbf{J}(\theta)$ [p. 11–17]

GIVEN pose (x, y, θ), wheel layout, $\dot{\boldsymbol{\varphi}}$ (forward) or wanted motion (inverse) FIND $\boldsymbol{\zeta} = (u, v, \dot{\theta})$ in $\{C\}$, $\boldsymbol{\nu} = (\dot{x}, \dot{y}, \dot{\theta})$ in $\{O\}$; or $\dot{\boldsymbol{\varphi}}$



[W10 lec p. 13] — $\{O\}$, $\{C\}$, (u, v) along chassis axes, heading θ

World ↔ chassis [p. 14–16]

$$\underbrace{\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix}}_{\boldsymbol{\nu}} = \underbrace{\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}}_{\mathbf{J}(\theta)} \underbrace{\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix}}_{\boldsymbol{\zeta}}$$

$$\boldsymbol{\zeta} = \mathbf{J}^{\top}\boldsymbol{\nu} = \begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix}$$

where $\{O\}$ = world, z up; $\{C\}$ on the robot, $\hat{x}_C$ forward, $\hat{y}_C$ left; θ = angle $\hat{x}_O \to \hat{x}_C$ (ccw +); u, v = velocity of C along $\hat{x}_C, \hat{y}_C$; $\dot{\theta}$ same in both frames. $\mathbf{J}^{-1} = \mathbf{J}^{\top}$, $\det \mathbf{J} = 1$. Slides call $\mathbf{J}^{-1}$ "$\mathbf{R}^{-1}(\theta)$" [p. 36, 68].

DERIVATION of J [added]

$\hat{x}_C = (\cos\theta, \sin\theta)$, $\hat{y}_C = (-\sin\theta, \cos\theta)$ in $\{O\}$; velocity of $C = u\hat{x}_C + v\hat{y}_C \Rightarrow \dot{x} = u\cos\theta - v\sin\theta$, $\dot{y} = u\sin\theta + v\cos\theta$. Orthonormal columns $\Rightarrow \mathbf{J}^{-1} = \mathbf{J}^{\top}$.

WORKED $\theta = 30°$ [added]

$\boldsymbol{\nu} = (0.5, 0.2, 0.3)$: $u = 0.8660(0.5) + 0.5(0.2) = \mathbf{0.5330}$, $v = -0.5(0.5) + 0.8660(0.2) = \mathbf{-0.0768}$ m/s. Back: $\dot{x} = 0.8660(0.5330) - 0.5(-0.0768) = 0.5$, $\dot{y} = 0.2$ ✓. Speed kept: 0.5385 m/s. [checked]

TRAP

$\mathbf{J}$ vs $\mathbf{J}^{\top}$ ($\theta \to -\theta$). Test: at $\theta = 90°$, $u > 0$ must move along $+y_O$.

Euler update: $x_{k+1} = x_k + \dot{x}_k \Delta t$ etc., $(\dot{x}, \dot{y})$ from $\mathbf{J}(\theta_k)$. **Two maps ([p. 17, 47]):** $\boldsymbol{\zeta} = \mathbf{W}\dot{\boldsymbol{\varphi}}$ ($3 \times n$) and $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$ ($n \times 3$, one row per wheel). Write $\mathbf{M}$ first; $\mathbf{W} = \mathbf{M}^{-1}$ if $n = 3$, $\det \neq 0$; $n = 2$: add $v = 0$.

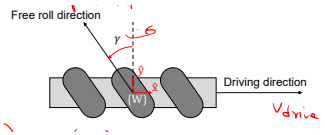
Wheel types and constraints [p. 18–20]

Wheel [p. 18]	what it is	constrains chassis?
Fixed standard	axle fixed, β_i const, $\gamma_i = 0$; rolls, *cannot slide sideways*	yes: roll + no-slide
Steerable standard	axle turns about vertical axis, $\beta_i(t)$, $\gamma_i = 0$	yes: roll + no-slide
Castor	offset steering axis, passive	no
Swedish (omni)	rollers free along γ_i from wheel-plane normal	no: roll only
Spherical	ball, rolls any way	no

[p. 19]: only standard wheels constrain the chassis (no lateral motion). [p. 20]: pure rolling, no slip: contact point moves at $v_{drive} = r\dot{\varphi}$.

Generalised wheel model [p. 21–36]

GIVEN position $({}^{C}x_{W_i}, {}^{C}y_{W_i})$ in $\{C\}$, drive direction β_i, roller angle γ_i, radius r_i, $\boldsymbol{\zeta}$ or $\boldsymbol{\nu}$ FIND wheel speed $\dot{\varphi}_i$



[W10 lec p. 22] — omni wheel in $\{W\}$: $\hat{x}_W$ drive, $\hat{y}_W$ plane normal, rollers free along γ from $\hat{y}_W$

DERIVATION omni wheel in its own frame [p. 23–25]

$\begin{bmatrix}v_x\\ v_y\end{bmatrix} = v_{drive}\begin{bmatrix}1\\ 0\end{bmatrix} + v_{free}\begin{bmatrix}-\sin\gamma\\ \cos\gamma\end{bmatrix}$: $v_y = v_{free}\cos\gamma$, $v_x = v_{drive} - \sin\gamma\, v_{free} \Rightarrow v_{drive} = v_x + v_y\tan\gamma$, so $\dot{\varphi} = \begin{bmatrix}\frac{1}{r} & \frac{\tan\gamma}{r}\end{bmatrix}\begin{bmatrix}v_x\\ v_y\end{bmatrix}$. $\gamma = 0$: $\dot{\varphi} = v_x/r$.

DERIVATION the step chain [p. 27–35]

1 ([p. 28]): wheel frame → $\{C\}$: ${}^{C}\mathbf{v}_{W_i} = \begin{bmatrix}\cos\beta_i & -\sin\beta_i\\ \sin\beta_i & \cos\beta_i\end{bmatrix}\begin{bmatrix}v_x\\ v_y\end{bmatrix}$. **2 ([p. 29–30]):** body point $= \mathbf{v}_C + \dot{\theta}\hat{z} \times \mathbf{p}$, $\dot{\theta}\hat{z} \times (x, y) = \dot{\theta}(-y, x)$: ${}^{C}\mathbf{v}_{W_i} = \begin{bmatrix}u - \dot{\theta}\,{}^{C}y_{W_i}\\ v + \dot{\theta}\,{}^{C}x_{W_i}\end{bmatrix} = \begin{bmatrix}1 & 0 & -{}^{C}y_{W_i}\\ 0 & 1 & {}^{C}x_{W_i}\end{bmatrix}\boldsymbol{\zeta}$. **3 ([p. 32, 34]):** equate, multiply by the transpose: $\begin{bmatrix}v_x\\ v_y\end{bmatrix} = \begin{bmatrix}\cos\beta_i & \sin\beta_i\\ -\sin\beta_i & \cos\beta_i\end{bmatrix}\begin{bmatrix}1 & 0 & -{}^{C}y_{W_i}\\ 0 & 1 & {}^{C}x_{W_i}\end{bmatrix}\boldsymbol{\zeta}$. **4:** $\dot{\varphi}_i = \frac{1}{r_i}[1 \;\; \tan\gamma_i](v_x, v_y)^{\top}$; world frame: $\boldsymbol{\zeta} = \mathbf{R}^{-1}(\theta)\boldsymbol{\nu}$ [p. 35].

Generalised model of a wheeled robot [p. 36]

$$\dot{\varphi}_i = \begin{bmatrix}\frac{1}{r_i} & \frac{\tan\gamma_i}{r_i}\end{bmatrix}\begin{bmatrix}\cos\beta_i & \sin\beta_i\\ -\sin\beta_i & \cos\beta_i\end{bmatrix}\begin{bmatrix}1 & 0 & -{}^{C}y_{W_i}\\ 0 & 1 & {}^{C}x_{W_i}\end{bmatrix}\boldsymbol{\zeta}$$

$$\boldsymbol{\zeta} = \underbrace{\begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}}_{\mathbf{R}^{-1}(\theta)=\mathbf{J}^{\top}}\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix}$$

where β_i = angle from $\hat{x}_C$ to v_{drive}; γ_i = angle between v_{free} and plane normal; $\dot{\varphi}_i > 0$ drives the robot along β_i. Given $\boldsymbol{\zeta}$ directly: drop $\mathbf{R}^{-1}$.

Multiplied out and stacked [added]

$$\dot{\varphi}_i = \frac{1}{r_i\cos\gamma_i}\Big[\cos(\beta_i+\gamma_i) \;\; \sin(\beta_i+\gamma_i) \;\; {}^{C}x_{W_i}\sin(\beta_i+\gamma_i) - {}^{C}y_{W_i}\cos(\beta_i+\gamma_i)\Big]\boldsymbol{\zeta}$$

$\gamma_i = 0$: $r_i\dot{\varphi}_i = \cos\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i})$. Stacked: $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$, $m_{i1} = \frac{\cos\beta_i - \tan\gamma_i\sin\beta_i}{r_i}$, $m_{i2} = \frac{\sin\beta_i + \tan\gamma_i\cos\beta_i}{r_i}$, $m_{i3} = -{}^{C}y_{W_i}m_{i1} + {}^{C}x_{W_i}m_{i2}$. World data: $\dot{\boldsymbol{\varphi}} = \mathbf{M}\mathbf{R}^{-1}(\theta)\boldsymbol{\nu}$. [checked]

STEPS wheel speeds of any wheeled robot

1. Draw $\{C\}$, every wheel; list $({}^{C}x_{W_i}, {}^{C}y_{W_i})$, r_i, β_i (ccw from $\hat{x}_C$), γ_i.
2. Write the row; substitute *each wheel on its own line.*
3. Standard wheel: add no-slide row $0 = -\sin\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \cos\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i})$.
4. Stack $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$; invert if $n = 3$, $\det \neq 0$.
5. World data: $\boldsymbol{\zeta} = \mathbf{J}^{\top}\boldsymbol{\nu}$ first; output $\boldsymbol{\nu} = \mathbf{J}\boldsymbol{\zeta}$.
6. Check: pure u and pure $\dot{\theta}$ give sensible signs.

Standard wheels: rolling and no-sliding [p. 37–42, 68]

$\gamma_i = 0$: fixed wheel β_i const [p. 39]; steerable $\beta_i(t)$, extra input $\dot{\beta}_i$ [p. 42]

Slide form: $\dot{\varphi}_i = \begin{bmatrix}\frac{1}{r_i} & 0\end{bmatrix}\begin{bmatrix}\cos\beta_i & \sin\beta_i\\ -\sin\beta_i & \cos\beta_i\end{bmatrix}\begin{bmatrix}1 & 0 & -{}^{C}y_{W_i}\\ 0 & 1 & {}^{C}x_{W_i}\end{bmatrix}\mathbf{R}^{-1}(\theta)\boldsymbol{\nu}$; v_y unused, must be *zero* [p. 68].

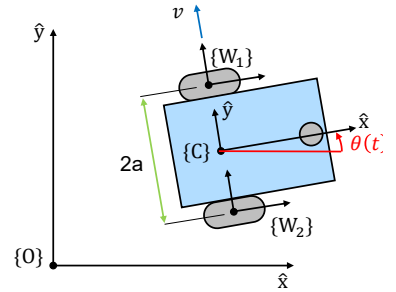
$$\text{rolling: } r_i\dot{\varphi}_i = \cos\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i}) = \begin{bmatrix}\cos\beta_i & \sin\beta_i & {}^{C}x_{W_i}\sin\beta_i - {}^{C}y_{W_i}\cos\beta_i\end{bmatrix}\mathbf{R}^{-1}\boldsymbol{\nu}$$

$$\text{no-sliding: } 0 = -\sin\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \cos\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i}) = \begin{bmatrix}-\sin\beta_i & \cos\beta_i & {}^{C}y_{W_i}\sin\beta_i + {}^{C}x_{W_i}\cos\beta_i\end{bmatrix}\mathbf{R}^{-1}\boldsymbol{\nu}$$

All standard wheels: $\mathbf{C}_1(\beta)\boldsymbol{\zeta} = \mathbf{0}$, rows $[-\sin\beta_k, \cos\beta_k, {}^{C}y_{W_k}\sin\beta_k + {}^{C}x_{W_k}\cos\beta_k]$. No-sliding rows are constraints on $\boldsymbol{\zeta}$: source of non-holonomy [p. 68].

Differential drive and ICR [p. 43–47, 61]

GIVEN $\dot{\varphi}_1$ (left, $y = +a$), $\dot{\varphi}_2$ (right, $y = -a$), radius r, half track a FIND $u, v, \dot{\theta}$, ICR; or the inverse



[W10 lec p. 45] — two fixed standard wheels at $(0, a)$, $(0, -a)$

wheel [p. 45]	r_i	γ_i	β_i	${}^{C}x_{W_i}$	${}^{C}y_{W_i}$
W_1	r	0	0	0	a
W_2	r	0	0	0	$-a$

DERIVATION rows [p. 45–47]

$\dot{\varphi}_1 = \begin{bmatrix}\frac{1}{r} & 0\end{bmatrix}\begin{bmatrix}1 & 0 & -a\\ 0 & 1 & 0\end{bmatrix}\boldsymbol{\zeta} = \frac{1}{r}(u - a\dot{\theta})$; $\dot{\varphi}_2 = \begin{bmatrix}\frac{1}{r} & 0\end{bmatrix}\begin{bmatrix}1 & 0 & a\\ 0 & 1 & 0\end{bmatrix}\boldsymbol{\zeta} = \frac{1}{r}(u + a\dot{\theta})$. Add, subtract: $\dot{\varphi}_1 + \dot{\varphi}_2 = 2u/r$, $\dot{\varphi}_2 - \dot{\varphi}_1 = 2a\dot{\theta}/r$. No-slide ($\beta = 0$, $x = 0$): $0 = v$.

$\dot{\varphi} = M\zeta$, $\zeta = W\dot{\varphi}$ [p. 47]

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\end{bmatrix} = \begin{bmatrix}\frac{1}{r} & 0 & -\frac{a}{r}\\ \frac{1}{r} & 0 & \frac{a}{r}\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \begin{bmatrix}\frac{r}{2} & \frac{r}{2}\\ 0 & 0\\ -\frac{r}{2a} & \frac{r}{2a}\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\end{bmatrix}$$

$u = \frac{r}{2}(\dot\varphi_1 + \dot\varphi_2)$, $\dot\theta = \frac{r}{2a}(\dot\varphi_2 - \dot\varphi_1)$ (ccw +), $v = 0$. Left wheel at $+a$ *slows* when turning left.

WORKED [checked]

$r = 0.05$, $a = 0.15$, $\dot{\boldsymbol\varphi} = (4, 6)$: $u = 0.05(10)/2 = \mathbf{0.25}$ m/s, $\dot\theta = 0.05(2)/0.3 = \mathbf{0.3333}$ rad/s. Back: $\dot\varphi_1 = (0.25 - 0.05)/0.05 = 4$ ✓, $\dot\varphi_2 = 6$ ✓.

ICR (practical Q3)

Instantaneous centre of rotation

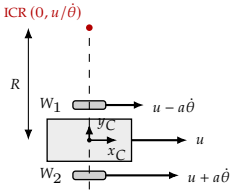
$$(x_{ICR}, y_{ICR}) = \left(-\frac{v}{\dot\theta}, \frac{u}{\dot\theta}\right)$$

diff. drive: $x_{ICR} = 0$, $y_{ICR} = \frac{u}{\dot\theta} = a\,\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1}$, $R = |y_{ICR}|$

$y_{ICR} > 0$ = left. Equal speeds: ∞ (line). $\dot\varphi_2 = -\dot\varphi_1$: ICR at C (spin). $\dot\varphi_1 = 0$: pivot on left wheel ($y = a$). r cancels.

DERIVATION [added]

Chassis point (x, y) moves at $(u - \dot\theta y,\ v + \dot\theta x) = 0$ at $y = u/\dot\theta$, $x = -v/\dot\theta$. Wheel rim speeds = $\dot\theta\times$ distance to ICR: $r\dot\varphi_1 = \dot\theta(y_{ICR} - a)$, $r\dot\varphi_2 = \dot\theta(y_{ICR} + a)$.



[drawn for this booklet] turning left: arrows $\propto \dot\theta\times$ distance to ICR (these are $r\dot\varphi_i$)

Diff-drive scenarios [added]

STEPS circle radius R, speed v_0, ccw

1. $u = v_0$, $\dot\theta = v_0/R$ (clockwise: −), $v = 0$.
2. $\dot\varphi_1 = \frac{v_0}{r}(1 - \frac{a}{R})$ left, $\dot\varphi_2 = \frac{v_0}{r}(1 + \frac{a}{R})$ right; lap $2\pi R/v_0$.
3. Check ICR = R; $R = a$: $\dot\varphi_1 = 0$; $R < a$: left wheel reverses.

WORKED $r = 0.05$, $a = 0.15$ [checked]

Circle $R = 1$, $v_0 = 0.4$: $\dot\theta = 0.4$; $\dot\varphi = (\mathbf{8.8}, \mathbf{9.2})$ rad/s; ICR $0.15 \cdot 16/2.4 = 1.0$ ✓; lap **15.71** s. Clockwise: $(9.2, 6.8)$. $R = a$: $(0, 16)$.
Spin $\omega = 0.5$: $\dot\varphi = \mp a\omega/r = (-1.5, +1.5)$; 90° in 3.142 s.
Rectangle 2×1, $u = 0.25$: straights $(5, 5)$ for 8 s and 4 s; corners $(-1.5, 1.5)$ for 3.142 s each; total $2(8 + 4) + 4(3.142) = \mathbf{38.57}$ s.
Wheel speeds to motion: $r = 0.1$, $a = 0.25$, $\dot{\boldsymbol\varphi} = (10, 2)$: $u = \mathbf{0.6}$, $\dot\theta = \mathbf{-1.6}$, $y_{ICR} = \mathbf{-0.375}$ m (right, beyond the slow wheel).

Exact pose, constant $(u, \dot\theta)$, $v = 0$ [added]

$$\theta_1 = \theta_0 + \dot\theta T,\quad x_1 = x_0 + \tfrac{u}{\dot\theta}(\sin\theta_1 - \sin\theta_0),\quad y_1 = y_0 - \tfrac{u}{\dot\theta}(\cos\theta_1 - \cos\theta_0)$$

Circle radius $u/\dot\theta$; radians. $u = 0.4$, $\dot\theta = 0.8$, $(0, 0, 30°)$, $T = 1.5$: $(0.2442, 0.5091)$ m. Constraint any pose: $-\dot x\sin\theta + \dot y\cos\theta = 0$. [checked]

Skid-steer and 4 Swedish wheels [p. 48–55]

GIVEN 4 standard wheels (left pair, right pair), radius r; or 4 Swedish wheels with given γ_i FIND $\dot{\boldsymbol\varphi} = M\boldsymbol\zeta$ (4×3) and the least-squares $\boldsymbol\zeta$

Skid-steer [p. 50, 52]: $W_1(-b, a), W_2(b, a), W_3(-b, -a), W_4(b, -a)$, $\beta = \gamma = 0$

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\\ \dot\varphi_4\end{bmatrix} = \begin{bmatrix}1/r & 0 & -a/r\\ 1/r & 0 & -a/r\\ 1/r & 0 & a/r\\ 1/r & 0 & a/r\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \begin{bmatrix}\frac{r}{4} & \frac{r}{4} & \frac{r}{4} & \frac{r}{4}\\ 0 & 0 & 0 & 0\\ -\frac{r}{4a} & -\frac{r}{4a} & \frac{r}{4a} & \frac{r}{4a}\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\\ \dot\varphi_4\end{bmatrix}$$

3×4 = least-squares inverse (average). Wheels 1,2 left, 3,4 right. $r = 0.1$, $a = 0.2$, $\dot{\boldsymbol\varphi} = (4, 4, 6, 6)$: $u = 0.5$, $\dot\theta = 0.5$. [checked]

Trap. Four no-slide rows $[0, 1, {}^C x_{W_i}]$, ${}^C x_{W_i} = \mp b$: rank 2 ($v = 0$, $\dot\theta = 0$), so a real skid-steer *cannot* turn without slip; the slide keeps $v = 0$ only [added].

4 Swedish wheels [p. 55], $\beta_i = 0$, wheels at $(\pm\ell, \pm w)$ [added]

$$\dot\varphi_i = \tfrac{1}{r}\begin{bmatrix}1 & \tan\gamma_i & \tan\gamma_i\,{}^C x_{W_i} - {}^C y_{W_i}\end{bmatrix}\boldsymbol\zeta$$

Pattern FL −45°, FR +45°, RL +45°, RR −45° (take signs from the question):

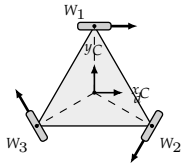
$$\begin{bmatrix}\dot\varphi_{FL}\\ \dot\varphi_{FR}\\ \dot\varphi_{RL}\\ \dot\varphi_{RR}\end{bmatrix} = \frac{1}{r}\begin{bmatrix}1 & -1 & -(\ell + w)\\ 1 & 1 & \ell + w\\ 1 & 1 & -(\ell + w)\\ 1 & -1 & \ell + w\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix},$$

$$\boldsymbol\zeta = \frac{r}{4}\begin{bmatrix}1 & 1 & 1 & 1\\ -1 & 1 & 1 & -1\\ -\frac{1}{\ell+w} & \frac{1}{\ell+w} & -\frac{1}{\ell+w} & \frac{1}{\ell+w}\end{bmatrix}\dot{\boldsymbol\varphi}$$

Rank 3, holonomic. $r = 0.05$, $\ell = 0.20$, $w = 0.15$: $v = 0.2 \to (-4, 4, 4, -4)$; $u = 0.3 \to (6, 6, 6, 6)$ rad/s. [checked]

Omni triangle: all steps [p. 53–54]

GIVEN 3 omni wheels at triangle vertices, distance d from C, rollers ⊥ drive, radius r FIND $\dot{\boldsymbol\varphi} = M\boldsymbol\zeta$, $\boldsymbol\zeta = W\dot{\boldsymbol\varphi}$, holonomic?



i	${}^C x_{W_i}$	${}^C y_{W_i}$	β_i	cos	sin
1	0	d	0°	1	0
2	$\frac{\sqrt3}{2}d$	$-\frac{d}{2}$	−120°	$-\frac12$	$-\frac{\sqrt3}{2}$
3	$-\frac{\sqrt3}{2}d$	$-\frac{d}{2}$	120°	$-\frac12$	$\frac{\sqrt3}{2}$

[drawn for this booklet] drive arrows tangential, clockwise (read from the slide figure, [W10 lec p. 54]); $\beta_i = \phi_i - 90°$, $\phi_i = 90°, -30°, 210°$; $\gamma_i = 0$. Anticlockwise: every $-d$ becomes $+d$ [added]

DERIVATION Step 2: each wheel's row from $r\dot\varphi_i = \cos\beta_i(u - \dot\theta\,{}^C y_{W_i}) + \sin\beta_i(v + \dot\theta\,{}^C x_{W_i})$ [added]

W1: $r\dot\varphi_1 = 1(u - \dot\theta d) + 0 = u - d\dot\theta$.

W2: $r\dot\varphi_2 = -\frac12(u + \frac{d}{2}\dot\theta) - \frac{\sqrt3}{2}(v + \frac{\sqrt3}{2}d\dot\theta) = -\frac12 u - \frac{\sqrt3}{2}v - (\frac14 + \frac34)d\dot\theta$.

W3: $r\dot\varphi_3 = -\frac12(u + \frac{d}{2}\dot\theta) + \frac{\sqrt3}{2}(v - \frac{\sqrt3}{2}d\dot\theta) = -\frac12 u + \frac{\sqrt3}{2}v - d\dot\theta$.

W2 by the slide chain [p. 36]: $\begin{bmatrix}\frac1r & 0\end{bmatrix}\begin{bmatrix}-\frac12 & -\frac{\sqrt3}{2}\\ \frac{\sqrt3}{2} & -\frac12\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{d}{2}\\ 0 & 1 & \frac{\sqrt3}{2}d\end{bmatrix} = \frac1r\begin{bmatrix}-\frac12 & -\frac{\sqrt3}{2} & -\frac{d}{4} - \frac{3d}{4}\end{bmatrix} = \frac1r\begin{bmatrix}-\frac12 & -\frac{\sqrt3}{2} & -d\end{bmatrix}$.

Full 3×3 matrix and inverse (Sample Q10) [p. 54]

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\end{bmatrix} = \frac1r\begin{bmatrix}1 & 0 & -d\\ -\frac12 & -\frac{\sqrt3}{2} & -d\\ -\frac12 & \frac{\sqrt3}{2} & -d\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \frac{r}{3}\begin{bmatrix}2 & -1 & -1\\ 0 & -\sqrt3 & \sqrt3\\ -\frac1d & -\frac1d & -\frac1d\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\end{bmatrix}$$

det (unscaled) = $\frac{3\sqrt3}{2}d \neq 0$; no no-slide rows ⇒ $u, v, \dot\theta$ all free: holonomic, $\delta_m = 3$. Check: spin ⇒ all $\dot\varphi_i = -d\dot\theta/r$; translation ⇒ $\sum\dot\varphi_i = 0$. [checked]

DERIVATION Step 3: inverse by elimination [added]

$a_i = r\dot\varphi_i$. Sum: $a_1 + a_2 + a_3 = -3d\dot\theta \Rightarrow \dot\theta = -\frac{a_1+a_2+a_3}{3d}$. Difference: $a_3 - a_2 = \sqrt3 v \Rightarrow v = \frac{a_3-a_2}{\sqrt3}$. Row 1: $u = a_1 + d\dot\theta = \frac{2a_1-a_2-a_3}{3}$. det along row 1: $1[\frac{\sqrt3}{2}d + \frac{\sqrt3}{2}d] - d[-\frac{\sqrt3}{4} - \frac{\sqrt3}{4}] = \sqrt3 d + \frac{\sqrt3}{2}d = \frac{3\sqrt3}{2}d$.

WORKED $d = 0.10$, $r = 0.05$ [checked]

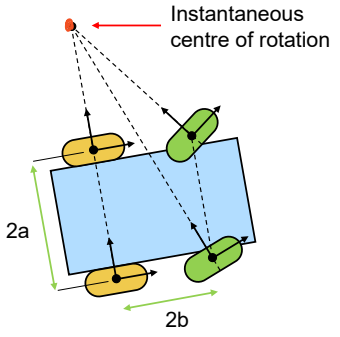
$u = 0.2$: $\frac{1}{0.05}(0.2, -0.1, -0.1) = (\mathbf{4}, \mathbf{-2}, \mathbf{-2})$ rad/s (W13 numbers). $v = 0.2$: $(0, -3.464, 3.464)$. $\dot\theta = 0.5$: $(-1, -1, -1)$. Heading 30°, $\boldsymbol\nu = (0.3, 0.1, 0)$: $u = 0.3098$, $v = -0.0634$; $\dot{\boldsymbol\varphi} = (6.196, -2.000, -4.196)$ (ignoring heading: $(6, -4.732, -1.268)$).

WORKED omni paths, $\dot\theta = 0$, $\theta = 0$: $\zeta = \nu$ [checked]

1 × 1 square, 0.2 m/s	(u, v)	$\dot\varphi_1$	$\dot\varphi_2$	$\dot\varphi_3$
east	$(0.2, 0)$	4	−2	−2
north	$(0, 0.2)$	0	−3.464	3.464
west	$(-0.2, 0)$	−4	2	2
south	$(0, -0.2)$	0	3.464	−3.464

5 s per side, 20 s total. Circle $R = 0.5$, heading fixed, $s = 0.2$: $\omega = 0.4$, period 15.71 s, $(u, v) = s(-\sin\omega t, \cos\omega t)$: $\dot\varphi_1 = -\frac{s}{r}\sin\omega t$, $\dot\varphi_2 = \frac{s}{r}\sin(\omega t - 60°)$, $\dot\varphi_3 = \frac{s}{r}\sin(\omega t + 60°)$, $s/r = 4$. Circle with heading following ($u = 0.2$, $\dot\theta = 0.4$, $v = 0$): $\dot{\boldsymbol\varphi} = (3.2, -2.8, -2.8)$; ICR $(0, 0.5)$.

Tricycle, car, Ackermann, bicycle [p. 56–61, 69]



[W10 lec p. 61] — Ackermann: all axle lines meet at one ICR

All rows from rolling + no-slide rows, $\gamma = 0$.

Tricycle [p. 58] (front wheel $(b, 0)$ driven and steered, rear axle through C): rear $v = 0$. Front contact velocity $(u, b\dot\theta) = r\dot\varphi_1(\cos\beta_1, \sin\beta_1) \Rightarrow \dot\varphi_1 = \frac1r\sqrt{u^2 + b^2\dot\theta^2}$, $\beta_1 = \text{atan2}(b\dot\theta, u)$, $\tan\beta_1 = b/R$. Rear wheels $(u \mp a\dot\theta)/r$. $b = 0.6$, $r = 0.1$, $u = 0.4$, $\dot\theta = 0.5$: $\dot\varphi_1 = \mathbf{5}$, $\beta_1 = \mathbf{36.87}°$, $R = 0.8$ [checked]

Car [p. 60] [added] (rear W_1, W_3 driven at $(-b, \pm a)$, front W_2, W_4 steered at $(b, \pm a)$): rear no-slide $v = b\dot\theta$; $r\dot\varphi_1 = u - a\dot\theta$, $r\dot\varphi_3 = u + a\dot\theta$; $\tan\beta_2 = \frac{2b\dot\theta}{u - a\dot\theta}$, $\tan\beta_4 = \frac{2b\dot\theta}{u + a\dot\theta}$. ICR $(-b, R)$. $a = 0.75$, $b = 1.25$, $r = 0.3$, $u = 2$, $R = 6$:

$\dot\theta = 0.3333$, $v = 0.4167$, rear $(5.833, 7.5)$, $\beta_2 = 25.46°$, $\beta_4 = 20.32°$ [checked]

Ackermann [p.61] (ICR on the rear-axle line): $\tan\beta_{in} = \frac{L}{R-t/2}$, $\tan\beta_{out} = \frac{L}{R+t/2}$, $\cot\beta_{out} - \cot\beta_{in} = \frac{t}{L}$ ($L = 2b$, $t = 2a$, R from mid rear axle). $L = 2.5$, $t = 1.5$, $R = 6$: $\mathbf{25.46}°$, $\mathbf{20.32}°$ [checked]

Bicycle [p.69] [added](rear fixed, driven, at C; front $(L, 0)$ steered β): rear $r\dot\varphi_r = u$, $v = 0$. Front no-slide $0 = -u\sin\beta + L\dot\theta\cos\beta$. $\dot\theta = \frac{u\tan\beta}{L}$, $R = \frac{L}{\tan\beta}$, $\dot x = u\cos\theta$, $\dot y = u\sin\theta$, $r_f\dot\varphi_f = \frac{u}{\cos\beta}$. $u = 2$, $\beta = 20°$, $L = 1.2$: $\dot\theta = \mathbf{0.6066}$, $R = \mathbf{3.297}$. $R = 3$: $\beta = 21.80°$, lap 9.425 s [checked]Jammed steering $\beta = 0$: front row $0 = v + L\dot\theta$ and $v = 0 \Rightarrow \dot\theta = 0$: rank $\mathbf{C}_1 = 2$, $\delta_m = 1$, $\delta_s = 0$ [added].

Manoeuvrability and holonomy [p.62–68]

$\delta_M = \delta_m + \delta_s$ [p.63]

δ_m = mobility (slide: "number of driven wheels"; precisely [added]$3 - \operatorname{rank}\mathbf{C}_1$); δ_s = steerability ("number of steerable wheels"; [added]rank of steered no-slide rows); $\delta_M \le 3$.

Robot [p.64]	δ_m	δ_s	δ_M
2 fixed std + castor (diff. drive)	2	0	2
2 fixed + 1 steerable std (tricycle, car)	1	1	2
2 steerable std + 1 Swedish	1	2	3
3 Swedish (omni)	3	0	3

Why [added]: diff. drive rows $[0, 1, 0]$ twice, rank 1, $\delta_m = 2$. Tricycle: rear $[0, 1, 0]$, front $[-\sin\beta, \cos\beta, b\cos\beta]$, rank 2, $\delta_m = 1$. Omni: $\mathbf{C}_1$ empty, $\delta_m = 3$. [checked]

Holonomic vs non-holonomic [p.66–68]

Holonomic constraint: position only, $f(\beta, \gamma, \varphi, \theta, x, y) = 0$. Non-holonomic: contains velocities, *not integrable*. Standard wheel no-sliding row $\Rightarrow$ non-holonomic; diff. drive: $-\dot x\sin\theta + \dot y\cos\theta = 0$. Omni/mecanum (no standard wheel): holonomic, $\delta_M = \delta_m = 3$.

Explain. Non-holonomic forbids sideways *velocity*, not poses: parking reaches every (x, y, θ) by manoeuvring. Holonomic would confine the robot to a surface in pose space. Test [added]: $\omega = -\sin\theta\,dx + \cos\theta\,dy$, $\boldsymbol\omega\cdot\nabla\times\boldsymbol\omega = -1 \neq 0$. [checked]

Decision table, slide notes, questions

If the question gives …	use …
wheel positions, drive directions	generalised row $r\dot\varphi_i = \cos\beta(u - \dot\theta y) + \sin\beta(v + \dot\theta x)$
two fixed wheels at $\pm a$; ICR?	$\dot\varphi_{1,2} = (u \mp a\dot\theta)/r$, $v = 0$; $y_{ICR} = u/\dot\theta$
circle R at v_0	$\dot\theta = v_0/R$, rows; lap $2\pi R/v_0$
steering angle, wheelbase	bicycle $\dot\theta = u\tan\beta/L$; Ackermann
3 Swedish wheels, d	rows $[\cos\beta, \sin\beta, -d]$; invert by sum/difference
world velocity given	$\boldsymbol\zeta = \mathbf{J}^\top\boldsymbol\nu$ first
"holonomic?"	standard wheel (no-slide row) exists?

SLIDE CHECK

No numerical error found. (1) [p.36] leaves the world→chassis matrix unnamed, [p.68] calls it $\mathbf{R}^{-1}$, [p.14] calls the inverse $\mathbf{J}$. (2) Omni β_i ([p.54]) not written: figure shows clockwise tangents; anticlockwise flips the sign of d. (3) [p.58]: $\beta_1 + 180°$ with $\dot\varphi_1 < 0$ is equally valid. (4) [p.63] δ_m, δ_s informal; [p.55], [p.60] figures only.

TRAP lost marks

Frame not stated; calculator in wrong angle mode; a is the *half* track; left wheel at $+a$ slows on a left turn; ICR side ($y_{ICR} > 0$ left); no-slide row written but unused ($v = 0$); "holonomic because it reaches every pose"; $\dot\varphi$ without the drive-direction convention.

W13 sample Q10 (omni triangle): solved on p. 42.

EXAM-STYLE QUESTION (practical Q2) [marks not published]

Omni robot above, $d = 0.10$, $r = 0.05$, $\dot{\boldsymbol\varphi} = (4, -1, -1)$ rad/s, start at origin, $\theta = 0$: find $(u, v, \dot\theta)$, the path and the pose after 2 s.

SOLUTION $u = \frac{r}{3}(2\dot\varphi_1 - \dot\varphi_2 - \dot\varphi_3) = \frac{0.05}{3}(10) = 0.1667$; $v = \frac{r}{\sqrt3}(\dot\varphi_3 - \dot\varphi_2) = 0$; $\dot\theta = -\frac{r}{3d}\sum\dot\varphi_i = -0.3333$. ICR $(-v/\dot\theta, u/\dot\theta) = (0, -0.5)$: clockwise circle, radius 0.5 m. $\theta = -0.6667$; $x = 0.5\sin 0.6667 = \boxed{0.3092}$ m, $y = -0.5(1 - \cos 0.6667) = \boxed{-0.1071}$ m. Check: 0.5 m from ICR ✓. [checked]

EXAM-STYLE QUESTION (practical Q3 and Q4) [marks not published]

(a) ICR of a diff. drive, $a = 0.3$, $\dot{\boldsymbol\varphi} = (2, 5)$. (b) Bicycle kinematics; holonomic? $\dot\theta$ for $u = 2$, $\beta = 20°$, $L = 1.2$.

SOLUTION (a) $u - \dot\theta y = 0$, $v + \dot\theta x = 0$, $v = 0$: $y_{ICR} = a\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1} = 0.3\cdot\frac{7}{3} = \boxed{0.7}$ m left. Check: equal → ∞, opposite → 0. (b) Rear: $v = 0$, $r\dot\varphi_r = u$. Front: $0 = -u\sin\beta + L\dot\theta\cos\beta \Rightarrow \dot\theta = \frac{u\tan\beta}{L} = \boxed{0.6066}$ rad/s ($R = 3.297$). $\dot x = u\cos\theta$, $\dot y = u\sin\theta$. Non-holonomic: $-\dot x\sin\theta + \dot y\cos\theta = 0$ restricts velocity, not integrable. [checked]

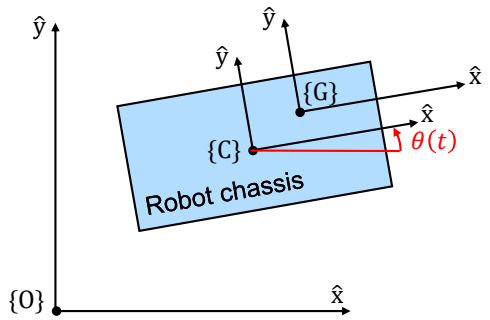
EXAM-STYLE QUESTION (Practice Q10.1) [marks assumed]

Omni wheels at $d = 0.12$, $\phi_i = 0°, 120°, 240°$, driving *anticlockwise* tangentially ($\beta_i = \phi_i + 90°$), $\gamma_i = 0$, $r = 0.04$. Derive $\dot\varphi_i$; speeds for $(u, v, \dot\theta) = (0.3, 0.1, 0.5)$.

SOLUTION $(x_i, y_i) = d(\cos\phi_i, \sin\phi_i)$, $\cos\beta_i = -\sin\phi_i$, $\sin\beta_i = \cos\phi_i$: $r\dot\varphi_i = -u\sin\phi_i + v\cos\phi_i + d\dot\theta$; rows $\frac{1}{r}[-\sin\phi_i, \cos\phi_i, d]$, $\det = \frac{3\sqrt3}{2}d \neq 0$. $\dot\varphi = \boxed{(4.000, -6.245, 6.745)}$ rad/s. Check: spin gives $+d\dot\theta/r$ on all; the clockwise (slide) layout has $-d$. [checked]

W11 Dynamics of wheeled robots

Frames, unknowns, the two problems [p. 9–12]



[W11 lec p. 12] — $\{O\}$ inertial, $\{C\}$ on chassis, $\{G\}$ at COM, same orientation as $\{C\}$

Setup

$\boldsymbol{\xi} = (x, y, \theta)$ pose of C in $\{O\}$; $\boldsymbol{\zeta} = (u, v, \dot\theta)$ chassis velocity in $\{C\}$; $\dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\zeta}$, $\mathbf{J} = {}^O\mathbf{R}_C = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$. $\mathbf{Q} = (f_x, f_y, \tau_z)$ net load applied at C (wheels+ground), ${}^C\mathbf{Q}$ or ${}^O\mathbf{Q}$: always name the frame. m; ${}^GI = {}^GI_{zz}$ about G; ${}^C\mathbf{P}_G = ({}^Cx_G, {}^Cy_G, 0)$ constant (both frames turn with the robot); $C = G$ simplifies everything.

GIVEN $\mathbf{Q}$, m, ${}^GI$, geometry FIND $(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$: simulate. **Inverse** [p. 11]: given the motion, find $\mathbf{Q}$ (control)

No control in W11 (it is W8–W9). Do not mix $\boldsymbol{\xi}$ (pose, $\{O\}$) with $\boldsymbol{\zeta}$ (velocity, $\{C\}$).

Velocity and acceleration of G [p. 18–24]

DERIVATION velocity [p. 19–20]

${}^O\mathbf{P}_G = {}^O\mathbf{P}_C + {}^O\mathbf{R}_C\,{}^C\mathbf{P}_G \Rightarrow {}^O\dot{\mathbf{P}}_G = {}^O\dot{\mathbf{P}}_C + {}^O\boldsymbol{\omega} \times {}^O\mathbf{R}_C\,{}^C\mathbf{P}_G$ (${}^C\dot{\mathbf{P}}_G = \mathbf{0}$, G fixed in chassis; $\dot{\mathbf{R}}\mathbf{p} = \boldsymbol{\omega} \times \mathbf{R}\mathbf{p}$). Rotations keep cross products, so factor ${}^O\mathbf{R}_C$:

Velocity of G [p. 19–20]

$$ {}^O\dot{\mathbf{P}}_G = {}^O\mathbf{R}_C\left[{}^C\dot{\mathbf{P}}_C + {}^C\boldsymbol{\omega}_C \times {}^C\mathbf{P}_G\right],\quad {}^C\dot{\mathbf{P}}_C = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix},\quad {}^C\boldsymbol{\omega}_C = \begin{bmatrix} 0 \\ 0 \\ \dot\theta \end{bmatrix} $$

$$ \Rightarrow\ {}^C\dot{\mathbf{P}}_G = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} + \begin{bmatrix} -\dot\theta\,{}^Cy_G \\ \dot\theta\,{}^Cx_G \\ 0 \end{bmatrix} = \begin{bmatrix} u - \dot\theta\,{}^Cy_G \\ v + \dot\theta\,{}^Cx_G \\ 0 \end{bmatrix} $$

$(0,0,k) \times (a,b,0) = (-kb,\ ka,\ 0)$: used on every line of this week.

WORKED velocity, ours

$G = (0.3, 0.2)$, $u = 1.0$, $v = 0.5$, $\dot\theta = 2$: $\{C\}$: $(1.0 - 0.4,\ 0.5 + 0.6) = (\mathbf{0.6}, \mathbf{1.1})$ m/s. At $\theta = 30°$: $\mathbf{J}(0.6, 1.1) = (\mathbf{-0.0304}, \mathbf{1.2526})$; speed 1.253 in both. [checked]

DERIVATION acceleration [p. 22]

Differentiate again (product rule); the term $\boldsymbol{\omega} \times {}^C\dot{\mathbf{P}}_G = 0$. With $\dot{\mathbf{R}}\mathbf{p} = \mathbf{R}(\boldsymbol{\omega} \times \mathbf{p})$ factor ${}^O\mathbf{R}_C$ and premultiply by its inverse:

Acceleration of G in $\{C\}$: four terms [p. 22–24]

$$ \boxed{{}^C\ddot{\mathbf{P}}_G = {}^C\ddot{\mathbf{P}}_C + {}^C\dot{\boldsymbol{\omega}}_C \times {}^C\mathbf{P}_G + {}^C\boldsymbol{\omega}_C \times {}^C\mathbf{P}_G + {}^C\boldsymbol{\omega}_C \times {}^C\dot{\mathbf{P}}_C + {}^C\boldsymbol{\omega}_C \times {}^C\boldsymbol{\omega}_C \times {}^C\mathbf{P}_G} $$

${}^C\ddot{\mathbf{P}}_C = (\dot u, \dot v, 0)$, ${}^C\dot{\boldsymbol{\omega}}_C = (0, 0, \ddot\theta)$. Using $[\boldsymbol{\omega}\times] = \begin{bmatrix} 0 & -\dot\theta & 0 \\ \dot\theta & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ ($\ddot\theta$ for $\dot{\boldsymbol{\omega}}$):

$$ \underbrace{\begin{bmatrix} -y_G\ddot\theta \\ x_G\ddot\theta \\ 0 \end{bmatrix}}_{\text{tangential}} + \underbrace{\begin{bmatrix} -v\dot\theta \\ u\dot\theta \\ 0 \end{bmatrix}}_{\text{Coriolis-like}} + \underbrace{\begin{bmatrix} -x_G\dot\theta^2 \\ -y_G\dot\theta^2 \\ 0 \end{bmatrix}}_{\text{centripetal}}, \qquad \begin{aligned} {}^C\ddot x_G &= \dot u - y_G\ddot\theta - v\dot\theta - x_G\dot\theta^2 \\ {}^C\ddot y_G &= \dot v + x_G\ddot\theta + \boxed{u}\dot\theta - y_G\dot\theta^2 \end{aligned} $$

SLIDE CHECK pp. 24–26: $v\dot\theta$ should be $u\dot\theta$ in y

Slides print $+v\dot\theta$ in ${}^C\ddot y_G$, ${}^CF_y$, ${}^Cf_y$. Correct: $+u\dot\theta$ (the slide's own cross product gives it; p. 26 matrix uses u; constant u on a circle needs $a_y = u\dot\theta$). The x row is right. [checked]

WORKED all four terms, ours

u=1.5, v=0.2, $\dot\theta$=0.8, $\dot u$=0.3, $\dot v$=−0.1, $\ddot\theta$=0.5, G=(0.2, 0.1): $\ddot{\mathbf{P}}_C(0.3, -0.1)$ + tangential $(-0.05, 0.10)$ + Coriolis-like $(-0.16, 1.20)$ + centripetal $(-0.128, -0.064) = \mathbf{a}_G = (\mathbf{-0.038}, \mathbf{1.136})$ m/s² (printed $+v\dot\theta$: a_y=0.096, wrong). [checked]

Newton–Euler in $\{C\}$: the equation of motion [p. 14–26]

Balance [p. 15–17]

Inertial force/torque at G: ${}^C\mathbf{F}_G = m\,{}^C\ddot{\mathbf{P}}_G$, ${}^CN_z = {}^GI\ddot\theta$ ($\boldsymbol{\omega} \times {}^GI\boldsymbol{\omega} = \mathbf{0}$ in the plane). Applied $({}^Cf_x, {}^Cf_y)$, ${}^Cn_z$ act at C; lever arm $G \to C$ is $-{}^C\mathbf{P}_G$:

$$ {}^CF_x = {}^Cf_x,\ {}^CF_y = {}^Cf_y, \quad {}^CN_z = {}^Cn_z + \underbrace{(-{}^C\mathbf{P}_G) \times {}^C\mathbf{f}}_{= y_Gf_x - x_Gf_y} $$

Newton–Euler equations [p. 25]

$$ {}^Cf_x = m(\dot u - y_G\ddot\theta - v\dot\theta - x_G\dot\theta^2), \quad {}^Cf_y = m(\dot v + x_G\ddot\theta + u\dot\theta - y_G\dot\theta^2) $$

$$ {}^Cn_z = {}^GI\ddot\theta - y_G\,{}^Cf_x + x_G\,{}^Cf_y $$

DERIVATION to matrix form [added]

Insert f_x, f_y in n_z; the $x_Gy_G\dot\theta^2$ terms cancel: ${}^Cn_z = -my_G\dot u + mx_G\dot v + \left({}^GI + m(x_G^2 + y_G^2)\right)\ddot\theta + m\dot\theta(x_Gu + y_Gv)$.

Matrix form $\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^C\mathbf{Q}$ [p. 26]

$$ \begin{bmatrix} m & 0 & -my_G \\ 0 & m & mx_G \\ -my_G & mx_G & {}^GI + m(x_G^2 + y_G^2) \end{bmatrix} \begin{bmatrix} \dot u \\ \dot v \\ \ddot\theta \end{bmatrix} + \begin{bmatrix} -m\dot\theta(v + x_G\dot\theta) \\ m\dot\theta(u - y_G\dot\theta) \\ m\dot\theta(x_Gu + y_Gv) \end{bmatrix} = \begin{bmatrix} {}^Cf_x \\ {}^Cf_y \\ {}^Cn_z \end{bmatrix} $$

M_{33} = inertia about C (parallel axis). Simulate: $\dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}({}^C\mathbf{Q} - c)$. $C = G$: $\mathbf{M} = \mathrm{diag}(m, m, {}^GI)$, $c = (-mv\dot\theta,\ mu\dot\theta,\ 0)$: $m(\dot u - v\dot\theta) = f_x$, $m(\dot v + u\dot\theta) = f_y$, ${}^GI\ddot\theta = n_z$.

STEPS Derive the EoM (W13 Q4 style)

1. Draw $\{O\}$, $\{C\}$, $\{G\}$, loads at C, inertial loads at G, (x_G, y_G).
2. ${}^C\dot{\mathbf{P}}_G = (u - \dot\theta y_G,\ v + \dot\theta x_G)$.
3. $\mathbf{a}_G$ from the four terms (expand each cross product).
4. $f_x = ma_x$, $f_y = ma_y$, $n_z = {}^GI\ddot\theta - (y_Gf_x - x_Gf_y)$.
5. Collect $\rightarrow \mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^C\mathbf{Q}$.
6. **Check:** $\mathbf{M}$ symmetric; $c = 0$ at rest; $C = G$ gives diag; units N, N m.

Lagrange alternative [p. 27], [added]: $T = \frac{1}{2}m|\dot{\mathbf{P}}_G|^2 + \frac{1}{2}{}^GI\dot\theta^2 = \frac{1}{2}\boldsymbol{\zeta}^\top\mathbf{M}\boldsymbol{\zeta}$, $\frac{d}{dt}\frac{\partial T}{\partial \dot{\boldsymbol{\xi}}} - \frac{\partial T}{\partial \boldsymbol{\xi}} = {}^O\mathbf{Q}$: same motion, no cross products [checked]

WORKED offset COM, at rest

m=8, ${}^GI$=0.3, G=(0.05, 0.02), ${}^C\mathbf{f}$=(6, 0), ${}^Cn_z$=0.5, c=**0**; $M_{33} = 0.3 + 8(0.0025 + 0.0004) = 0.3232$:

$$ \begin{bmatrix} 8 & 0 & -0.16 \\ 0 & 8 & 0.4 \\ -0.16 & 0.4 & 0.3232 \end{bmatrix} \begin{bmatrix} \dot u \\ \dot v \\ \ddot\theta \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0.5 \end{bmatrix} \Rightarrow \ddot\theta = \mathbf{2.067},\ \dot v = \mathbf{-0.103},\ \dot u = \mathbf{0.791} $$

($M_{33} = {}^GI$ alone gives 2.24: wrong.) [checked]

From $\{C\}$ to $\{O\}$ [p. 28–34]

Kinematic link [p. 30]

$$ \dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\zeta}, \qquad \dot{\boldsymbol{\zeta}} = \mathbf{J}^{-1}\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] $$

$$ \dot{\mathbf{J}} = \dot\theta \begin{bmatrix} -s\theta & -c\theta & 0 \\ c\theta & -s\theta & 0 \\ 0 & 0 & 0 \end{bmatrix},\ \mathbf{J}^{-1}\dot{\mathbf{J}} = \dot\theta \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} $$

Loads, power invariance [p. 32]

${}^O\mathbf{Q}^\top\dot{\boldsymbol{\xi}} = {}^C\mathbf{Q}^\top\boldsymbol{\zeta} \Rightarrow {}^O\mathbf{Q}^\top\mathbf{J}\boldsymbol{\zeta} = {}^C\mathbf{Q}^\top\boldsymbol{\zeta}\ \forall\boldsymbol{\zeta} \Rightarrow \boxed{{}^C\mathbf{Q} = \mathbf{J}^\top{}^O\mathbf{Q}}$, so ${}^O\mathbf{Q} = \mathbf{J}^C\mathbf{Q}$ (rotation: $\mathbf{J}^\top = \mathbf{J}^{-1}$).

EoM in $\{O\}$ [p. 34]

$$ \boxed{{}^O\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^Oc = {}^O\mathbf{Q}},\quad {}^O\mathbf{M} = (\mathbf{J}^\top)^{-1}\mathbf{M}\mathbf{J}^{-1},\quad {}^Oc = (\mathbf{J}^\top)^{-1}\left[c - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] $$

$C = G$: $c = \mathbf{M}[\boldsymbol{\omega}\times]\boldsymbol{\zeta}$, so ${}^O\mathbf{M} = \mathrm{diag}(m, m, {}^GI)$, ${}^Oc = \mathbf{0}$: $m\ddot x = f_x$, $m\ddot y = f_y$, ${}^GI\ddot\theta = \tau_z$. The c in $\{C\}$ exists only because the axes turn. [checked]

SLIDE CHECK p. 30 and p. 34

p. 30 "$\mathbf{J}(\eta)$" should be $\mathbf{J}(\boldsymbol{\xi})$. p. 34 prints ${}^Oc$ without the $\mathbf{J}^{-1}$ in front of $\dot{\mathbf{J}}$ (test C=G, m=10, θ=90°, u=1, v=0, $\dot\theta$=2: correct **0**, printed $(-20, 20, 0)$). [checked]

WORKED loads between frames, ours

θ=120°, ${}^O\mathbf{Q}$=(4, −2, 0.7): ${}^C\mathbf{Q} = \mathbf{J}^\top{}^O\mathbf{Q} = (\mathbf{-3.732}, \mathbf{-2.464}, 0.7)$, $|\mathbf{f}|$=4.472 in both ($\mathbf{J}$ instead: wrong). [checked]

Euler integration step [p. 35]

Slide recipe [p. 35]

$$ \dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}\left[{}^C\mathbf{Q} - c(\boldsymbol{\zeta})\right],\quad \boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta} + \Delta t\,\dot{\boldsymbol{\zeta}},\quad \dot{\boldsymbol{\xi}} = \mathbf{J}(\boldsymbol{\xi})\boldsymbol{\zeta},\quad \boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \Delta t\,\dot{\boldsymbol{\xi}} $$

STEPS One Euler step

1. $c(\boldsymbol{\zeta}_k)$ ($C=G$: $(-m\dot{\theta}v,\ m\dot{\theta}u,\ 0)$); ${}^{C}\mathbf{Q}$ from wheels.
2. No slip: add the lateral reaction so $\dot{v}=0$.
3. $\dot{\boldsymbol{\zeta}}=\mathbf{M}^{-1}({}^{C}\mathbf{Q}-c)$; $\boldsymbol{\zeta}_{k+1}=\boldsymbol{\zeta}_k+\Delta t\dot{\boldsymbol{\zeta}}$.
4. $\dot{\boldsymbol{\xi}}=\mathbf{J}(\theta_k)\boldsymbol{\zeta}_{k+1}$; $\boldsymbol{\xi}_{k+1}=\boldsymbol{\xi}_k+\Delta t\dot{\boldsymbol{\xi}}$.
5. **Check:** radians; $v=0$ stays for diff drive.

WORKED one step, ours

$C=G$, m=10, ${}^{G}I$=0.25, r=0.1, a=0.2; $\boldsymbol{\xi}$=(1, 2, π/6), $\boldsymbol{\zeta}$=(1, 0, 1.6); τ_L=0.4, τ_R=0.6; Δt=0.1. F=(4, 6), ${}^{C}\mathbf{Q}$=(10, 0, 0.4), c=(0, 16, 0). Free body $\dot{\boldsymbol{\zeta}}$=(1, −1.6, 1.6); no slip (${}^{C}f_y$=16): $\dot{\boldsymbol{\zeta}}$=(**1**, **0**, **1.6**). $\boldsymbol{\zeta}_1$=(1.1, 0, 1.76); $\dot{\boldsymbol{\xi}}$=(1.1c30°, 1.1s30°, 1.76)=(0.9526, 0.55, 1.76); $\boldsymbol{\xi}_1$=(**1.0953**, **2.055**, **0.6996**). (Old $\boldsymbol{\zeta}_k$ in $\dot{\boldsymbol{\xi}}$: (1.0866, 2.05, 0.6836); state your order.) [checked]

Differential drive: wheel torques to chassis load [p. 36–37]

Differential drive [p. 37]

$$\begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}, \quad F_i = \frac{\tau_i}{r}$$

$$\dot{u} = \frac{F_1+F_2}{m}, \quad \ddot{\theta} = \frac{a(F_2-F_1)}{{}^{G}I} \quad (C{=}G,\ \text{rest})$$

F_1 left wheel at $(0,+a)$, F_2 right at $(0,-a)$, both along $+\hat{x}$; $2a$ = track; column = $(d_x, d_y, x_i d_y - y_i d_x)$. Row 2 is zero: the sideways force is a no-slip reaction (not stated in the deck; say "no slip ⇒ v=$\dot{v}$=0"). Power: $\boldsymbol{\tau}\cdot\dot{\boldsymbol{\varphi}} = {}^{C}\mathbf{Q}\cdot\boldsymbol{\zeta}$.

STEPS Torques → $\dot{u}, \ddot{\theta}$

1. $F_i=\tau_i/r$. 2. $f_x=F_1+F_2$, $n_z=a(F_2-F_1)$.
2. $\dot{u}=f_x/m$, $\ddot{\theta}=n_z/{}^{G}I$, $v=0$; $R={}^{G}If_x/(mn_z)$. **Check:** right wheel harder ⇒ left turn.

WORKED ours

m=10, ${}^{G}I$=0.25, r=0.1, a=0.2, τ_L=0.4, τ_R=0.6: F=(4, 6), f_x=10, n_z=0.4; $\dot{u}$=**1.0**, $\ddot{\theta}$=**1.6**, R=0.625. At t=1: $\dot{\varphi}_{L,R}=(1\mp 0.32)/0.1$=6.8, 13.2 rad/s; power 0.4(6.8)+0.6(13.2)=10.64 W=10(1)+0.4(1.6). Holding v=0 needs $mu\dot{\theta}$=16 N sideways. [checked]

TRAP

$2a$ as lever arm (doubles $\ddot{\theta}$); wrong sign of the n_z row; forgetting $F=\tau/r$; $\mathbf{J}$ instead of $\mathbf{J}^{\top}$ for loads; calling a simulation from given forces "inverse" dynamics; $M_{33}={}^{G}I$ for an offset G.

Scenario: force to drive a circle [added]

STEPS Circle of radius R at speed u

1. $\dot{\theta}=u/R$, $v=0$, $\dot{u}=\ddot{\theta}=0$.
2. ${}^{C}\mathbf{Q}=c=(0,\ mu\dot{\theta},\ 0)=(0,\ mu^2/R,\ 0)$: force mu^2/R sideways, towards the centre; supplied by the no-slip reaction, not the drive ($f_x=n_z=0$).
3. $\dot{\varphi}_{L,R}=(u\mp a\dot{\theta})/r$.
4. **Check:** $R\to\infty$ gives 0; double speed gives 4× force.

WORKED ours

m=12, u=0.6, R=1.5: $\dot{\theta}$=0.4, f_y=**2.88** N. a=0.2, r=0.1: $\dot{\varphi}_L$=**5.2**, $\dot{\varphi}_R$=**6.8** rad/s. [checked]

Exam-style questions

W13 sample Q4 (derive EoM), **Q10** (wheels): solved on p. 42. Marks not published; 1 per step.

EXAM-STYLE QUESTION 1: offset COM [6 steps]

m=8 kg, ${}^{G}I$=0.3 kg m^2, G at (${}^{C}x_G$, ${}^{C}y_G$)=(0.05, 0.02) m. At C: ${}^{C}\mathbf{f}$=(6, 0) N, ${}^{C}n_z$=0.5 N m, at rest. (a) Derive $\mathbf{M}$. (b) Find $\ddot{\theta}$, $\dot{v}$.

SOLUTION (a) At rest $\mathbf{a}_G=(\dot{u}-y_G\ddot{\theta},\ \dot{v}+x_G\ddot{\theta})$, $c=\mathbf{0}$. Newton: $f_x=m(\dot{u}-y_G\ddot{\theta})$, $f_y=m(\dot{v}+x_G\ddot{\theta})$; Euler: $n_z={}^{G}I\ddot{\theta}-y_Gf_x+x_Gf_y$. Substitute:

$$\mathbf{M}=\begin{bmatrix} m & 0 & -my_G \\ 0 & m & mx_G \\ -my_G & mx_G & {}^{G}I+m(x_G^2+y_G^2) \end{bmatrix}=\begin{bmatrix} 8 & 0 & -0.16 \\ 0 & 8 & 0.4 \\ -0.16 & 0.4 & 0.3232 \end{bmatrix}.$$ (b)

$\dot{v}=-0.05\ddot{\theta}$, $\dot{u}=0.75+0.02\ddot{\theta}$; row 3: $-0.12+0.3\ddot{\theta}=0.5$.

$$\boxed{\ddot{\theta}=2.067\ \text{rad/s}^2,\quad \dot{v}=-0.103,\quad \dot{u}=0.791\ \text{m/s}^2}$$

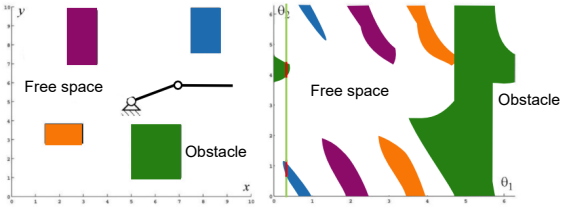
Check: $\ddot{\theta}$ coefficient is exactly ${}^{G}I=0.3$ after elimination. [checked]

W12 Planning: C-space, graph search, sampling-based planners

Tie-break throughout: equal priority ⇒ alphabetical; grid neighbours up, right, down, left. State your own rule at the top of an answer. Planning [p. 5–9]: given map, kinematics, $\mathbf{q}_{start}$, $\mathbf{q}_{goal}$, find a collision-free continuous path or "none".

Configuration space, inflation [p. 12–24]

Task space [p. 12]: frame of the task. **C-space** C [p. 14]: generalised coordinates (joint values; x, y, θ); a configuration $\mathbf{q}$ is a point, the robot is a point there. $C_{free} = \{\mathbf{q} \in C \mid \mathbf{q} \cap C_{obs} = \varnothing\}$ [p. 15] (slide writes ∈; intended ⊂). Planning = a curve in C_{free}. Rotary joints wrap ($q \equiv q + 2\pi$): RR arm = torus.



[W12 lec p. 19] — 2R arm: task space (left), C-space (θ_1, θ_2) (right); a point is two C-space points (elbow up/down)

C-space image of a point (x, y) [added]

$$\cos\theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}, \quad \theta_2 = \pm\arccos(\cdot)$$

$$\theta_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin\theta_2, L_1 + L_2\cos\theta_2)$$

Use: $|\cos\theta_2| > 1$: *out of reach, no C-space point.*

WORKED $L_1 = L_2 = 1, P = (1, 1)$
$\cos\theta_2 = 0 \Rightarrow \theta_2 = \pm 90°$; $\theta_1 = 45° \mp 45° = 0°$ or $90°$: points $(0°, 90°)$, $(90°, -90°)$; FK of each gives $(1, 1)$. [checked]

Inflate [p. 20–24]: disc robot of radius r → point; obstacle grows by r (Minkowski sum). Corridor width w: free width $w - 2r < 0 \Rightarrow$ blocked.

Grown polygon area [added]

$$A_{grown} = A + P\,r + \pi r^2$$

Use: A area, P perimeter of the convex polygon, r robot radius.

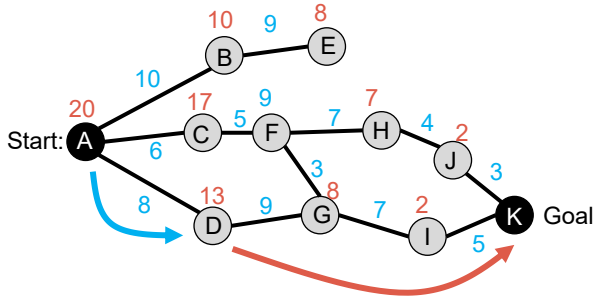
Triangle $(5,3),(6,3),(5,1)$, $r = 0.5$: $A = 1$, $P = 5.236$, $A_{grown} = 1 + 2.618 + 0.785 = 4.403\ \text{m}^2$. [checked]

Roadmap, graph search overview [p. 25–49]

Visibility graph [p. 28–36]: join start, goal and vertices whose segment is free; shortest path, hugs corners. **Voronoi** [p. 37–40]: equidistant from obstacles; max clearance, longer. **Cells** [p. 41–46]: exact (trapezoids), approximate (uniform grid), adaptive. Practical map: visibility optimum via corner $(8, 6)$: $7.906 + 3.162 = 11.068$ to the goal-disc centre, 10.068 to its edge. [checked]

Method [p. 49]	Frontier	Uses
BFS	queue (FIFO)	number of edges
DFS	stack (LIFO)	nothing (dives)
Dijkstra	priority on g	edge costs ≥ 0
A*	priority on $f = g + h$	costs + heuristic
PRM, RRT	roadmap / tree	collision checker

Frontier = discovered, not expanded; discovered ≠ expanded; path read back through the parent table. **Grid** [p. 64]: free cell = node; 4-connected (up/right/down/left) or 8-connected (diagonal $\sqrt{2}$).



[W12 lec p. 59] — the slide graph, start A, goal K; blue g = edge weights, red h (A* slide; none printed at F, K). Shared by samples 1–4

BFS and DFS [p. 50–54]

STEPS BFS
1. Queue = [start], visited = {start}.
2. Pop the *front* n; stop if goal.
3. Each unvisited neighbour m (alphabetical): parent(m) = n, push to the *back*, mark visited.
4. Path = parents from goal back. **Check:** fewest edges; costs ignored. Time $O(|V| + |E|)$ [p. 65].

WORKED SAMPLE 1: BFS, A to K
Edges: A–B, B–E, A–C, C–F, F–H, H–J, J–K, A–D, D–G, F–G, G–I, I–K. Queue front at left.

step	pop	new (parent)	queue after	visited
init	–	–	A	A
1	A	B(A), C(A), D(A)	B, C, D	A, B, C, D
2	B	E(B)	C, D, E	A, B, C, D, E
3	C	F(C)	D, E, F	A, B, C, D, E, F
4	D	G(D)	E, F, G	A, B, C, D, E, F, G
5	E	–	F, G	A, B, C, D, E, F, G
6	F	H(F)	G, H	A, B, C, D, E, F, G, H
7	G	I(G)	H, I	A, B, C, D, E, F, G, H, I
8	H	J(H)	I, J	A, B, C, D, E, F, G, H, I, J
9	I	K(I)	J, K	A, B, C, D, E, F, G, H, I, J, K
10	J	–	K	A, B, C, D, E, F, G, H, I, J, K
11	K	–	–	A, B, C, D, E, F, G, H, I, J, K

Parents: B←A, C←A, D←A, E←B, F←C, G←D, H←F, I←G, J←H, K←I. Path **A–D–G–I–K**, 4 edges, cost 8 + 9 + 7 + 5 = **29** (fewest edges, not cheapest: 25). [checked]

STEPS DFS (explicit stack)
1. Mark visited when *popped*; push neighbours in reverse alphabetical order. Stack = [start].
2. Pop the *top*; skip if visited; else mark, record parent (the pusher).
3. Stop if goal; else push unvisited neighbours. **Check:** *a* path, not shortest; depth limit [p. 54].

WORKED SAMPLE 2: DFS, A to K
Top of stack at right.

step	pop	action	stack after
init	–	push A	A
1	A	push D, C, B	DCB
2	B	push E	DCE
3	E	nothing new	DC
4	C	push F	DF
5	F	push H, G	DHG
6	G	push I, D	DHID
7	D	nothing new	DHI
8	I	push K	DHK
9	K	GOAL: stop	DH

Parents: B←A, E←B, C←A, F←C, G←F, D←G, I←G, K←I. Path **A–C–F–G–I–K**, cost 6 + 5 + 3 + 7 + 5 = **26** (not 25, not fewest edges). [checked]

TRAP
Queue for DFS (that is BFS); visited-on-push changes DFS order; "DFS is complete" (not on infinite/deep graphs). Equal edge costs ⇒ BFS = Dijkstra (unit costs: A–D–G–I–K, cost 4). [checked]

Dijkstra [p. 55–58]

Relaxation [p. 56]

$$\text{if } g(n) + d(n,m) < g(m): \quad g(m) \leftarrow g(n) + d(n,m), \quad \text{parent}(m) \leftarrow n$$

Use: g = cheapest known cost start→node (∞ if unreached); n = unexpanded node with smallest g; m its neighbour. Weights ≥ 0.

STEPS Dijkstra
1. g(start) = 0, others ∞. 2. Expand smallest-g unexpanded n (alphabetical ties).
2. Relax each non-expanded neighbour (strict <); never change an expanded node.
3. **Stop when the goal is expanded**, not discovered. **Check:** g(goal) = sum of path edges; expansions in non-decreasing g.

WORKED SAMPLE 3: Dijkstra, A to K
Row = table after expanding the node (its g in brackets); cell g/parent; **bold** = expanded; underline = changed; – = ∞.

# expand	A	B	C	D	E	F	G	H	I	J	K
1 A (0)	**0**	10/A	6/A	8/A	–	–	–	–	–	–	–
2 C (6)	**0**	10/A	**6/A**	8/A	–	11/C	–	–	–	–	–
3 D (8)	**0**	10/A	**6/A**	**8/A**	–	11/C	17/D	–	–	–	–
4 B (10)	**0**	**10/A**	**6/A**	**8/A**	19/B	11/C	17/D	–	–	–	–
5 F (11)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	14/F	18/F	–	–	–
6 G (14)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	**14/F**	18/F	21/G	–	–
7 H (18)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	**14/F**	**18/F**	21/G	22/H	–
8 E (19)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	21/G	22/H	–
9 I (21)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	22/H	26/I
10 J (22)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	**22/H**	25/J
11 K (25)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	**22/H**	**25/J**

Order A, C, D, B, F, G, H, E, I, J, K (11 expansions). K first discovered at 26 (via I), improves to 25 (via J): stopping at discovery is wrong. Path **A–C–F–H–J–K**, cost 6 + 5 + 7 + 4 + 3 = **25**. [checked]

A* [p. 59–62]

A* priority [p. 59]

$$f(n) = g(n) + h(n)$$

$$\text{admissible: } h(n) \le d(n, \text{goal})\ \forall n$$

$$\text{consistent: } h(n) \le d(n,m) + h(m)\ \forall \text{ adjacent } n, m$$

Use: h = optimistic cost to go. Dijkstra with priority $g \to g + h$. Admissible ⇒ optimal; consistent ⇒ admissible, popped f non-decreasing. $h = 0$ is Dijkstra.

STEPS A*

1. Write h at each node; check admissible and consistent (every edge, both directions).
2. Pop smallest f (ties: smaller h, then alphabetical). Relax neighbours; $f(m) = g(m) + h(m)$ uses the *child's* h.
3. Stop when the goal is popped. **Check:** $f(\text{goal}) = g(\text{goal}) = \text{path cost}$.

SLIDE CHECK pp. 56–60

Slide prints no h at F, K (a stray blue 9 above F): we use $h(\text{K}) = 0$, $h(\text{F}) = 9$; any $h(\text{F}) \in [0, 13]$ gives the same order and cost. Slide tables pp. 57, 60 are blank (fills are ours). Slide asks for admissible and consistent; its h is not consistent.

WORKED SAMPLE 4a: A*, A to K

Open list: node $g{+}h{=}f$ (parent), sorted by f.

#	expand	g	h	f	open list after expanding
1	A	0	20	20	B 10+10=20 (A); D 8+13=21 (A); C 6+17=23 (A)
2	B	10	10	20	D 8+13=21 (A); C 6+17=23 (A); E 19+8=27 (B)
3	D	8	13	21	C 6+17=23 (A); G 17+8=25 (D); E 19+8=27 (B)
4	C	6	17	23	F 11+9=20 (C); G 17+8=25 (D); E 19+8=27 (B)
5	F	11	9	20	G 14+8=22 (F); H 18+7=25 (F); E 19+8=27 (B)
6	G	14	8	22	I 21+2=23 (G); H 18+7=25 (F); E 19+8=27 (B)
7	I	21	2	23	H 18+7=25 (F); K 26+0=26 (I); E 19+8=27 (B)
8	H	18	7	25	J 22+2=24 (H); K 26+0=26 (I); E 19+8=27 (B)
9	J	22	2	24	K 25+0=25 (J); E 19+8=27 (B)
10	K	25	0	25	E 19+8=27 (B)

Order A B D C F G I H J K (10 expansions; E discovered, never expanded). Path A–C–F–H–J–K, cost **25** = f(K). Popped f drops twice: signature of inconsistent h. [checked]

WORKED SAMPLE 4b: admissible? consistent?

True $d(n, \text{K})$ from backwards Dijkstra.

node	h	$d(n,\text{K})$	$d - h$	$h \le d$?
A	20	25	5	yes
B	10	35	25	yes
C	17	19	2	yes
D	13	21	8	yes
E	8	44	36	yes
F	9	14	5	yes
G	8	12	4	yes
H	7	7	0	yes
I	2	5	3	yes
J	2	3	1	yes
K	0	0	0	yes

All $h \le d$: **admissible** (tight at H, K). Edges, $h(n) \le w + h(m)$ both ways:

edge (w)	first direction	second direction
A–B (10)	A→B: 20≤10+10=20	B→A: 10≤10+20=30
A–C (6)	A→C: 20≤6+17=23	C→A: 17≤6+20=26
A–D (8)	A→D: 20≤8+13=21	D→A: 13≤8+20=28
B–E (9)	B→E: 10≤9+8=17	E→B: 8≤9+10=19
C–F (5)	C→F: **17>5+9=14 FAIL**	F→C: 9≤5+17=22
D–G (9)	D→G: 13≤9+8=17	G→D: 8≤9+13=22
F–G (3)	F→G: 9≤3+8=11	G→F: 8≤3+9=12
F–H (7)	F→H: 9≤7+7=14	H→F: 7≤7+9=16
G–I (7)	G→I: 8≤7+2=9	I→G: 2≤7+8=15
H–J (4)	H→J: **7>4+2=6 FAIL**	J→H: 2≤4+7=11
I–K (5)	I→K: 2≤5+0=5	K→I: 0≤5+2=7
J–K (3)	J→K: 2≤3+0=3	K→J: 0≤3+2=5

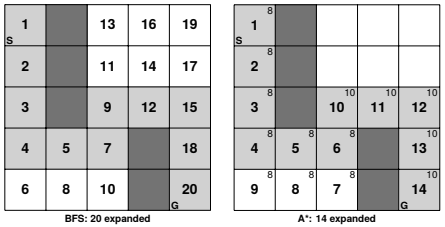
Fails: C→F ($17 > 5 + 9$), H→J ($7 > 4 + 2$): **not consistent** (no h(F) repairs it). A* still optimal since admissible. [checked]

TRAP

f with the parent's h; testing consistency one way only; "admissible means $h < d$" (it is ≤); overestimating h (e.g. $h(\text{F}) = 30$) breaks optimality: A–D–G–I–K, cost 29. Manhattan on an 8-connected grid overestimates (offset $(7, 5)$: $12 > 5\sqrt{2} + 2 = 9.071$); use Euclidean (8.602). [checked]

Grid sample, dimensionality [p. 63–65]

GIVEN 5×5 grid, dark = obstacle, 4-connected, unit moves, S = $(0, 0)$ top-left, G = $(4, 4)$, cells (row,col) FIND BFS and A* (h = Manhattan) expansions and path



[drawn for this booklet] expansion order; small number = f; shaded = the 10-move path. Order up, right, down, left

BFS expands 20 of 20 free cells; A* 14. $h(\text{S}) = 8$, true cost 10: f stays 8 down column 0 and along row 3, rises to 10 after the bottom dead end $(4, 0), (4, 1), (4, 2)$. Path (0,0) → (1,0) → (2,0) → (3,0) → (3,1) → (3,2) → (2,2) → (2,3) → (2,4) → (3,4) → (4,4), **10 moves**, both optimal. [checked]

Grid size in d dimensions [p. 65]

$$|V| = N^d, \quad \text{neighbours} = 2d \text{ or } 3^d - 1 \text{ (diagonals)}$$
$$|E|_{m\times n} = m(n-1) + n(m-1)$$

Use: N *cells per dimension,* d *C-space dimension. 6D at* $N = 100$: 10^{12} *nodes, 12 or 728 neighbours [added].* 40×25: $960 + 975 = 1935$ *edges; 5D,* $N = 50$: 312 500 000 *nodes.*

[checked]

Sampling-based planners [p. 66–83]

Idea [p. 66,69]: no explicit C_{free}; sampler + distance function + local planner + collision detector. PRM (1996), RRT (1998) [p. 68].

STEPS PRM [p. 70–72]

1. Learning: sample $\mathbf{q}$ uniformly, discard if in C_{obs} (point-in-polygon: ray to $+x$, odd crossings = inside); add to V.
2. Connect to vertices within r (or k nearest) if the segment is free (segment–edge test).
3. Query: add start, goal; A*/Dijkstra on the roadmap. **Check:** valid, rarely shortest; disconnected ⇒ sample more.

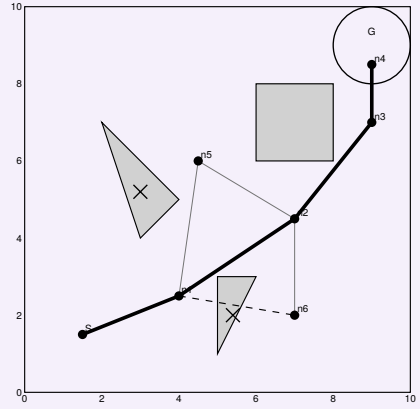
Hit probability [added]

$$p = \frac{A_{goal}}{A_C}, \quad E[N] = \frac{1}{p}, \quad P(\ge 1 \text{ hit in } N) = 1 - (1-p)^N$$

Use: practical map: $p = \pi/100 = 0.0314$, $E[N] = 31.8$.

WORKED SAMPLE 6a: PRM, practical map, $r = 4$

$C = [0, 10]^2$. T1 $(2, 7), (4, 5), (3, 4)$; R $(6, 6), (6, 8), (8, 8), (8, 6)$; T2 $(5, 3), (6, 3), (5, 1)$. $S = (1.5, 1.5)$, goal disc $(9, 9)$, radius 1. Nodes: n1 $(4, 2.5)$, n2 $(7, 4.5)$, n3 $(9, 7)$, n4 $(9, 8.5)$ (in disc), n5 $(4.5, 6)$, n6 $(7, 2)$. Rejected: $(3, 5.2)$ in T1, $(5.4, 2)$ in T2, $(7, 7)$ in R.



edge ($\le r$)	length	segment test
S–n1	2.693	free
n1–n2	3.606	free
n1–n5	3.536	free
n2–n3	3.202	free
n2–n5	2.915	free
n2–n6	2.500	free
n3–n4	1.500	free
n1–n6	3.041	**rejected**: crosses T2

S–G crosses R. Dijkstra (order, g): S (0.000), n1 (2.693), n5 (6.228), n2 (6.298), n6 (8.798), n3 (9.500), n4 (11.000). Path **S–n1–n2–n3–n4**, $2.693 + 3.606 + 3.202 + 1.500 = \mathbf{11.00}$ (10.068 optimum: S–n2 was > r, never tried). [checked]

RRT extension [p. 73–76]

$$\mathbf{q}_{new} = \mathbf{q}_{near} + \delta \frac{\mathbf{q}_{rand} - \mathbf{q}_{near}}{\|\mathbf{q}_{rand} - \mathbf{q}_{near}\|} \quad (\text{if} \le \delta : \ \mathbf{q}_{new} = \mathbf{q}_{rand})$$

Use: $\mathbf{q}_{near}$ *= nearest tree node;* δ *step; keep* $\mathbf{q}_{new}$ *(parent* $\mathbf{q}_{near}$*) only if the segment is free. Goal bias [added]:* $\mathbf{q}_{rand} = \mathbf{q}_{goal}$ *with prob. 5–10 %. Stop in goal region.*

WORKED SAMPLE 6b: RRT, practical map, $\delta = 1$

$\mathbf{q}_{rand} = (6, 2)$, tree $\{S\}$: $\|\cdot\| = \sqrt{4.5^2 + 0.5^2} = 4.528$, $\mathbf{q}_{new} = (1.5 + 0.994, 1.5 + 0.110) = (\mathbf{2.494}, \mathbf{1.610})$ free. Next $\mathbf{q}_{rand} = (4, 5)$: nearest n1 (3.709, S is 4.301), $\mathbf{q}_{new} = (\mathbf{2.900}, \mathbf{2.524})$. From $(4.5, 2.5)$ towards $(6.5, 2)$: $(5.470, 2.257)$ in T2, discarded. [checked]

Non-holonomic [p. 77–79]: straight lines assume sideways motion. Sample controls (v, ω), simulate the kinematic model for time T, keep the end pose closest to the target; every candidate obeys $\dot{y}\cos\theta - \dot{x}\sin\theta = 0$.

TRAP

"RRT finds the shortest path" (no); PRM builds first then queries (RRT stops at the goal); testing the endpoint but not the *segment*.

Choosing a planner [p. 80–83]

Complete: finds a path if one exists, else reports failure. *Resolution complete*: at the chosen grid resolution. *Probabilistically complete*: $P(\text{found}) \to 1$ as samples $\to \infty$; may not terminate if no path [p. 81].

Planner	Complete?	Optimal?	Use when
BFS	resolution (finite graph)	fewest edges; cost only if equal costs	equal moves
DFS	not on deep graphs	no	any path, low memory
Dijkstra	resolution	cheapest, costs ≥ 0	costs, no goal estimate
A*	resolution	cheapest if h admissible	costs + distance h; 2D grids
Visibility	complete	shortest	2D polygons (inflate)
Voronoi	complete	max clearance	range-sensor following
PRM	probabilistic	no	static map, many queries, high DoF
RRT	probabilistic	no	single query, differential constraints

STEPS "Which planner?" [added]

1. $d \leq 3$ and known map: grid/visibility graph; large d: grid needs N^d nodes $\Rightarrow$ sample.
2. Car/diff-drive: RRT with sampled controls. Static map, many queries: PRM; one query: RRT.
3. Costs + good guess: A*; costs only: Dijkstra; equal costs: BFS. Name the method *and* its guarantee.

EXAM-STYLE QUESTION W12 practical sheet [marks not published]

Q1 maze (our reading: 5 columns, entrance cell for S): (a) draw the graph; (b) BFS and DFS routes; (c) compare; (d) upgrade to A*. Q2 $C = [0, 10]^2$, $S = (1.5, 1.5)$, goal disc radius 1 at $(9, 9)$: PRM components and one path.

SOLUTION **Q1a** each open cell a node, each opening an undirected edge (weight 1): 36 open cells, 38 edges; BFS 8 moves (26 cells popped); DFS 16 moves (19 popped); A* 8 moves (16 popped). **b** BFS: fewest moves (8); DFS (up, right, down, left) finds a longer route (16). **c** BFS expands in rings, guaranteed shortest; DFS depends on neighbour order; goal is 4 cells away by Manhattan but a wall forces an 8-move detour. **d** g = cost per move, h = Manhattan/Euclidean distance to G (admissible, walls only lengthen), priority queue on $f = g + h$: same 8-move path, fewer cells.

Q2 (a) graph = vertex list + weighted edge dict (segment lengths), S first. (b) sampler uniform $x, y \in [0, 10]$; $p = \pi/100$, about 32 samples. (c) point test: ray to $+x$, odd crossings = inside. (d) nearest vertex Euclidean. (e) segment ab meets edge cd iff c, d on opposite sides of ab and a, b on opposite sides of cd (orientation signs). (f) goal check $\|\mathbf{q} - (9, 9)\| \leq 1$. (g) path S–n1–n2–n3–n4, 11.00 (sample 6a). [checked]

EXAM-STYLE QUESTION (W12) Graph search [indicative 8]

Undirected: S–A 2, S–B 5, A–C 4, A–D 7, B–D 3, C–G 6, D–G 2. h: S 7, A 6, B 4, C 5, D 2, G 0. (a) A* S to G: table, path, cost. (b) Nodes Dijkstra expands. (c) h admissible? consistent?

SOLUTION Ties: lowest f, then alphabetical.

#	expand	g	h	f	open after
1	S	0	7	7	A 2+6=8 (S); B 5+4=9 (S)
2	A	2	6	8	B 5+4=9 (S); D 9+2=11 (A); C 6+5=11 (A)
3	B	5	4	9	D 8+2=10 (B); C 6+5=11 (A)
4	D	8	2	10	G 10+0=10 (D); C 6+5=11 (A)
5	G	10	0	10	C 6+5=11 (A)

(a) Order S A B D G; D improves $9 \to 8$ (via B). Path **S–B–D–G**, cost $5 + 3 + 2 = \boxed{10} = f(G)$. [4] (b) Dijkstra: S (0), A (2), B (5), C (6), D (8), G (10), 6 nodes (A* 5; also expands C). [1] (c) True costs to G: S 10, A 9, B 5, C 6, D 2, G 0; all $h \leq d$: admissible; every edge $h(n) \leq d(n, m) + h(m)$ both ways: consistent. [3] [checked]

EXAM-STYLE QUESTION (W12) Sampling-based [indicative 6]

(a) Uniform samples in a 10×10 m map, goal disc radius 0.8 m: p and expected samples to first hit. (b) $\mathbf{q}_{near} = (1.5, 1.5)$, $\mathbf{q}_{rand} = (6, 2)$, $\delta = 1$: $\mathbf{q}_{new}$. (c) Planner for a 6-DoF arm in a static workshop; meaning of its guarantee?

SOLUTION (a) $p = \pi(0.8)^2/100 = 0.0201$, $E[N] = 1/p = 49.7$. [2] (b) $\|\cdot\| = 4.528$; $\mathbf{q}_{new} = (1.5, 1.5) + (4.5, 0.5)/4.528 = \boxed{(2.494, 1.610)}$, then test the segment. [2] (c) PRM (static, repeated queries) or RRT (single query); grid needs 10^{12} nodes. Probabilistically complete: $P(\text{path}) \to 1$ as samples grow, no termination guarantee if none exists, path not shortest. [2] [checked]

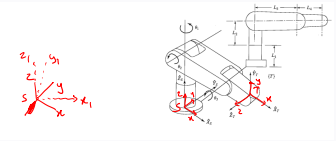
SAMPLE W13 sample questions, solved

The ten W13 questions, in order. Marks not published; indicative, ~1 per step. Method and traps: see the week.

Sample Q1 (W2, p. 11)

EXAM-STYLE QUESTION W13 Q1: FK of an RRR arm by ETS [p. 4–5 of W13]

Write ${}^S\mathbf{T}_T$ for the RRR arm; show intermediate frames. (L_1, L_2 vertical, L_3 upper arm, L_4 forearm; $L = (0.6, 0.5, 0.4, 0.3)$ m.)



[W13 lec p. 4] — Q1 arm, lecturer's frames

SOLUTION $\{S\}$ on base, z_S up. Joints 2, 3 lift the arm = *negative* about $+y$. At $\theta = \mathbf{0}$: x_T along forearm, y_T up. ETS (W13 p.4); $c_{23} = \cos(\theta_2{+}\theta_3)$:

$$
\begin{aligned}
{}^S\mathbf{T}_T &= {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2\,{}^2\mathbf{T}_3\,{}^3\mathbf{T}_T \\
&= \mathbf{R}_z(\theta_1)\,t_z(L_1{+}L_2)\,\mathbf{R}_y(-\theta_2)\,t_x(L_3)\,\mathbf{R}_y(-\theta_3)\,t_x(L_4)\,\mathbf{R}_x(90^\circ)
\end{aligned}
$$

EXAM-STYLE QUESTION Q1 (cont.): the four elementary frames

$$
{}^S\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\quad
{}^1\mathbf{T}_2 = \begin{bmatrix} c_2 & 0 & -s_2 & 0 \\ 0 & 1 & 0 & 0 \\ s_2 & 0 & c_2 & L_1{+}L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^2\mathbf{T}_3 = \begin{bmatrix} c_3 & 0 & -s_3 & L_3 \\ 0 & 1 & 0 & 0 \\ s_3 & 0 & c_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\quad
{}^3\mathbf{T}_T = \begin{bmatrix} 1 & 0 & 0 & L_4 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

EXAM-STYLE QUESTION Q1 (cont.): product, result, check

$$
{}^S\mathbf{T}_3 = {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2\,{}^2\mathbf{T}_3 = \begin{bmatrix} c_1c_{23} & -s_1 & -c_1s_{23} & c_1c_2L_3 \\ s_1c_{23} & c_1 & -s_1s_{23} & s_1c_2L_3 \\ s_{23} & 0 & c_{23} & L_1{+}L_2{+}s_2L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

(${}^S\mathbf{T}_2$ has rotation $\mathbf{R}_z\mathbf{R}_y(-\theta_2)$ at $(0, 0, L_1{+}L_2)$.) Last factor: $\mathbf{R}_x(90^\circ)$ sends column 2 to +col 3, col 3 to −col 2; $t_x(L_4)$ adds L_4·col 1. With $r = L_3c_2 + L_4c_{23}$:

$$
{}^S\mathbf{T}_T = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1r \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1r \\ s_{23} & c_{23} & 0 & L_1{+}L_2{+}L_3s_2{+}L_4s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

EXAM-STYLE QUESTION Q1 (cont.): check, numbers

Check $\theta = \mathbf{0}$: tip $(L_3{+}L_4, 0, L_1{+}L_2)$, $x_T = x_S$, $y_T = z_S$. $\theta_2 = 90^\circ, \theta_3 = 0$: arm straight up. **Numbers** $\theta = (30, 20, 40)^\circ$: $\{2\}$ at $(0, 0, 1.1)$, $\{3\}$ at $(0.3255, 0.1879, 1.2368)$,

$$
{}^S\mathbf{T}_T = \begin{bmatrix} 0.4330 & -0.7500 & 0.5000 & 0.4554 \\ 0.2500 & -0.4330 & -0.8660 & 0.2629 \\ 0.8660 & 0.5000 & 0 & 1.4966 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

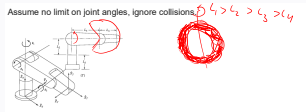
[checked]

Marks lost: $\mathbf{R}_y(+\theta_2)$; dropping $\mathbf{R}_x(90^\circ)$; no intermediate frames.

Sample Q2 (W3, p. 14)

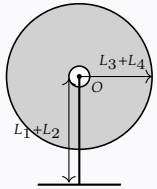
EXAM-STYLE QUESTION W13 Q2: sketch the workspace

Sketch the workspace of the Q1 arm; no joint limits, ignore collisions.



[W13 lec p. 6] — lecturer's "donut": ring in the side view

SOLUTION Shoulder $O = (0, 0, L_1{+}L_2)$. Joints 2, 3 = planar RR (links L_3, L_4) in a vertical plane through z_S; cosine rule: $\rho = \sqrt{L_3^2 + L_4^2 + 2L_3L_4\cos\theta_3} \in [\,|L_3{-}L_4|, L_3{+}L_4]$ from O; θ_2 free ⇒ full annulus; joint 1 spins the plane about $z_S \ni O$.
Result: spherical shell centred on O, radii $|L_3{-}L_4| \le \rho \le L_3{+}L_4$ ($0.1 \le \rho \le 0.7$ m); L_1, L_2 only lift the centre. $L_3 = L_4$: solid ball. Dexterous workspace empty.



side view (x_Sz_S): ring about O

Top view: disc of radius $L_3{+}L_4$. **Check:** $\theta_2 = \theta_3 = 0$: tip $(L_3{+}L_4, 0, L_1{+}L_2)$ on the outer sphere; $\theta_3 = \pi$: $\rho = |L_3{-}L_4|$ (500 random poses: $\rho \in [0.1, 0.7]$). [checked]

Sample Q3 (W3, p. 14)

EXAM-STYLE QUESTION W13 Q3: IK of the Q1 arm, given (x_T, y_T, z_T)

SOLUTION FK (Q1): $\mathbf{p}_T = (c_1r,\ s_1r,\ L_1{+}L_2{+}L_3s_2{+}L_4s_{23})$, $r = L_3c_2 + L_4c_{23}$.
Top view: $\theta_1 = \text{atan2}(y_T, x_T)$, $r = +\sqrt{x_T^2 + y_T^2}$; branch 2: $\theta_1 + \pi$, $r = -\sqrt{x_T^2 + y_T^2}$ (atan2, not $\tan^{-1}$).

EXAM-STYLE QUESTION Q3 (cont.): side view, branches, numbers

Side view: planar RR, $z' = z_T - L_1 - L_2$: $r = L_3c_2 + L_4c_{23}$, $z' = L_3s_2 + L_4s_{23}$. Square, add:

$$
\cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2L_3L_4}, \qquad \theta_3 = \text{atan2}\left(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3\right)
$$

$r = (L_3{+}L_4c_3)c_2 - (L_4s_3)s_2$, $z' = (L_4s_3)c_2 + (L_3{+}L_4c_3)s_2$:

$$
\theta_2 = \text{atan2}(z', r) - \text{atan2}(L_4\sin\theta_3,\ L_3 + L_4\cos\theta_3)
$$

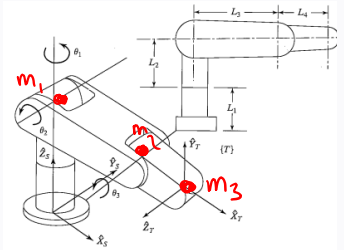
$2 \times 2 = 4$ solutions; none if $|\cos\theta_3| > 1$ (outside the Q2 shell); on the surface the elbow branches merge; $x_T = y_T = 0$: θ_1 free.
Numbers: $L = (0.6, 0.5, 0.4, 0.3)$, $\mathbf{p}_T = (0.4554, 0.2629, 1.4966)$: $\theta_1 = 30^\circ$, $r = 0.5259$, $z' = 0.3966$; $\cos\theta_3 = \frac{0.2766+0.1573-0.16-0.09}{0.24} = 0.7660$, $\theta_3 = \pm 40^\circ$; $\text{atan2}(z', r) = 37.024^\circ$, $\mu = \pm 17.024^\circ$. $(\theta_1, \theta_2, \theta_3)$ in deg:

$(30, 20, 40)$, $(30, 54.047, -40)$, $(-150, 125.953, 40)$, $(-150, 160, -40)$

Each reproduces $\mathbf{p}_T$ by FK (first = Q1 pose). $(0.5, 0.5, 1.8)$ is $0.995 > 0.7$ m from the shoulder: no solution. [checked]

Sample Q4 (W7, p. 26)

EXAM-STYLE QUESTION W13 Q4 (a): equations of motion, 3D RRR arm



[W13 lec p. 8] — W13 p. 8: red dots = m_1 shoulder, m_2 elbow, m_3 tool

Mass, inertias, COM positions given; derive the equations of motion.
SOLUTION **Set-up.** z_S up, g along $-z_S$; θ_1 about z_S; arm plane vertical; shoulder height $H = L_1{+}L_2$; θ_2 = upper-arm elevation, forearm elevation $\theta_2{+}\theta_3$ (angles from vertical: swap $c \leftrightarrow s$). Point masses first; m_1 on the axis adds only a constant to P. ρ = distance from z_S, h = height.
Positions. $\rho_2 = L_3c_2$, $h_2 = H + L_3s_2$; $\rho_3 = L_3c_2 + L_4c_{23}$, $h_3 = H + L_3s_2 + L_4s_{23}$.
Velocity. $v^2 = \dot\rho^2 + \dot h^2 + \rho^2\dot\theta_1^2$. $\dot\rho_2^2 + \dot h_2^2 = L_3^2\dot\theta_2^2$; $\dot\rho_3^2 + \dot h_3^2 = L_3^2\dot\theta_2^2 + L_4^2(\dot\theta_2{+}\dot\theta_3)^2 + 2L_3L_4c_3\dot\theta_2(\dot\theta_2{+}\dot\theta_3)$.

$$
\begin{aligned}
K &= \tfrac{1}{2}(m_2\rho_2^2 + m_3\rho_3^2)\dot\theta_1^2 + \tfrac{1}{2}\left[(m_2{+}m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3\right]\dot\theta_2^2 \\
&\quad + m_3(L_4^2 + L_3L_4c_3)\dot\theta_2\dot\theta_3 + \tfrac{1}{2}m_3L_4^2\dot\theta_3^2 \\
P &= m_2g(H{+}L_3s_2) + m_3g(H{+}L_3s_2{+}L_4s_{23}), \quad \mathcal{L} = K - P
\end{aligned}
$$

$M_{11} = m_2\rho_2^2 + m_3\rho_3^2$; $M_{22} = (m_2{+}m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3$; $M_{23} = m_3(L_4^2 + L_3L_4c_3)$; $M_{33} = m_3L_4^2$; $M_{12} = M_{13} = 0$.

SAMPLE

EXAM-STYLE QUESTION Q4 (b): Euler–Lagrange

Joint 1: $\partial\mathcal{L}/\partial\theta_1 = 0$; $\frac{d}{dt}(M_{11}\dot\theta_1) = M_{11}\ddot\theta_1 + \dot M_{11}\dot\theta_1$, $\dot M_{11} = 2m_2\rho_2\dot\rho_2 + 2m_3\rho_3\dot\rho_3$, $\dot\rho_2 = -L_3 s_2\dot\theta_2$, $\dot\rho_3 = -L_3 s_2\dot\theta_2 - L_4 s_{23}(\dot\theta_2+\dot\theta_3)$:

$$Q_1 = M_{11}\ddot\theta_1 - 2\dot\theta_1\left[m_2\rho_2 L_3 s_2\dot\theta_2 + m_3\rho_3(L_3 s_2\dot\theta_2 + L_4 s_{23}(\dot\theta_2+\dot\theta_3))\right]$$

Joint 2: $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot\theta_2} = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 - m_3L_3L_4 s_{3}\dot\theta_3(2\dot\theta_2+\dot\theta_3)$; $\frac{\partial\mathcal{L}}{\partial\theta_2} = \frac12\dot\theta_1^2\frac{\partial M_{11}}{\partial\theta_2} - g[(m_2+m_3)L_3c_2 + m_3L_4c_{23}]$, $\frac12\frac{\partial M_{11}}{\partial\theta_2} = -[m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2+L_4s_{23})]$:

$$Q_2 = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 + \dot\theta_1^2[m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2+L_4s_{23})] - m_3L_3L_4s_3(2\dot\theta_2\dot\theta_3+\dot\theta_3^2) + g[(m_2+m_3)L_3c_2 + m_3L_4c_{23}]$$

Joint 3: $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot\theta_3} = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 - m_3L_3L_4s_3\dot\theta_3\dot\theta_2$; $\frac{\partial K}{\partial\theta_3} = -m_3\rho_3L_4s_{23}\dot\theta_1^2 - m_3L_3L_4s_3(\dot\theta_2^2+\dot\theta_2\dot\theta_3)$; $\frac{\partial P}{\partial\theta_3} = m_3gL_4c_{23}$; $\dot\theta_2\dot\theta_3$ terms cancel:

$$Q_3 = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 + m_3\rho_3L_4s_{23}\dot\theta_1^2 + m_3L_3L_4s_3\dot\theta_2^2 + gm_3L_4c_{23}$$

EXAM-STYLE QUESTION Q4 (c): M, C, g, general links, numbers

$$\mathbf{M} = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33}\end{bmatrix},\ \mathbf{g} = \begin{bmatrix} 0 \\ g[(m_2+m_3)L_3c_2 + m_3L_4c_{23}] \\ gm_3L_4c_{23}\end{bmatrix}$$

$\mathbf{C} = (V_1, V_2, V_3)^\top$ = velocity terms of $Q_{1,2,3}$. Centrifugal: squares; Coriolis: products ($\dot\theta_1\dot\theta_{2,3}$ in V_1, $\dot\theta_2\dot\theta_3$ in V_2). Gravity loads joints 2, 3 only.

EXAM-STYLE QUESTION Q4 (d): general links (inertias, COM positions)

General links (COM at l_2 from shoulder, l_3 from elbow; $I_2, I_3 \perp$ link through COM, I_1 column about z_S): $L_3 \to l_2$ in link 2's terms, $L_4 \to l_3$ in link 3's (keep L_3 in ρ_3), $\rho_2 = l_2c_2$, $\rho_3 = L_3c_2 + l_3c_{23}$:

$$M_{11} = I_1 + m_2\rho_2^2 + m_3\rho_3^2 + I_2c_2^2 + I_3c_{23}^2$$
$$M_{22} = m_2l_2^2 + I_2 + m_3(L_3^2+l_3^2+2L_3l_3c_3) + I_3$$
$$M_{23} = m_3(l_3^2 + L_3l_3c_3) + I_3, \quad M_{33} = m_3l_3^2 + I_3$$
$$\mathbf{g} = \begin{bmatrix} 0 \\ g[(m_2l_2+m_3L_3)c_2 + m_3l_3c_{23}] \\ gm_3l_3c_{23}\end{bmatrix}$$

$$V_1 = -2\dot\theta_1\left[m_2\rho_2l_2s_2\dot\theta_2 + m_3\rho_3(L_3s_2\dot\theta_2 + l_3s_{23}(\dot\theta_2+\dot\theta_3)) + I_2c_2s_2\dot\theta_2 + I_3c_{23}s_{23}(\dot\theta_2+\dot\theta_3)\right]$$

$$V_2 = \dot\theta_1^2\left[m_2\rho_2l_2s_2 + m_3\rho_3(L_3s_2+l_3s_{23}) + I_2c_2s_2 + I_3c_{23}s_{23}\right] - m_3L_3l_3s_3(2\dot\theta_2\dot\theta_3+\dot\theta_3^2)$$

$$V_3 = \dot\theta_1^2\left[m_3\rho_3l_3s_{23} + I_3c_{23}s_{23}\right] + m_3L_3l_3s_3\dot\theta_2^2$$

Red-dot case: $l_2 = L_3$, $l_3 = L_4$, $I_2 = I_3 = 0$.

EXAM-STYLE QUESTION Q4 (e): numbers

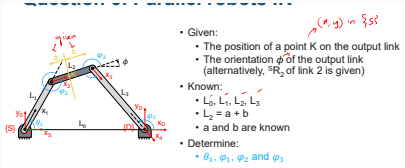
Numbers (point masses) $m_2 = 2$, $m_3 = 1$ kg, $L_3 = 0.6$, $L_4 = 0.4$ m, $\boldsymbol\theta = (20, 30, -40)°$, $\dot{\boldsymbol\theta} = (1, 0.5, -0.5)$, $\ddot{\boldsymbol\theta} = (0.2, 1, 0.5)$: $\rho_2 = 0.5196$, $\rho_3 = 0.9135$ m,

$$\mathbf{M} = \begin{bmatrix} 1.3746 & 0 & 0 \\ 0 & 1.6077 & 0.3439 \\ 0 & 0.3439 & 0.1600\end{bmatrix},\ \mathbf{C} = \begin{bmatrix} -0.5858 \\ 0.4838 \\ -0.1020\end{bmatrix},\ \mathbf{g} = \begin{bmatrix} 0 \\ 19.157 \\ 3.864\end{bmatrix}$$

$Q_1 = \boxed{-0.311}$; $Q_2 = \boxed{21.420}$; $Q_3 = \boxed{4.186}$ N m. At rest $\mathbf{Q} = \mathbf{g}$; stretched horizontal: $(0, 21.582, 3.924)$ N m. **Checks:** $\mathbf{M}$ symmetric positive definite; $\dot{\boldsymbol\theta}^\top\mathbf{Q} = 8.306$ W $= \dot K + \dot P$. [checked]

Sample Q5 (W4, p. 17)

EXAM-STYLE QUESTION W13 Q5: four-bar IK



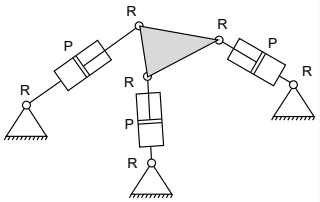
[W13 lec p. 9] — W13 p. 9: K and ϕ given; find $\theta_1, \varphi_1, \varphi_2, \varphi_3$

$L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$; coupler $a = 0.2$, $b = 0.25$ m; K $= (0.3710, 0.4678)$ m in S; $\phi = -0.600°$. Find $\theta_1, \varphi_1, \varphi_2, \varphi_3$.

SOLUTION S at left pivot, D at $(L_0, 0)$, angles ccw from $+x$. Left chain: $x = L_1c_{\theta_1} + a\cos(\theta_1+\varphi_2)$, $y = L_1s_{\theta_1} + a\sin(\theta_1+\varphi_2)$, $\theta_1 + \varphi_2 = \phi$. So $L_1\cos\theta_1 = x - a\cos\phi$, $L_1\sin\theta_1 = y - a\sin\phi$: $\theta_1 = \text{atan2}(0.4699, 0.1710) = \boxed{70.0°}$; $\varphi_2 = \phi - \theta_1 = \boxed{289.4°}$ $(-70.6°)$.
Right chain, $\varphi_1 + \varphi_3 = \phi + \pi$: $L_3\cos\varphi_1 = x - L_0 + b\cos\phi = -0.3790$, $L_3\sin\varphi_1 = y + b\sin\phi = 0.4652$: $\varphi_1 = \text{atan2}(0.4652, -0.3790) = \boxed{129.2°}$; $\varphi_3 = \phi + \pi - \varphi_1 = \boxed{50.2°}$.
Check: $\|(-0.379, 0.4652)\| = 0.600 = L_3$; $\theta_1 + \varphi_2 = 359.4° = \varphi_1 + \varphi_3 + 180°$. Each leg is solved alone with atan2. The slide's printed $\varphi_3 = \varphi_1 - \phi + \pi$ gives 309.8°: K misses by 0.4 m.

Sample Q6 (W1, p. 9)

EXAM-STYLE QUESTION W13 Q6: mobility of the planar 3-RPR



[W13 lec p. 13] — W13 pp.12–13: links 1 ground, 2 platform, 3–4, 5–6, 7–8 legs

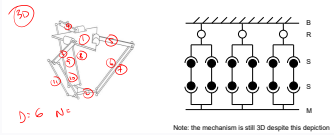
Find the mobility of the planar parallel manipulator (three R–P–R legs).
SOLUTION $M = D(N-J-1) + \sum F_i$, planar $D = 3$. $N = 8$ (ground, platform, 2 links per leg: 3,4; 5,6; 7,8). $J = 9$ (R,P,R per leg), $\sum F_i = 9$.

$$M = 3(8-9-1) + 9 = \boxed{3} \qquad (3(8-1) - 2\cdot 9 = 3)$$

Platform x, y, rotation, driven by 3 actuators. Check: forgetting ground ($N = 7$) gives 0. [checked]

Sample Q7 (W4, p. 17)

EXAM-STYLE QUESTION W13 Q7: delta mobility, plain vs extended



[W13 lec p. 14] — W13 p. 14: delta and joint graph (B, R, S, S, M)

Delta: base, 3 R joints, each followed by a parallelogram of 2 bars with S joints at both ends, moving platform. (a) Plain G–K. (b) Platform has 3 translational DoF: reconcile with extended G–K.
SOLUTION (a) $D = 6$, $N = 1 + 3 + 6 + 1 = 11$, $J = 3 + 3\cdot 4 = 15$, $\sum F_i = 3(1 + 4\cdot 3) = 39$:

$$M = 6(11 - 15 - 1) + 39 = \boxed{9}$$

(b) Each of the 6 bars spins about its own axis (S joints at both ends): $R_M = 6$; no repeated constraint, $R_C = 0$:

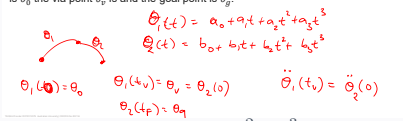
$$M = 6(11 - 15 - 1) + 39 + 0 - 6 = \boxed{3}$$

Plain G–K counts the 6 internal bar spins as motion; 3 = 3 motors = 3 platform translations.

Sample Q8 (W5, p. 20)

EXAM-STYLE QUESTION W13 Q8: symbolic via-point cubics

Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 the via point θ_v is and the goal point is θ_g.



[W13 lec p. 18] — W13 p. 18: $\theta_1 = a_0 + a_1t + a_2t^2 + a_3t^3$, $\theta_2 = b_0 + \ldots + b_3t^3$

SOLUTION **Assume:** rest at start and end, each segment on its own clock, equal durations t_f.
8 conditions: $\theta_1(0) = \theta_0$, $\dot\theta_1(0) = 0$, $\theta_1(t_f) = \theta_v$, $\theta_2(0) = \theta_v$, $\theta_2(t_f) = \theta_g$, $\dot\theta_2(t_f) = 0$, $\dot\theta_1(t_f) = \dot\theta_2(0)$, $\ddot\theta_1(t_f) = \ddot\theta_2(0)$.
1. $a_0 = \theta_0$, $a_1 = 0$, $b_0 = \theta_v$; $w = \dot\theta_2(0) = \dot\theta_1(t_f)$.
2. Segment 1 (end speeds 0, w; $\Delta = \theta_v - \theta_0$): $a_2 = \frac{3\Delta - wt_f}{t_f^2}$, $a_3 = \frac{-2\Delta + wt_f}{t_f^3}$.
Segment 2 (speeds $w, 0$; $\Delta' = \theta_g - \theta_v$): $b_1 = w$, $b_2 = \frac{3\Delta' - 2wt_f}{t_f^2}$, $b_3 = \frac{-2\Delta' + wt_f}{t_f^3}$.

SAMPLE

EXAM-STYLE QUESTION Q8 (cont.): acceleration match, results

3. $\ddot{\theta}_1(t_f) = 2a_2 + 6a_3 t_f = \frac{-6\Delta+4wt_f}{t_f^2}$, $\ddot{\theta}_2(0) = 2b_2 = \frac{6\Delta'-4wt_f}{t_f^2}$; equal:

$$\boxed{w = \frac{3(\theta_g - \theta_0)}{4t_f}}.$$

4. Substitute:

$$\boxed{\begin{aligned} a_2 &= \tfrac{12\theta_v-3\theta_g-9\theta_0}{4t_f^2}, \quad a_3 = \tfrac{-8\theta_v+3\theta_g+5\theta_0}{4t_f^3}, \\ b_1 &= \tfrac{3(\theta_g-\theta_0)}{4t_f}, \quad b_2 = \tfrac{-12\theta_v+6\theta_g+6\theta_0}{4t_f^2}, \; b_3 = \tfrac{8\theta_v-5\theta_g-3\theta_0}{4t_f^3} \end{aligned}}$$

Check $\theta = (10, 40, 20)°$, $t_f = 1$: $w = 7.5$, $\theta_1 = 10 + 82.5t^2 - 52.5t^3$, $\theta_2 = 40 + 7.5t - 75t^2 + 47.5t^3$; $\theta_1(1) = 40$, $\theta_2(1) = 20$, $\ddot{\theta}_1(1) = -150 = \ddot{\theta}_2(0)$. Variant $(0, 40, 30)°$: $w = 22.5$, $97.5t^2 - 57.5t^3$, $40 + 22.5t - 75t^2 + 42.5t^3$; peak $55.11°/\text{s}$ at 0.565 s. [checked]

Slips: one cubic for both; $a_1 \neq 0$; total time for t_f; no acceleration condition (w undetermined).

Sample Q9 (W5, p. 20)

EXAM-STYLE QUESTION W13 Q9: cubic for a prismatic joint

A prismatic joint, initially at rest, is programmed to move from an initial position d=0.02m to a resting goal position for d=0.15m in 0.8 seconds. Calculate the coefficients of a cubic spline that accomplishes this motion.

[W13 lec p. 19] — W13 p. 19

SOLUTION $q = d$ (m): $d(0) = 0.02$, $d(0.8) = 0.15$, $\dot{d}(0) = \dot{d}(0.8) = 0$; $\Delta d = 0.13$, $t_F = 0.8$ s. $a_0 = 0.02$, $a_1 = 0$, $a_2 = \frac{3(0.13)}{0.8^2} = 0.609375$, $a_3 = \frac{-2(0.13)}{0.8^3} = -0.5078125$.

$$\boxed{d = 0.02 + 0.609375t^2 - 0.5078125t^3 \text{ m}}$$

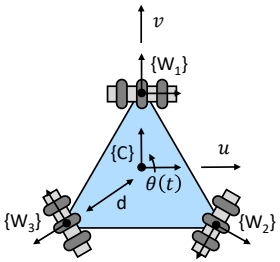
$$\dot{d} = 1.21875t - 1.5234375t^2 \text{ m/s}$$

Check: $d(0.8) = 0.02 + 0.39 - 0.26 = 0.15$; $\dot{d}(0.8) = 0.975 - 0.975 = 0$. Peak $1.5\Delta d/t_F = 0.24375$ m/s at 0.4 s ($d = 0.085$ m); end accelerations ± 1.21875 m/s^2. [checked]

Slips: a_3/t_F^2; cm/m mixed; no a_0.

Sample Q10 (W10, p. 34)

EXAM-STYLE QUESTION W13 Q10: omnidirectional 3-wheel robot



[W13 lec p. 20] — W13 p. 20

Three driven omni wheels (rollers $\perp$ drive), equilateral triangle, centre to each vertex d. Derive $(\dot{\varphi}_1, \dot{\varphi}_2, \dot{\varphi}_3)$ vs chassis $(u, v, \dot{\theta})$. Holonomic? Wheel speeds for $u = 0.2$ m/s, $d = 0.10$, $r = 0.05$ m.

SOLUTION **Setup.** $\{C\}$ at centre, $\hat{x}_C$ right, $\hat{y}_C$ up. $W_1(0, d)$, $W_2(\frac{\sqrt{3}}{2}d, -\frac{d}{2})$, $W_3(-\frac{\sqrt{3}}{2}d, -\frac{d}{2})$; drive directions (clockwise as drawn) $\beta_1 = 0°$, $\beta_2 = -120°$, $\beta_3 = 120°$; $\gamma_i = 0$.

Slide [p. 36]: $\dot{\varphi}_i = \left[\frac{1}{r} \;\; \frac{\tan\gamma_i}{r}\right] \mathbf{R}(\beta_i)^\top \begin{bmatrix} 1 & 0 & -{}^C y_{W_i} \\ 0 & 1 & {}^C x_{W_i} \end{bmatrix} \boldsymbol{\zeta}$; $\gamma_i = 0$: $r\dot{\varphi}_i = \cos\beta_i(u - \dot{\theta}\,{}^C y_{W_i}) + \sin\beta_i(v + \dot{\theta}\,{}^C x_{W_i})$.

EXAM-STYLE QUESTION Q10 (cont.): rows, matrix, inverse, numbers

Rows: $r\dot{\varphi}_1 = u - d\dot{\theta}$; $r\dot{\varphi}_2 = -\frac{1}{2}(u + \frac{d}{2}\dot{\theta}) - \frac{\sqrt{3}}{2}(v + \frac{\sqrt{3}}{2}d\dot{\theta}) = -\frac{1}{2}u - \frac{\sqrt{3}}{2}v - d\dot{\theta}$; $r\dot{\varphi}_3 = -\frac{1}{2}u + \frac{\sqrt{3}}{2}v - d\dot{\theta}$.

$$\boxed{\begin{bmatrix} \dot{\varphi}_1 \\ \dot{\varphi}_2 \\ \dot{\varphi}_3 \end{bmatrix} = \frac{1}{r}\begin{bmatrix} 1 & 0 & -d \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & -d \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & -d \end{bmatrix}\begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}}$$

$\det = \frac{3\sqrt{3}}{2}d \neq 0$; $\boldsymbol{\zeta} = \frac{r}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & -\sqrt{3} & \sqrt{3} \\ -\frac{1}{d} & -\frac{1}{d} & -\frac{1}{d} \end{bmatrix}\dot{\boldsymbol{\varphi}}$; world frame $\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$.

Holonomic: Swedish wheels give no no-slide row: $u, v, \dot{\theta}$ free, $\delta_m = 3$.

Numbers: $\dot{\varphi} = \frac{1}{0.05}(0.2, -0.1, -0.1) = \boxed{(4, -2, -2) \text{ rad/s}}$.

Check: wheel 1 carries all of u; wheels 2, 3 see $-\frac{1}{2}u$; pure spin: $-d\dot{\theta}/r$ on each; pure translation: speeds sum to 0. [checked]

SAMPLE